



ID1042 - Connect'API Issuing

Other Solution

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Africa

HPS - Casablanca Nearshore Park, Shore 1
1100, boulevard Al Qods - Sidi Maârouf
20270 Casablanca - Morocco

Tel: +212 529 045 000

Middle East & Asia

HPS - 209 / 210 Building 12, 2nd Floor,
P.O. Box 500308
Dubai Internet City, Dubai

Tel: +971 4390 0750

Europe & Americas

HPS - 65-67 rue du Faubourg St
Honoré 4th Floor
75008 Paris - France

Tel: +33 1 58 18 01 34

HPS - 805 av Guillibert
Gauthier de la Lauzière
13290 Aix en Provence - France

Tel: +33 4 42 91 42 99

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1 How to read this document

This document is intended to inform diverse readership about the specification of web service interface between PowerCARD and external systems.

The right approach to reading this document will depend upon your purpose, interests and level of knowledge about PowerCARD Web Services. We recognize that not everyone will want to read this document from end to end. Readers will usually look at the table of content to understand the structure of this document and identify the areas where more detail is required for their particular situation.

This document consists of 6 main sections, each one with different sub sections:

- Section 1 provides an overview of the web service interface and describes the intended or potential audience of this document.
- Section 2 with several sub sections motivates the purpose of every web service from a functional perspective (e.g., overview of operations performed on participating systems, data flows, and major processing steps).

The later sub section of each part of this section assumes that the reader is familiar with the broad context in which the PowerCARD web service operates. It includes the version of the concerned web service and describes later the different fields present in the request message as well as the response message. Finally, a sub section is dedicated to list all the error codes that can be generated by PowerCARD during the processing.

Please note that each service version is constituted as followed X.Y.Z, where:

- 'X' refers to the endpoint version.
- 'Y' is intended to refer to a set of web services that constitute the interface. 0 for instance.
- 'Z' is the web service version, for instance 1,2...
- Section 3 provides a general protocol description used to call web services, along with request and response general format.
- Section 4 is designed to contain the general error codes that could be part of the response message. Those codes are common to all the PowerCARD web services.
- Section 5 contains the entire field types (Base, Simple and complex) used as attributes of the data fields present in the request and response messages.
- Section 6 this section of the document is dedicated to example implementation. It provides some sample request and response messages that could help developers and testers during their quality test

2 Introduction

2.1 Overview

This document provide a general overview of all PowerCARD Web Services used in the project, it is intended to be used as guidance for the understanding at a business as well as a technical level of the entire 'PowerCARD Issuing API's'.

The document is described at a level of detail that would enable HPS customers to understand how PowerCARD is designed. For every web service, what are the controls from the PowerCARD perspectives that are put in place, business rules that describe the activity and the process workflow and finally the input/output data exchanged within the request and response messages with their properties, occurrence and field description.

2.2 Audience

This document is designed for use by HPS customers involved in the following activities:

- **Business analysis:** Analyze the information needed to effectively define your business requirements and give development and testing teams directions on how to build interfaces between PowerCARD and external systems.
- **Architectural design:** Determine what aspects of the interface are pertinent to the design. Analyze the variability of interface specification and look for data assets to reuse into similar systems.
- **Development:** Identify the appropriate data sources, and develop any processes needed to extract and deliver that data in the form the target system requires.
- **Testing:** Understand what is required by the interface. Build test harness to monitor system behavior under varying conditions and inputs/outputs.

3 Technical protocol description

3.1 For an easier reading

Each request and response message may consist of several simple or complex fields. The following table provides a comprehensive description of the attributes for each field:

Field Attributes	Description
Sequence	Identifies the field's sequence number within the interface file.
Field Name	Name of the field.
Data type	Specifies the type of the field.
Parent	Parent fields are used to determine the level of each field in the hierarchy.
Occurrence	With 2 parts: First part indicates whether the field is mandatory or optional The second part of the column, refers to the number of time the field can appear in the message
Field Description	Explains the intended use of a specific field.

3.1.1 When to version

PowerCARD APIs only need to be up-versioned when a change is made. The changes cover:

- Structure change in the request/response message such as adding/removing fields, aggregates
- Type change in the request/response message.

URI version Alias	Initially released in	Deprecated?
v2	R5.1	No

3.1.2 Deprecated version

Deprecated category contains API that have been replaced by other new API or by another version of the same API. HPS recommends to do not use deprecated API in created applications.

A deprecated API is a version that HPS is planning to remove in a future release. In case of deprecated API, HPS will send you updated version of the API specifications that mentioned the API to use.

An API is deprecated when it is superseded by a newer version, or if there is a planned obsolescence in a future release. An API in this state is still supported and becomes obsolete in a later release.

Note: You must use the active version of the API.(HPS will send you the updated version of the API specifications that mentioned which API to use)

3.1.3 Showing the message hierarchy

The following example shows you how the message content becomes nodes :

#	Field name	Data type	Parent	Occurrence	Field description
-	CardDetails	Complex	-	Optional (max: 1)	This aggregate contains the Card information
-	CardDemog	Complex	CardDetails	Optional (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
5	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits code that identifies the cardholder branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.

The parent column allows to establish the hierarchy between the aggregates. So the CardDetails element is a parent aggregate of the CardDemog aggregate. The CardDemog aggregate is also the parent of the branchCode field.

3.1.4 PowerCARD resource modeling & design

PowerCARD resource modeling

The fundamental concept in Connect'API RESTful Design is the PowerCARD Resource. A resource is a primary data representation with a set of methods that operate on it corresponding to the standard HTTP GET, POST, PUT, PATCH and DELETE methods.

For example, "Clients" is a collection resource and "Client" is a singleton resource (in a banking domain). We can identify "Clients" collection resource using the URI "/Clients". We can identify a single "Client" resource using the URI "/Clients/{ClientCode}".

A PowerCARD resource can contain sub-collection resources also. For example, sub-collection resource "Accounts" of a particular "Client" can be identified using the URI "/Clients/{ClientCode}/Accounts"

For each API, the supported methods are listed in the "Consuming a RESTful API" section.

PowerCARD Resource URI	Method	URI params
/Clients/ClientCode	GET	Optional: ClientCode=[alphanumeric]

PowerCARD Connect'API returns a response code some status code and error code, the following table lists HTTP response status codes and their meanings:

Status code	Description	Success response	Error response
200 OK	Successful	x	
401 unauthorized	The PowerCard user passed in the token is invalid. Client should refresh the token and then try again.		x

Status code	Description	Success response	Error response
404 not found	PowerCARD resource not found.		x
405 method not allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 internal server error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 service unavailable	Service unavailable.		x
415 unsupported media type	Returned when the format you requested is not supported for the PowerCARD resource.		x
504 gateway timeout	A service required to fulfill the request was not available.		x

When the request is complete, each API will provide a final response message containing the aggregate responseInfo :

```
"responseInfo": {
  "keyValues": [],
  "responseUID": "",
  "resultID": "",
  "errorCode": "",
  "errorDescription": "",
  "keyValuesLength": 0
}
```

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWithSuccess: Call was ended successfully. - ProceedWithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.

#	Field Name	Data type	Parent	Occurrence	Field Description
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

3.1.5 Consuming PowerCARD API

Working with POST HTTP

The general syntax for making a POST looks like :

url: POST { dictionary that contains the key-value pair of the information we want to send to Connect'API }

Working with HTTP Methods

Connect'API guidelines suggest to use specific HTTP method on specific type of call made to server i.e. HTTP GET, PUT, POST, DELETE or PATCH

3.2 Connect'API security

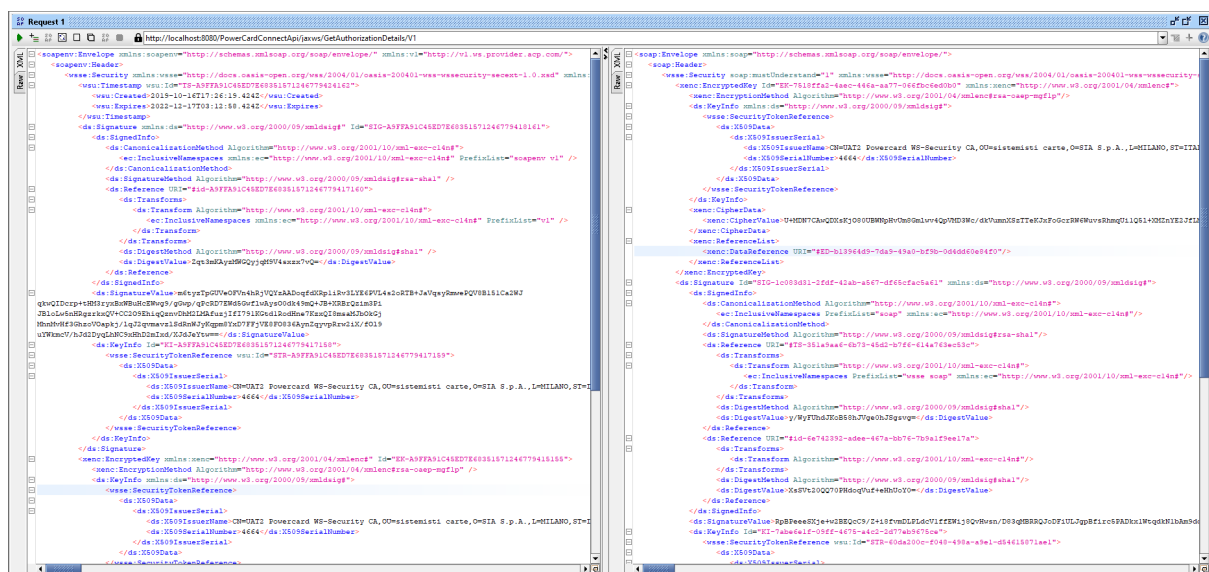
3.2.1 Protect SOAP Connect'API

WS-Security is standard that is based on securing SOAP messages through XML digital signature, confidentiality through XML encryption, and credential propagation through security tokens.

PowerCARD Connect'API describes 4 main mechanisms:

- To enforce authentication on request, token is used for service authentication.
- To maintain confidentiality, a request is encrypted at its endpoint.
- To maintain integrity, the request is verified for a valid signature.

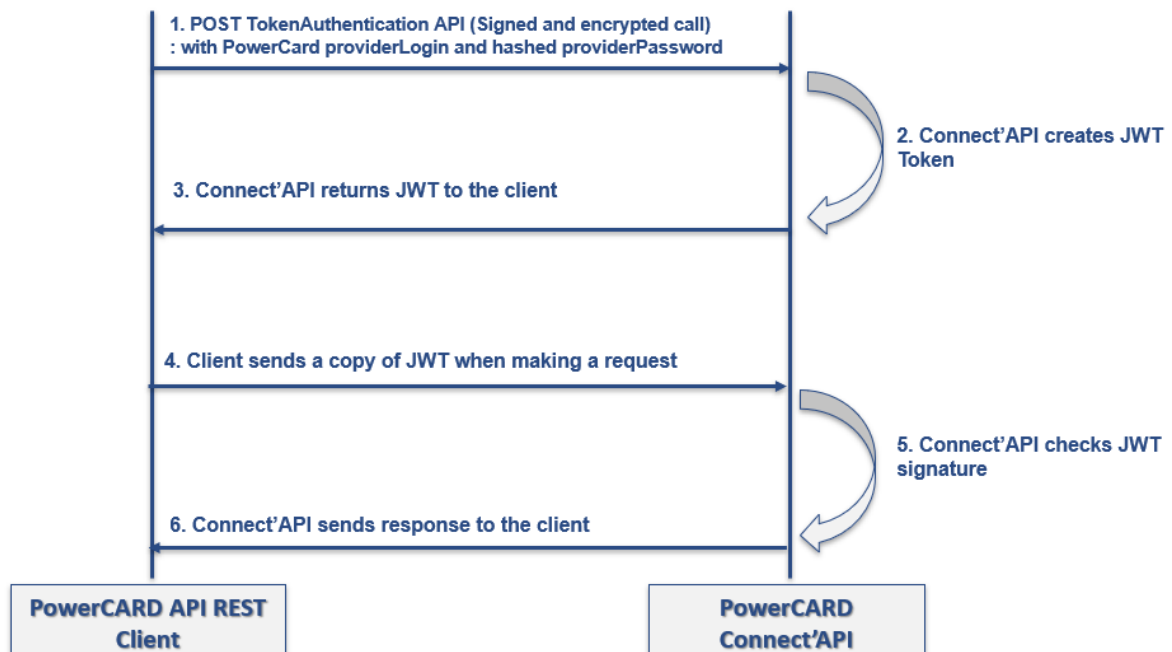
To track the time of the request, a timestamp is inserted in the request.



3.2.2 Protect RESTful Connect'API

PowerCARD Connect'API uses JWT authentication for securing the exposed REST APIs, the user will be able to consume API TokenAuthentication only if it has a valid Token.

The Connect'API authenticates all requests using the token available in the request itself, If such an authentication fails, Connect'API simply sends an HTTP code 401 (Unauthorized).

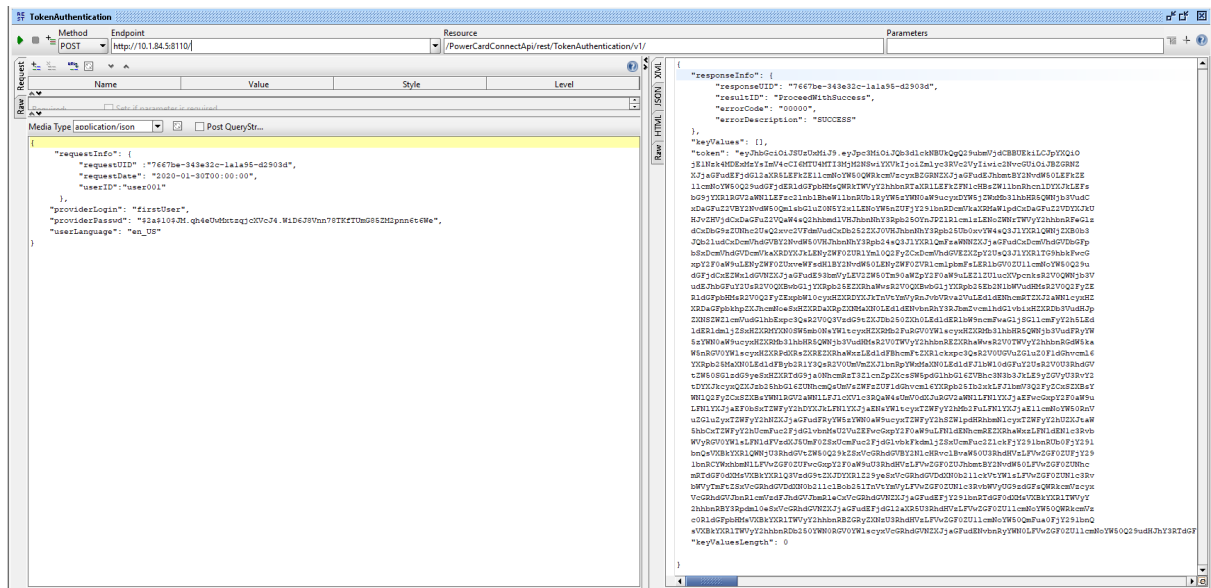


First, the client sends its PowerCARD credentials (a providerLogin and providerPassword) to an Connect'API that returns an access token. below shows the decomposition of a JWT Token into its three components:

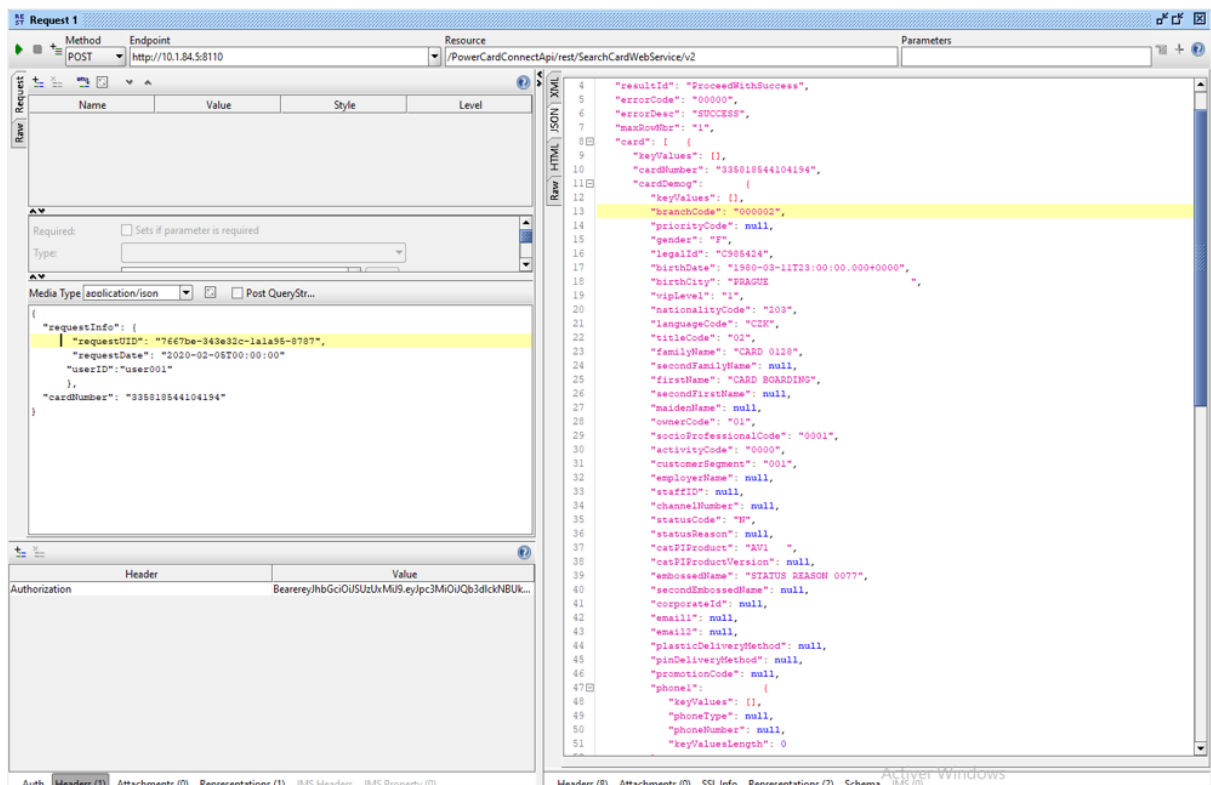
Header : contains metadata about the type of JWT token and the cryptographic algorithms used to secure its contents

Payload: contains verifiable security statements, such as the identity of the user and the PowerCARD permissions they are allowed.

Signature: used to validate that the token is trustworthy and has not been tampered with.



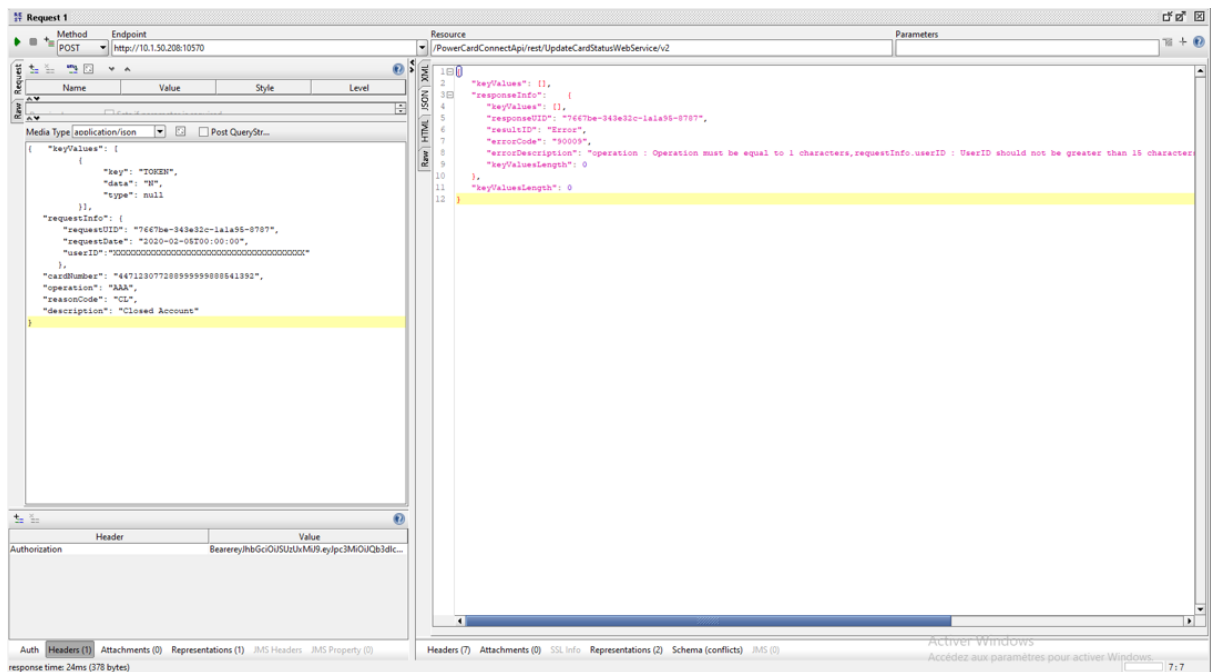
Second, the client sends a request to the API with that access token and the API verifies it and either authorizes the call or rejects it with a 401 Unauthorized response.



3.2.3 Validating resource data

Connect'API supports the Bean Validation to verify resource data. This mechanism allows to trigger client side validation providing immediate feedback to users and apply secure coding practices by performing server side validation early in the request process.

When the validation fails, Connect'API return a meaningful error code 90009 to the client in order to allow him to display a helpful error message, Connect'API return a data structure that contains an error message for each API validation that failed.

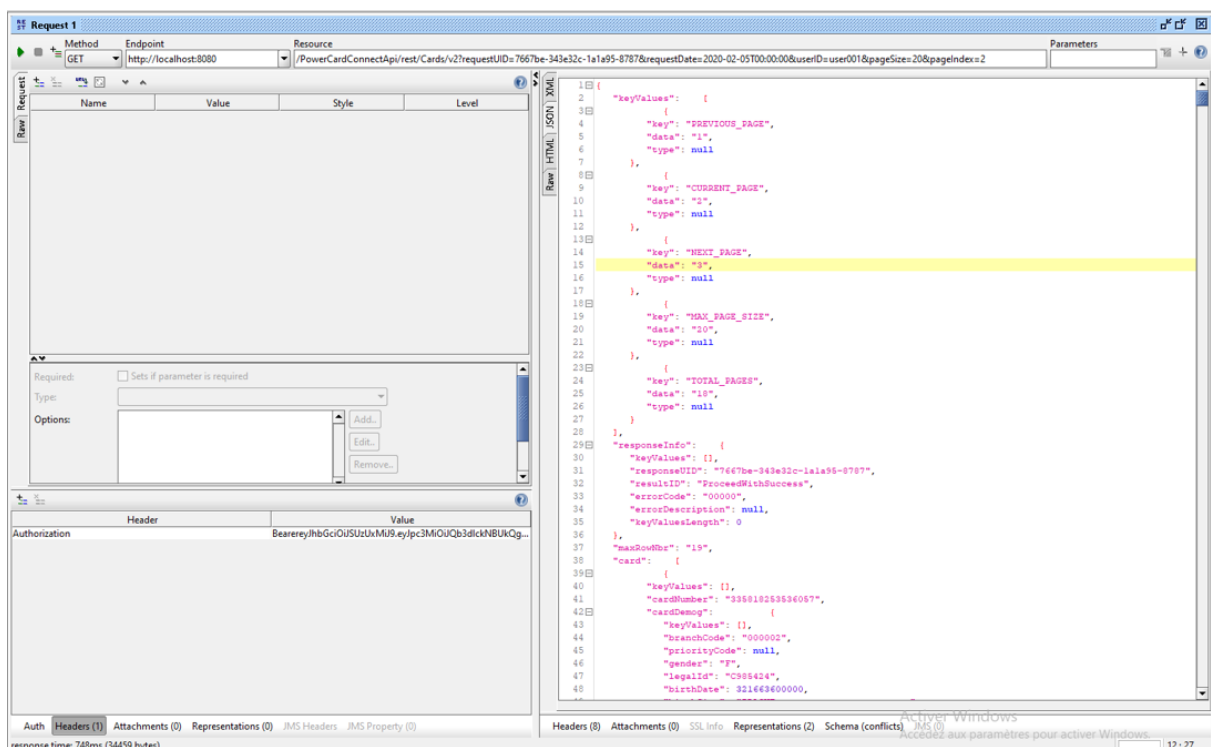


3.2.4 Paging in Connect'API

A lot of the time, when you're making calls to the search API in the Connect'API, there'll be a lot of results to return. For that reason, Connect'API can paginate the results to make sure responses are easier to handle.

Use the `pageSize` (a positive value) parameter to control how many resource representations to display per page.

Use the `pageIndex` (a positive value) to display the current page's number in the pager sequence.



3.2.5 Protect data with tokenization

For a tokenization purpose, the card number can be replaced by a random token value. For every web service request that contains the card number as input, the client software need to replace the card number by the Token value or the "Alias PAN". PowerCARD will then make the appropriate matching between the Alias PAN sent in the request and the card number as stored in PowerCARD.

The client must indicate to the PowerCARD Connect'API that the Alias PAN is being sent instead of the card number. To do that the client software is obliged to include a key TOKEN with value Y in the request message indicating to PowerCARD that the value transmitted in the CardNumber field is a Alias PAN and not a card number.

Any request containing an invalid Alias PAN not pre-configured in the system will be definitely rejected with a specific error code.

Here after a sample of request containing such key, developers shall use the same for any request where a Alias PAN is transmitted in the request message instead of the clear card number.

The screenshot displays a REST client interface with the following details:

- Method:** POST
- Endpoint:** http://10.1.84.5:8110
- Resource:** /PowerCardConnectApi/rest/SearchCardWebService/v2
- Request Body (JSON):**

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": null
    }
  ],
  "requestInfo": {
    "requestUID": "7667be343e32c-1a1a55-8787",
    "requestDate": "2020-02-05T00:00:00",
    "userId": "user001"
  },
  "cardNumber": "A6518B329CC194"
}
```
- Response Body (JSON):**

```
{
  "resultId": "ProceedWithSuccess",
  "errorCode": "00000",
  "errorDesc": "SUCCESS",
  "maxRowNb": "1",
  "card": {
    "keyValues": [],
    "cardNumber": "335018544104194",
    "cardDemo": {
      "keyValues": [],
      "branchCode": "000003",
      "priorityCode": null,
      "gender": "F",
      "legalId": "C985424",
      "birthDate": "1980-03-11T23:00:00.000+0000",
      "birthCity": "PARIS",
      "nigLevel": "1",
      "nationalityCode": "203",
      "languageCode": "CER",
      "titleCode": "03",
      "familyName": "CARD 0128",
      "secondFamilyName": null,
      "firstName": "CARD BORRONG",
      "secondFirstName": null,
      "maidenName": null,
      "ownerCode": "01",
      "socioProfessionalCode": "0001",
      "activityCode": "0000",
      "customerSegment": "001",
      "employeeName": null,
      "staffID": null,
      "channelNumber": null,
      "statusCode": "H",
      "statusReason": null,
      "catProduct": "AV1",
      "catProductVersion": null,
      "embossedName": "STATUS REASON 0077",
      "secondEmbossedName": null,
      "corporateId": null,
      "email": null,
      "email2": null,
      "plasticDeliveryMethod": null,
      "pinDeliveryMethod": null,
      "promotionCode": null,
      "phone": {
        "keyValues": [],
        "phoneType": null,
        "phoneNumber": null,
        "keyValuesLength": 0
      }
    }
  }
}
```

In the other hand, PowerCARD Connect'API will replace any card number information present in the response messages by the corresponding token value. By doing so, PowerCARD will ensure that in no case the card number will be sent in a clear format to the third party system making the request to PowerCARD and by the way will not be stored in elsewhere.

4 PowerCARD Connect'API

4.1 getCardDetails API

API overview

PowerCARD offers a selection of web services with regards to cards maintenance, the get card details web services is one of those services that can be used by both agents and customers in order to retrieve more details regarding the concerned card being sent in the request message.

Card details can include the cardholder's full name, mailing address, phone number, date of birth, the card type, card number, expiration date.

Functional description

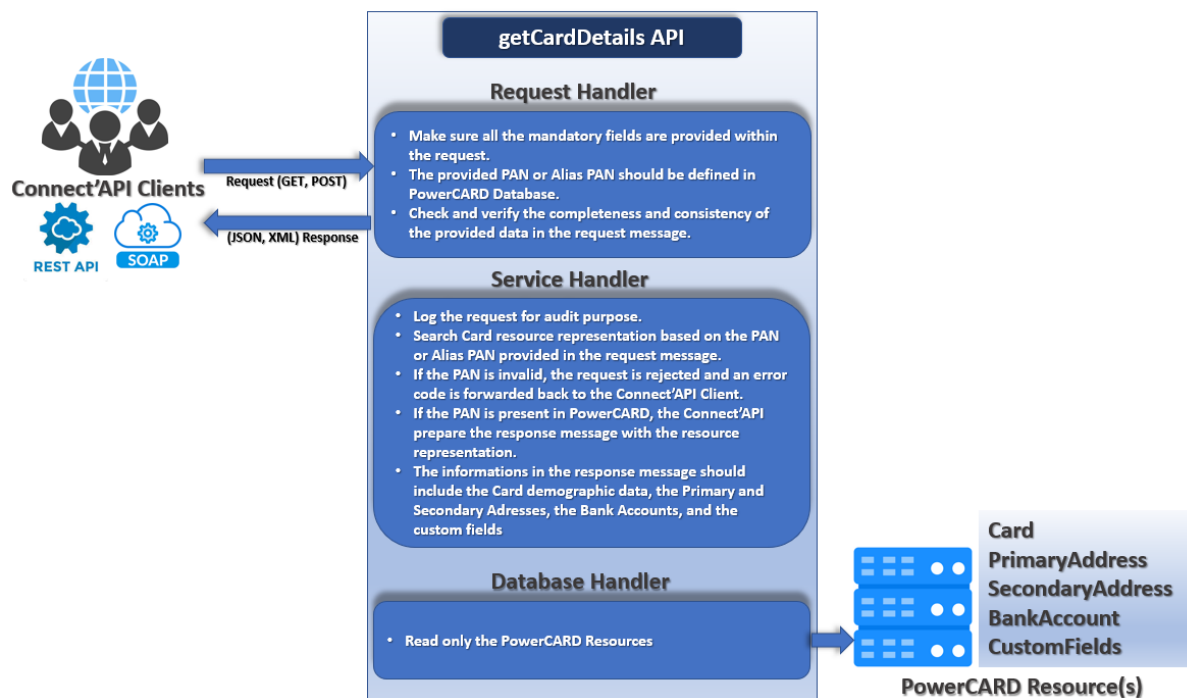


Figure 1: getCardDetails process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required(max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842-7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex		Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one

#	Field Name	Data type	Parent	Occurrence	Field Description
					sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	CardDetails	Complex	-	Optional (max: 1)	This aggregate contains the Card information
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as

#	Field Name	Data type	Parent	Occurrence	Field Description
					names, card product type, plastic type...
5	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
6	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD EMBOSSING PRIORITY table under the card parameters section.
7	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
8	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
9	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
10	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
11	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARDVIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
12	Nationality Code	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999"

#	Field Name	Data type	Parent	Occurrence	Field Description
					in case of value not available
13	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
14	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
15	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.
16	Second FamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
17	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
18	Second FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
19	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
20	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. Business/Corporate cards.
21	Socio Professional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
22	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
23	Customer Segment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
24	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
25	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.
26	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
27	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the last one is also not available.
28	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.
29	CatPIProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code to be assigned to the card.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
30	CatPIProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
31	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.
32	Second EmbossedName	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
33	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
34	ContactlessFlag	C(1)	CardDemog	Optional (max: 1)	Contactless Activation Flag indicator - Y: Active - N: Not Active
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the

#	Field Name	Data type	Parent	Occurrence	Field Description
					cardholder can be reached. There is no propagation of this aggregate from client to card.
35	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
36	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
37	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.

#	Field Name	Data type	Parent	Occurrence	Field Description
38	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
39	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
40	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate from client to card.
41	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
42	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
43	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no propagation of this field from client to card.
44	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
45	Plastic Delivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
46	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
47	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.
	PrimaryAddress	Complex	-	Optional (max: 1)	To provide the primary address of the card Required In case of a successful response.
48	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
49	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
50	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of

#	Field Name	Data type	Parent	Occurrence	Field Description
					Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
51	LinkFlag	C (1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
52	LinkEntityCode	C (2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
53	LinkAddress Type	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
54	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field is mandatory in case of address not linked.
55	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
56	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
57	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
58	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
59	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
60	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
61	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
62	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
63	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of the cardholder's address. It should be the ISO alphanumeric value (e.g. ITA for Italy, DEU for Germany)

#	Field Name	Data type	Parent	Occurrence	Field Description
					This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Secondary Address	Colmplex	-	Optional infinite list	To provide the secondary address of the cardholder.
64	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
65	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRange Date	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
66	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
67	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
68	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...

#	Field Name	Data type	Parent	Occurrence	Field Description
69	LinkFlag	C(1)	Secondary Address	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
70	LinkEntityCode	C(2)	Secondary Address	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
71	LinkAddress Type	C(2)	Secondary Address	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
72	AddressLine1	C (64)	Secondary Address	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
73	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked

#	Field Name	Data type	Parent	Occurrence	Field Description
74	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
75	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
76	ZipCode	C (10)	Secondary Address	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
77	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
78	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise,

#	Field Name	Data type	Parent	Occurrence	Field Description
					it can be provided only when the free capture of city is allowed at bank level
79	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
80	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
81	CountryCode	C (3)	Secondary Address	Required (max: 1)	The region code of the cardholder's address. Refers to the ISO country code of the cardholder's address. It should be the ISO alphanumeric value (e.g. ITA for Italy, DEU for Germany) This value must be present in the PowerCARD COUNTRY table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Not required when the address is linked.
-	AddressPurpose	Complex	-	Optional (max: 1)	To specify the purpose (card delivery, pin delivery, statement....) of the provided addresses either the primary one or the secondary ones
82	Address PurposeID	C (1)	Address Purpose	Required (max: 1)	Specifies the possible address usage values, namely: • S: Statement address • C: Card address • P: Pin address • L: Litigation address. • G: Gift address
83	AddressType	C (2)	Address Purpose	Required (max: 1)	Address type based on the intended purpose of use. This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This code must be already provided as part of the card Primary Address or Secondary Address
84	SendingMode	C (1)	Address Purpose	Required (max: 1)	To be set to « M » - Mail.
-	Auxiliary Product	Complex	-	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)

#	Field Name	Data type	Parent	Occurrence	Field Description
85	Product Identifier	C (6)	Auxiliary Product	Required (max: 1)	ID of the auxiliary product linked to the card. This code must be defined in the PowerCARD LINKED PRODUCT 1DESC table under the cardholder parameters section.
86	PANAuxProduct	C (6)	Auxiliary Product	Required (max : 1)	It is a 6 digit code identifying the PAN auxiliary product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = I
87	PANAux ProductVersion	C (10)	Auxiliary Product	Required (max : 1)	It is a 10 character fields identifying the PAN auxiliary version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = I
88	Contract Number	C (16)	Auxiliary Product	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.
89	Membership Date	Date	Auxiliary Product	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD
90	BeneficiaryAge	Long (3)	Auxiliary Product	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.

#	Field Name	Data type	Parent	Occurrence	Field Description
91	Beneficiary Name	C (40)	Auxiliary Product	Optional (max: 1)	Full name of the product's beneficiary.
92	StatusCode	C (1)	Auxiliary Product	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Canceled Products with a 'Canceled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
93	StatusDate	Date	Auxiliary Product	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
94	Insurance BranchCode	C (6)	Auxiliary Product	Optional (max: 1)	Applies to insurance products solely. Specifies the branch code of the insurance company. Must be defined in the PowerCARD BRANCH table under the general parameters section.
95	InsuranceAge	Long (4)	Auxiliary Product	Optional (max: 1)	Specifies in number of years the amount of time that has elapsed since the membership date. Only positive value will be accepted.
96	AuxiliaryFee Amount	Decimal (18,3)	Auxiliary Product	Optional (max: 1)	Fees amount charged to the cardholder as remuneration for the auxiliary product.
97	PolicyNumber	C (32)	Auxiliary Product	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents

#	Field Name	Data type	Parent	Occurrence	Field Description
					to the insurance company.
-	BankAccount	Complex	-	Optional infinite list	To provide information regarding bank accounts to which the card is linked. Mandatory for debit cards. Not required for credit cards.
98	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT _TYPE_LIST table under the general parameter section.
99	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
100	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
101	BankCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level

#	Field Name	Data type	Parent	Occurrence	Field Description
102	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
103	BranchCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
104	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
105	Account Currency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
106	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and

#	Field Name	Data type	Parent	Occurrence	Field Description
					then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
107	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
-	CardCustom Fields	Complex	-	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
108	FieldName	C (30)	CardCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
109	FieldValue	C (256)	CardCustom Fields	Required (max: 1)	The value of the user defined field.

Specific error codes

#	Error code	Error Description
1	21301	Invalid Card Number/Token Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCardDetailsWebService/V2	POST	
/Cards/V2/:aliasPAN	GET	Optional: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": ""
}
```

4.2 createDebitCard API

Overview

A debit card allows the cardholder access to a source of funding that is linked to his/her card, typically a saving or a checking account. Users may use this service to send a request to PowerCARD in order to ask for the creation of a client and its debit card.

Many details with regards to the card are shared within the request. Those data includes information related to the card. All are then checked by the system before it will proceed by the creation of the debit card.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

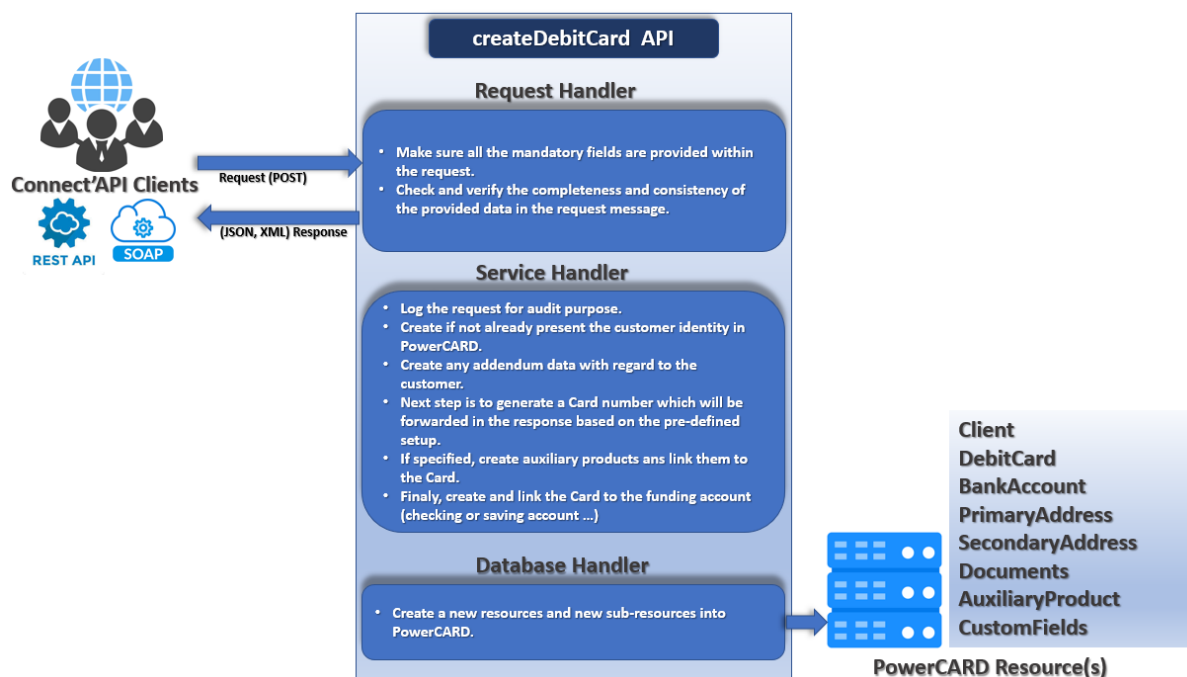


Figure 2: createDebitCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required(max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXX Example: 6948DF80 -14BD -4E04-8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
-	CustomerDetails	Complex	-	Required (max: 1)	It must contain the details information of the cardholder.
-	ExistingClient	Complex	CustomerDetails	Optional (max: 1)	To provide the existing client code to which all the entities within the boarding request should be linked. When the sub block "ExistingClient" is not provided, the sub blocks ClientDemog, PrimaryAddress and Documents will become mandatory
4	BankCode	C (6)	ExistingClient	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in

#	Field Name	Data type	Parent	Occurrence	Field Description
					the PowerCARD BANK table.
5	ClientCode	C (24)	ExistingClient	Required (max: 1)	The Client Code of the existing client
-	ClientDemog	Complex	CustomerDetails	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if ExistingClient block is not provided
6	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
7	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client. Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.
8	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs. This value will be propagated at card level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
9	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
10	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support. This field doesn't apply to corporate cardholders. Only positive value will be accepted.
11	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable. Defaulted to 'N' when not provided.
12	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
13	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by

#	Field Name	Data type	Parent	Occurrence	Field Description
					the card applicant at application time
14	TaxRegistryOffice	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
15	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
16	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
17	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
18	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP LIST table under the general parameter section.
19	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value

#	Field Name	Data type	Parent	Occurrence	Field Description
					"N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
20	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
21	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
22	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
23	SecondFamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
24	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
25	SecondFirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if compound names are in use.
26	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name

#	Field Name	Data type	Parent	Occurrence	Field Description
27	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section
28	SocioProfessionalCode	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
29	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
30	CustomerSegment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
31	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
32	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
33	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
34	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner. • D: Dependent.
35	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
36	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
37	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
38	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
39	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					[no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
40	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
41	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
42	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					2: Home phone. 3: Personal mobile phone. 4: Others.
43	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
44	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
45	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address. There is no propagation of this field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder Becomes mandatory when sub block ExistingClient is not provided.
46	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
47	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.

#	Field Name	Data type	Parent	Occurrence	Field Description
48	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
49	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
50	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
51	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
52	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.
53	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
54	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
55	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's

#	Field Name	Data type	Parent	Occurrence	Field Description
					city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
56	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
57	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
58	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					#2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	SecondaryAddress	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
59	AddressType	C (2)	SecondaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
60	SendingMode	C (1)	SecondaryAddress	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	SecondaryAddress	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
61	StartDt	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity Start date
62	EndDt	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity End date
63	CareOf	C (64)	SecondaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
64	AddressLine1	C (64)	SecondaryAddress	Required	Address Line 1

#	Field Name	Data type	Parent	Occurrence	Field Description
				(max: 1)	1st line of the recipient address.
65	AddressLine2	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
66	AddressLine3	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
67	AddressLine4	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.
68	ZipCode	C (10)	SecondaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
69	CityCode	C (5)	SecondaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
70	CityFreeText	C (32)	SecondaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
71	RegionCode	C (3)	SecondaryAddress	Optional (max: 1)	The region code of the cardholder's address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
72	RegionFreeText	C(30)	SecondaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
73	CountryCode	C (3)	SecondaryAddress	Required (max: 1)	Country code. Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport

#	Field Name	Data type	Parent	Occurrence	Field Description
					details, civil ID details ... Becomes mandatory when sub block ExistingClient is not provided.
74	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
75	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
76	DocumentIssueDate	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
77	DocumentExpiryDate	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD
78	DocumentIssueCity	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
79	Document IssueCountry	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	ClientCustomFields	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
80	FieldName	C (30)	ClientCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank

#	Field Name	Data type	Parent	Occurrence	Field Description
					level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
81	FieldValue	C (256)	ClientCustomFields	Required (max: 1)	The value of the user defined field.
-	CardDetails	Complex	-	Optional (max: 1)	It must contain the details information of card and bank accounts data of the requested card. Not mandatory in case of One Shot card
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
82	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
83	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD EMBOSsing PRIORITY table under the card parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
84	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
85	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
86	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the Client if not provided at card level.
87	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
88	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
89	NationalityCode	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
90	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
91	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or

#	Field Name	Data type	Parent	Occurrence	Field Description
					academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
92	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.
93	SecondFamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
94	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the Client if not provided at card level.
95	SecondFirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
96	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
97	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. Business/Corporate cards.
98	SocioProfessionalCode	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
99	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
100	CustomerSegment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER

#	Field Name	Data type	Parent	Occurrence	Field Description
					SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
101	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
102	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.
103	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
104	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the

#	Field Name	Data type	Parent	Occurrence	Field Description
					last one is also not available.
105	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.
106	CatPIProduct	C (6)	CardDemog	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
107	CatPIProductVersion	C (10)	CardDemog	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
108	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name

#	Field Name	Data type	Parent	Occurrence	Field Description
					fields as a default value.
109	Second EmbossedName	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
110	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
111	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
112	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
113	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
114	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
115	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					3: Personal mobile phone. 4: Others.
116	PhoneNumber	C(30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
117	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
118	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
119	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no

#	Field Name	Data type	Parent	Occurrence	Field Description
					propagation of this field from client to card.
120	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
121	Plastic DeliveryMethod	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
122	PinDeliveryMethod	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
123	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	PrimaryAddress	Complex	CardDetails	Required (max: 1)	To provide the primary address of the card
124	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
125	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
126	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
127	LinkFlag	C(1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
128	LinkEntityCode	C(2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.

#	Field Name	Data type	Parent	Occurrence	Field Description
129	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
130	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
131	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
132	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
133	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
134	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
135	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
136	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
137	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
138	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
139	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This 3 digit code must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	SecondaryAddress	Colmplex	CardDetails	Optional infinite list	To provide the secondary address of the cardholder.
140	AddressType	C (2)	SecondaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
141	SendingMode	C (1)	SecondaryAddress	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	SecondaryAddress	Optional (max: 1)	An address can be temporary and

#	Field Name	Data type	Parent	Occurrence	Field Description
					restricted to a range date Valid only in case of Address Type "Temporary"
142	StartDt	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity Start date
143	EndDt	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity End date
144	CareOf	C (64)	SecondaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
145	LinkFlag	C(1)	SecondaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: 'Y': Yes 'N': No This field should be used for card addresses only.
146	LinkEntityCode	C(2)	SecondaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: 'CL': Client 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
147	LinkAddressType	C(2)	SecondaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
148	AddressLine1	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
149	AddressLine2	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
150	AddressLine3	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
151	AddressLine4	C (64)	SecondaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
152	ZipCode	C (10)	SecondaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
153	CityCode	C (5)	SecondaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
154	CityFreeText	C (32)	SecondaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
155	RegionCode	C (3)	SecondaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
156	RegionFreeText	C(30)	SecondaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					It can be provided only when the free capture of region is allowed at bank level
157	CountryCode	C (3)	SecondaryAddress	Required (max: 1)	The region code of the cardholder's address. Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	AddressPurpose	Complex	CardDetails	Optional (max: 1)	To specify the purpose (card delivery, pin delivery, statement....) of the provided addresses either the primary one or the secondary ones
158	AddressPurposeID	C (1)	AddressPurpose	Required (max: 1)	Specifies the possible address usage values, namely: • S: Statement address • C: Card address • P: Pin address • L: Litigation address. • G: Gift address

#	Field Name	Data type	Parent	Occurrence	Field Description
159	AddressType	C (2)	AddressPurpose	Required (max: 1)	Address type based on the intended purpose of use. This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This code must be already provided as part of the card PrimaryAddress or SecondaryAddress
160	SendingMode	C (1)	AddressPurpose	Required (max: 1)	To be set to « M » - Mail.
-	AuxiliaryProduct	Complex	CardDetails	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)
161	ProductIdentifier	C (6)	AuxiliaryProduct	Required ((max: 1)	ID of the auxiliary product linked to the card. This code must be defined in the PowerCARD LINKED PRODUCT 1DESC table under the cardholder parameters section.
162	PANAuxProduct	C (6)	AuxiliaryProduct	Required (max : 1)	It is a 6 digit code identifying the PAN auxiliary product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = I
163	PANAuxProductVersion	C (10)	AuxiliaryProduct	Required (max : 1)	It is a 10 character fields identifying the PAN auxiliary version.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = I
164	ContractNumber	C (16)	AuxiliaryProduct	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.
165	MembershipDate	Date	AuxiliaryProduct	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD
166	BeneficiaryAge	Long (3)	AuxiliaryProduct	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.
167	BeneficiaryName	C (40)	AuxiliaryProduct	Optional (max: 1)	Full name of the product's beneficiary.
168	StatusCode	C (1)	AuxiliaryProduct	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Cancelled Products with a 'Cancelled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
169	StatusDate	Date	AuxiliaryProduct	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
170	InsuranceBranchCode	C (6)	AuxiliaryProduct	Optional	Applies to insurance products solely.

#	Field Name	Data type	Parent	Occurrence	Field Description
				(max: 1)	Specifies the branch code of the insurance company. Must be defined in the PowerCARD BRANCH table under the general parameters section.
171	InsuranceAge	Long (4)	AuxiliaryProduct	Optional (max: 1)	Specifies in number of years the amount of time that has elapsed since the membership date. Only positive value will be accepted.
172	AuxiliaryFeeAmount	Decimal (18,3)	AuxiliaryProduct	Optional (max: 1)	Fees amount charged to the cardholder as remuneration for the auxiliary product.
173	PolicyNumber	C (32)	AuxiliaryProduct	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents to the insurance company.
-	BankAccount	Complex	CardDetails	Required infinite list	To provide information regarding bank accounts to which the card is linked.
174	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT_TYPE_LIST table under the general parameter section.
175	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of

#	Field Name	Data type	Parent	Occurrence	Field Description
					transaction posting rules.
176	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
177	BankCodeFreertext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
178	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
179	BranchCodeFreertext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
180	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
181	AccountCurrency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It

#	Field Name	Data type	Parent	Occurrence	Field Description
					should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
182	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
183	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
184	DefaultFlag	C(1)	BankAccount	Required (max:1)	Flag to indicate the default bank account: • Y : Default bank account • N : Not the default bank account

#	Field Name	Data type	Parent	Occurrence	Field Description
-	CardCustomFields	Complex	CardDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
185	FieldName	C (30)	CardCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
186	FieldValue	C (256)	CardCustomFields	Required (max: 1)	The value of the user defined field.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With Success MC: Call made successfully,

#	Field Name	Data type	Parent	Occurrence	Field Description
					however the request will be inserted in a Maker Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CardNumber	C (22)	-	Optional (max: 1)	Represents the Card Number or the Card Token Id generated by PowerCARD in case of successful processing. The Card Token ID is returned if the complex KeyValue with key "TOKEN" is included in the WS call.

Specific error codes

#	Error code	Error Description
1	29604	Invalid Client Code
2	29605	Error while generating the Client Code
3	29606	Invalid Bank Code in Customer Details segment
4	29607	Invalid Branch Code in Customer Details Segment

#	Error code	Error Description
5	29608	Invalid Family Status
6	29610	Invalid Cardholder's Gender
7	29614	Invalid Cardholder's Birth Date
8	29617	Invalid VIP Level
9	29618	Invalid Nationality Code
10	29619	Invalid Language Code
11	29620	Invalid Title Code
12	29626	Invalid Owner Code
13	29627	Invalid Socio-Professional Code
14	29628	Invalid Activity Code
15	29629	Invalid Customer Segment
16	29633	Invalid Lodging Mode
17	29634	Invalid Channel Number
18	29635	Invalid Phone Type 1
19	29637	Invalid Phone Type 2
20	29639	Invalid Phone Type 3
21	29641	Invalid Phone Type 4
22	29645	Invalid Primary Address Type
23	29646	Invalid Primary Address Sending Mode
24	29655	Invalid Client Primary Address City Code
25	29657	Invalid Client Primary Address Region Code
26	29659	Invalid Client Primary Address Country Code
27	29667	City Code or City Free Text is required for Card Primary Address
28	29668	Region Code or Region Free Text is required for Card Primary Address
29	29675	Invalid Client Secondary Address Type
30	29676	Invalid Client Secondary Address Sending Mode
31	29685	Invalid Client Secondary Address City Code
32	29687	Invalid Client Secondary Address Region Code
33	29689	Invalid Client Secondary Address Country Code

#	Error code	Error Description
34	29690	Validity Start Date must not be in the past in case Client Secondary Address
35	29691	Validity Start Date cannot be greater than Validity End Date in case Client Secondary Address
36	29692	Validity End Date must not be in the past in case Client Secondary Address
37	29695	City Code or City Free Text is required for Card Secondary Address
38	29696	Region Code or Region Free Text is required for Card Secondary Address
39	29703	Invalid Client Document Code
40	29711	Invalid Card Branch Code
41	29712	Invalid Card Priority Code
42	29713	Invalid Card Gender
43	29715	Invalid Card Birth Date
44	29717	Invalid Card VIP Level
45	29718	Invalid Card Nationality Code
46	29719	Invalid Card Language Code
47	29720	Invalid Card Title Code
48	29726	Invalid Card Owner Code
49	29727	Invalid Card Socio Professional Code
50	29728	Invalid Card Activity Code
51	29729	Invalid Card Customer Segment
52	29730	Invalid Card Product Code
53	29731	Invalid Plastic Type
54	29733	Invalid Card Status Code
55	29734	Invalid Card Status Reason
56	29736	Invalid Card Card Fees Code
57	29737	Invalid Card Velocity Limit Index
58	29741	Invalid Card Corporate Id
59	29742	Invalid Card Phone Type 1
60	29744	Invalid Card Phone Type 2
61	29746	Invalid Card Phone Type 3
62	29748	Invalid Card Phone Type 4
63	29752	Invalid Card Plastic Delivery Method
64	29753	Invalid Card Pin Delivery Method

#	Error code	Error Description
65	29754	Invalid Card Promotion Code
66	29759	Invalid Card Primary Address Type
67	29760	Invalid Card Primary Address Sending Mode
68	29761	Invalid Card Primary Address Purpose ID
69	29769	Invalid Card Primary Address City Code
70	29771	Invalid Card Primary Address Region Code
71	29773	Invalid Card Primary Address Country Code
72	29781	City Code or City Free Text is required for Card Secondary Address
73	29782	Region Code or Region Free Text is required for Card Secondary Address
74	29789	Invalid Card Secondary Address Type
75	29790	Invalid Card Secondary Address Sending Mode
76	29791	Invalid Card Secondary Address Purpose ID
77	29799	Invalid Card Secondary Address City Code
78	29801	Invalid Card Secondary Address Region Code
79	29803	Invalid Card Secondary Address Country Code
80	29804	Validity Start Date must not be in the past in case Card Secondary Address
81	29805	Validity Start Date cannot be greater than Validity End Date in case Card Secondary Address
82	29806	Validity End Date must not be in the past in case Card Secondary Address
83	29809	City Code or City Free Text is required for Card Secondary Address
84	29810	Region Code or Region Free Text is required for Card Secondary Address
85	29815	Region Free Text is required in case of Region not valued for Client Secondary Address
86	29825	Invalid AuxiliaryProduct Identifier
87	29832	Invalid Insurance Branch Code

#	Error code	Error Description
88	29836	Invalid Account Type
89	29838	Invalid Bank Code in Bank Account Segment
90	29840	Invalid Branch Code in Bank Account Segment
91	29844	BranchCode is missing when AccountNumber is a valued field
92	29845	AccountNumber is missing when BranchCode is a valued field
93	29846	IBAN is missing when SwiftCode is a valued field
94	29848	Invalid Account Number
95	29849	Invalid Account Currency
96	29852	Invalid Card Field Name
97	29853	Invalid Card Field Value
98	29854	Duplicate AddressType, more Address with the same AddressType
99	29855	Duplicate AddressPurpose, more AddressPurposeID with the same purpose
100	29856	An error occurred while inserting or updating a Bank Account
101	29858	Client Demog is Mandatory when Existing Client is not provided
102	29859	Primary Address is mandatory when Existing Client is not provided
103	29860	Invalid Link Flag
104	29861	Invalid Link Entiy Code
105	29862	Invalid Link Address Type
106	29863	Invalid Purpose Address Type
107	29864	Invalid Purpose Address Type Sending Mode
108	29865	Document Expiry Date must not be in the past
109	29866	Document Expiry Date cannot be lesser than document issue date
110	29867	Document Issue Date can not be in the future
111	29868	Invalid Notification Type
112	29869	Invalid Inclusion Indicator

RESTful response codes

Status Code	Description	Success Response	Error Response
201 Created	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/CreateDebitCardWebService/V2	POST	
/DebitCards/V2	POST	

REST Sample Call

```
{
  "keyValues": [{
    "key": "",
    "data": "",
    "type": ""
  }],
  "requestInfo": {
    "requestUID": "",
    "userID": "",
    "requestDate": ""
  },
  "customerDetails": [{
    "existingClient": {
      "bankCode": "",
      "clientCode": ""
    }
  },
  "clientDemog": {
```

```

"bankCode": "",
"clientCode": "",
"branchCode": "",
"familyStatus": "",
"numberOfDependents": "",
"gender": "",
"legalId": "",
"taxNumber": "",
"taxRegistryOffice": "",
"birthDate": "",
"birthCountry": "",
"birthCity": "",
"vipLevel": "",
"nationalityCode": "",
"languageCode": "",
"titleCode": "",
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"secondFamilyName": "",
"firstName": "",
"secondFirstName": "",
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"customerSegment": "",
"takingOnDate": "",
"employerName": "",
"staffID": "",
"lodgingMode": "",
"channelNumber": "",
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  "phoneType": "",
  "phoneNumber": ""
},
"phone2": {
  "phoneType": "",
  "phoneNumber": ""
},
"phone3": {
  "phoneType": "",
  "phoneNumber": ""
},
"phone4": {
  "phoneType": "",
  "phoneNumber": ""
},
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"email2": ""
},
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  "sendingMode": "",
  "careOf": "",
  "addressLine1": "",
  "addressLine2": "",
  "addressLine3": "",
  "addressLine4": "",
  "zipCode": "",
  "cityCode": "",
  "cityFreeText": "",
  "regionCode": "",
  "regionFreeText": "",
  "countryCode": ""
},
"secondaryAddress": [{
  "addressType": "",
  "sendingMode": "",
  "validityRangeDate": {
    "startDt": "",
    "endDt": ""
  },
  "careOf": "",
  "linkFlag": "",
  "linkEntityCode": "",
  "linkAddressType": "",
  "addressLine1": "",

```



```

        "addressLine2": "",
        "addressLine3": "",
        "addressLine4": "",
        "zipCode": "",
        "cityCode": "",
        "cityFreeText": "",
        "regionCode": "",
        "regionFreeText": "",
        "countryCode": ""
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        "documentIssueDate": "",
        "documentExpiryDate": "",
        "documentIssueCity": "",
        "documentIssueCountry": ""
    }],
    "clientCustomFields": [{
        "fieldName": "",
        "fieldValue": ""
    }]
},
"cardDetails": {
    "cardDemog": {
        "branchCode": "",
        "priorityCode": "",
        "gender": "",
        "legalId": "",
        "birthDate": "",
        "birthCity": "",
        "vipLevel": "",
        "nationalityCode": "",
        "languageCode": "",
        "titleCode": "",
        "familyName": "",
        "secondFamilyName": "",
        "firstName": "",
        "secondFirstName": "",
        "maidenName": "",
        "ownerCode": "",
        "socioProfessionalCode": "",
        "activityCode": "",
        "customerSegment": "",
        "employerName": "",
        "staffID": "",
        "channelNumber": "",
        "statusCode": "",
        "statusReason": "",
        "catPIProduct": "",
        "catPIProductVersion": "",
        "embossedName": "",
        "secondEmbossedName": "",
        "corporateId": "",
        "phone1": {
            "phoneType": "",
            "phoneNumber": ""
        },
        "phone2": {
            "phoneType": "",
            "phoneNumber": ""
        },
        "phone3": {
            "phoneType": "",
            "phoneNumber": ""
        },
        "phone4": {
            "phoneType": "",
            "phoneNumber": ""
        },
        "email1": "",
        "email2": "",
        "plasticDeliveryMethod": "",
        "pinDeliveryMethod": "",
        "promotionCode": ""
    },

```

```

"primaryAddress": {
  "addressType": "",
  "sendingMode": "",
  "careOf": "",
  "linkFlag": "",
  "linkEntityCode": "",
  "linkAddressType": "",
  "addressLine1": "",
  "addressLine2": "",
  "addressLine3": "",
  "addressLine4": "",
  "zipCode": "",
  "cityCode": "",
  "cityFreeText": "",
  "regionCode": "",
  "regionFreeText": "",
  "countryCode": ""
},
"secondaryAddress": {
  "addressType": "",
  "sendingMode": "",
  "validityRangeDate": {
    "startDt": "",
    "endDt": ""
  },
  "careOf": "",
  "linkFlag": "",
  "linkEntityCode": "",
  "linkAddressType": "",
  "addressLine1": "",
  "addressLine2": "",
  "addressLine3": "",
  "addressLine4": "",
  "zipCode": "",
  "cityCode": "",
  "cityFreeText": "",
  "regionCode": "",
  "regionFreeText": "",
  "countryCode": ""
},
"addressPurpose": {
  "addressPurposeID": "",
  "addressType": "",
  "sendingMode": ""
},
"auxiliaryProduct": [{
  "productIdentifier": "",
  "pANAAuxProduct": "",
  "pANAAuxProductVersion": "",
  "contractNumber": "",
  "membershipDate": "",
  "beneficiaryAge": "",
  "beneficiaryName": "",
  "statusCode": "",
  "statusDate": "",
  "insuranceBranchCode": "",
  "insuranceAge": "",
  "auxiliaryFeeAmount": "",
  "policyNumber": ""
}],
"bankAccount": [{
  "acctType": "",
  "acctSequence": "",
  "bankCode": "",
  "bankCodeFreetext": "",
  "branchCode": "",
  "branchCodeFreetext": "",
  "accountNumber": "",
  "accountCurrency": "",
  "iban": "",
  "swiftCode": "",
  "defaultFlag": ""
}],
"cardCustomFields": [{
  "fieldName": "",
  "fieldValue": ""
}]

```

```

    }
  }
}

```

4.3 setCardDetails API

API overview

The setCardDetails web service can be used by both agents and customers in order to update details regarding the concerned card being sent in the request message.

The setCardDetails web service includes the cardholder's full name, embossed name, phones, etc.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

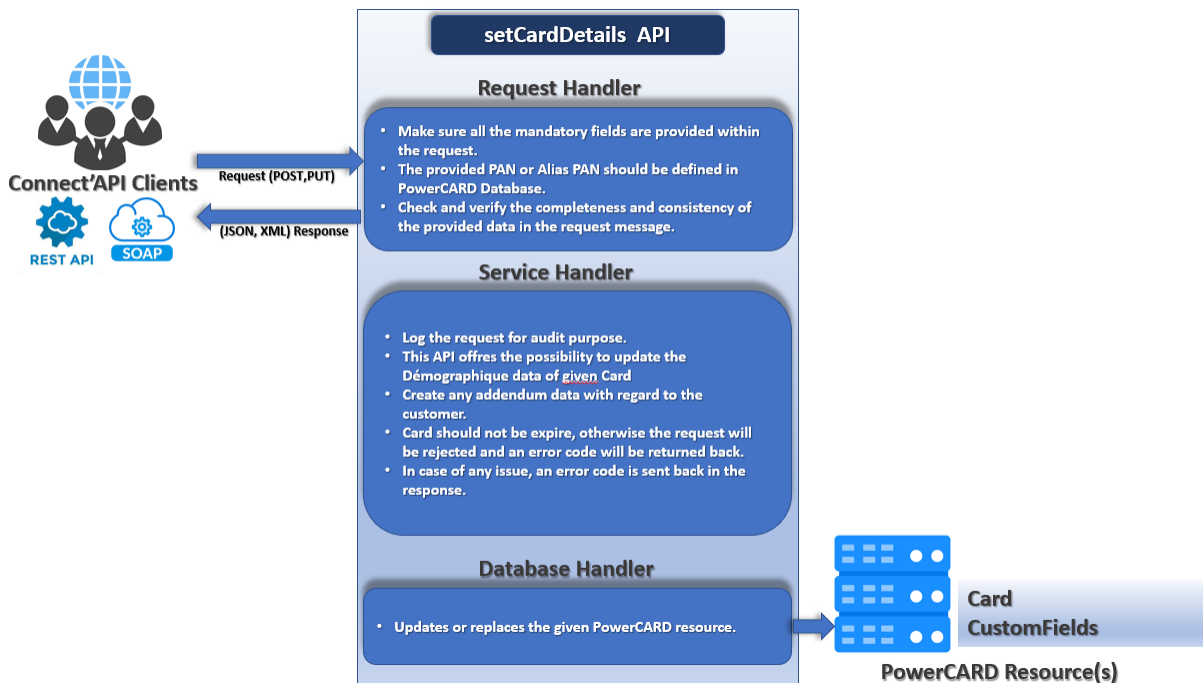


Figure 3: setCardDetails process

Technical specification

Version 1

URI version Alias	Initially released in	Deprecated ?
v1	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required(max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
-	CardDemog	Complex	-	Optional (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
5	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
6	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					EMBOSSING PRIORITY table under the card parameters section.
7	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
8	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
9	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the Client if not provided at card level.
10	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
11	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
12	Nationality Code	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's

#	Field Name	Data type	Parent	Occurrence	Field Description
					citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two scenario might occurred, based on how Client data were set: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
13	Language Code	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
14	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
15	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last

#	Field Name	Data type	Parent	Occurrence	Field Description
					name. It will be inherited from the Client if not provided at card level.
16	Second FamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
17	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the Client if not provided at card level.
18	Second FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
19	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
20	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. SIA configuration: - IN "INDIVIDUAL" from retail cards - BU "BUSINESS" for Business/ Corporate cards.
21	Socio Professional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based

#	Field Name	Data type	Parent	Occurrence	Field Description
					on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
22	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
23	Customer Segment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
24	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
25	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
26	Channel Number	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
27	Embossed Name	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.
28	Second Embossed Name	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
29	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
30	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format:

#	Field Name	Data type	Parent	Occurrence	Field Description
					00 xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
31	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
32	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00 xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
33	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
34	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00 xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
35	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
36	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00 xxxxxxxxxx (xx based on the country, e.g 39 – Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					[no format check is foreseen in WS processing]
37	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no propagation of this field from client to card.
38	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
39	Plastic Delivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
40	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
41	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	CardCustom Fields	Complex	-	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
42	FieldName	C (30)	CardCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
43	FieldValue	C (256)	CardCustom Fields	Required (max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call

#	Field Name	Data type	Parent	Occurrence	Field Description
					was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	22001	Invalid Card Number/Token Id
2	22002	Invalid Card Branch Code
3	22003	Invalid Card Priority Code
4	22004	Invalid Card Gender
5	22005	Invalid Card Birth Date
6	22006	Invalid Card VIP Level
7	22007	Invalid Card Nationality Code
8	22008	Invalid Card Language Code
9	22009	Invalid Card Title Code
10	22010	Invalid Card Owner Code

#	Error code	Error Description
11	22011	Invalid Card Socio Professional Code
12	22012	Invalid Card Activity Code
13	22013	Invalid Card Customer Segment
14	22014	Invalid Channel Number
15	22015	Invalid Card Status Code
16	22016	Invalid Card Status Reason
17	22017	Invalid Card Phone Type 1
18	22018	Invalid Card Phone Type 2
19	22019	Invalid Card Phone Type 3
20	22020	Invalid Card Phone Type 4
21	22021	Invalid Card Plastic Delivery Method
22	22022	Invalid Card Pin Delivery Method
23	22023	Invalid Card Promotion Code
24	22024	Invalid Card Field Name
25	22025	Invalid Card Field Value

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SetCardDetailsWebService/v1	POST	
/Cards/V2	PUT	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "cardDemog": {
    "branchCode": "",
    "priorityCode": "",
    "gender": "",
    "legalId": "",
    "birthDate": "",
    "birthCity": "",
    "vipLevel": "",
    "nationalityCode": "",
    "languageCode": "",
    "titleCode": "",
    "familyName": "",
    "secondFamilyName": "",
    "firstName": "",
    "secondFirstName": "",
    "maidenName": "",
    "ownerCode": "",
    "socioProfessionalCode": "",
    "activityCode": "",
    "customerSegment": "",
    "employerName": "",
    "staffID": "",
    "channelNumber": "",
    "embossedName": "",
    "secondEmbossedName": "",
    "email1": "",
    "email2": "",
    "plasticDeliveryMethod": "",
    "pinDeliveryMethod": "",
    "promotionCode": "",
    "phone1": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone2": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone3": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone4": {
      "phoneType": "",
      "phoneNumber": ""
    }
  },
  "CardCustomFields": [{
    "FieldName": "",

```



```

    "FieldValue": ""
  }
}

```

4.4 searchCard API

API overview

When a search card web service is used, agents and customers can include several options in the query message. PowerCARD is then responsible of looking for all the cards in the cardholder database that meets the combination of the input criteria being sent within the request.

Input criteria can include for instance card number, customer identification, card status, card product code, cardholder name...

Functional description

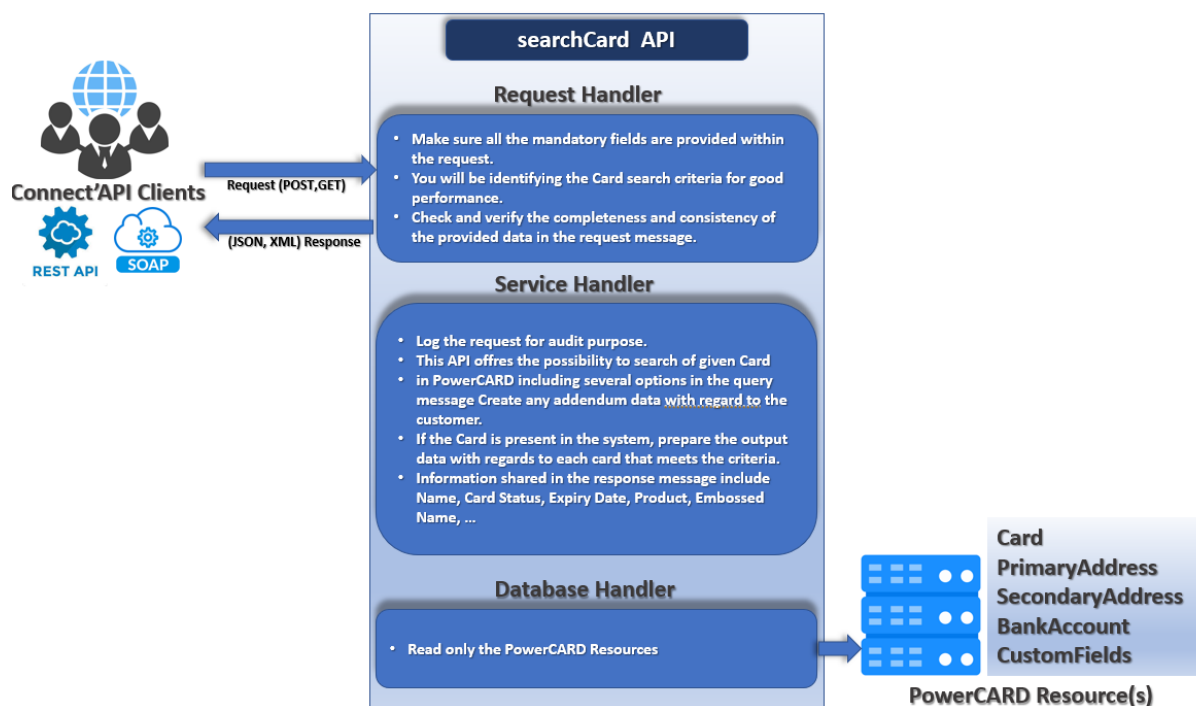


Figure 4: searchCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required(max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	ClientCode	C (24)	-	Optional (max : 1)	Represent the customer identifier. This field is required if Corporate Id and Card Number are empty.
5	CardNumber	C (22)	-	Optional (max : 1)	The card number. This field is required if ClientCode and Corporateld are empty.
6	Corporateld	C (15)	-	Optional (max : 1)	The corporate identifier. This field is required if ClientCode and CardNumber are empty.
7	CatPIProduct	C (6)	-	Optional (max : 1)	It is a 6 digit code identifying the payment instrument

#	Field Name	Data type	Parent	Occurrence	Field Description
					product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
8	CatPIProduct Version	C (10)	-	Optional (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
9	CardStatus	ClosedEnum	-	Optional (max : 1)	The card Status This field refers to the card status as stored in the system. Possible values are: • N: Normal • M: Monitoring • S: Suspend • R: Replaced • C: Canceled • P: Prospect
10	FirstName	C (80)	-	Optional (max : 1)	The first name or given name.
11	FirstName2	C (80)	-	Optional (max : 1)	The first name , part 2 , if compound names are in use
12	LastName	C (80)	-	Optional (max : 1)	The last name
13	LastName2	C (80)	-	Optional (max : 1)	The last name 2

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
5	MaxRowNbr	Long	-	Optional (max : 1)	This will depict the maximum number of row that can be included in the response message. Note that if the result exceeds this number, only this number of cards will be returned. This mechanism is used to limit the number of cards that can be fetched in one call.
-	CardDetails	Complex	-	Optional (max: 1)	It must contain the details information of card and bank accounts data of the requested card. Not mandatory in case of One Shot card
6	CardNumber	C (22)	-	Optional (max : 1)	The card number of the requested card. It represent the unique identifier.
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
7	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
8	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be

#	Field Name	Data type	Parent	Occurrence	Field Description
					previously defined in PowerCARD EMBOSSING PRIORITY table under the card parameters section.
9	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
10	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
11	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the Client if not provided at card level.
12	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
13	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined

#	Field Name	Data type	Parent	Occurrence	Field Description
					in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
14	NationalityCode	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
15	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
16	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to

#	Field Name	Data type	Parent	Occurrence	Field Description
					the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
17	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.
18	Second FamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
19	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the Client if not provided at card level.
20	Second FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
21	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
22	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate,

#	Field Name	Data type	Parent	Occurrence	Field Description
					and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. Business/Corporate cards.
23	Socio Professional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
24	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
25	Customer Segment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
26	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
27	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.
28	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
29	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If

#	Field Name	Data type	Parent	Occurrence	Field Description
					not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the last one is also not available.
30	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.
31	CatPIProduct	C (6)	CardDemog	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
32	CatPIProduct Version	C (10)	CardDemog	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
33	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic.

#	Field Name	Data type	Parent	Occurrence	Field Description
					If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.
34	Second Embossed Name	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
35	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
36	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
37	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					[no format check is foreseen in WS processing]
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
38	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
39	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
40	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					2: Home phone. 3: Personal mobile phone. 4: Others.
41	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
42	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
43	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
44	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first

#	Field Name	Data type	Parent	Occurrence	Field Description
					email address. There is no propagation of this field from client to card.
45	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
46	Plastic Delivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
47	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
48	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list

#	Field Name	Data type	Parent	Occurrence	Field Description
					defined at bank level.
-	PrimaryAddress	Complex	CardDetails	Required (max: 1)	To provide the primary address of the card
49	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.
50	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
51	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
52	LinkFlag	C(1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
53	LinkEntityCode	C(2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field becomes mandatory when LinkFlag is set to 'Y'.
54	LinkAddress Type	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
55	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
56	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
57	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
58	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
59	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked.

#	Field Name	Data type	Parent	Occurrence	Field Description
					It should not be validated.
60	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
61	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
62	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
63	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
64	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This 3 digit code must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Secondary Address	Complex	CardDetails	Optional infinite list	To provide the secondary address of the cardholder.
65	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_ TYPE_ LIST table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
66	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	Validity RangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
67	StartDt	DateTime	Validity RangeDate	Optional (max: 1)	The Validity Start date
68	EndDt	DateTime	Validity RangeDate	Optional (max: 1)	The Validity End date
69	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
70	LinkFlag	C(1)	Secondary Address	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
71	LinkEntityCode	C(2)	Secondary Address	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.

#	Field Name	Data type	Parent	Occurrence	Field Description
72	LinkAddress Type	C(2)	Secondary Address	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and Link Entity Code = 'CL'. This address type must be already present as part of the client addresses.
73	AddressLine1	C (64)	Secondary Address	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
74	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
75	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
76	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
77	ZipCode	C (10)	Secondary Address	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
78	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
79	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
80	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or Region FreeText should be provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
81	RegionFree Text	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
82	CountryCode	C (3)	Secondary Address	Required (max: 1)	The region code of the cardholder's address. Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	AddressPurpose	Complex	CardDetails	Optional (max: 1)	To specify the purpose (card delivery, pin delivery, statement....) of the provided addresses either the primary one or the secondary ones
83	Address PurposeID	C (1)	Address Purpose	Required (max: 1)	Specifies the possible address usage values, namely: S: Statement address

#	Field Name	Data type	Parent	Occurrence	Field Description
					• C: Card address • P: Pin address • L: Litigation address. • G: Gift address
84	AddressType	C (2)	Address Purpose	Required (max: 1)	Address type based on the intended purpose of use. This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This code must be already provided as part of the card PrimaryAddress or Secondary Address
85	SendingMode	C (1)	Address Purpose	Required (max: 1)	To be set to « M » - Mail.
-	Auxiliary Product	Complex	CardDetails	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)
86	PANAux Product	C (6)	Auxiliary Product	Required (max : 1)	It is a 6 digit code identifying the PAN auxiliary product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = I
87	PANAux ProductVersion	C (10)	Auxiliary Product	Required (max : 1)	It is a 10 character fields identifying the PAN auxiliary version. Valid values must be retrieved from PowerCARD CATALOGUE

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PRODUCT table with product type = I
88	Contract Number	C (16)	Auxiliary Product	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.
89	Membership Date	Date	Auxiliary Product	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD
90	BeneficiaryAge	Long (3)	Auxiliary Product	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.
91	Beneficiary Name	C (40)	Auxiliary Product	Optional (max: 1)	Full name of the product's beneficiary.
92	StatusCode	C (1)	Auxiliary Product	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Canceled Products with a 'Canceled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
93	StatusDate	Date	Auxiliary Product	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
94	Insurance BranchCode	C (6)	Auxiliary Product	Optional (max: 1)	Applies to insurance products solely. Specifies the branch code of the insurance company.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Must be defined in the PowerCARD BRANCH table under the general parameters section.
95	InsuranceAge	Long (4)	Auxiliary Product	Optional (max: 1)	Specifies in number of years the amount of time that has elapsed since the membership date. Only positive value will be accepted.
96	AuxiliaryFeeAmount	Decimal (18,3)	Auxiliary Product	Optional (max: 1)	Fees amount charged to the cardholder as remuneration for the auxiliary product.
97	PolicyNumber	C (32)	Auxiliary Product	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents to the insurance company.
-	BankAccount	Complex	CardDetails	Required infinite list	To provide information regarding bank accounts to which the card is linked.
98	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT_TYPE_LIST table under the general parameter section.
99	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.

#	Field Name	Data type	Parent	Occurrence	Field Description
100	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
101	BankCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
102	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
103	BranchCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
104	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
105	Account Currency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value

#	Field Name	Data type	Parent	Occurrence	Field Description
					(e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
106	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
107	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally – it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
108	DefaultFlag	C(1)	BankAccount	Required (max:1)	Flag to indicate the default bank account: - Y: Default bank account - N: Not the default bank account
-	CardCustom Fields	Complex	CardDetails	Optional infinite list	To provide User defined fields. User

#	Field Name	Data type	Parent	Occurrence	Field Description
					defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
109	FieldName	C (30)	CardCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
110	FieldValue	C (256)	CardCustom Fields	Required (max: 1)	The value of the user defined field.

Specific error codes

#	Error code	Error Description
1	20501	Invalid Card Number/Token Id
2	20502	One of the three fields (CardNumber, ClientCode, CorporateId) must be completed
3	20503	Invalid Customer Id M20503_INVALID_CUSTOMER_ID
4	20504	Invalid Catalogue Payment Instrument Product
5	20505	Invalid Catalogue Payment Instrument Product Version
6	20506	Invalid Corporate Id
7	20507	Invalid Card Status Code
8	20508	No data found

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	

Status Code	Description	Success Response	Error Response
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchCardWebService/V2	POST	
/Cards/V2? clientCode={clientCode}&aliasPAN={aliasPAN}&corporateId={corporateId}& catPIProduct={catPIProduct}&catPIProductVer={ catPIProductVer}&cardStatus={cardStatus}&firstName={ firstName}&lastName={lastN	GET	clientCode=[C(24)] cardNumber=[C(22)] corporateId=[C(15)] catPIProduct=[C(6)] catPIProductVersion=[C(10)] cardStatus=[N/M/S/R/C/P] firstName=[C(80)] firstName2=[C(80)] lastName=[C(80)] lastName2=[C(80)] pageSize=[PositiveInteger] pageIndex=[PositiveInteger]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "N"
    },
    {
      "key": "pageSize",
      "data": ""
    },
    {
      "key": "pageIndex",
      "data": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "clientCode": "",
  "corporateId": "",
  "catPIProduct": "",
  "catPIProductVersion": "",
  "cardNumber": "",
  "firstName": "",
  "firstName2": "",
  "lastName": "",
  "lastName2": ""
}
```

4.5 updateCardStatus API

API overview

This web services was introduced to allow changing the status of an existing card in the system. Agent and customers can ask to change the status of a card by selecting a reason for that operation from a pre-defined list of reasons. It is important to note that the web service requires the card number and the desired operation as input. As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

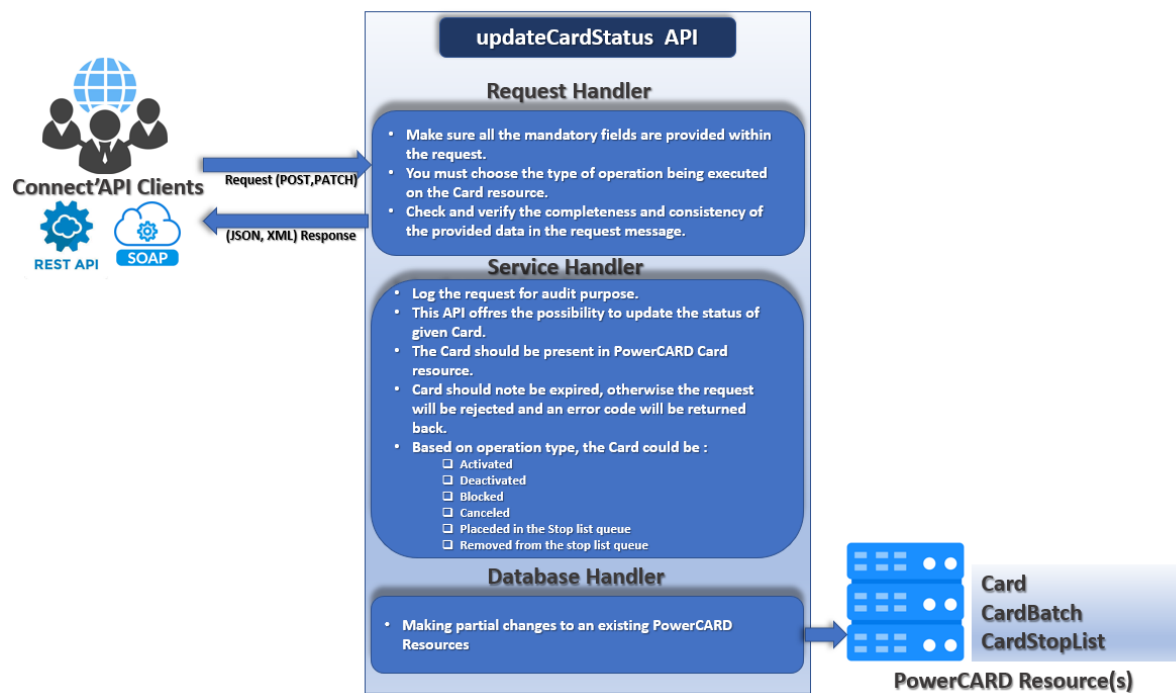


Figure 5: updateCardStatus process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required(max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX- XXX XX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	The card number.
5	Operation	ClosedEnum	-	Required (max : 1)	Must indicate the type of operation being executed on the card. Possible values are: • A: Activate the card • L: Deliver the card • E: Activate & Deliver the card • D: Deactivate the card • B: Block the card • U: Unblock the card • S: Place the card in stop list queue. • R: Remove the card from the Stop list queue. • C: Cancel the card • T: Stop renewing the card
6	ReasonCode	C (2)	-	Required (max : 1)	Reason code Refers to the reason why the agent or the customer is requesting to

#	Field Name	Data type	Parent	Occurrence	Field Description
					change the status of an existing card, required only if Operation is set to B, U, S and C. If operation is set to B, C and U, this code must be a valid value that must previously defined in the STATUS _REASON _LIST table under the general parameter section. If operation is set to S, the reason code field must match one of the codes present in the STOP _LIST _REASON _CODE table Finally, If operation is set to R, the reason code field must match of the codes present in the WITHDRAWAL _REASON _CODE table. Note that reasons must be retrieved from PowerCARD
7	Description	C (300)	-	Optional (max : 1)	Additional description. May contain an additional comments or description of the operation that describes briefly the operation.
-	RangeDate	Complex	-	Optional (max : 1)	The range date. Requested only in case the operation field is set to S. this aggregate is then used to transmit the period where the card must be placed in stop list.
8	StartDt	DateTime	RangeDate	Optional (max : 1)	Start date

#	Field Name	Data type	Parent	Occurrence	Field Description
9	EndDt	DateTime	RangeDate	Optional (max : 1)	End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program

#	Field Name	Data type	Parent	Occurrence	Field Description
					was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	02001	Card Does not Exist
2	02002	Wrong Card Number
3	02003	Invalid Operation code
4	02004	Card Expired
5	02005	Invalid Range Date
6	02006	Operation not permitted, replaced card
7	02007	Operation not permitted, cancelled card
8	02008	Operation not permitted, expired card
9	02009	Card in the stop list
10	02010	Invalid Reason Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is		x

Status Code	Description	Success Response	Error Response
	probably valid but needs to be requested again later.		
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCardStatusWebService/V2	POST	
/Cards/V2/:aliasPAN/ Status/:cardStatus/ StatusReason/:cardStatusReason/ Description/:description/ RangeStartDate/:rangeStartDate/ RangeEndDate/:rangeEndDate	PATCH	Required cardStatus=[C(1)] Optional aliasPAN=[C(22)] cardStatusReason=[C(2)] rangeStartDate=[Date] rangeEndDate=[Date] description=[C(300)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"cardNumber": "",
"operation": "",
"reasonCode": "",
"description": ""
}

```

4.6 renewCard API

API overview

Usually a card is automatically renewed within the month it expires. However, for some reasons, as if when the customer is going to be out of the town during the renewal process, he may request to renew his card before its expiration date. For that, PowerCARD offers the possibility to use this web service. The service is mainly used to store the request that will be processed during the end of the day.

It is important to mention here that PowerCARD makes multiple controls on the card before proceeding to perform the renewal action. Those controls are all defined in the system and can be activated or deactivated by the financial institution using the PowerCARD setup screens.

Functional description

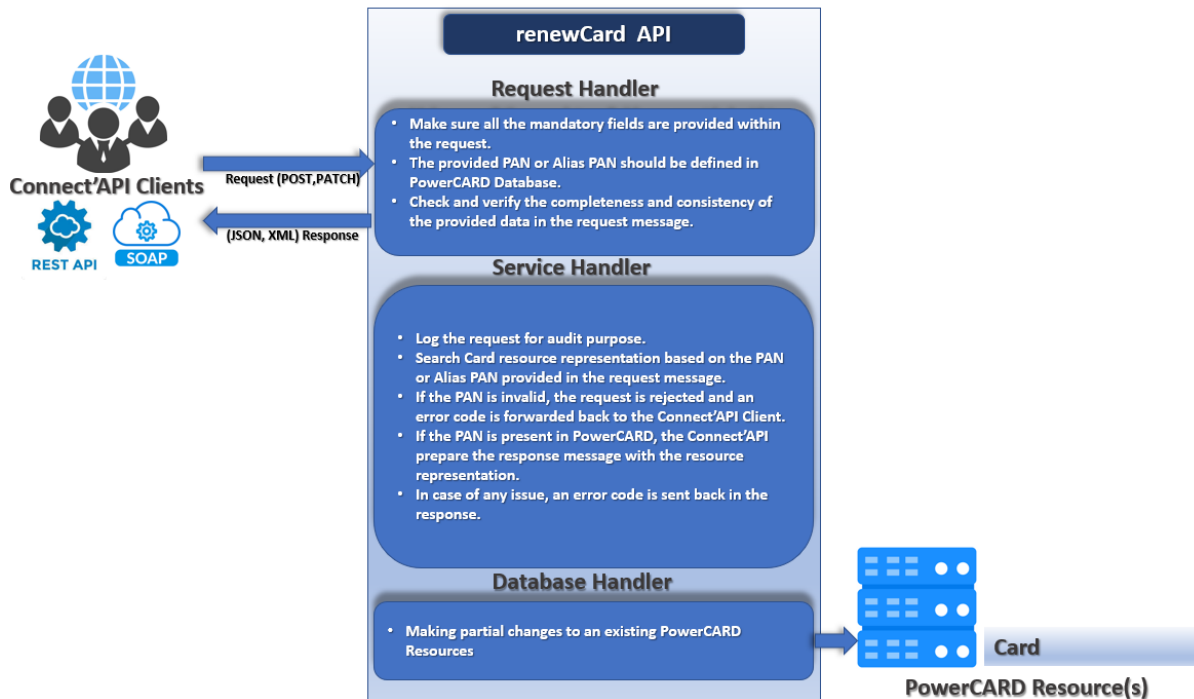


Figure 6: renewCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The generation mask should be the following: XXXXX XXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	The card number.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: Proceed With Success: Call

#	Field Name	Data type	Parent	Occurrence	Field Description
					was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	ExpiryDt	DateTime	-	Optional (max : 1)	Refers to the expiry date of the card being returned back in the response message.

Specific error codes

#	Error code	Error Description
1	12701	Invalid card number
2	12702	Invalid Card product
3	12703	Operation not allowed
4	12704	Operation not allowed for Canceled card
5	12705	Operation not allowed for Replaced card

#	Error code	Error Description
6	12706	Card blocked for renewal
7	12707	Invalid Expiration date
8	12708	The card is in production process, Contact your administrator
9	12709	Card is in the stop list, Contact your administrator
10	12710	The card is in Pin request process, Contact your administrator
11	12711	The card is in Replacement process
12	12712	The card is in Renewal process
13	12713	The card do not belong this this bank, Contact your administrator
14	12714	Card not allowed renewal

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCardStatusWebService/V2	POST	
/Cards/V2/:aliasPAN/Renew	POST	Optional aliasPAN=[C(22)]

REST sample call

```

{
  "keyValues": [
    {
      "key": "",
      "data": "",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": ""
}

```

4.7 replaceCard API

API overview

In PowerCARD, lost or damaged cards can be replaced in the system using this web service. Agents as well as customers can ask for the replacement of an existing card by specifying the reason. Based on that reason, the system can either replace the card by keeping the same card number or generating a new card number for the cardholder.

The service allows also applying or not some fees amount while replacing a card. Fees are then charged to the account with a specific transaction code.

Functional description

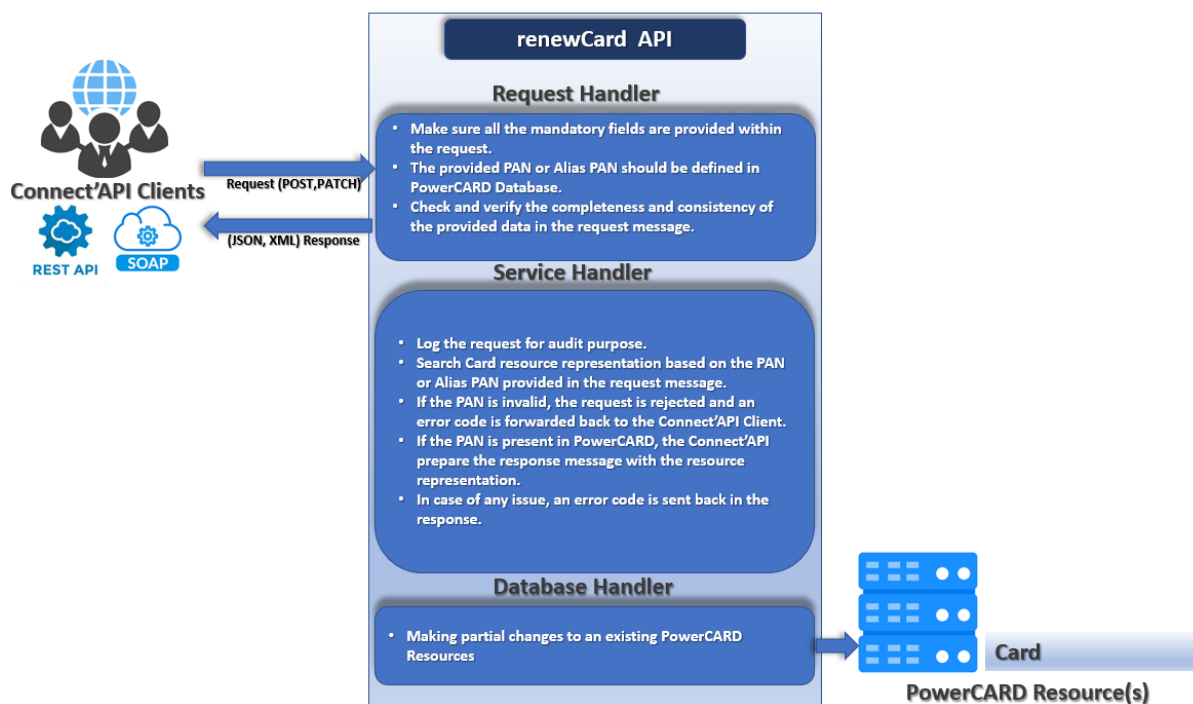


Figure 7: replaceCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX -XXXXX XXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	The card number.
5	ReasonCode	C (2)	-	Required (max : 1)	Must contain the replacement reason code. This code must be previously defined in the PowerCARD REPLACEMENT_REASON_CODE table under the cardholder parameter section. Valid values must be downloaded from PowerCARD
6	DeliveryBranch	C (15)	-	Optional (max : 1)	The delivery branch code. 6 digits code of the branch to which the new card will be delivered if the customer has requested so. Valid values must be downloaded from PowerCARD BRANCH table
7	Priority Indicator	C (3)	-	Optional (max: 1)	Refer to the priority of card embossing The value must exist in the table PRIORITY_INDICATOR

#	Field Name	Data type	Parent	Occurrence	Field Description
8	Plastic Delivery Method	C (3)	-	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product. Should be defined in the table ELIGIBLE_DELIVERY_METHOD
9	FeesOption	C (1)	-	Required (max : 1)	The fees option used to set fees during the replacement of the card. Possible values are : - Y: Apply fees while replacing the card. - N: Do not apply fees on replace card.
-	FeeAmount	Complex	-	Optional (max : 1)	Used only if the FeesOption field is set to Y and the requester would like to set a different replacement fee than the one defined on the card product level.
10	Amt	Long ()	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	NC (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue

#	Field Name	Data type	Parent	Occurrence	Field Description
					for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CardNumber	C (22)	-	Optional (max : 1)	Card Number Represents the card number generated by PowerCARD in case of successful replacement with a new card number..
6	ExpiryDt	DateTime	-	Optional (max : 1)	Refers to the expiry date of the card being returned back in the response message.

Specific error codes

#	Error code	Error Description
1	00530	Invalid Card number
	00531	Invalid replacement reason code
3	00533	Invalid fee option
4	00534	Invalid fee amount
5	00535	Invalid delivery branch
6	00536	Invalid Card number
7	00537	Expired Card
9	00538	Invalid Priority Indicator

#	Error code	Error Description
10	00539	Invalid Plastic Delivery Method

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCardStatusWebService/V2	POST	
/Cards/V2/:aliasPAN/Replace	POST	Optional aliasPAN=[C(22)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardnumber": "",
  "reasonCode": "",
  "deliveryBranch": ""
}
```



```

    "priorityIndicator": "",
    "plasticDeliveryMethod": "",
    "feesOption": "",
    "feeAmount": {
      "amt": "",
      "curCode": ""
    }
  }

```

4.8 changeCardProduct API

API overview

PowerCARD offers a service to downgrade or upgrade a card. This operation is done by changing the card product of the card. Moving a card from product to a new one can be performed by providing a set of data including the card number, the product code as well as some other additional fields like fee amount...

A product change lets you maintain your existing account, while enjoying the benefits and terms of a different plastic that could have different limits, services... the new card will be produced then shipped to the address specified by the customer.

Functional description

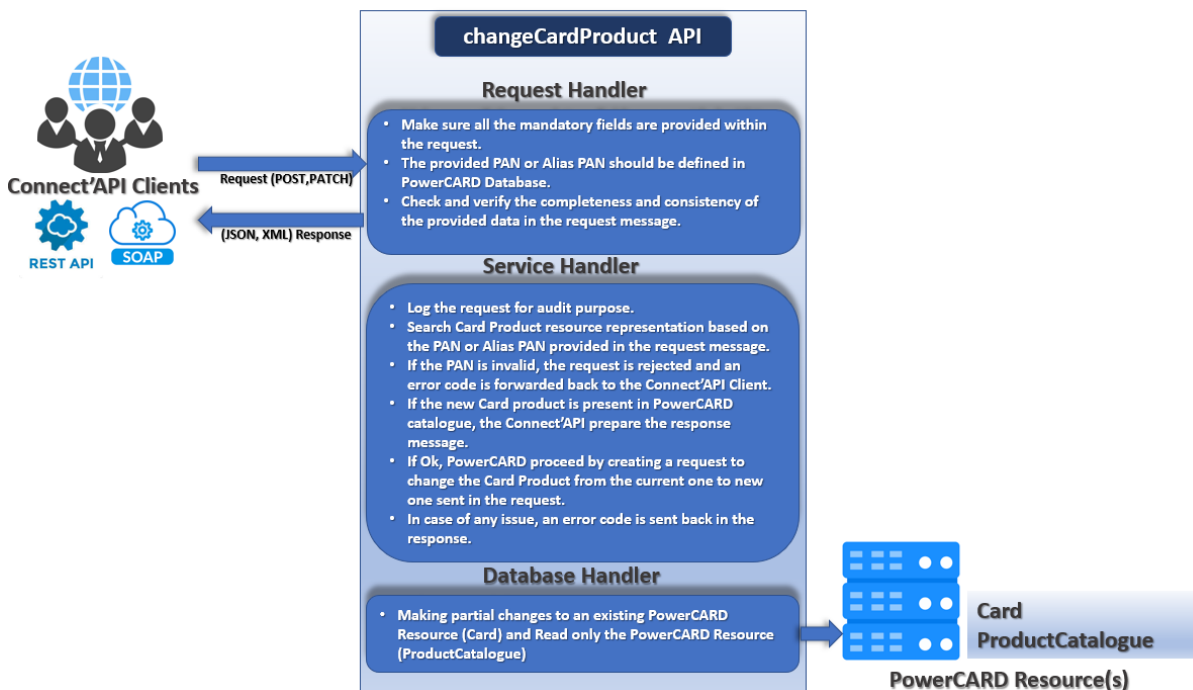


Figure 8: changeCardProduct process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	The card number.
5	NewCatPI Product	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code to be assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P

#	Field Name	Data type	Parent	Occurrence	Field Description
6	NewCatPI Product Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
7	NewCatACC Product	C (6)	-	Optional (max : 1)	It is a 6 digit code identifying the account product code to be assigned to the card's account. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A Required in case of Credit/Prepaid/ Charge Card update.
8	NewCatACC Product Version	C (10)	-	Optional (max : 1)	It is a 10 character fields identifying the account version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A Required in case of Credit/Prepaid/ Charge Card update.
9	ExpiryDt	DateTime	-	Optional (max : 1)	Refers to the new expiry date of the card
10	Delivery Branch	C (15)	-	Optional (max : 1)	The delivery branch code. 6 digits code of the branch to which the new card will be delivered if

#	Field Name	Data type	Parent	Occurrence	Field Description
					the customer has requested so. Valid values must be downloaded from PowerCARD BRANCH table
11	FeesOption	C (1)	-	Required (max : 1)	The fees option used to set fees during the PIN request. Possible values are: - Y: Apply fees - N: Do not apply fees
-	FeeAmount	Complex	-	Optional (max : 1)	Used only if the FeesOption field is set to Y. This aggregate contains the fee amount and currency
12	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
13	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the

#	Field Name	Data type	Parent	Occurrence	Field Description
					description of the error encountered

Specific error codes

#	Error code	Error Description
1	17101	Invalid card number
2	17102	Invalid Expiry date
3	17105	Invalid Fee amount/Currency
4	17106	Invalid Fee option
5	17107	Operation is not permitted for the target product
6	17108	An error occurred while inserting a product change request
7	17119	Invalid Catalogue Payment Instrument Product
8	17120	Invalid Catalogue Payment Instrument Product Version
9	17121	Invalid Catalogue Account Product
10	17122	Invalid Catalogue Account Product Version

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		X
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		X
404 Not Found	PowerCARD Resource not found.		X
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		X
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		X
503 Service Unavailable	Service Unavailable.		x

Status Code	Description	Success Response	Error Response
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ChangeCardProductWebService/V2	POST	
/Cards/V2/:aliasPAN/ CatPIProduct/:catPIProduct/ CatPIProductVer/:catPIProductVer/ CatAccProduct/:catAccProductVer/ ExpiryDate/:expiryDate/ DeliveryBranch/:deliveryBranch/ FeesOption/:feesOption/ FeeAmount/:feeAmount/ FeeAmountCur/: feeAmountCur	PATCH	Required aliasPAN=[C(22)] feesOption=[C(1)] Optional catPIProduct=[C(6)] catPIProductVer=[C(10)] catAccProduct=[C(6)] catAccProductVer=[C(10)] expiryDate=[Date] deliveryBranch=[C(15)] feeAmount=[Decimal(18,3)] feeAmountCur=[C(3)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "newCatPIProduct": "",
  "newCatPIProductVersion": "",
  "newCatAccProduct": "",
  "newCatAccProductVersion": "",
  "expiryDT": "",
  "cardNumber": "",
  "feesOption": "",
  "feeAmount": {
    "amt": "",
    "curCode": ""
  }
}
```

4.9 requestPIN API

API overview

The PIN request API allows the bank cardholders to request for the PIN .A PIN request is usually called when the cardholder request from the bank to send him the PIN associated

with his card. The cardholder has the ability to choose the appropriate mode of PIN re-sending.

Note that the cardholder PIN is never logged, not even in an encrypted form; the system will generate it again based on some data.

Functional description

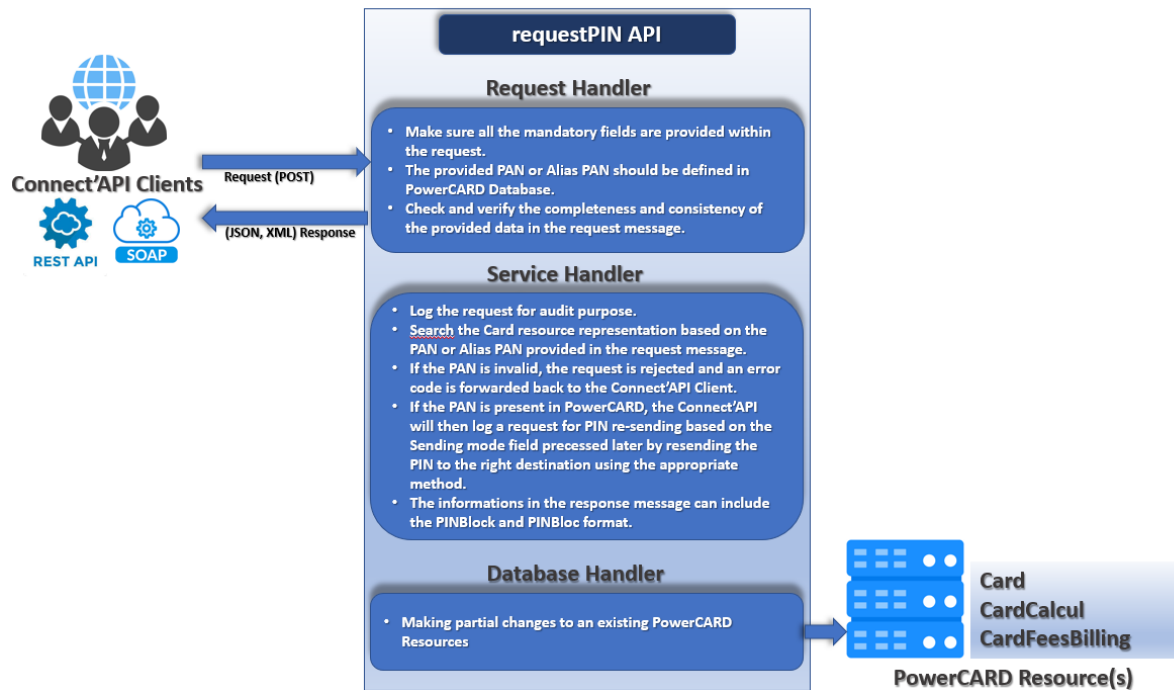


Figure 9: requestPIN process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The generation mask should be the following: XXXXX XXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	The card number.
5	SendingMode	ClosedEnum	-	Required (max : 1)	Mode of re-sending the PIN Select one value from the following list: - WBS: Returns the new PIN value in the response - VEW: Returns the old PIN (For future use) - EML: Send the PIN using the customer email address (For future use) - SMS: Send the PIN by SMS (For future use) - PML: Send the PIN by Postal address - BRC: Send the PIN mailer to the branch (For future use)

#	Field Name	Data type	Parent	Occurrence	Field Description
					For the time being, only the options WBS and PML are supported.
6	FeesOption	C (1)	-	Required (max : 1)	The fees option used to set fees during the PIN request. Possible values are: - Y: Apply fees - N: Do not apply fees
-	FeeAmount	Complex	-	Optional (max : 1)	Used only if the FeesOption field is set to Y. This aggregate contains the fee amount and currency
7	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999. 99 should be sent as 9999 999.
8	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the

#	Field Name	Data type	Parent	Occurrence	Field Description
					description of the error encountered
5	PIN	C (256)	-	Optional (max : 1)	This field contains the current PIN. It is encrypted and formatted as a block of 16 hexadecimal digits. The PIN Block is transmitted encrypted under the ZPK (Zone PIN key). This key should be shared between PowerCARD and the other party. This field is reserved for future use.
6	pinBlock Format	C (2)	-	Required (max : 1)	PIN Block Format This field indicates the format of the PIN block Select one valid value from the list: 01: Format is based on the PIN, the PIN length, the selected rightmost digits of the account number, and the pad characters 0 and F —combined through an exclusive OR operation 03: Format is based on the PIN and the F pad character

Specific error codes

#	Error code	Error Description
1	02201	Invalid Card number
2	02202	Invalid Sending Mode
3	02203	Invalid Fees option
8	02204	Invalid Product
9	02205	Replace without fees ex
10	02206	Replace with fees ex

#	Error code	Error Description
11	02207	Renewal ex
12	02208	Manual renewal ex
13	02209	Pin calculate with fees ex
14	02210	Pin calculate ex
15	02211	Pin not managed
16	02212	Card in process

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/RequestPIN/WebService/V2	POST	
/Cards/V2/:aliasPAN/RequestPIN/ SendingMode/:sendingMode/ FeesOption/:feesOption/ FeeAmount/:feeAmount/ CurCode/:curCode	GET	Required aliasPAN=[C(22)] sendingMode=[C(3)] feesOption=[C(1)]

PowerCARD Resource URI	Method	URI Params
		Optional feeAmount=[Decimal(18,3)] curCode=[C(3)]

RESTful sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "sendingMode": "",
  "feesOption": "",
  "feeAmount": {
    "amt": "",
    "curCode": ""
  }
}
```

4.10 getCardCVC2

API overview

This web service is mainly used to retrieve the CVC2 when providing a PAN or PAN Alias. PowerCARD is in fact a tokenization system that accepts as entry token value and proceeds by the reverse process of redeeming a token for its associated PAN value.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

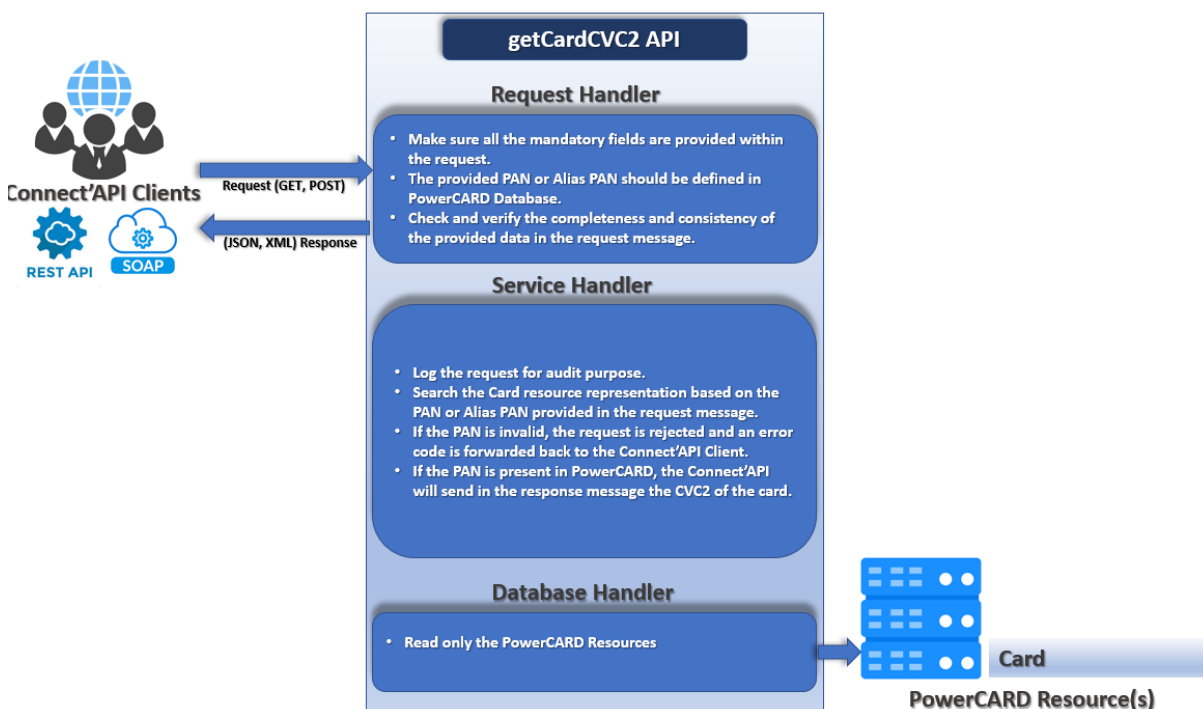


Figure 10: getCardCVC2 process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
5	ExpiryDate	Date	-	Required (max: 1)	Refers to the expiry date of the card

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CVC2	C (3)	-	Optional (max: 1)	The three-digit security code that is printed on the back of credit cards. Clear Value will be provided, no encryption.

Specific error codes

#	Error code	Error Description
1	20901	Invalid Card Number/Token Id

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCardCVC2WebService/V2	POST	
/Cards/V2/:aliasPAN/CVC2	GET	Optional: aliasPAN=[alphanumeric]

RESTful sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "expiryDate": ""
}
```

4.11 getCustomerDetails API

API overview

Get or find contact information is a web service that might be used by both customers and agents with the aim of getting more information details regarding an existing contact in the system.

The service could be used to get more details regarding a card member based on the card number.

As output the requester could get information that can include details regarding: The customer full name, address, marital status, branch, VIP level, revenue...

Functional description

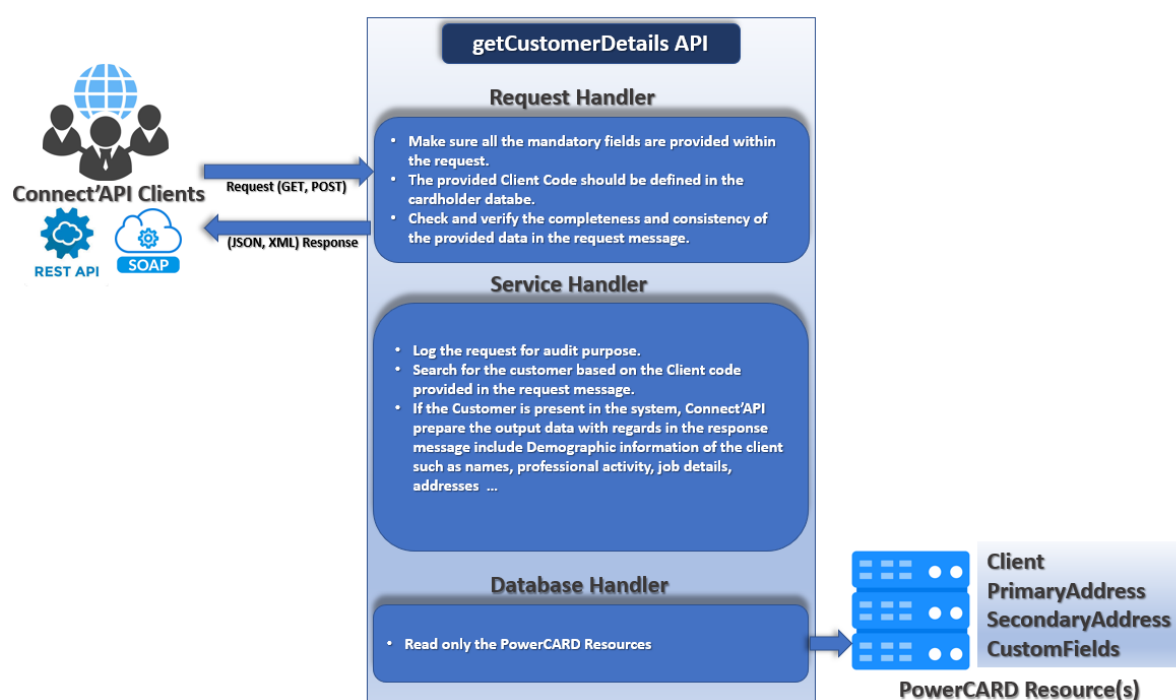


Figure 11: getCustomerDetails process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX-XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	ClientCode	C (22)	-	Required (max : 1)	Should contain the customer identification number in PowerCARD. This value must be previously defined in the PowerCARD CLIENT table.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Required if Corporateld is empty.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	ClientDemog	Complex	Customer Details	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if Existing Client block is not provided
5	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
6	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client. Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.
7	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs. This value will be propagated at card

#	Field Name	Data type	Parent	Occurrence	Field Description
					level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
8	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
9	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support. This field doesn't apply to corporate cardholders. Only positive value will be accepted.
10	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable. Defaulted to 'N' when not provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
11	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
12	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
13	TaxRegistry Office	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
14	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
15	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
16	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
17	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP LIST table under the general parameter section.
18	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
19	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
20	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
21	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
22	Second FamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
23	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
24	Second FirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if

#	Field Name	Data type	Parent	Occurrence	Field Description
					compound names are in use.
25	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
26	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section
27	Socio Professional Code	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
28	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
29	Customer Segment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
30	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
31	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
32	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
33	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner. • D: Dependent.
34	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate from client to card.
35	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
36	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
37	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
38	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
39	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
40	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
41	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
42	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
43	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
44	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address. There is no propagation of this field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder Becomes mandatory when sub block ExistingClient is not provided.
45	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
46	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
47	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
48	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
49	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
50	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
51	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.
52	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
53	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or

#	Field Name	Data type	Parent	Occurrence	Field Description
					CityFreeText should be provided.
54	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
55	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
56	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
57	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported:

#	Field Name	Data type	Parent	Occurrence	Field Description
					#1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Secondary Address	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
58	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.
59	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	Validity RangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
60	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
61	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
62	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer,

#	Field Name	Data type	Parent	Occurrence	Field Description
					a merchant, a chain...
63	AddressLine1	C (64)	Secondary Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
64	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
65	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
66	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
67	ZipCode	C (10)	Secondary Address	Required (max: 1)	Recipient's postal code. This field should not be validated.
68	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
69	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is

#	Field Name	Data type	Parent	Occurrence	Field Description
					allowed at bank level.
70	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
71	RegionFreeText	C (30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
72	CountryCode	C (3)	Secondary Address	Required (max: 1)	Country code. Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport details, civil ID details ... Becomes mandatory when sub block ExistingClient is not provided.
73	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passeport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
74	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
75	Document IssueDate	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
76	Document ExpiryDate	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD
77	Document IssueCity	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
78	Document IssueCountry	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	ClientCustom Fields	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that

#	Field Name	Data type	Parent	Occurrence	Field Description
					can be provided by a given bank.
79	FieldName	C (30)	ClientCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
80	FieldValue	C (256)	ClientCustomFields	Required (max: 1)	The value of the user defined field.

Specific error codes

#	Error code	Error Description
1	20301	Invalid Client Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCustomerDetailsWebService/V2	POST	
/Clients/V2/:clientCode	GET	Optional: clientCode=[C(22)]

RESTful sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "clientCode": ""
}
```

4.12 setCustomerDetail API

API overview

The setCustomerDetails web service can be used by both agents and customers in order to update details regarding the concerned customer being sent in the request message. It also allows the creation of a new ones.

The setCustomerDetails web service includes the demographic information of the client such as names, professional activity, job details ...

As output the web service sends back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

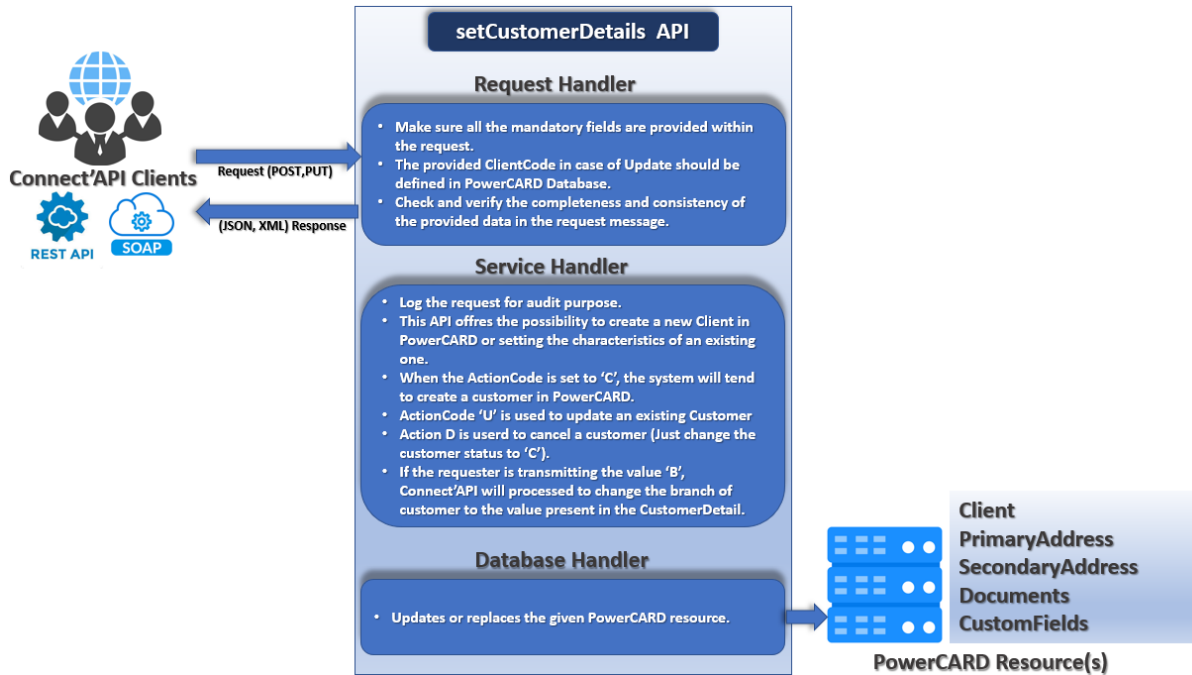


Figure 12: setCustomerDetail process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	ActionCode	ClosedEnum	-	Required (max : 1)	Action Code Valid values are : • C: Create • U: Update • D: Cancel • B: Assign New branch to customer (Valid only in case CustomerDetail)
-	ClientDemog	Complex	CustomerDetails	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if ExistingClient block is not provided
5	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
6	ClientCode	C (24)	ClientDemog	Optional (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD. Should be generated for new client. Mandatory when Action code is set to U, D & B.
7	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs. This value will be propagated at card level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
8	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
9	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support. This field doesn't apply to corporate cardholders. Only positive value will be accepted.
10	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable. Defaulted to 'N' when not provided.
11	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
12	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
13	TaxRegistry Office	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
14	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
15	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
16	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
17	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant

#	Field Name	Data type	Parent	Occurrence	Field Description
					special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP LIST table under the general parameter section.
18	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
19	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
20	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional

#	Field Name	Data type	Parent	Occurrence	Field Description
					designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
21	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
22	SecondFamily Name	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
23	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
24	SecondFirst Name	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if compound names are in use.
25	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
26	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section
27	Socio Professional Code	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
28	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
29	Customer Segment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
30	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
31	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
32	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
33	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: - N: Not reported or not applicable. - R: Renter.

#	Field Name	Data type	Parent	Occurrence	Field Description
					- O: Owner. - D: Dependent.
34	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
35	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
36	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate from client to card.
37	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
38	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
39	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
40	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
41	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
42	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. [no format check is foreseen in WS processing]
43	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
44	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder Becomes mandatory when sub block ExistingClient is not provided.
45	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
46	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
47	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
48	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
49	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
50	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
51	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.

#	Field Name	Data type	Parent	Occurrence	Field Description
52	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
53	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
54	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
55	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or

#	Field Name	Data type	Parent	Occurrence	Field Description
					RegionFreeText should be provided.
56	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
57	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Secondary Address	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
58	AddressType	C (2)	SecondaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.
59	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ValidityRange Date	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
60	StartDt	DateTime	Validity RangeDate	Optional (max: 1)	The Validity Start date
61	EndDt	DateTime	Validity RangeDate	Optional (max: 1)	The Validity End date
62	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
63	AddressLine1	C (64)	Secondary Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
64	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
65	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
66	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
67	ZipCode	C (10)	Secondary Address	Required (max: 1)	Recipient's postal code. This field should not be validated.
68	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
69	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
70	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
71	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
72	CountryCode	C (3)	Secondary Address	Required (max: 1)	Country code. Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport details, civil ID details ... Becomes mandatory when sub block ExistingClient is not provided.
73	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passeport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
74	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
75	Document IssueDate	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
76	Document ExpiryDate	Date	Documents	Optional (max: 1)	Date when the document will expire.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Format: YYYYMMDD
77	Document IssueCity	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
78	Document IssueCountry	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	ClientCustom Fields	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
79	FieldName	C (30)	ClientCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
80	FieldValue	C (256)	ClientCustom Fields	Required (max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.

#	Field Name	Data type	Parent	Occurrence	Field Description
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CustomerId	C (22)	-	Optional (max : 1)	Customer identifier

#	Field Name	Data type	Parent	Occurrence	Field Description
					Refers to the unique customer identification auto generated by the system in case of PersonId aggregate not passed in the request.

Specific error codes

#	Error code	Error Description
1	10101	Invalid Action Code
2	10102	Invalid Customer details
3	10103	Invalid Corporate details
4	10104	Customer does not exists
5	10105	Invalid Owner Code
6	10106	Error while cancelling the customer
7	10107	Error while updating the Customer branch
8	10108	Invalid Socio-professional code
9	10109	Invalid Bank Code
10	10110	Error while inserting a customer record
11	10111	Error while inserting a customer address record
12	10112	Error inserting the customer secondary Address
13	10113	Error while updating a customer record
14	10114	Error while updating customer address record
15	10115	Error updating the customer secondary Address
16	10116	Error updating the corporate status
17	10117	Error inserting a new corporate
18	10118	Error updating a corporate record in the database
19	10119	Empty Set of contact Information
20	10120	Customer/Organisation operation are mutually exclusive
21	10121	Corporate already exists in the system with the same ID

#	Error code	Error Description
21	10121	Corporate already exists in the system with the same ID
22	10122	Invalid Branch Code in Customer Details Segment
23	10123	Invalid Family Status
24	10124	Invalid Cardholder's Gender
25	10125	Invalid Cardholder's Birth Date
26	10126	Invalid VIP Level
27	10127	Invalid Nationality Code
28	10128	Invalid Language Code
29	10129	Invalid Title Code
30	10130	Invalid Socio-Professional Code
31	10131	Invalid Activity Code
32	10132	Invalid Customer Segment
33	10133	Invalid Lodging Mode
34	10134	Invalid Channel Number
35	10135	Invalid Phone Type 1
36	10136	Invalid Phone Type 2
37	10137	Invalid Phone Type 3
38	10138	Invalid Phone Type 4
39	10139	Invalid Primary Address Type
40	10140	Invalid Primary Address Sending Mode
41	10141	Invalid Client Primary Address City Code
42	10142	Invalid Client Primary Address Region Code
43	10143	Invalid Client Primary Address Country Code
44	10144	City Code or City Free Text is required for Card Primary Address
45	10145	Region Code or Region Free Text is required for Card Primary Address
46	10146	Invalid Client Secondary Address Type
47	10147	Invalid Client Secondary Address Sending Mode
48	10148	Invalid Client Secondary Address City Code
49	10149	Invalid Client Secondary Address Region Code

#	Error code	Error Description
50	10150	Invalid Client Secondary Address Country Code
51	10151	Validity Start Date must not be in the past in case Client Secondary Address
52	10152	Validity Start Date cannot be greater than Validity End Date in case Client Secondary Address
53	10153	Validity End Date must not be in the past in case Client Secondary Address
54	10154	City Code or City Free Text is required for Card Secondary Address
55	10155	Region Code or Region Free Text is required for Card Secondary Address
56	10156	Invalid Client Document Code
57	10157	Invalid Client Field Name
58	10158	Invalid Client Field Value
59	10159	Wrong Format For Client Field Value

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested		x

Status Code	Description	Success Response	Error Response
	is not supported for the PowerCARD Resource.		
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SetCustomerDetailsWebService/V2	POST	
/Customers/V2/	POST	
/Customers/V2/	PUT	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "actionCode": "",
  "clientDemog": {
    "bankCode": "",
    "clientCode": "",
    "branchCode": "",
    "familyStatus": "",
    "numberOfDependents": "",
    "gender": "",
    "legalId": "",
    "taxNumber": "",
    "taxRegistryOffice": "",
    "birthDate": "",
    "birthCountry": "",
    "birthCity": "",
    "vIPLevel": "",
    "nationalityCode": "",
    "languageCode": "",
    "titleCode": "",
    "familyName": "",
    "secondFamilyName": "",
    "firstName": "",
    "secondFirstName": "",
    "maidenName": "",
    "ownerCode": "",
    "socioProfessionalCode": "",
    "activityCode": "",
    "customerSegment": "",
    "takingOnDate": "",
    "employerName": "",
    "staffID": "",
    "lodgingMode": "",
    "channelNumber": "",
    "phone1": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone2": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone3": {
      "phoneType": "",
      "phoneNumber": ""
    },
    "phone4": {
```

```

    "phoneType": "",
    "phoneNumber": "",
  },
  "email1": "",
  "email2": ""
},
"primaryAddress": {
  "addressType": "",
  "sendingMode": "",
  "careOf": "",
  "addressLine1": "",
  "addressLine2": "",
  "addressLine3": "",
  "addressLine4": "",
  "zipCode": "",
  "cityCode": "",
  "cityFreeText": "",
  "regionCode": "",
  "regionFreeText": "",
  "countryCode": ""
},
"secondaryAddress": {
  "addressType": "",
  "sendingMode": "",
  "validityRangeDate": {
    "startDt": "",
    "endDt": ""
  },
  "careOf": "",
  "addressLine1": "",
  "addressLine2": "",
  "addressLine3": "",
  "addressLine4": "",
  "zipCode": "",
  "cityCode": "",
  "cityFreeText": "",
  "regionCode": "",
  "regionFreeText": "",
  "countryCode": ""
},
"documents": [
  {
    "documentCode": "",
    "documentId": "",
    "documentIssueDate": "",
    "documentExpiryDate": "",
    "documentIssueCity": "",
    "documentIssueCountry": ""
  }
],
"clientCustomFields": [
  {
    "fieldName": "",
    "fieldValue": ""
  }
]
}

```

4.13 feeInquiry API

API overview

This API is intended to be used whenever a customer or an agent would like to get further details with regards to fees amount linked to several card operations, fees on transactions as well as detail of a statement fees previously defined in the system...

Functional description

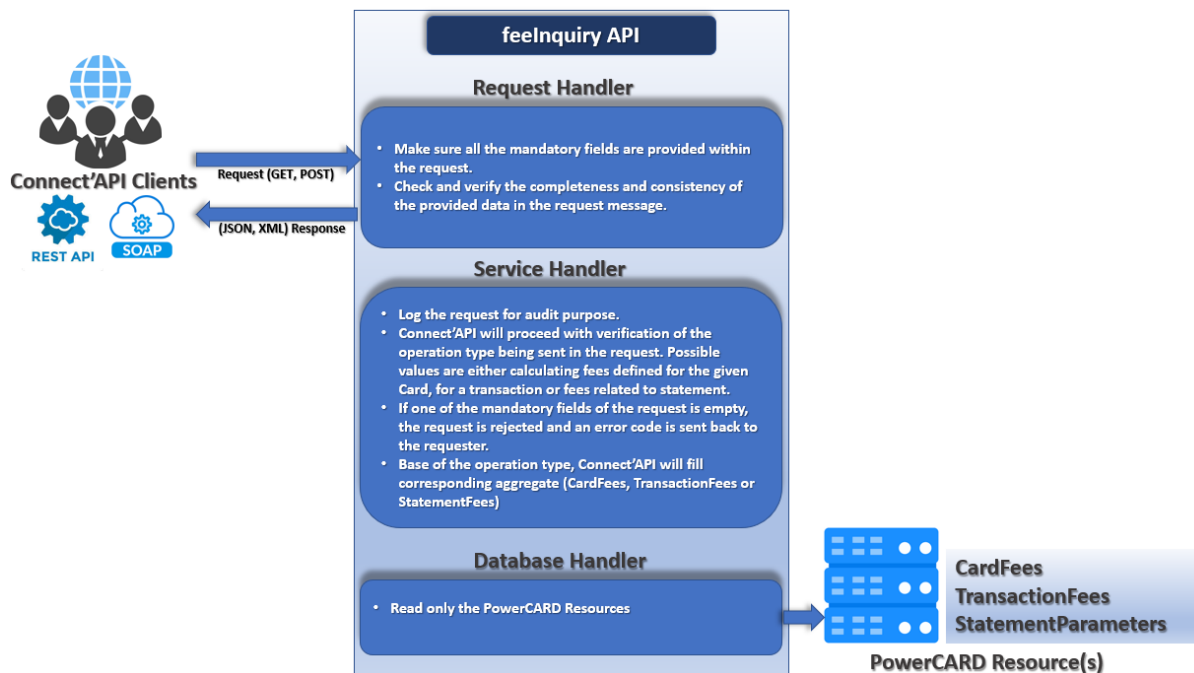


Figure 13: feelInquiry process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	OperationType	ClosedEnum	-	Required (max : 1)	This field indicates the type of action that the agent would like to run. Possible values are the following: • CF: Calculate the Card Fees. • TF: Calculate the Transaction Fees. • SF: Calculate the Statement Fees.
-	CardFees	Complex	-	Optional (max : 1)	It refers to card fees calculation request. Required only in case the operation type is set to CF. This aggregate is then used to transmit the card number, card product code or/and fee code.
5	OperationEvent	ClosedEnum	CardFees	Required (max : 1)	This field indicates the type of event that the agent would like to . Possible values are the following: • GS: General card fee Settings • FI: Card issuing Fees.

#	Field Name	Data type	Parent	Occurrence	Field Description
					- RF: Renewal Fees - RP: Replacement Fees - PI: Pin Fees
6	Replacement Reason	ClosedEnum	CardFees	Optional (max : 1)	Must indicate the reason of the replacement of the card. Possible values are: - L: Lost the card - D: Damaged the card - E: Emergency replacement of the card - S: Stolen card - R: Erroneous card - C: Counterfeit card - O: Other. This field is mandatory when OperationEvent is set to RP
7	CardNumber	C (22)	CardFees	Optional (max : 1)	The card number. Must contain the primary card number of the customer. The number should be valid and previously present in the PowerCARD CARD table under the cardholder section. This field becomes mandatory when the card product code and Fee code is not provided.
8	CatPIProduct	C (6)	CardFees	Optional (max : 1)	It is a 6 digit code identifying the payment instrument product code. Valid values must be retrieved from PowerCARD CATALOGUE

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PRODUCT table with product type = P
9	CatPIProduct Version	C (10)	CardFees	Optional (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
-	TransactionFees	Complex	-	Optional (max : 1)	It refers to Transaction fees calculation request. Required only in case the operation type is set to TF. This aggregate is then used to transmit the profile code, transaction code and reversal flag.
10	CardNumber	C (22)	TransactionFees	Optional (max : 1)	The card number. Must contain the primary card number of the customer. The number should be valid and previously present in the PowerCARD CARD table under the cardholder section. This field becomes mandatory when the card product code is not provided.
11	CatAccProduct	C (6)	TransactionFees	Optional (max : 1)	It is a 6 digit code identifying the account product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A

#	Field Name	Data type	Parent	Occurrence	Field Description
12	CatAccProduct Version	C (10)	TransactionFees	Optional (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
13	TransactionCode	C (20)	TransactionFees	Required (max : 1)	The transaction code, as defined in the CMS. Valid values must be retrieved from PowerCARD TRANSACTIONS _DEFINITION table
14	ReversalFlag	ClosedEnum	TransactionFees	Required (max : 1)	Indicates whether it is a normal or reversal transaction. • N: Normal. • R: Reversal.
15	CatPIProduct	C (6)	TransactionFees	Optional (max : 1)	It is a 6 digit code identifying the payment instrument product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
16	CatPIProduct Version	C (10)	TransactionFees	Optional (max : 1)	It is a 10 character fields identifying the payment instrument product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
-	Transaction Amount	Complex	TransactionFees	Optional (max : 1)	The transaction amount of the operation.
17	Amt	Long ()	Transaction Amount	Optional (max : 1)	Amount.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
18	CurCode	NC (3)	Transaction Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	StatementFees	Complex	-	Optional (max : 1)	It refers to statement fees calculation request. Required only in case the operation type is set to SF. This aggregate is then used to transmit the account number, profile code and minimum due.
19	CardNumber	C (22)	StatementFees	Optional (max : 1)	The card number. Must contain the card number. The number should be valid and previously present in the PowerCARD CARD table under the cardholder section. This field becomes mandatory when the card product code is not provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
20	AccountNumber	C (40)	StatementFees	Optional (max : 1)	It refers to financial account number. Mandatory in case of Statement fees for credit card & prepaid card
21	CatAccProduct	C (6)	StatementFees	Optional (max : 1)	It is a 6 digit code identifying the account product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A using the Web Service: getParameterLi st
22	CatAccProduct Version	C (10)	StatementFees	Optional (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
23	CatPIProduct	C (6)	StatementFees	Optional (max : 1)	It is a 6 digit code identifying the payment instrument product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
24	CatPIProduct Version	C (10)	StatementFees	Optional (max : 1)	It is a 10 character fields identifying the payment instrument product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					with product type = P

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell

#	Field Name	Data type	Parent	Occurrence	Field Description
					you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	CardFeesRes	Complex	-	Optional (max : 1)	It refers to card fees calculation response. Requested only in case the operation type is set to CF.
-	FeeAmt	Complex	CardFeesRes	Required (max : 1)	It refers to fee amount that will be charged for the operation outlined in the request.
5	Amt	Long	FeeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
6	CurCode	C (3)	FeeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	CardFees	Complex	CardFeesRes	Optional (max : 1)	Refers to the card fees details in case of the

#	Field Name	Data type	Parent	Occurrence	Field Description
					OperationEvent are 'GS' outlined in the request.
7	BankCode	C (22)	CardFees	Required (max : 1)	The bank code. This field refers to the financial institution code for which the fees code is being defined. This code must be previously defined in the PowerCARD BANK table under the general parameter section Valid values must be retrieved from PowerCARD
8	CardFeesCode	C (15)	CardFees	Required (max : 1)	This code is used to identify the card fees that will be assigned to the card. Valid values must be retrieved from PowerCARD CARD_FEES table
9	CurrencyCode	C (3)	CardFees	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	FirstIssuance	Complex	CardFees	Optional (max : 1)	The First Issuance aggregate include details about the first Issuance fees.
-	SubscFee	Complex	FirstIssuance	Optional (max : 1)	Subscription fee is paid only once for the first issuance of main card. It is not charged on renewals and on supplementary.

#	Field Name	Data type	Parent	Occurrence	Field Description
10	Amt	Long	SubscFee	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	C (3)	SubscFee	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	FeeAmount	Complex	FirstIssuance	Optional (max : 1)	Periodic fees amount to be paid during the first validity period of the card. This amount is also added initially to the subscription fee. This amount is charged at first issuance.
12	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
13	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
14	PlasticFees	Complex	CardFees	Optional (max : 1)	The PlasticFees aggregate include details about the plastic fees.
15	FeeFlag	ClosedEnum	PlasticFees	Optional (max : 1)	Indicates if the plastic fees are to be included in the card fees or are to be generated as a separate transaction. List of values: 0: Fees included in the card fees 1: Fees billed separately
-	FeeAmount	Complex	PlasticFees	Optional (max : 1)	It is possible to enter in the system the cost of the card. This is appreciated in term of accounting since the system will be able to provide the card fees plastic cost excluded. These fees are added to the different amounts each time a card is produced.
16	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n

#	Field Name	Data type	Parent	Occurrence	Field Description
					digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
17	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	PhotoFees	Complex	CardFees	Optional (max : 1)	The Photo Fees aggregate include details about photo fees.
18	FeeFlag	ClosedEnum	PhotoFees	Optional (max : 1)	Indicates if the photo fees are to be included in the card fees or are to be generated as a separate transaction. List of values: 0: Fees included in the card fees 1: Fees billed separately. It is possible to enter the cost of putting a photo on a card. It is also possible to hide the cost to the customer or in the contrary show it on the customer statement.
-	FeeAmount	Complex	PhotoFees	Optional (max : 1)	Photo fees amount of the card. This amount may be added to the different fee amount each time a card is produced.

#	Field Name	Data type	Parent	Occurrence	Field Description
19	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
20	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	RenewalFees	Complex	CardFees	Optional (max : 1)	The Renewal Fees aggregate include details about the renewal fees.
-	RenewAmount	Complex	RenewalFees	Optional (max : 1)	Amount that cardholder pays for the first renewal of the card. At first renewal, the system will charge this amount.
21	Amt	Long	RenewAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
22	CurCode	C (3)	RenewAmount	Optional (max : 1)	Currency code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	SubseqAmount	Complex	RenewalFees	Optional (max : 1)	Amount for subsequent renewals of the card. It is valid for the second renewal and more.
23	Amt	Long	SubseqAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
24	CurCode	C (3)	SubseqAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Replacement Fees	Complex	CardFees	Optional (max : 1)	The Replacement Fees aggregate include details about the replacement fees.
-	LostAmount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card lost case.
-	Amt	Long	LostAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the

#	Field Name	Data type	Parent	Occurrence	Field Description
					currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
25	CurCode	C (3)	LostAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	StolenAmount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card stolen case.
26	Amt	Long	StolenAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
27	CurCode	C (3)	StolenAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Emergency Amount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is

#	Field Name	Data type	Parent	Occurrence	Field Description
					replaced due to a card emergency case.
28	Amt	Long	Emergency Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
29	CurCode	C (3)	Emergency Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Damaged Amount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card damaged case.
30	Amt	Long	Damaged Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
31	CurCode	C (3)	Damaged Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY _TABLE table under the general parameters section.
-	Erroneous Amount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card erroneous case.
32	Amt	Long	Erroneous Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
33	CurCode	C (3)	Erroneous Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Counterfeit Amount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card counterfeit case.
34	Amt	Long	Counterfeit Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
35	CurCode	C (3)	Counterfeit Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	OtherAmount	Complex	Replacement Fees	Optional (max : 1)	Fees amount posted to the cardholder in case the card is replaced due to card undeclared case.
36	Amt	Long	OtherAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
37	CurCode	C (3)	OtherAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	PINFees	Complex	CardFees	Optional (max : 1)	The PIN Fees aggregate include details about the PIN fees.
-	FeeAmount	Complex	PINFees	Optional (max : 1)	Fees amount generated for the replacement of the card PIN. Charges to be applied for PIN calculation only.
38	Amt	Long	FeeAmount	Optional (max : 1)	Amount.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
39	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	PromotionFees	Complex	CardFees	Optional (max : 1)	The Promotion Fees aggregate include details about the promotion fees.
40	StartDate	Date	PromotionFees	Optional (max : 1)	Promotion beginning date.
41	EndDate	Date	PromotionFees	Optional (max : 1)	Promotion end date.
-	FeeAmount	Complex	PromotionFees	Optional (max : 1)	Promotion amount.
42	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
43	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY

#	Field Name	Data type	Parent	Occurrence	Field Description
					_TABLE table under the general parameters section.
-	Transaction FeesRes	Complex	-	Optional (max : 1)	It refers to transaction fees calculation response. Requested only in case the operation type is set to TF.
-	FeeAmount	Complex	Transaction FeesRes	Optional (max : 1)	Fees amount of the transaction.
44	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
45	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
46	Transaction Code	C (20)	Transaction FeesRes	Optional (max : 1)	The transaction code, as defined in the CMS. Valid values must be retrieved from PowerCARD TRANSACTIONS_DEFINITION table
-	StatementFees Res	Complex	-	Optional (max : 1)	It refers to statement fees calculation response. Requested only in case the operation type is set to SF.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	FeeAmount	Complex	Statement FeesRes	Optional (max : 1)	Fees amount of the transaction.
47	Amt	Long	FeeAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
48	CurCode	C (3)	FeeAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
49	Transaction Code	C (20)	Statement FeesRes	Optional (max : 1)	The transaction code, as defined in the CMS. Valid values must be retrieved from PowerCARD TRANSACTIONS_DEFINITION table

Specific error codes

#	Error code	Error Description
1	20201	Invalid Card Number/Token Id

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x

Status Code	Description	Success Response	Error Response
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/FeeInquiryWebService/V2	POST	
/Cards/V2/ Fees? aliasPAN={aliasPAN}&operationEvent={operationEvent}&replaceReason={replaceReason}&catPIProduct={catPIProduct}&catPIProductVer={catPIProductVer} Or /Transactions/V2/Fees? aliasPAN={aliasPAN}&catAccProduct={catAccProduct}&catAccProductVer={catAccProductVer}&transactionCode={transactionCode}&reverFlag={reversalFlag}&catPIProduct={catPIProduct}&catPIProductVer={catPIProductVer}&transAmount={transAmount}&transCurCode={transCurCode} Or /Statements/V2/Fees? aliasPAN={aliasPAN}&accountNumber={accountNumber}&catAccProduct={catAccProduct}&catAccProductVer={catAccProductVer}&catPIProductVer={catPIProductVer}	GET	Optional: aliasPAN=[aliasPAN] operationEvent=[operationEvent] replaceReason=[replaceReason] catPIProduct=[catPIProduct] catPIProductVer=[catPIProductVer] catAccProduct=[catAccProduct] catAccProductVer=[catAccProductVer] transactionCode=[transactionCode] reverFlag=[reversalFlag] transAmount=[transAmount] transCurCode=[transCurCode]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "operationType": "",
  "cardFees": {
    "operationEvent": "",
```

```

    "replacementReason": "",
    "cardNumber": "",
    "catPIProduct": "",
    "catPIProductVersion": ""
  },
  "transactionFees": {
    "cardNumber": "",
    "catAccProduct": "",
    "catAccProductVersion": "",
    "transactionCode": "",
    "reversalFlag": "",
    "catPIProduct": "",
    "catPIProductVersion": ""
  },
  "Transaction Amount": {
    "amt": "",
    "curCode": ""
  },
  "Transaction Amount": {
    "Amt": "",
    "CurCode": ""
  },
  "statementFees": {
    "cardNumber": "",
    "accountNumber": "",
    "catAccProduct": "",
    "catAccProductVersion": "",
    "transactionCode": "",
    "reversalFlag": "",
    "catPIProduct": "",
    "catPIProductVersion": ""
  }
}

```

4.14 updateCustomerPhoneNumber API

API overview

This web service serves to update the phone number of an existing principal customer. The phone numbers entered by the agent or the customer must be formatted properly and could be entered in international format. More details are given in the request specification section.

Functional description

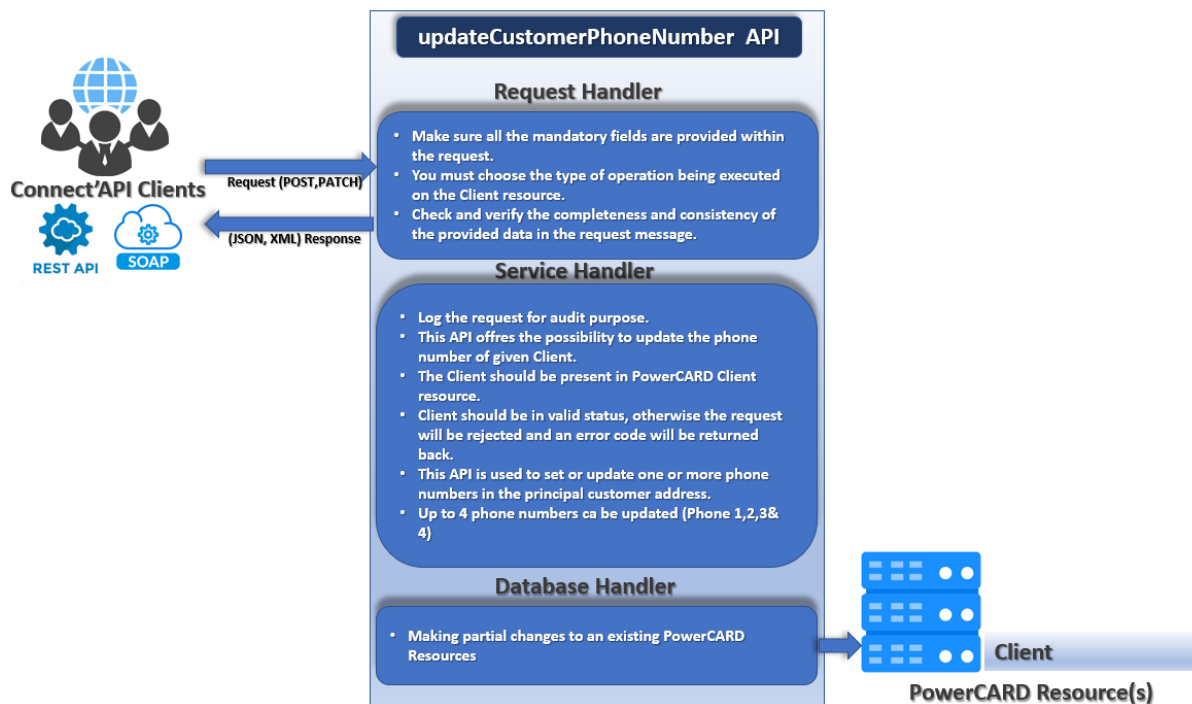


Figure 14: updateCustomerPhoneNumber process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CustomerId	C (22)	-	Required (max : 1)	Used to transmit the customer identification code in PowerCARD. If provided only data belonging to this customer will be returned. If present in the message, this code must be already defied in the PowerCARD CLIENT table
-	PhoneNum	Complex	-	Required infinite list	Phone number list being sent for update in PowerCARD.
5	PhoneType	C (1)	PhoneNum	Optional (max : 1)	Phone type Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
6	Phone	C (30)	PhoneNum	Required (max : 1)	Phone Number Where the contact can be reached at. The phone number

#	Field Name	Data type	Parent	Occurrence	Field Description
					must be formatted correctly based on the country policies. Value must match the following pattern: [+][0-9]*

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system

#	Field Name	Data type	Parent	Occurrence	Field Description
					error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	00801	Invalid Customer Id
2	00802	Invalid Phone Number
3	00803	Invalid Phone type

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested		x

Status Code	Description	Success Response	Error Response
	is not supported for the PowerCARD Resource.		
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/V2	POST	
/Clients/V2/:clientCode/PhoneType/:phoneType/Phone/:phone/PhoneType/:phoneType/Phone/:phone.../PhoneType/:phoneType/Phone/:phone	PATCH	Required clientCode=[C(22)]Optional phoneType=[C(1)] phone=[C(30)]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "customerId": "",
  "phoneNum": {
    "phoneType": "",
    "phone": ""
  }
}
```

4.15 updateCustomerEmail API

API overview

The aim of this web service is to set or update the customer email address part of the principal address. The email address supplied by the agent of the customer himself must be a valid address. It must include both the alias (the part before the '@' sign) and the domain portion (i.e.: @yahoo.com, or @verizon.com, or @gmail.com...).

The request could be rejected if the provided email address fails to comply with what was explained before.

Functional description

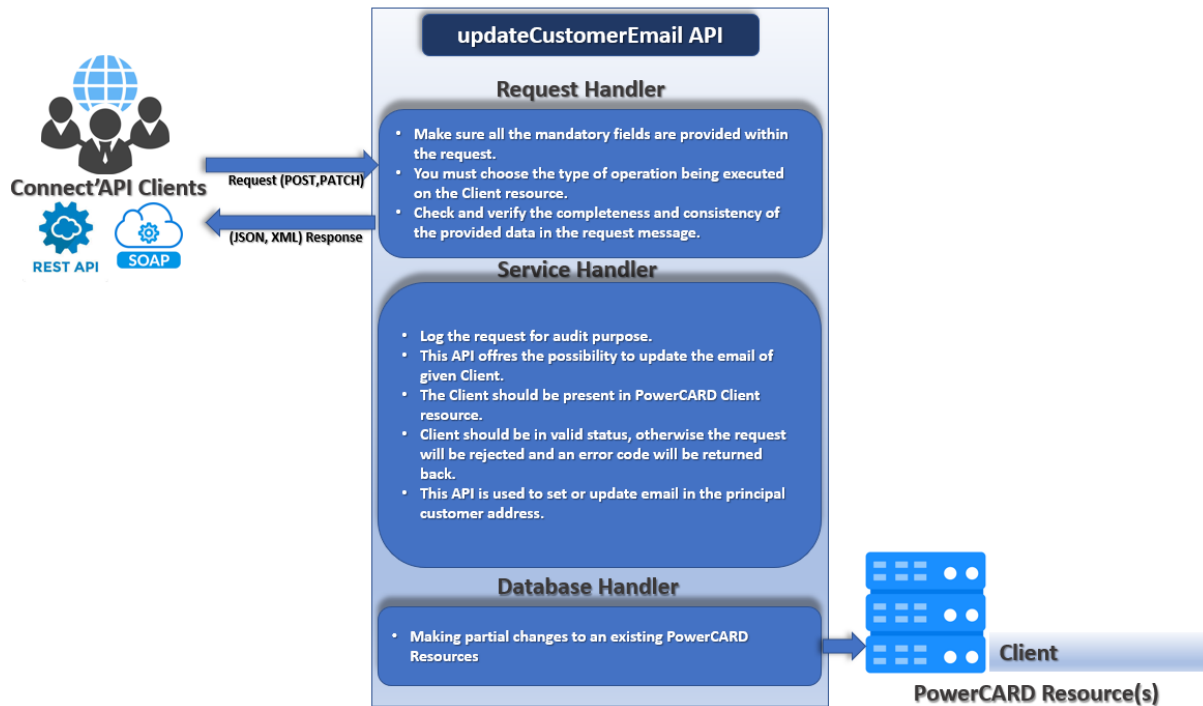


Figure 15: updateCustomerEmail process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX- XXXX-XXXX-XXXX-XXXX XXXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CustomerId	C (22)	-	Required (max : 1)	Used to transmit the customer identification code in PowerCARD. If provided only data belonging to this customer will be returned. If present in the message, this code must be already defied in the PowerCARD CLIENT table
23	Email	C (128)	-	Required (max : 1)	Must contain a valid Email address. It must include two parts (the part before the '@' sign) and the domain portion (i.e.: @yahoo.com, or @verizon.com, or @gmail.com...)

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	02501	Invalid Customer Id
2	02502	Invalid Email address

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerEmailWebService/V2	POST	
/Clients/V2/:clientCode/Email/:email	PATCH	Required clientCode=[C(22)] Optional email=[C(128)]

REST sample call

```

{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "customerId": "",
  "email": ""
}

```

4.16 searchTransactions API

API overview

This web services is used to search for transactions of an existing customer in the system. Agent and customers may use different criteria to search and display transactions posted to the customer account. Criteria's could include amount ranges, dates, Account id, transaction types, network...

Functional description

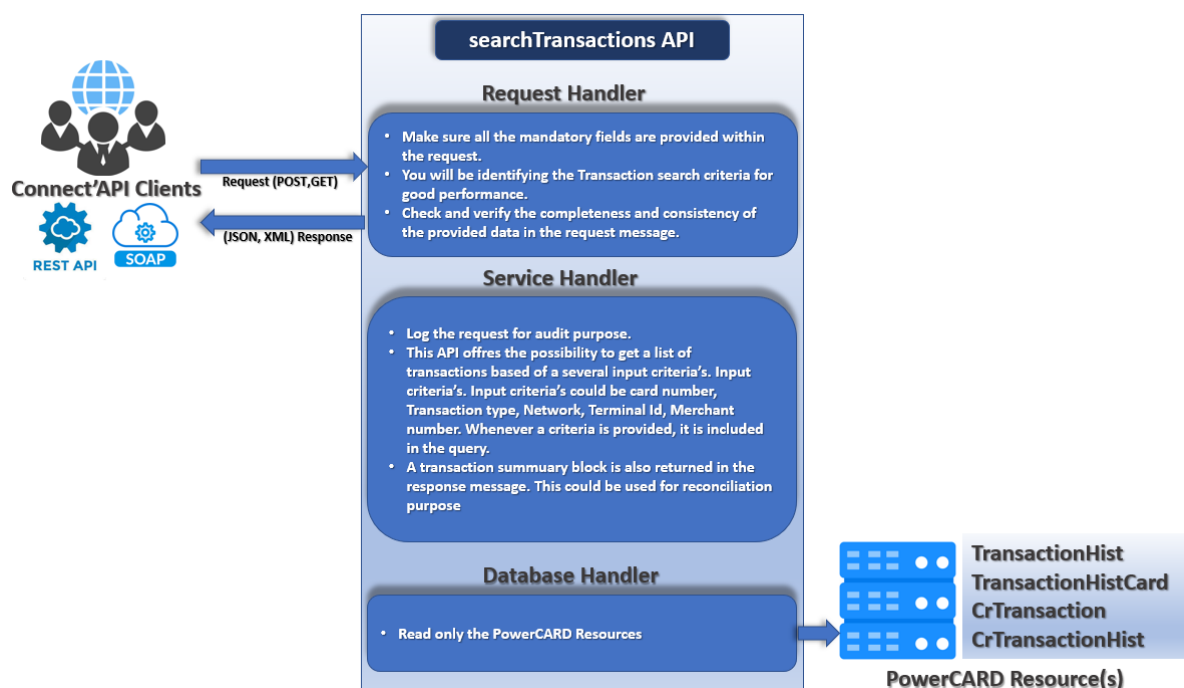


Figure 16: searchTransactions process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Optional (max : 1)	Refers to the PowerCARD card number. This value corresponds to the card to which the transaction is posted.
5	ClientCode	C (24)	-	Optional (max : 1)	Represent the customer identifier.
-	RangeDate	Complex	-	Required (max : 1)	This aggregate is mandatory and should contain the range of the dates when the

#	Field Name	Data type	Parent	Occurrence	Field Description
					transactions happened. The aggregate is constituted of two items (Start Date & End Date)
6	StartDt	DateTime	RangeDate	Optional (max : 1)	Start date
7	EndDt	DateTime	RangeDate	Optional (max : 1)	End date
-	AmountFrom	Complex	-	Optional (max : 1)	The requester is allowed to specify the minimum amount of the transactions made by the customer. This aggregate contains the amount as well as the transaction currency.
8	Amt	Long	AmountFrom	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
9	CurCode	C (3)	AmountFrom	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	AmountTo	Complex	-	Optional (max : 1)	Like for the minimum amount, a maximum amount could be specified in the request message. If present, only transactions with amounts less or

#	Field Name	Data type	Parent	Occurrence	Field Description
					equal to that amount will be returned in the result.
10	Amt	Long	AmountTo	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	C (3)	AmountTo	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved.
12	TrnType	C (3)	-	Optional (max : 1)	This field refers to the grouping criteria of transactions codes that should be returned. If empty, any transaction can be returned. Note that PowerCARD transactions can be grouped in different categories for reporting purpose. This code must be previously defined in the PowerCARD TRANSACTIONS_DEFINITION table under the general parameter section.

#	Field Name	Data type	Parent	Occurrence	Field Description
13	Corporateld	C (15)	-	Optional (max : 1)	Refers to the corporate identification. Valid values must be populated from the PowerCARD CORPORATE table under the cardholder section
14	Network	C (2)	-	Optional (max : 1)	The network field allows you to specify the network where transactions are made. It is to mention that this code should be previously defined in the NETWORK table under the general parameter section.
15	Territoriality	ClosedEnum	-	Optional (max : 1)	Valid values : • ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. • NAT: National, apply only to transactions made in the same country as the issuer country. • INT: International, apply only to transactions that took place outside the cardholder country.
16	Acquirer Country	C (3)	-	Optional (max : 1)	Refers to the ISO code of the country where the transaction took place. It is 3 alphanumeric characters

#	Field Name	Data type	Parent	Occurrence	Field Description
					The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing.

#	Field Name	Data type	Parent	Occurrence	Field Description
					SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	MaxRowNbr	Long	-	Optional (max : 1)	This will depict the maximum number of row that can be included in the response message. Note that if the result exceeds this number, only this number of cards will be returned. This mechanism is used to limit the number of cards that can be fetched in one call.
-	Transaction	Complex	-	Optional infinite list	The transaction aggregate (list). Must contain the list of transactions that meet the different criteria outlined in the request.
-	transactionCommon	Complex	Transaction	Required (max : 1)	This aggregate contains data which may be used in both contexts (Issuing and acquiring). For instance the transaction date, Amount, Card number, VAT Amount... Present all the time.

#	Field Name	Data type	Parent	Occurrence	Field Description
6	TransId	C (23)	transactionCommon	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
7	RefSeq	N (4)	transactionCommon	Required (max : 1)	Refers to the transaction sequence Number
-	TrnType	Complex	transactionCommon	Required (max : 1)	This aggregate must contain the transaction code as defined in PowerCARD TRANSACTIONS_DEFINITION table as well as the description of the transaction.
8	Code	C (20)	TrnType	Required (max : 1)	Must contain the code of the parameter being shared
9	Label	C (52)	TrnType	Required (max : 1)	The label of the parameter being shared
10	TransSign	C (1)	transactionCommon	Required (max : 1)	Transaction sign Contains a code that indicates the sign of the transaction. Possible values are: • D: Debit • C: Credit
11	TransDate	DateTime	transactionCommon	Required (max : 1)	The date when the transaction was made.
12	TransPostDate	Date	transactionCommon	Required (max : 1)	Refers to the date when the concerned transaction was posted to the cardholder account
13	TransDateEvent	DateTime	transactionCommon	Optional (max : 1)	The date when the transaction was proceed by the card management system.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	TransAmt	Complex	transactionCommon	Required (max : 1)	Identified the transaction amount in the currency used during the transaction. The entry must be a numeric greater than zero.
14	Amt	Long	TransAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
15	CurCode	C (3)	TransAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Cashback	Complex	transactionCommon	Optional (max : 1)	The cash back amount of a transaction.
16	Amt	Long	Cashback	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
17	CurCode	C (3)	Cashback	Optional (max : 1)	Currency code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Authorization	Complex	transactionCommon	Optional (max : 1)	Authorization aggregate Contains the Authorization code and date. Mandatory only in case an issuer, an authorizing processor, or Stand-In Processing (STIP) provides an approval of a transaction.
18	AuthCode	C (6)	Authorization	Optional (max : 1)	Authorization code (approval code of the transaction), (Field 38) It refers to a code that an issuer, its authorizing processor, or Stand-In Processing (STIP) provides to indicate approval of a transaction. The code is returned in the authorization Response and is usually recorded on the Transaction Receipt. The field must contain a six-position Authorization Code. Allowed entries are: • Spaces • A through Z • 0 through 9
19	AuthDate	Date	Authorization	Optional (max : 1)	Authorization date The actual date that the request for the

#	Field Name	Data type	Parent	Occurrence	Field Description
					authorization was made.
20	ReferenceNumber	C (12)	Authorization	Optional (max : 1)	Contains the reference number as defined in ISO-8583 norm (Field 37)
21	PosData	C (12)	Authorization	Optional (max : 1)	POS Data as defined in ISO-8583 norm (Field 60)
-	InterchangeAmt	Complex	transactionCommon	Optional (max : 1)	The interchange fee amount and currency aggregate.
22	Amt	Long	InterchangeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
23	CurCode	C (3)	InterchangeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
24	ConversionRate	C (3)	transactionCommon	Optional (max : 1)	Rate of conversion.
25	CardNumber	C (22)	transactionCommon	Optional (max : 1)	The card number used during the transaction.
-	Country	Complex	transactionCommon	Required (max : 1)	Country where the transaction took place.
26	CountryCode	NC (3)	Country	Required (max : 1)	Identify the country code ISO-3166 1-numeric

#	Field Name	Data type	Parent	Occurrence	Field Description
27	CountryName	C (250)	Country	Optional (max : 1)	Identify the country name Both the country code and the country name are downloaded from the PowerCARD COUNTRY table present under the general parameter section.
-	transactionAcquiring	Complex	Transaction	Optional (max : 1)	Data which are more related to the acquiring context are shared in this aggregate. For instance, Discount Amount, Tips, POS Entry mode,
28	Network	ClosedEnum	transactionAcquiring	Required (max : 1)	The network field allows you to specify the network where transactions are made. It is to mention that this code should be previously defined in the NETWORK table under the general parameter section.
-	MCC	Complex	transactionAcquiring	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the PowerCARD MCC table
29	MccCode	C (4)	MCC	Optional (max : 1)	MCC code or Merchant category code. This code must be previously defined in the PowerCARD MCC table under the general parameter section.

#	Field Name	Data type	Parent	Occurrence	Field Description	
30	MccLabel	C (100)	MCC	Optional (max : 1)	MCC description as stored in PowerCARD.	
-	TrnStatus	Complex	transactionAcquiring	Optional (max : 1)	Transaction status, possible values are:	
					Code	Label
					LTR	Loaded transaction
					RTR	Rejected transaction
					TTR	Transaction to be settled
					STR	Transaction settled
					BTR	Blocked transaction
					CTR	Canceled transaction
					PTR	Processed transaction
					VTR	Pending for validation
31	Code	C (20)	TrnStatus	Required (max : 1)	Must contain the code of the parameter being shared	
32	Label	C (52)	TrnStatus	Required (max : 1)	The label of the parameter being shared	
33	TipsAmt	Complex	transactionAcquiring	Optional (max : 1)	This field contains the tips amount.	
34	Amt	Long	TipsAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10n; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.	

#	Field Name	Data type	Parent	Occurrence	Field Description
35	CurCode	C (3)	TipsAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	DiscountAmt	Complex	transactionAcquiring	Required (max : 1)	This field contains the discount amount of a transaction.
36	Amt	Long	DiscountAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
37	CurCode	C (3)	DiscountAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD using the Web Service: getParameterList
38	TaxesAmt	Complex	transactionAcquiring	Optional (max : 1)	This field provides the cumulated amount of tax for the concerned transaction.
39	Amt	Long	TaxesAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent

#	Field Name	Data type	Parent	Occurrence	Field Description
					of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
40	CurCode	C (3)	TaxesAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
41	LinkupCode	Long	transactionAcquiring	Required (max : 1)	A linkup code is an identification used to group transactions within a remittance or a batch. For instance PowerCARD may allocate a different linkup code to each remittance type (Pos, Manal, ATM, E Commerce...)
42	BatchNumber	Long	transactionAcquiring	Required (max : 1)	A batch is a collection of transactions, usually a single day's worth. Batches can be set at the terminal level as manual or automatic to send at a certain time during the day. This number can be used to group transactions.
43	RemittanceSequence	Long	transactionAcquiring	Required (max : 1)	Refers to the remittance Identification number as stored internally in PowerCARD.
44	PosEntryMode	ClosedEnum	transactionAcquiring	Required (max : 1)	Field indicating the method by which a point-of-transaction

#	Field Name	Data type	Parent	Occurrence	Field Description
					terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : • 00: Unknown or terminal not used • 01: Manual (key entry) • 02: Magnetic stripe read • 03: Integrated circuit card read • 04: Contactless device-read
45	TransRejectCode	C (6)	transactionAcquiring	Optional infinite list	Mandatory if the transaction was rejected. This field refers to the rejection reason.
46	ReversalInd	C (1)	transactionAcquiring	Optional (max : 1)	It indicates whether the transaction is reversed or not. Valid values are: • 0: Not reversed • 1: Fully reversed • 2: Partially reversed
47	MerchantAcronym	C (40)	transactionAcquiring	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or member's imprinter plate.
48	IndicatorStatus	ClosedEnum	transactionAcquiring	Required (max : 1)	This indicator is used to specify the whether the transaction was subject to a reversal,

#	Field Name	Data type	Parent	Occurrence	Field Description	
					chargeback, retrieval required... Possible values are as follow:	
					#	Value
					0	Normal
					1	Reversed
					2	Subject to retrieval request
					3	Subject to retrieval request cancellation
					4	Subject retrieval request response
					5	Subject to retrieval request response reversal
					6	Subject to a chargeback
					7	Subject to a chargeback reversal
					8	Subject to a second presentment
					9	Subject to a second presentment reversal
					A	Subject to a second chargeback
					B	Subject to a second chargeback reversal
					C	Subject to a third chargeback
					D	Subject to a third

#	Field Name	Data type	Parent	Occurrence	Field Description
					chargeback reversal E Subject to a third presentment F Subject to a third presentment reversal
49	SettlementDate	DateTime	transactionAcquiring	Required (max : 1)	The merchant settlement date, date when the transaction has been paid to the merchant
-	transactionIssuing	Complex	Transaction	Optional (max : 1)	Data which are more related to the issuing context are shared in this aggregate. Note that the transaction aggregate must contain the transactionCommon aggregate and one of the two other aggregate (transactionAcquiring or transactionIssuing)
50	Description	C (40)	transactionIssuing	Optional (max : 1)	Short description of the merchant where the transaction occurred.
51	Territoriality	ClosedEnum	transactionIssuing	Optional (max : 1)	Valid values : - ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. - NAT: National, apply only to transactions made in the same country

#	Field Name	Data type	Parent	Occurrence	Field Description
					as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
-	BillCurAmt	Complex	transactionIssuing	Optional (max : 1)	This aggregate will contain the billing amount in the issuer's billing currency.
52	Amt	Long ()	BillCurAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
53	CurCode	NC (3)	BillCurAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
54	TransAccountId	C (24)	transactionIssuing	Optional (max : 1)	The Account Number where the transaction is posted.
55	ClearingTrxAmt	Complex	transactionIssuing	Optional (max : 1)	This aggregate is used to identify the cleared amount of a transaction.
56	Amt	Long ()	ClearingTrxAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent

#	Field Name	Data type	Parent	Occurrence	Field Description
					of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
57	CurCode	NC (3)	ClearingTrxAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	TransactionTotal	Complex	-	Optional infinite list	This aggregate contains a summary of the transactions returned above. It might be used for reconciliation.
58	NbTransactions	Long	TransactionTotal	Required (max : 1)	Indicates the number of transactions returned.
-	TotalCredit	Complex	TransactionTotal	Optional (max : 1)	Refers to the sum total of transaction amounts of transactions with Credit sign.
59	Amt	Long ()	TotalCredit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
60	CurCode	NC (3)	TotalCredit	Optional (max : 1)	Currency code. ISO currency code of the transaction

#	Field Name	Data type	Parent	Occurrence	Field Description
					amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	TotalDebit	Complex	TransactionTotal	Optional (max : 1)	Refers to the sum total of transaction amounts of transactions with debit sign.
61	Amt	Long ()	TotalDebit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
62	CurCode	C (3)	TotalDebit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

Specific error codes

#	Error code	Error Description
1	02701	Invalid Card Number
2	02702	Invalid Range Date
3	02703	Invalid Amount Criteria
4	02704	Invalid Transaction Type
5	02705	Invalid Network Code
6	02706	Invalid Acquirer country code
7	02707	Invalid card product
8	02708	Label Translation Error

#	Error code	Error Description
9	02709	Internal error - error translating the transaction type to processing code
10	02710	Internal error - error translating the processing code to transaction type
11	02711	Invalid client code
12	02712	Card number and Client Code are missing in the request
13	02713	Invalid Territoriality
14	02714	Invalid Corporate Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consulting a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchTransactionsWebService/V2	POST	
/Transactions/V2?clientCode={clientCode}&aliasPAN={aliasPAN}&corporateld={corporateld}&network={network} &	GET	clientCode=[C(24)] cardNumber=[C(22)]

PowerCARD Resource URI	Method	URI Params
territoriality={territoriality}&acquirerCountry={acquirerCountry}&trnType={trnType}&startDt={startDt}&endDt={endDt}&pageSize={pageSize}&pageIndex={pageIndex}		corporatId=[C(15)] network=[C(3)] territoriality=[C(3)] acquirerCountry=[C(3)] trnType=[C(80)] startDt=[Date] endDt=[Date] pageSize=[PositiveInteger] pageIndex=[PositiveInteger]

REST sample call

```
{
  "requestInfo": {
    "requestUID": " ",
    "requestDate": "",
    "userID": ""
  },
  "CardNumber": "",
  "ClientCode": "",
  "RangeDate": {
    "StartDt": "",
    "EndDt": ""
  },
  "AmountFrom": {
    "Amt": "",
    "CurCode": ""
  },
  "AmountTo": {
    "Amt": "",
    "CurCode": ""
  },
  "TrnType": "",
  "CorporateId": "",
  "Network": "",
  "Territoriality": "",
  "AcquirerCountry": ""
}
```

4.17 updateBankAccount API

API overview

This web services is aimed to update bank account information. Agent as well as customers can ask to add, edit or delete one or multiple bank account in the system. It is important to note that the web service requires either the card number or the credit account number for which the adjustment will be performed. As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code and description for more details.

Functional description

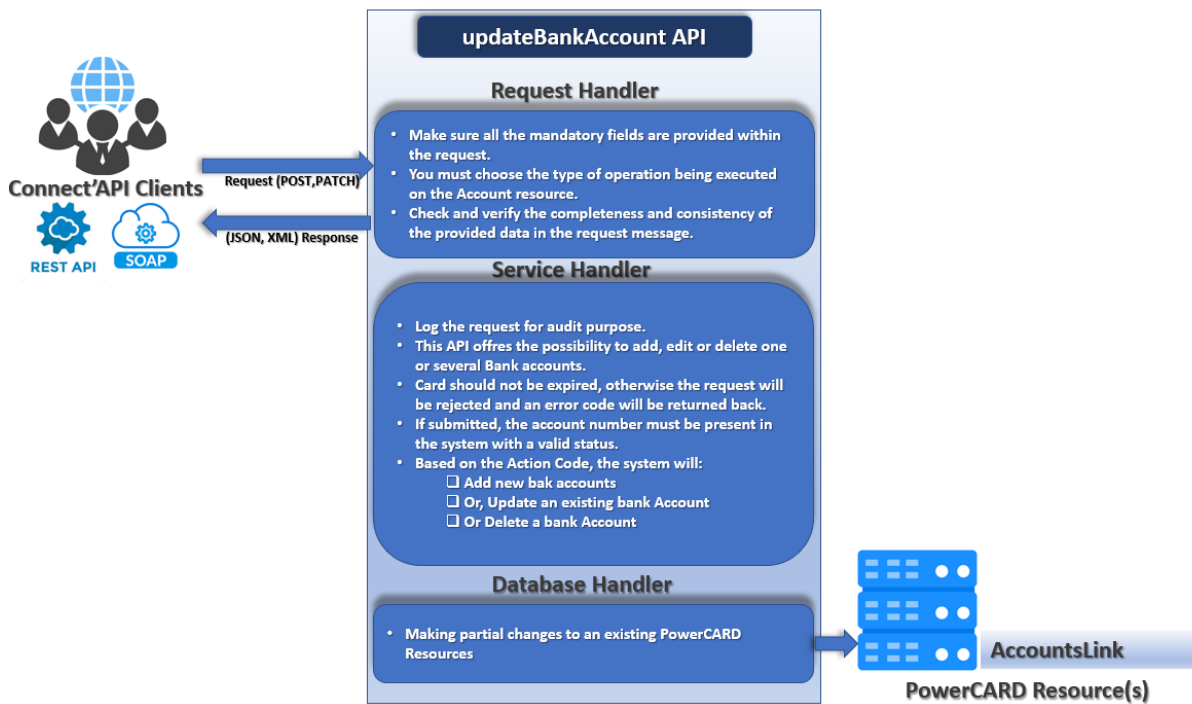


Figure 17: updateBankAccount process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	ActionCode	ClosedEnum	-	Required (max : 1)	Refers to the desired action, possible values are: • A: Add Bank Account • U: Update Bank Account • D: Delete Bank Account
-	BankAcctRec	Complex	-	Required infinite list	Should contain the concerned bank account records.
5	AccountNumber	C (40)	BankAcctRec	Required (max : 1)	It refers to the bank account number part of the IBAN.
6	IBAN	C (34)	BankAcctRec	Optional (max : 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty five characters for the bank account number.
7	AcctCcy	C (3)	BankAcctRec	Required (max : 1)	Currency code as defined by ISO-4217 Possible values must be downloaded from the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY_TABLE table
8	AcctStatus	ClosedEnum	BankAcctRec	Required (max : 1)	Account Status Refers to the status of the bank Account. Valid values are: • N = Normal • C = Canceled • S = Suspended • W = Waiting for validation • V = To be sent for validation • R = Rejected.
-	AcctAutRangeDt	Complex	BankAcctRec	Optional (max : 1)	Account Authorization start/ end date Refers to the opening/closing date of the bank account.
9	StartDt	DateTime	AcctAutRangeDt	Optional (max : 1)	Start date
10	EndDt	DateTime	AcctAutRangeDt	Optional (max : 1)	End date
11	AcctBankCode	C (6)	BankAcctRec	Required (max : 1)	Code identifying the bank where the account is opened. It is a 6 digit code must be stored in the PowerCARD BANK table under the general parameter section.
12	AcctBankAgency	C (15)	BankAcctRec	Required (max : 1)	Code of the branch where the account is opened. This code must be stored in the PowerCARD BRANCH table under the general parameter section.
13	AcctType	C (40)	BankAcctRec	Required (max : 1)	Account type Refers to the type of the account. Possible values can be downloaded from the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					ACCOUNT _TYPE_LIST
14	CardNumber	C (22)	-	Optional (max : 1)	Card Number This field must hold the card number to which, 1) the bank account will be added, 2) the bank account information ned to be updated, or 3) the bank account must be deleted.
15	AccountId	C (24)	-	Optional (max : 1)	Account Number This field must hold the credit account number to which, 1) the bank account will be added, 2) the bank account information ned to be updated, or 3) the bank account must be deleted. Note that, one of the two fields, Card number and Account number is mandatory.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	02901	Invalid Currency Code
2	02902	Invalid Action Code
3	02903	Invalid Card Number
4	02904	Invalid Bank Code
5	02905	Invalid Account number
6	02906	Invalid card number or account id
7	02907	Invalid account status

#	Error code	Error Description
8	02908	Invalid account Type
9	02909	Error Adding new account
10	02910	Error while updating account
11	02911	Error updating account status
12	02912	Invalid Account Bank Agency

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateBankAccountWebService/V2	POST	
/BankAccount/V2/	PATCH	

REST sample call

```
{
  "requestInfo": {
    "requestUID": " ",

```

```

    "requestDate": "",
    "userID": ""
  },
  "ActionCode": "",
  "BankAcctRec": [
    {
      "AccountNumber": "",
      "IBAN": "",
      "AcctCcy": "",
      "AcctStatus": "",
      "AcctAutRangeDt": {
        "StartDt": "",
        "EndDt": ""
      },
      "AcctBankCode": "",
      "AcctBankAgency": "",
      "AcctType": ""
    }
  ],
  "CardNumber": "",
  "AccountId": ""
}

```

4.18 createAccountTransaction API

API overview

Users (Agents or bank customers) can use this web service to create and post a transaction to an existing account in the system.

For every single transaction that the user would like to create, he needs to specify at least the following information in the request message: Card Number or Account number, Transaction code, Amount, Date. Transaction is then created in the system but will wait for the End of day processing to be completely posted to the account.

We should note that both debit and credit transactions could be created using this web service.

Functional description

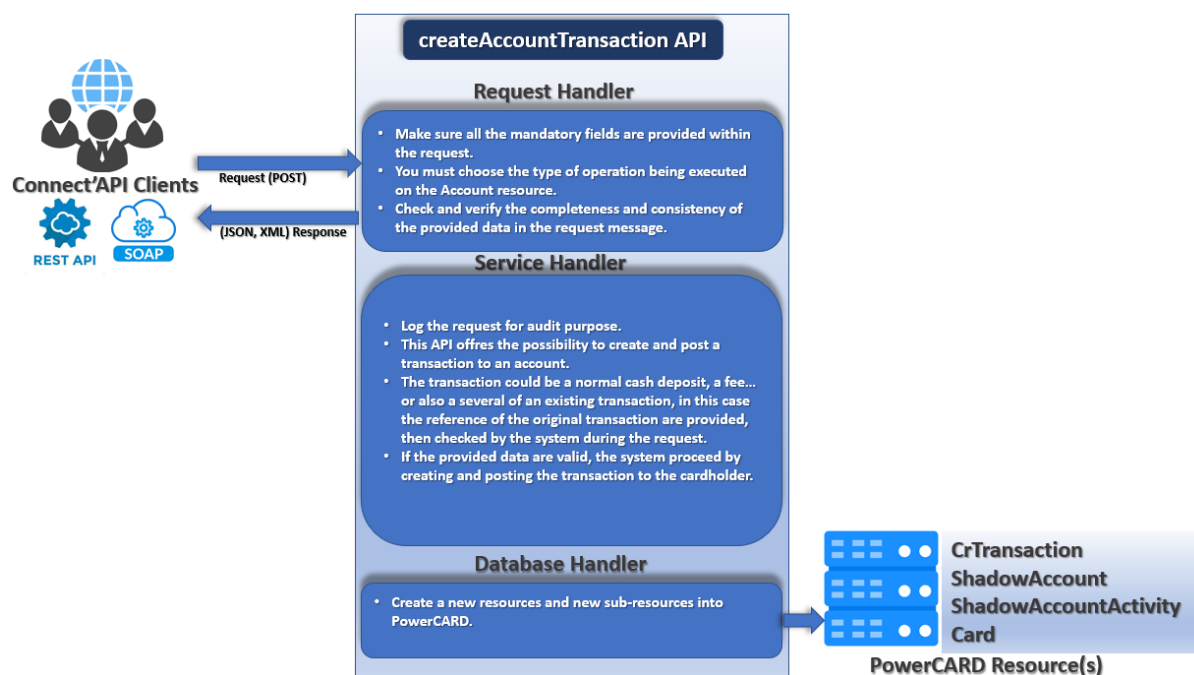


Figure 18: createAccountTransaction process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
-	AccountTransaction	Complex	-	Required (max : 1)	Should include data if the transaction that is going to be created or cancelled. More details are shared in the aggregate definition.
4	AccountId	C (40)	Account Transaction	Required (max : 1)	The account identifier: It refers

#	Field Name	Data type	Parent	Occurrence	Field Description
					to the internal account number. This value should be already defined in the PowerCARD SHADOW_ACCOUNT table. This field is mandatory if the card number is not provided.
5	CardNumber	C (22)	Account Transaction	Optional (max : 1)	The card number. Must contain the primary card number of the customer. The number should be valid and previously present in the PowerCARD CARD table under the cardholder section. This field becomes mandatory when the shadow account number is not provided.
6	TransactionCode	C (2)	Account Transaction	Required (max : 1)	The transaction code, as defined in the CMS. Valid values must be retrieved from PowerCARD TRANSACTIONS_DEFINITION table
7	ReversalFlag	ClosedEnum	Account Transaction	Required (max : 1)	Indicates whether it is a normal or reversal transaction. - N: Normal. - R: Reversal.
8	CaptureReason Code	ClosedEnum	Account Transaction	Required (max : 1)	This field holds the transaction capture reason code, or transaction reversal reason code. Values assigned to this field must be defined in the TRANS_REASON_CODE table under the PowerCARD general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	TransactionAmount	Complex	Account Transaction	Required (max : 1)	The transaction amount and currency. Amount and currency of the transaction that will be posted to the customer account.
9	Amt	Long ()	Transaction Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
10	CurCode	NC (3)	Transaction Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
11	TransactionDate	DateTime	Account Transaction	Required (max : 1)	The transaction date. Date and time when the transaction took place.
12	OriginalRef Number	C (23)	Account Transaction	Optional (max : 1)	Contains the original Acquirer Reference Number of the transaction being reversed (Original debit, Or Original Credit). This field is mandatory when the reversal flag is set to 'R'. (Reversal).
13	OriginalRefSeq	C (4)	Account Transaction	Optional (max : 1)	This field should contain the original ARN sequence of

#	Field Name	Data type	Parent	Occurrence	Field Description
					the transaction being reversed (Original debit, Or Original Credit). This field is mandatory when the reversal flag is set to 'R'. (Reversal).
14	ValueDate	DateTime	Account Transaction	Optional (max : 1)	The transaction value date. Date when the transaction is posted to the shadow account.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue

#	Field Name	Data type	Parent	Occurrence	Field Description
					for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	11001	Invalid Account Number
2	11002	Invalid Account Status
3	11003	Invalid Transaction Code
4	11004	Invalid Transaction Amount
5	11005	Invalid Transaction date
6	11006	Invalid Card number
7	11007	Card number and Account number are both missing
8	11008	Card account link not found
9	11009	Multiple Card account links found
10	11010	Invalid Reversal Flag
11	11011	Original transaction not found
12	11012	Invalid value date
13	11013	Invalid Reason Code
14	11014	Invalid Currency Code

RESTful response codes

Status Code	Description	Success Response	Error Response
201 Created	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/CreateAccountTransactionWebService/V2	POST	
/AccountTransaction/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": " ",
    "requestDate": "",
    "userID": ""
  },
  "accountTransaction": {
    "accountId": "",
    "cardNumber": "",
    "transactionCode": "",
    "reversalFlag": "",
    "captureReasonCode": "",
    "transactionAmount": {
      "amt": "",

```

```

    "curCode": ""
  },
  "transactionDate": "",
  "originalRefNumber": "",
  "originalRefSeq": "",
  "valueDate": ""
}
}

```

4.19 createLoyaltyAccount API

API overview

For any customer to whom the financial institution is aiming to create a loyalty account, users can use this web service to proceed with first the creation, then the linking of the loyalty account to that customer. The service requires to specify the loyalty program subscribed by the customer. As soon as the account is created in the system, and customers making purchases, the customer will start collecting loyalty points based on the setup of the chosen loyalty program.

Functional description

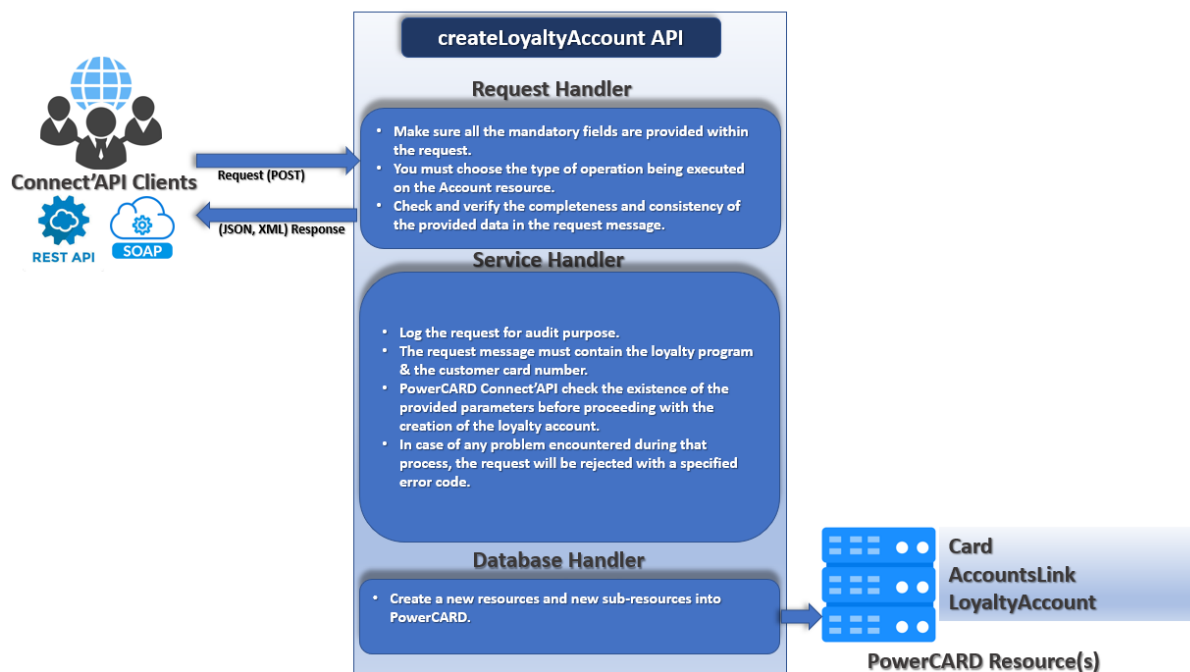


Figure 19: createLoyaltyAccount process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all

#	Field Name	Data type	Parent	Occurrence	Field Description
					the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX-XXXX-XXXX -XXXX XXXXXXXX Example: 6948DF80-14BD-4E04-8842-7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	BankCode	C(6)	-	Required (max : 1)	Represent the bank for the account owner.
5	AccountOwner Type	C (2)	-	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: - CH = Cardholder. - SA = Shadow Account The value transmitted in the next field is based on the value of that one.

#	Field Name	Data type	Parent	Occurrence	Field Description
6	AccountOwnerId	C (24)	-	Required (max : 1)	It refers to the identification of the entity that owns the loyalty account. Based on the value in the previous field, this field may contain : The shadow account (SA) The card number (CH)
7	CatalogLoyalty Product	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loyalty product code. Valid values must be retrieved from PowerCARD CATALOGUE_PRODUCT table with product type = Y
8	CatalogLoyalty ProductVersion	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loyalty product version. Valid values must be retrieved from PowerCARD CATALOGUE_PRODUCT table with product type = Y

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one

#	Field Name	Data type	Parent	Occurrence	Field Description
					sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	LoyaltyAccount Nbr	C (22)	-	Required (max : 1)	This field refers to a unique internal identification of the loyalty account linked to the card, channel.... Returned back in the response only if the

#	Field Name	Data type	Parent	Occurrence	Field Description
					response code is Succes.

Specific error codes

#	Error code	Error Description
1	13001	Invalid Account Owner Type
2	13002	Invalid Account Owner Id
3	13003	Invalid Loyalty Product
	13004	Invalid Bank Code

RESTful response codes

Status Code	Description	Success Response	Error Response
201 Created	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ CreateAccountTransactionWebService/V2	POST	

PowerCARD Resource URI	Method	URI Params
/LoyaltyAccount/V2	POST	

REST sample call

```
{
  "keyValues": {
    "key": "",
    "data": "",
    "type": ""
  },
  "requestInfo": {
    "requestUID": "",
    "userID": "",
    "requestDate": ""
  },
  "bankCode": "",
  "accountOwnerType": "",
  "accountOwnerId": "",
  "catalogLoyaltyProduct": "",
  "catalogLoyaltyProductVersion": ""
}
```

4.20 getLoyaltyAccounts API

API overview

In loyalty programs, for each dollar amount spent, the cardholder can earn a number of points. He can for instance use those points as credit against future purchases. This web service was introduced to help customers as well as agent to display all the loyalty accounts linked to a given card. It also helps checking their loyalty account balance. The response will include for reach loyalty account, the name of the subscribed loyalty program, the number of outstanding points in the account, and some more details. Needless to mention that, the loyalty balance takes into consideration the eared points and also the redeemed ones.

Functional description

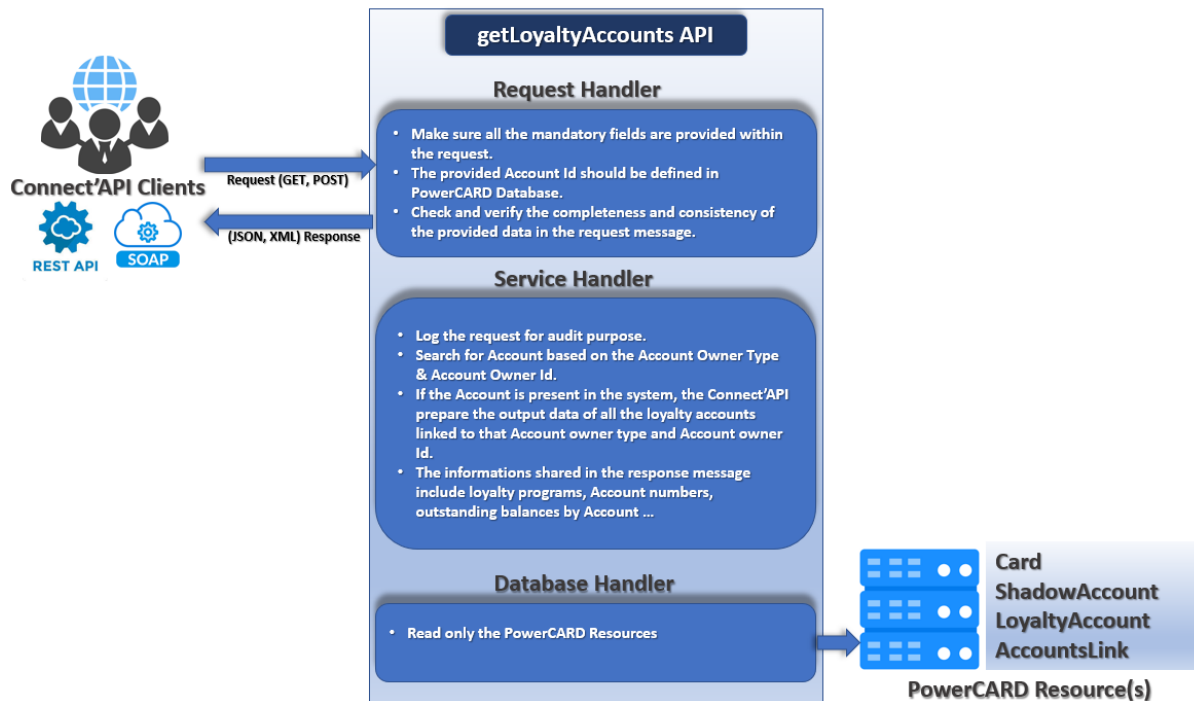


Figure 20: getLoyaltyAccounts process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
8	BankCode	C(6)	-	Required (max : 1)	Represent the bankcode of the account owner.
9	AccountOwner Type	C (2)	-	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: • CH = Cardholder. • SA = Shadow Account The value transmitted in the next field is based on the value of that one.
10	AccountOwner Id	C(22)	-	Required (max : 1)	It refers to the identification of the entity that owns the loyalty account. Based on the value in the previous field, this field may contain : The shadow account (SA) The card number (CH)

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the

#	Field Name	Data type	Parent	Occurrence	Field Description
					description of the error encountered
-	AccountLoyalty	Complex	-	Optional (Infinite list)	This aggregate must contain the list of loyalty accounts linked to the card.
5	BankCode	C(6)	-	Required (max : 1)	Represents the loyalty account owner bank code.
6	Loyalty AccountNbr	C (24)	AccountLoyalty	Required (max : 1)	This field refers to a unique internal identification of the loyalty account linked to the card, channel....
7	CatalogLoyalty Product	C (6)	AccountLoyalty	Required (max : 1)	It is a 6 digit code identifying the Loyalty product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = Y
8	CatalogLoyalty ProductVersion	C (10)	AccountLoyalty	Required (max : 1)	It is a 10 character fields identifying the loyalty product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = Y
9	AccountOwner Type	C (2)	AccountLoyalty	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: • CH = Cardholder. • SA = Shadow Account The value transmitted in the next field is based

#	Field Name	Data type	Parent	Occurrence	Field Description
					on the value of that one.
10	AccountOwner Id	C (24)	AccountLoyalty	Required (max : 1)	It refers to the identification of the entity that owns the loyalty account. Based on the value in the previous field, this field may contain : The card number (CH) The shadow account number (SA) At the time being only the first type is managed in this aggregate. Other types are reserved for future use.
11	BalancePoint	Decimal (18,3)	AccountLoyalty	Required (max : 1)	This field should contain the outstanding loyalty points earned by the customer.
12	LastTransaction Date	Date	AccountLoyalty	Optional (max : 1)	Indicates the date of the last transaction that earned a point in the system.
13	LastRedemption Date	Date	AccountLoyalty	Optional (max : 1)	Indicates the date of the last transaction used to redeem points from the loyalty account.
14	Status	C(1)	AccountLoyalty	Optional (max : 1)	Represents the status of the loyalty account.

Specific error codes

#	Error code	Error Description
1	12801	Invalid Account Owner Type
2	12802	Invalid Account Owner Id
3	12803	No loyalty account defined for this entity
	12804	Invalid Bank Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCardDetailsWebService/V2	POST	
/LoyaltyAccounts/V2? bankCode={bankCode}& accountOwnerType={accountOwnerType} & accountOwnerId={accountOwnerId}	GET	Required: bankCode=[C(6)] accountOwnerType=[C(2)] accountOwnerId=[C(2)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "bankCode": "",
  "accountOwnerType": "",
  "accountOwnerId": ""
}
```

4.21 getAccountBalance API

API overview

For an existing card or account in the system, Agent as well as customers may request to have more details regarding card account balances (Available & Pending). All these information could be made available for that purpose using this web service.

Functional description

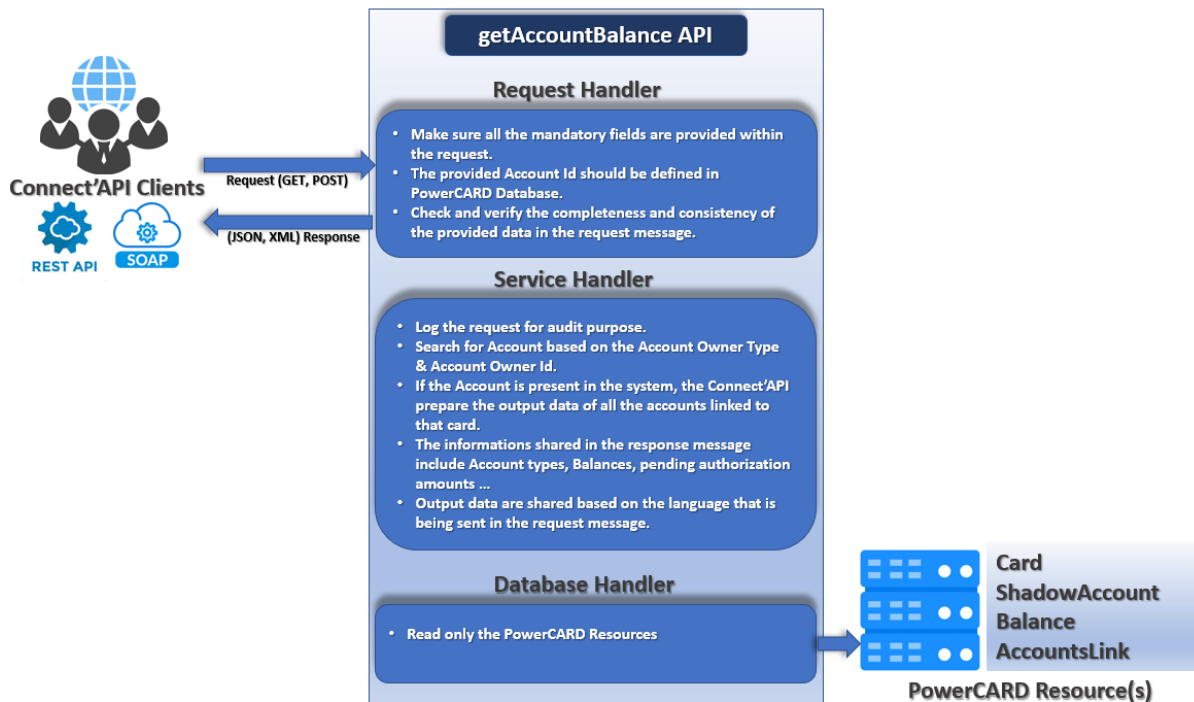


Figure 21: getAccountBalance process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to

#	Field Name	Data type	Parent	Occurrence	Field Description
					match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Optional (max : 1)	The card number If present in the message, this card number must be present in PowerCARD cardholder database.
5	ClientCode	C(24)	-	Optional (max : 1)	The client code present in the table CLIENT in PowerCARD.
6	AccountId	C (24)	-	Optional (max : 1)	The Account identifier as defined in PowerCard
7	AccountType	C (40)	-	Optional (max : 1)	Account type Refers to the type of the account. Possible values can be downloaded from the PowerCARD ACCOUNT _TYPE_LIST table.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	AccountBalance	Complex	-	Optional infinite list	This aggregate contains data related to the balance of the accounts linked to the card transmitted in the request message.
5	CMSAccountId	C (24)	AccountBalance	Required (max : 1)	The CMS account identifier as defined in PowerCARD.
6	AccountType	C (40)	AccountBalance	Required (max : 1)	Account type Refers to the type of the account. Possible values can be downloaded from the PowerCARD ACCOUNT _TYPE _LIST table
7	ClientCode	C(24)	AccountBalance	Optional (max : 1)	The client code present in the table CLIENT in PowerCARD.
8	ClientHostID	C(24)	AccountBalance	Optional (max : 1)	External Identified of the customer in the host
-	CrdAcctInfo	Complex	AccountBalance	Optional (max : 1)	The credit account information This aggregate contain information with regards to the credit account profile, account credit limit... Valid only in case of credit card account.
9	CatAccount Product	C (6)	CrdAcctInfo	Required (max : 1)	It is a 6 digit code identifying the account product code assigned to the account. Valid values must be retrieved from PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CATALOGUE _PRODUCT table with product type = A
10	CatAccount ProductVersion	C (10)	CrdAcctInfo	Required (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
11	CreditLimit	Long ()	CrdAcctInfo	Optional (max : 1)	Sets the maximum amount that can be borrowed on the cardholder's account. The value assigned to this field can be used to set initial credit limits for newly created cards in the PowerCARD system. In order to be valid, the value of this field should not exceed the ceiling amount defined in the credit account profile. If empty, the system will inherit the default value defined in the credit account profile.
12	CashLimit	Long ()	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of cash that can be obtained on the cardholder's account (ATM withdrawal, cash advance).
13	Loanlimit	Long ()	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of loan transactions that can be performed

#	Field Name	Data type	Parent	Occurrence	Field Description
					on the cardholder's account. The value of this field should not exceed the total credit limit permitted on the cardholder's account.
14	paymentMode AutoDebitInd	C (1)	CrdAcctInfo	Required (max : 1)	Should contain the automatic debit payment mode indicator: • 'M': Automatic debit is mandatory. • 'P': Automatic debit is allowed. • 'F': Automatic debit is not allowed.
15	CycleCode	C (20)	CrdAcctInfo	Optional (max : 1)	The credit cycle code The cycle is the period of time that elapses between 2 consecutive billing statements. The value of this field indicates the code that refers to the new billing cycle that should be assigned to the account. Possible values must be retrieved from the PowerCARD CYCLE _CUTOFF _PARAMETERS table
16	DefaultRate	Decimal	CrdAcctInfo	Optional (max : 1)	It refers to the default interest rate that could be charged by credit card issuer. Unless it is specified in the credit account profile, this rate will be used by

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD for all type of account balances
17	StatementCode	C (2)	CrdAcctInfo	Optional (max : 1)	Refers to the setting of statement parameters in PowerCARD that enables to specify all the data relative to the statement that will be sent to the customer. It includes periodicity, fees amount... This setting is defined as an index that must be retrieved from PowerCARD STATEMENT_PARAMETERS table using
-	Requested CreditAmount	Complex	CrdAcctInfo	Optional (max : 1)	The requested credit amount. The captured limit is made by PowerCARD.
18	Amt	Long ()	Requested CreditAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
19	CurCode	NC (3)	Requested CreditAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD
-	DebitOrder Amount	Complex	CrdAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
20	Amt	Long ()	DebitOrder Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
21	CurCode	NC (3)	DebitOrder Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	DebitOrder Percentage	Complex	CrdAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
22	Amt	Long ()	DebitOrder Percentage	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
23	CurCode	NC (3)	DebitOrder Percentage	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	CurrentBalance	Complex	AccountBalance	Required (max : 1)	Must include the amount value of the account balance. - For deposit Accounts, it refers to Current available balance (typically, ledger balance less outstanding authorizations. - For Credit Card Accounts, it refers to the cardholder outstanding balance during the cycle present cycle.
24	Amt	Long ()	CurrentBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
25	CurCode	NC (3)	CurrentBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction

#	Field Name	Data type	Parent	Occurrence	Field Description
					amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
26	BalanceStatus Date	DateTime	AccountBalance	Optional (max : 1)	Refers to the last update date of the concerned balance.
-	CurrentDebit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the debit transactions performed by the cardholder during the present cycle. Valid only for credit accounts
27	Amt	Long ()	CurrentDebit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
28	CurCode	NC (3)	CurrentDebit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	CurrentCredit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the credit transactions

#	Field Name	Data type	Parent	Occurrence	Field Description
					performed by the cardholder during the present cycle. (payment, refund...) Valid only for credit accounts
29	Amt	Long ()	CurrentCredit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
30	CurCode	NC (3)	CurrentCredit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	ClosingBalance	Complex	AccountBalance	Optional (max : 1)	The closing balance is the positive or negative outstanding amount in an account at the conclusion of the last cycle. Valid only for credit accounts
31	Amt	Long ()	ClosingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n

#	Field Name	Data type	Parent	Occurrence	Field Description
					digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
32	CurCode	NC (3)	ClosingBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	MinimumDue	Complex	AccountBalance	Optional (max : 1)	The last minimum amount calculated by the system that the cardholder must pay by the due date. Valid only for credit accounts
33	Amt	Long ()	MinimumDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
34	CurCode	NC (3)	MinimumDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD
35	LastCutoff	DateTime	AccountBalance	Optional (max : 1)	Refers to the last Account statement date Valid only for credit accounts
36	NextDueDate	DateTime	AccountBalance	Optional (max : 1)	It refers to the date the minimum monthly payment is due Valid only for credit accounts
37	Status	ClosedEnum	AccountBalance	Optional (max : 1)	This field must contain the status of the account: - For Deposit accounts, Valid values are : - N = Normal - C = Canceled - S = Suspended - W = Waiting for validation - V = To be sent for validation - R = Rejected. - For Credit Card Accounts: 0 = Normal 1 = Supervision 2 = Pre legal 3 = Legal 4 = Insurance management 5 = Lost / Closed

#	Field Name	Data type	Parent	Occurrence	Field Description
					• 6 = Lost / Death • 7 = Closed / Canceled • 8 = Write Off • 9 = Untraceable
-	withdrawal Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the account that may be withdrawn.
38	Amt	Long ()	withdrawal Available	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
39	CurCode	NC (3)	withdrawal Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	purchase Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the account that may be used for purchase transactions.
40	Amt	Long ()	purchase Available	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the

#	Field Name	Data type	Parent	Occurrence	Field Description
					currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
41	CurCode	NC (3)	purchase Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	withdrawalLimit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of cash that can be obtained on the cardholder's account (ATM withdrawal, cash advance).
42	Amt	Long ()	withdrawalLimit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
43	CurCode	NC (3)	withdrawalLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section. Valid values must be retrieved from PowerCARD
-	purchaseLimit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of money that can be used by the cardholder to make purchase transactions.
44	Amt	Long ()	purchaseLimit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
45	CurCode	NC (3)	purchaseLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	PendingBalance	Complex	AccountBalance	Optional (max : 1)	This is the total amount of all pending authorizations.
46	Amt	Long	PendingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n

#	Field Name	Data type	Parent	Occurrence	Field Description
					digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
47	CurCode	C (3)	PendingBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD

Specific error codes

#	Error code	Error Description
1	02601	Invalid Card number
2	02602	Invalid Customer Id
3	02603	Invalid Internal Account Id
4	02604	Card product not defined in the system
5	02605	Invalid Credit Account profile
6	02606	Invalid Account type
7	02607	Limits not defined
8	02608	Error getting the account available balance

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x

Status Code	Description	Success Response	Error Response
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetAccountBalanceWebService/V2	POST	
/AccountBalance/V2? aliasPAN={aliasPAN}&clientCode={clientCode}&accountId={accountId}&accountType={accountType}	GET	Optional: CardNumber=[C(22)] ClientCode=[C(22)] AccountId=[C(22)] AccountType=[C(2)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "clientCode": "",
  "accountId": "",
  "accountType": ""
}
```

4.22 getLoyaltyAccountTransactions API

API overview

Loyalty programs can help financial institution to increase their volume of card transaction because it builds a profitable relationship between the financial institutions and their customers. By using its card for shopping the customers can earn points that he can redeem later for gifts, miles, discounts, cash-back, coupons, etc...

The current service is intended to send back the list of points earned by the customer within a range of date. Each transaction must contain information's with regards to the shopping made by the customers as well as the number of points earned in that program.

Functional description

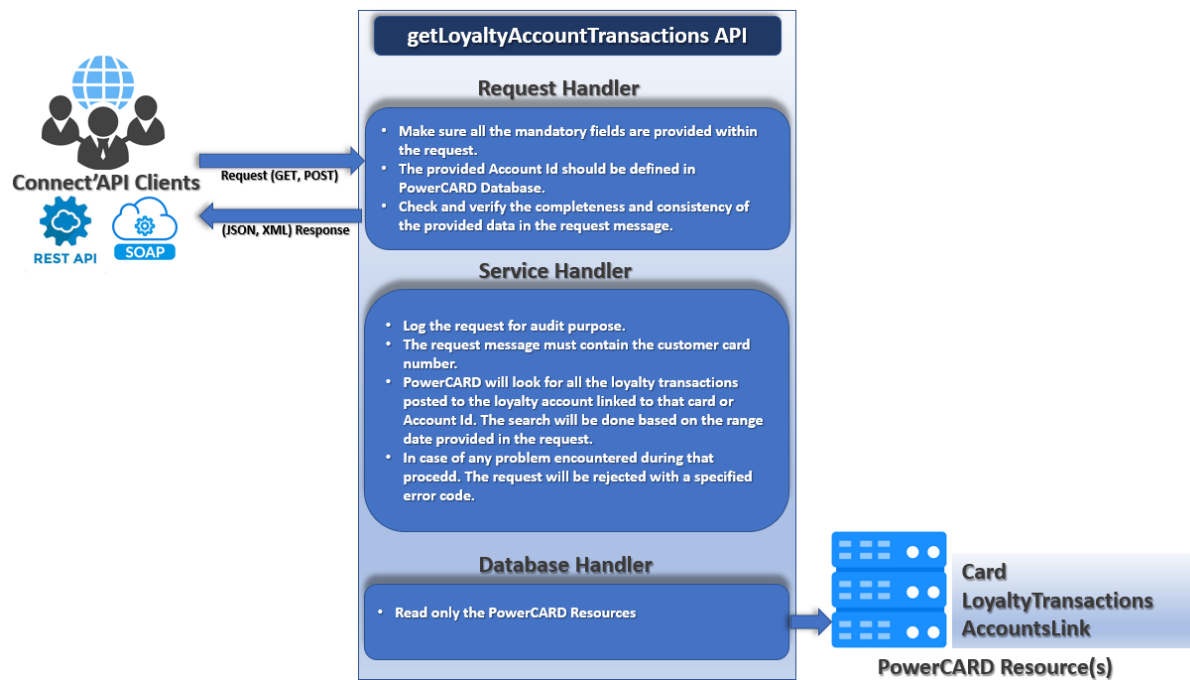


Figure 22: getLoyaltyAccountTransaction process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	BankCode	C(6)	-	Required	Represents the bank code of the loyalty account owner.
5	AccountOwner Type	AccountOwner Type_Type	-	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: • CH = Cardholder. • SA = Shadow Account The value transmitted in the next field is based on the value of that one.
6	Account OwnerId	AccountOwner Id_Type	-	Required (max : 1)	It refers to the identification of the entity that owns the loyalty account. Based on the value in the previous field, this field may contain: • The shadow account (SA) • The card number (CH)
7	LoyaltyAccount Nbr	C (24)	-	Required (max : 1)	This field refers to a unique internal identification of the loyalty account linked to the card, channel....

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field must contain the loyalty account subject to this cancellation.
-	SelRangeDt	Complex	-	Required (max : 1)	The selection range date aggregate. It must contain the range date used as criteria to select the loyalty transactions posted to the customer loyalty account.
8	StartDt	DateTime	SelRangeDt	Optional (max : 1)	Start date
9	EndDt	DateTime	SelRangeDt	Optional (max : 1)	End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker

#	Field Name	Data type	Parent	Occurrence	Field Description
					Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	MaxRowNbr	Long	-	Optional (max : 1)	This will depict the maximum number of row that can be included in the response message. Note that if the result exceeds this number, only this number of loyalty transactions will be returned. This mechanism is used to limit the number of transactions that can be fetched in the request.
-	Loyalty Transactions	Complex	-	Required infinite list	This aggregate contains information with regards to the loyalty transactions posted to the customer loyalty account.
6	TransId	C (23)	Loyalty Transactions	Required (max : 1)	The transaction ID A 23-digit identification number

#	Field Name	Data type	Parent	Occurrence	Field Description
					assigned by the system to each transaction. The TransId is unique by transaction.
7	RefSeq	N (4)	Loyalty Transactions	Required (max : 1)	Refers to the transaction sequence Number
8	AccountOwner Type	C (2)	-	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: • CH = Cardholder. • SA = Shadow Account The value transmitted in the next field is based on the value of that one.
9	AccountOwner Id	C(22)	-	Required (max : 1)	It refers to the identification of the entity that owns the loyalty account. Based on the value in the previous field, this field may contain : The shadow account (SA) The card number (CH)
10	Loyalty AccountNbr	C (24)	Loyalty Transactions	Required (max : 1)	This field refers to a unique internal identification of the loyalty account linked to the card, channel.... This field must contain the loyalty account subject to this cancellation.
-	TransAmt	Complex	Loyalty Transactions	Required (max : 1)	Identified the transaction amount in the currency used during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					transaction. The entry must be a numeric greater than zero.
11	Amt	Long ()	TransAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
12	CurCode	C (3)	TransAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
13	TransSign	C (1)	Loyalty Transactions	Required (max : 1)	Transaction sign Contains a code that indicates the sign of the transaction. Possible values are: • D: Debit • C: Credit
14	TransDate	DateTime	Loyalty Transactions	Required (max : 1)	The date when the transaction was made.
-	OriginalTrans	Complex		Required (max : 1)	This aggregate contains the transaction code and the description of the original transaction.
15	Code	C (2)	OriginalTrans	Required (max : 1)	Must contain the code of the parameter being shared

#	Field Name	Data type	Parent	Occurrence	Field Description
16	Label	C (52)	OriginalTrans	Required (max : 1)	The label of the parameter being shared
17	OriginalReversal Flag	C (1)	Loyalty Transactions	Optional (max : 1)	Nature of the original transaction. - N: Normal - R: Reversal
18	IssuerBankCode	C (6)	Loyalty Transactions	Optional (max : 1)	Bank code of the cardholder.
19	Loyalty TransDate	Date	Loyalty Transactions	Optional (max : 1)	Date of the loyalty transaction.
20	PointExpiryDate	Date	Loyalty Transactions	Optional (max : 1)	Expiry date of the gained points.
21	ValuePoin tGained	Long ()	Loyalty Transactions	Optional (max : 1)	Number of points accrued.
-	ValueAmout Gained	Complex	Loyalty Transactions	Required (max : 1)	This aggregate contains the monetary equivalent of the accrued points.
22	Amt	Long ()	ValueAmout Gained	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
23	CurCode	C (3)	ValueAmout Gained	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
24	ValuePoint Redemption	Long ()	-		Number of points that have been redeemed / used.
-	ValueAmount Redemption	Complex	Loyalty Transactions	Required (max : 1)	This aggregate contains the monetary equivalent of the redeemed points.
25	Amt	Long ()	ValueAmount Redemption	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	C (3)	ValueAmount Redemption	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
27	GiftCode	C (6)	Loyalty Transactions	Optional (max : 1)	Id identifying the category of the ticket / voucher requested as part of the redemption process.

Specific error codes

#	Error code	Error Description
1	13101	Invalid Account Owner Type
2	13102	Invalid Account Owner Id
3	13103	Invalid range date

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetLoyaltyAccountTransactionsWebService/V2	POST	
/LoyaltyAccounts/V2? bankCode={bankCode}& accountOwnerType={accountOwnerType} & accountOwnerId={accountOwnerId} &LoyaltyAccountNbr={ LoyaltyAccountNbr} &rangeStartDt={rangeStartDt }&rangeEndDt={rangeEndDt}	GET	Required: bankCode=[C(6)] accountOwnerType=[C(2)] accountOwnerId=[C(2)] loyaltyAccountNbr=[C(24)] rangeStartDt=[Date] rangeEndDt=[Date]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "bankCode": "",
  "accountOwnerType": "",
```



```

"accountOwnerId": "",
"loyaltyAccountNbr": "",
"selRangeDt": {
  "startDt": "",
  "endDt": ""
}
}

```

4.23 updateLoyaltyAccountStatus API

API overview

The loyalty account can be cancelled in the system for any reason. For that, PowerCARD offers this web service that requires the loyalty account as input in the request message as well as the cancellation reason. PowerCARD will then process to the cancellation of the loyalty account only if that account has no outstanding balance. If this is the case, the system will inform the requester who may need to make a manual adjustment on the account before proceeding to the account cancellation.

Functional description

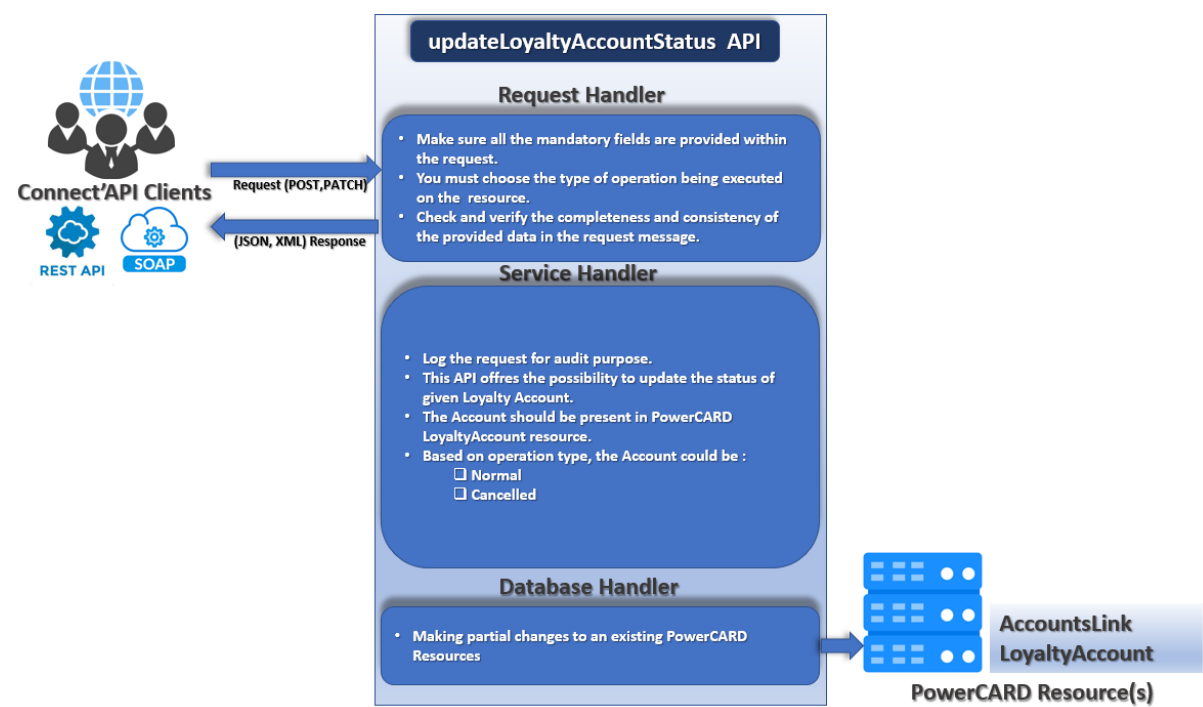


Figure 23: updateLoyaltyAccountStatus process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information

#	Field Name	Data type	Parent	Occurrence	Field Description
					necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
14	BankCode	C(6)	-	Required (max : 1)	Represent the bank code of the account owner.
15	AccountOwner Type	C (2)	-	Required (max : 1)	Indicates the type of the entity that owns the loyalty account. Possible values are the following: - CH = Cardholder. - SA = Shadow Account The value transmitted in the next field is based on the value of that one.
16	AccountOwner Id	C (22)	-	Required	It refers to the identification of the

#	Field Name	Data type	Parent	Occurrence	Field Description
				(max : 1)	entity that owns the loyalty account. Based on the value in the previous field, this field may contain: - The shadow account (SA) - The card number (CH)
17	LoyaltyAccount Nbr	C (24)	-	Required (max : 1)	This field refers to a unique internal identification of the loyalty account linked to the card, channel.... This field must contain the loyalty account subject to this cancellation.
18	StatusCode	C (2)	-	Required (max : 1)	This field must contain the account status code. Possible values are the following: - N = Normal. - C = Cancelled.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: Proceed With Success: Call

#	Field Name	Data type	Parent	Occurrence	Field Description
					was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	12901	Invalid reason code
2	12902	Account already cancelled.
3	12903	Account with outstanding balance. Need to adjust the balance before proceeding to cancellation.
4	12904	Invalid loyalty Account number
5	12905	Invalid Loyalty Program Identification
	12906	Invalid Status Code
	12907	Invalid Account Owner Type
	12908	Invalid Bank Code
	12909	Loyalty Account Not Founded

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateLoyaltyAccountStatusWebService/V2	POST	
/LoyaltyAccountStatus/V2/	PATCH	bankCode=[C(6)] accountOwnerType=[C(2)] accountOwnerId=[C(22)] loyaltyAccountNbr=[C(22)] statusCode=[C(1)]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "bankCode": "",
}
```

```

"accountOwnerType": "",
"accountOwnerId": "",
"loyaltyAccountNbr": "",
"statusCode": ""
}

```

4.24 updateCardLimits API

API overview

The purpose of the current service is to give the possibility to any allowed third party to update the velocity limits of a given card in PowerCARD the services requires as input the card number, the periodicity (daily, monthly...) the transaction type and then the amount that need to be set on the card level. Any transaction exceeding that velocity limits should be suspended later by the system if the velocity limits control is activated for that card.

Functional description

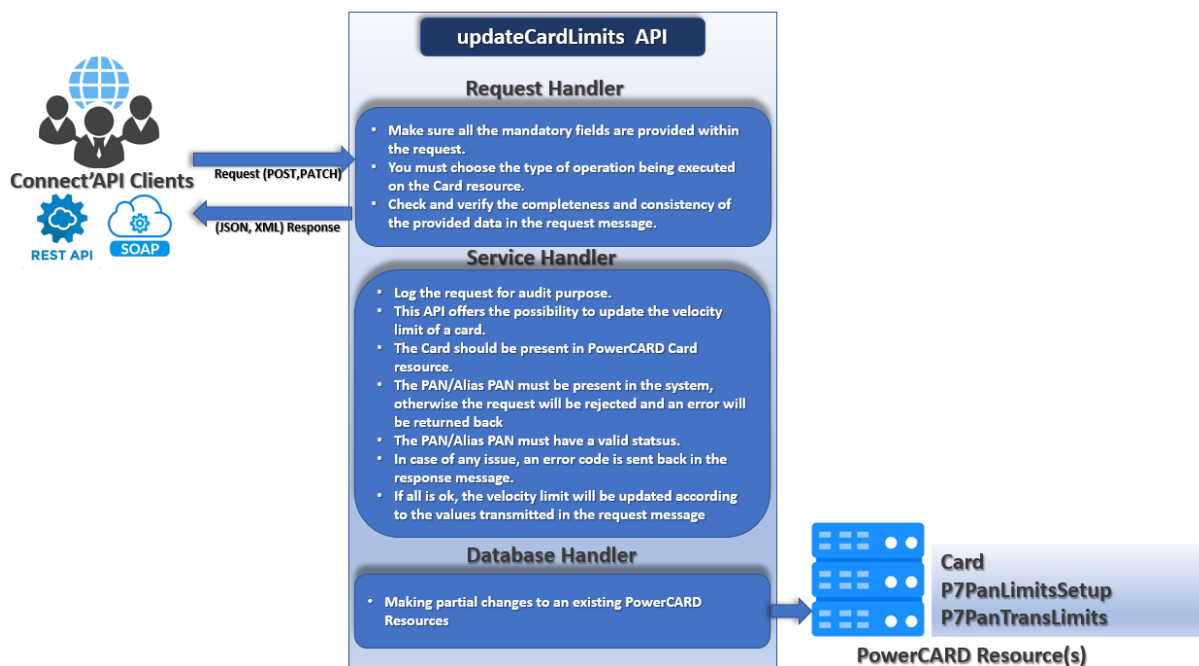


Figure 24: updateCardLimits process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	Refers to the card number for which the requester is asking to update the limit
-	CardLimit	Colmplex	-	Optional infinite list	The limit that should be set on the card.
5	LimitID	C (4)	CardLimit	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.
6	Limit Description	C (64)	CardLimit	Optional (max : 1)	Description of the limit
7	LimitAbv Wording	C (32)	CardLimit	Optional (max : 1)	A brief description of the limit
-	ValidityRange Date	Complex	CardLimit	Required (max: 1)	A limit can be temporary and restricted to a range date

#	Field Name	Data type	Parent	Occurrence	Field Description
8	StartDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
9	EndDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
-	Period	Colmplex	-	Optional (max: 4)	The Period of limit that should be set on the card.
10	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: Fixed period - W: Rolling period
11	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off
12	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: for daily - W: for Weekly - M: for Monthly - Y: for Yearly
	Territoriality	Complex	Period	Optional (max: 3)	
13	TerritorialityType	C (3)	Territoriality	Required (max: 1)	Valid values: - ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. - NAT: National, apply only to transactions made in the same country as the issuer country. - INT: International, apply only to transactions that took place outside the cardholder country.
14	Transaction Amount	Decimal (18,3)	Territoriality	Required (max : 1)	Must contain the target amount of the

#	Field Name	Data type	Parent	Occurrence	Field Description
					limit that should be set on the card.
15	NumberOf Transactions	Number (4)	Territoriality	Required (max : 1)	Must contain the target the number of transactions of the limit.
16	Total Transaction Amount	Decimal (18,3)	Period	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
17	TotalNumber OfTransactions	Number (4)	Period	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	Transaction Criteria	Colmplex	-	Optional (max : 1)	This aggregate is used to define the transaction criteria
18	SequenceNumber	Number (9)	Transaction Criteria	Optional (max : 1)	Should contain Sequence Number
19	ProcessingCode	C (2)	Transaction Criteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
20	CountryCode	C (3)	Transaction Criteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
21	MCC	C (4)	Transaction Criteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of

#	Field Name	Data type	Parent	Occurrence	Field Description
					services or products it provides. The valid MCC's are stored in the PowerCARD MCC table
22	PaymentType	C (2)	Transaction Criteria	Optional (max : 1)	Should contain the payment type
23	CurrencyCode	C (3)	Transaction Criteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
24	MaxAmount PerTrx	Long	Transaction Criteria	Optional (max : 1)	Used to set the maximum amount per transaction.
25	CardEntryMode	ClosedEnum	Transaction Criteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are: • 0: Unknown or terminal not used • 1: Manual (key entry) • 2: Magnetic stripe read • 3: Integrated circuit card read • 4: Contactless device-read

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the

#	Field Name	Data type	Parent	Occurrence	Field Description
					description of the error encountered

Specific error codes

#	Error code	Error Description
1	12001	Invalid Card number
2	12002	Invalid Start Date
3	12003	Invalid End Date
4	12004	Invalid Period Type
5	12005	Invalid Periodicity Flag
6	12006	Invalid Territoriality Type
7	12007	Invalid Processing Code
8	12008	Invalid Country Code
9	12009	Invalid MCC
10	12010	Invalid Payment Type
11	12011	Invalid Currency Code
12	12012	Invalid Card Entry Mode
13	12013	Size of period must be less than 4
14	12014	Size of Territoriality must be less than 3

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x

Status Code	Description	Success Response	Error Response
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCardStatusWebService/V2	POST	
/Cards/V2/:aliasPAN/Limits	POST	Required aliasPAN=[C(22)]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "limitDescription": "",
  "limitAbvWording": "",
  "ValidityRange Date": {
    "startDate": "",
    "endDate": ""
  },
  "cardLimit": [
    {
      "limitID": "",
      "territoriality": "",
      "periodicityFlag": "",
      "periodType": "",
      "periodDayOff": "",
      "transactionAmount": "",
      "numberOfTransactions": "",
      "transactionCriteria": {
        "processingCode": "",
        "countryCode": "",
        "mCC": "",
        "paymentType": "",
        "currencyCode": "",
        "maxAmountPerTrx": "",
        "cardEntryMode": ""
      }
    }
  ]
}
```

4.25 getCardLimits API

API overview

For risk management purpose, PowerCARD introduces the mechanism of card limits. Card limits are the maximum amount that cardholders can either withdraw in an ATM machine or also use to make a purchase via internet, terminal pos... Those limits are defined in PowerCARD as a package identified by a unique index. That index could be either linked to a particular card or also linked a range or a product and by the way the entire cards in this range or having such product inherit the characteristics of that index.

The index is constituted from a set of setup, each setup is intended to refer to a different transaction types. Within each setup, a daily and periodic limit amounts are shared to give an overview of the limits that should not be exceeded.

Functional description

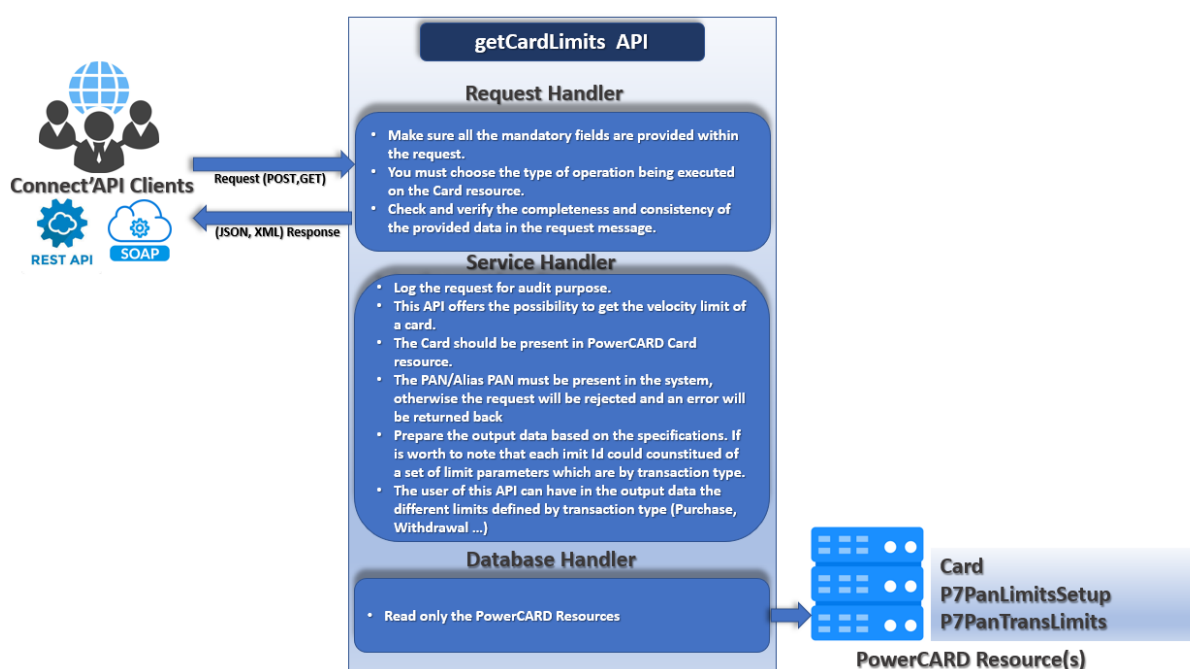


Figure 25: getCardLimitsprocess

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one

#	Field Name	Data type	Parent	Occurrence	Field Description
					sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	CardLimit	Colmplex	-	Optional infinite list	The limit that should be set on the card.
5	LimitID	C (4)	CardLimit	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.

#	Field Name	Data type	Parent	Occurrence	Field Description
6	LimitDescription	C (64)	CardLimit	Optional (max : 1)	Description of the limit
7	LimitAbv Wording	C (32)	CardLimit	Optional (max : 1)	A brief description of the limit
-	ValidityRange Date	Complex	CardLimit	Required (max: 1)	A limit can be temporary and restricted to a range date
8	StartDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
9	EndDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
-	Period	Complex	-	Optional (max: 4)	The Period of limit that should be set on the card.
10	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: • D: Fixed period • W: Rolling period
11	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off
12	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: • D: for daily • W: for Weekly • M: for Monthly • Y: for Yearly
	Territoriality Online	Colmplex	Period	Optional (max: 3)	
13	Territoriality Type	C (3)	Territoriality Online	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside

#	Field Name	Data type	Parent	Occurrence	Field Description
					the cardholder country.
14	Transaction Amount	Decimal (18,3)	Territoriality Online	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
15	NumberOfTransactions	Number (4)	Territoriality Online	Required (max : 1)	Must contain the target the number of transactions of the limit.
	Territoriality Offline	Complex	Period	Optional (max: 4)	
16	Territoriality Type	C (3)	Territoriality Offline	Required (max: 1)	Valid values: • ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. • NAT: National, apply only to transactions made in the same country as the issuer country. • INT: International, apply only to transactions that took place outside the cardholder country.
17	Transaction Amount	Decimal (18,3)	Territoriality Offline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
18	NumberOf Transactions	Number (4)	Territoriality Offline	Required (max : 1)	Must contain the target the number of transactions of the limit.
19	TotalTransaction AmountOnline	Decimal (18,3)	Period	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
20	TotalNumberOf TransactionsOnline	Number (4)	Period	Optional (max : 1)	Must contain total of the target the number of

#	Field Name	Data type	Parent	Occurrence	Field Description
					transactions of the limit.
21	TotalTransaction AmounOffline	Decimal (18,3)	Period	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
22	TotalNumberOf TransactionsOffline	Number (4)	Period	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	Transaction Criteria	Complex	-	Optional (max : 1)	This aggregate is used to define the transaction criteria
23	Sequence Number	Number (9)	Transaction Criteria	Optional (max : 1)	Should contain Sequence Number
24	ProcessingCode	C (2)	Transaction Criteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING _LIST table under the general parameter section.
25	CountryCode	C (3)	Transaction Criteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
26	MCC	C (4)	Transaction Criteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD MCC table
27	PaymentType	C (2)	Transaction Criteria	Optional (max : 1)	Should contain the payment type
28	CurrencyCode	C (3)	Transaction Criteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
29	MaxAmount PerTrx	Long	Transaction Criteria	Optional (max : 1)	Used to set the maximum amount per transaction.
30	CardEntryMode	ClosedEnum	Transaction Criteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are: • 00: Unknown or terminal not used • 01: Manual (key entry) • 02: Magnetic stripe read • 03: Integrated circuit card read • 04: Contactless device-read

Specific error codes

#	Error code	Error Description
1	21101	Invalid Card Number/Token Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	

Status Code	Description	Success Response	Error Response
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCardLimitsWebService/V2	POST	
/Cards/V2/:aliasPAN/Limits	GET	Optional: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": ""
}
```

4.26 getCardDomain API

API overview

The aim if this current web service is sharing the list of Domain control for a card. This list is sent back in the response message.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code and description for more details.

Functional description

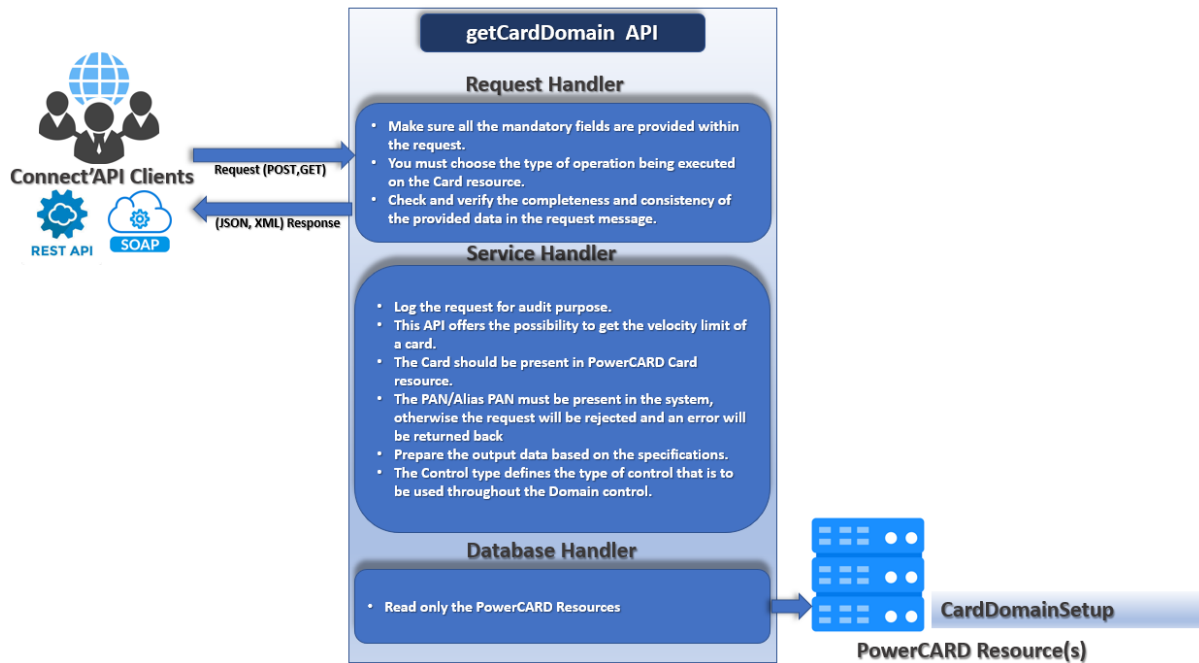


Figure 26: getCardDomain process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will

#	Field Name	Data type	Parent	Occurrence	Field Description
					be inserted in a Maker Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	DomainControl	Complex	-	Optional infinite list	This aggregate is used to specify the detail of the card services to be activated or not
5	ControlType	ClosedEnum	DomainControl	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: - CTR: Country - MCC: MCC - PRS: Processing code - MER: Merchant - CUR: Currency
6	ControlOperator	ClosedEnum	DomainControl	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or

#	Field Name	Data type	Parent	Occurrence	Field Description
					multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValueList
-	ControlValue List	Complex	DomainControl	Optional infinite list	Contains various values, such as country codes, processing codes ...
7	ControlValue	C (15)	ControlValue List	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section: COUNTRY table for ControlType = 'CTR'. MCC table for ControlType = 'MCC'. PROCESSING_LIST table for ControlType = 'PRS'. MERCHANT table for ControlType = 'MER'. CURRENCY_TABLE table for ControlType = 'CUR'. If not, the request will be rejected.

Specific error codes

#	Error code	Error Description
1	16701	Invalid card number/Token
2	16702	Process request is failed

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh		x

Status Code	Description	Success Response	Error Response
	the token and then try again.		
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Specific error codes

#	Error code	Error Description
1	16701	Invalid card number/Token
2	16702	Process request is failed

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested		x

Status Code	Description	Success Response	Error Response
	is not supported for the PowerCARD Resource.		
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCardDomainWebService/V2	POST	
/Cards/V2/:aliasPAN/Domain	GET	Optional: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": ""
}
```

4.27 setCardDomain API

API overview

This web services was introduced to allow users to restrict the scope of use of the card according to criteria defined by the bank such as Merchant, MCC, Acquiring country, Processing code, Currency.

Agent and customers can ask to change the Domain control by specifying the operation from a list and whether the service will be allowed or not.

It is important to note that the web service requires the card number and the desired operation as input. As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

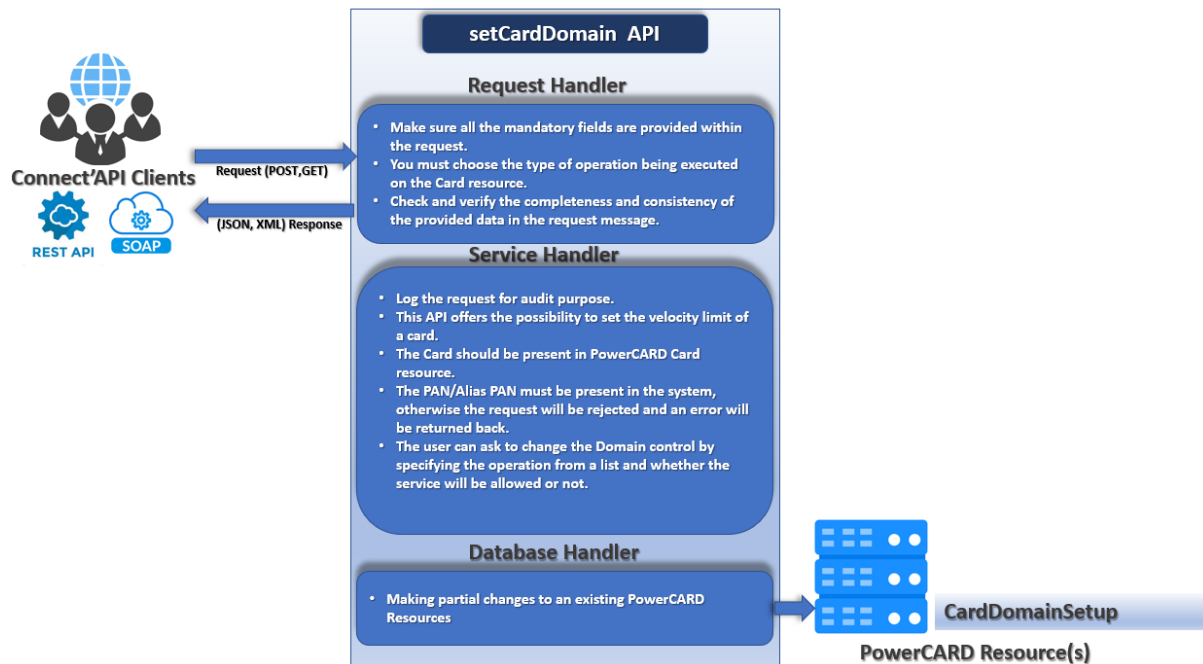


Figure 27: setCardDomain process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
-	DomainControl	Complex	-	Optional infinite list	This aggregate is used to specify the detail of the card services to be activated or not
5	ControlType	ClosedEnum	DomainControl	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: - CTR: Country - MCC: MCC - PRS: Processing code - MER: Merchant - CUR: Currency
6	ControlOperator	ClosedEnum	DomainControl	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValue List
-	ControlValue List	Complex	DomainControl	Optional infinite list	Contains various values, such as

#	Field Name	Data type	Parent	Occurrence	Field Description
					country codes, processing codes ...
7	ControlValue	C (15)	ControlValue List	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section: - COUNTRY table for ControlType = 'CTR'. - MCC table for ControlType = 'MCC'. - PROCESSING_LIST table for ControlType = 'PRS'. - MERCHANT table for ControlType = 'MER'. - CURRENCY_TABLE table for ControlType = 'CUR'. If not, the request will be rejected.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	16601	Invalid card number/Token
2	16602	Invalid Control Type
3	16603	Invalid Control Operator
4	16604	Invalid Country Code
5	16605	Invalid Currency Code
6	16606	Invalid Processing Code
7	16607	Invalid Merchant Id

#	Error code	Error Description
8	16608	Invalid MCC Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SetCardDomainWebService/V2	POST	
/Cards/V2/:aliasPAN/Domain	POST	Optional: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "domainControl": [
    {
```



```

    "controlType": "",
    "controlOperator": ""
  },
  "controlValueList": [
    {
      "controlValue": ""
    }
  ]
}

```

4.28 changePIN API

API overview

The PIN change web Service allows the any cardholder to change its PIN. The cardholder is required to enter the new PIN, as well as the old PIN for verification.

The same service is also used to set a pin value for the customer.

It is to mention that the new and old PINs are encrypted and formatted as a block of 16 hexadecimal digits. (The new PIN is chosen by the cardholder to replace the current PIN when the cardholder does not remember his current PIN, or the current PIN is compromised, or just wants a new PIN to be set for his card.)

A cardholder PINs (Old and new) are never logged, not even in an encrypted form.

Functional description

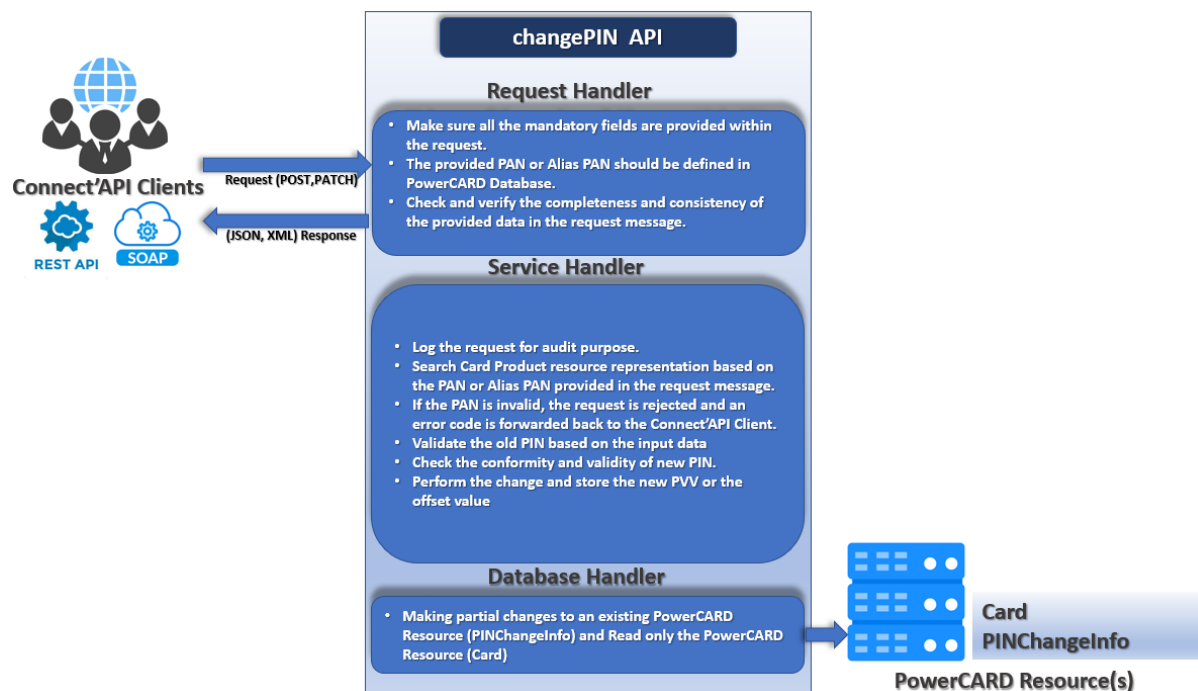


Figure 28: changePIN process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
5	pinBlockFormat	C (2)	-	Required (max : 1)	PIN Block Format This field indicates the format of the PIN block for the two PIN blocks (Old and new). Valid values are: 01: Format is based on the PIN, the PIN length,

#	Field Name	Data type	Parent	Occurrence	Field Description
					the selected rightmost digits of the account number, and the pad characters 0 and F—combined through an exclusive OR operation 03: Format is based on the PIN and the F pad character
6	pinMethod	C (3)	-	Required (max : 1)	Method of PIN validation This operation is performed within a Host Security Module (HSM) Valid values are: - PVV: Pin Verification Value - OFS: Offset
7	currentPIN	C (256)	-	Optional (max : 1)	This field contains the current PIN. It is encrypted and formatted as a block of 16 hexadecimal digits. The PIN Block is transmitted encrypted under the ZPK (Zone PIN key). This key should be shared between PowerCARD and the other party. This field is optional
8	currentPvv	C (4)	-	Optional (max : 1)	Refers to the current PVV/Offset value that resides in the track 2 of the card. This value is mandatory only when the customer is aiming to change

#	Field Name	Data type	Parent	Occurrence	Field Description
					the existing PIN by a new one. This field is used for PIN verification.
9	newPIN	C (256)	-	Required (max : 1)	This field contains a new PIN to replace an existing PIN. It is encrypted and formatted as a block of 16 hexadecimal digits. (A new PIN is chosen to replace the current PIN when the cardholder does not remember the current PIN, or the current PIN is compromised, or just wants a new PIN.). The PIN block encrypted under the ZPK (Zone PIN key) and must conform to the PIN Block Format 01 The ZPK key should be shared previously between PowerCARD and the other party.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	02101	Invalid Card number
5	02105	Invalid Pin Block Format
6	02106	Invalid Pin method
7	02107	Invalid New PIN
8	02108	Card do not belong to any range in the system
9	02109	Error when translating the Pin Block

#	Error code	Error Description
10	02110	Zone Pin Key not defined in the system
11	02111	Error generating the Visa PVV
12	02112	Error generating IBM Offset
13	02113	Error updating PVV value in the database
14	02114	Error while getting PVK Keys
15	02115	Error Verify Visa PIN
16	02116	Error Verify IBM PIN

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ChangePINWebService/V2	POST	
/Cards/V2/:aliasPAN/ChangePIN	POST	Optional:

PowerCARD Resource URI	Method	URI Params
		aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "pinBlockFormat": "",
  "pinMethod": "",
  "currentPIN": "",
  "currentPvv": "",
  "newPIN": ""
}
```

4.29 verifyPIN API

API overview

The verify PIN web service is used to verify if the Personal Identification Number (PIN) introduced is valid or not. Users must transmit different details in the request that are used internally by PowerCARD to communicate with the HSM to validate the PIN captured by the customer. Depending on the validation processes, the response message that will be returned will either contain a Yes or No. Yes to specify that the PIN captured is valid, and No in the opposite case.

Functional description

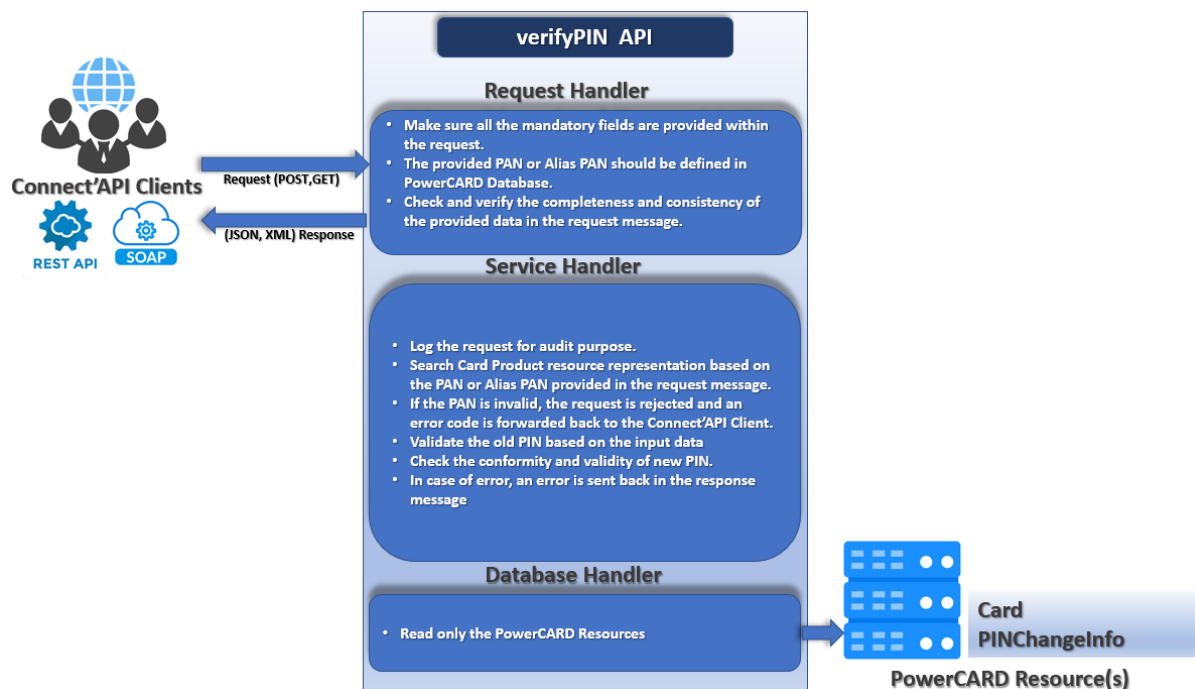


Figure 29: verifyPIN process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
5	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
6	pinBlockFormat	C (2)	-	Required (max : 1)	PIN Block Format This field indicates the format of the PIN block for the two

#	Field Name	Data type	Parent	Occurrence	Field Description
					PIN blocks (Old and new). Valid values are: 01: Format is based on the PIN, the PIN length, the selected rightmost digits of the account number, and the pad characters 0 and F—combined through an exclusive OR operation 03: Format is based on the PIN and the F pad character
7	pinMethod	C (3)	-	Required (max : 1)	Method of PIN validation This operation is performed within a Host Security Module (HSM) Valid values are: - PVV: Pin Verification Value - OFS: Offset
8	currentPIN	C (256)	-	Required (max : 1)	This field contains the current PIN. It is encrypted and formatted as a block of 16 hexadecimal digits. The PIN Block is transmitted encrypted under the ZPK (Zone PIN key). This key should be shared between PowerCARD and the other party.
9	currentPvv	C (4)	-	Required (max : 1)	Refers to the current PVV/Offset value

#	Field Name	Data type	Parent	Occurrence	Field Description
					that resides in the track 2 of the card.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell

#	Field Name	Data type	Parent	Occurrence	Field Description
					you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	IsVerified	ClosedEnum	-	Required (max : 1)	If the result of the verification of the PIN is returned back in this field specifying whether the PIN is valid or not. Possible values are: - Y: Yes, PIN is valid - N: No, PIN is invalid.

Specific error codes

#	Error code	Error Description
1	18001	Invalid Card number
2	18002	Invalid PIN Block
3	18003	Invalid Pin method
4	18004	Card do not belong to any range in the system
5	18005	Error while getting PVK Keys
6	18006	Error while Verifying Visa PIN
7	18007	Error while Verifying IBM PIN

Response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x

Status Code	Description	Success Response	Error Response
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/VerifyPINWebService/V2	POST	
/Cards/V2/:aliasPAN/VerifyPIN	POST	Optional: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "pinBlockFormat": "",
  "pinMethod": "",
  "currentPIN": "",
  "currentPvv": ""
}
```

4.30 getPINHistory API

API overview

The PIN History web service is used to remember a user's PIN history based on PowerCARD's historical reporting dates. Input criteria can include for the date range filter to select your desired start and end date.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

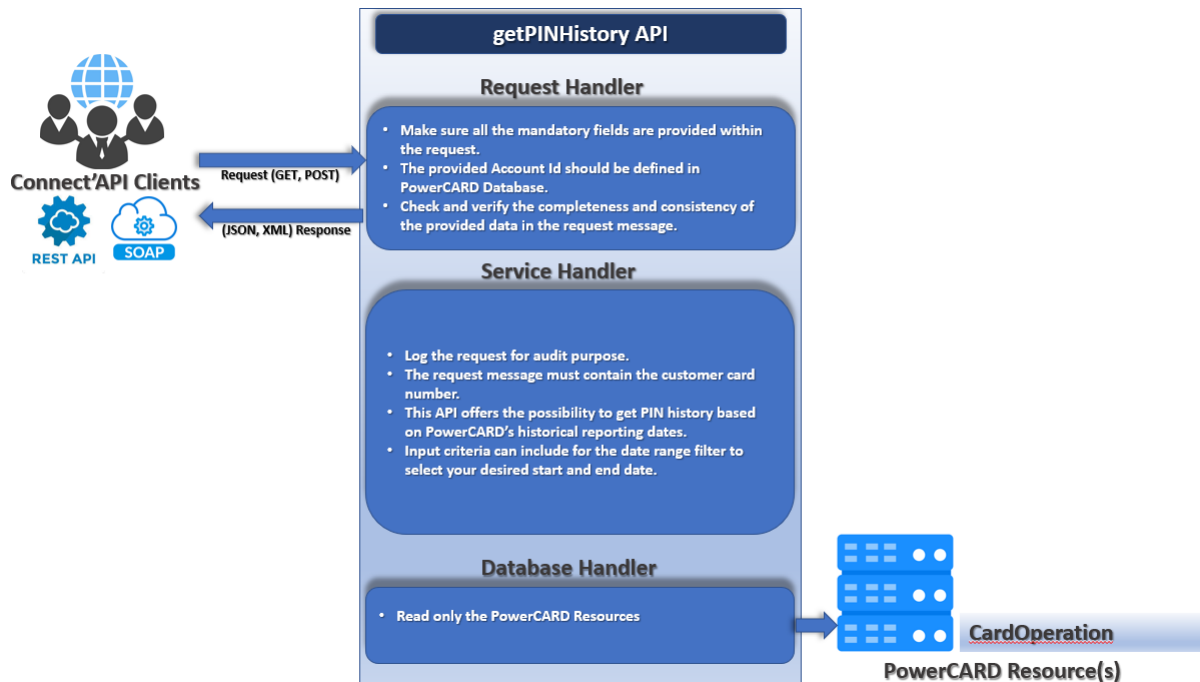


Figure 30: getPINHistory process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
-	RangeDate	Complex	-	Optional (max : 1)	The range date. This aggregate is used to transmit the period to select the desired start and end date.
8	StartDate	DateTime	RangeDate	Optional (max : 1)	Start date
9	EndDate	DateTime	RangeDate	Optional (max : 1)	End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	CardDetails	Complex	-	Optional (max: 1)	This aggregate contains the Card information
5	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.

Specific error codes

#	Error code	Error Description
1	20401	Invalid Card Number/Token Id
2	20402	Start Date cannot be greater than End Date

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetPINHistoryWebService/V2	POST	
/Cards/V2/:aliasPAN/PINHistory?startDate={startDate}&endDate={endDate}	GET	Optional: aliasPAN=[alphanumeric] startDate=[Date] endDate=[Date]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "rangeDate": {
    "startDate": "",
    "endDate": ""
  }
}
```

4.31 updateAddress API

API overview

The aim of this web service is to update the address of an existing customer or card, this web service can be invoked by customers and agents. The change will take place immediately in the system and all the next communication with that customer or card will take the new address being shared in the request into consideration.

Functional description

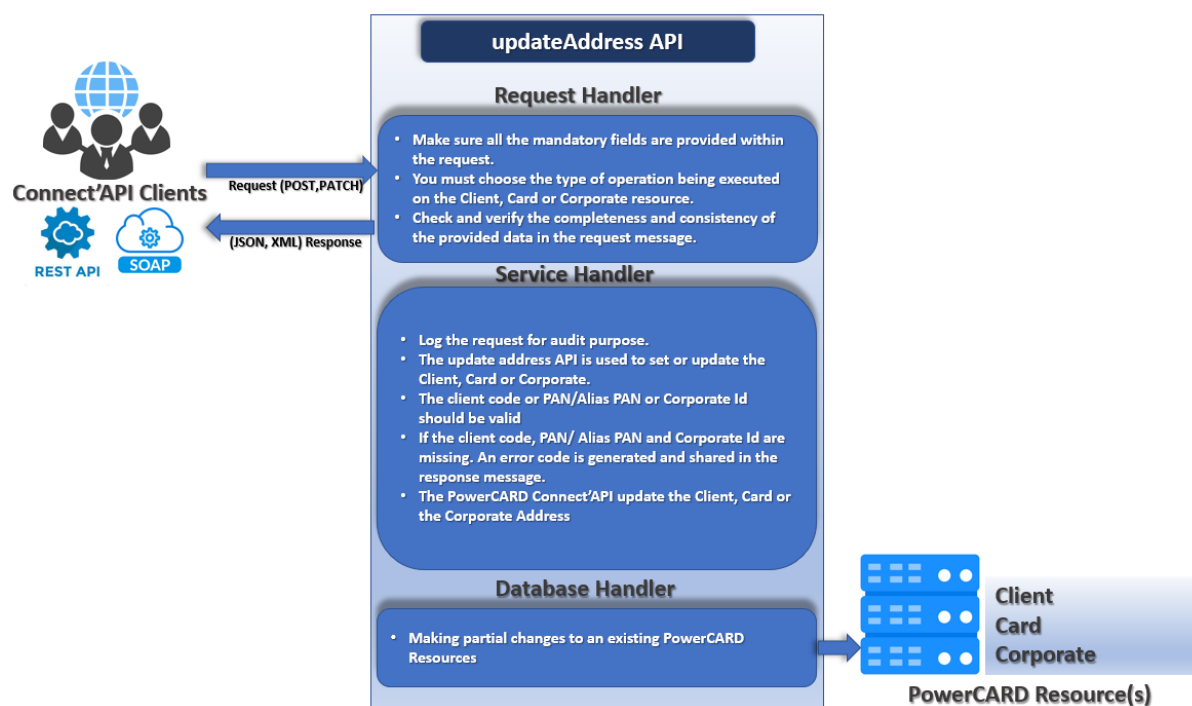


Figure 31: updateAddress process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
11	EntityType	C (2)	-	Required (max: 1)	Concerned entity code. Possible values are: • CH: Card • CL: Client • CO: Corporate
12	ClientCode	C (24)	-	Optional (max: 1)	To provide the client code. This field is mandatory in case of EntityType is 'CL'
13	CardNumber	C (22)	-	Optional (max: 1)	To provide the CardNumber or the Token ID.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field is mandatory in case of EntityType is 'CH'
14	Corporateld	C(15)		Optional (max: 1)	To provide the Corporate Id. This field is mandatory in case of EntityType is 'CO'
-	Address	Complex	-	Required (max: 1)	To provide the address of the entity.
14	AddressIndicator	C (1)	Address	Required (max: 1)	Used to indicate whether the address being sent is a principal or a secondary address. Possible values are as follow: - P: Principal address, each entity must have only one Principal address - S: Secondary address. It is to mention that there is no limitation as to the number of secondary addresses that an entity (Customer, cardholder, merchant, chain...) could have in the system
15	AddressType	C (2)	Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.
16	SendingMode	C (1)	Address	Required (max: 1)	Default Sending mode

#	Field Name	Data type	Parent	Occurrence	Field Description
					Describes what each entity's address type will be used for: '01': Mail. '02': Loyalty. '03': Dispute. '04': Statement.
-	ValidityRange Date	Complex	Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
17	StartDt	Date	ValidityRange Date	Optional (max: 1)	The Validity Start date for Secondary Address only
18	EndDt	Date	ValidityRange Date	Optional (max: 1)	The Validity End date for Secondary Address only
19	CareOf	C (64)	Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
20	AddressLine1	C (64)	Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
21	AddressLine2	C (64)	Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
22	AddressLine3	C (64)	Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
23	AddressLine4	C (64)	Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
24	ZipCode	C (10)	Address	Required (max: 1)	Recipient's postal code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field is mandatory in case of address not linked. It should not be validated..
25	CityCode	C (5)	Address	Optional (max: 1)	The city code of the cardholder address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
26	CityFreeText	C (32)	Address	Optional (max: 1)	A free text to capture the cardholder city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
27	RegionCode	C (3)	Address	Optional (max: 1)	The region code of the cardholder address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of

#	Field Name	Data type	Parent	Occurrence	Field Description
					region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
28	RegionFreeText	C(30)	Address	Optional (max: 1)	A free text to capture the cardholder region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
29	CountryCode	C (3)	Address	Required (max: 1)	Refers to the ISO country code of the cardholder address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section.
-	AddressPurpose	Complex	-	Optional (max: 1)	To specify the purpose (card delivery, pin delivery, statement....) of the provided addresses either the primary one or the secondary ones This aggregate can be set only in case of Secondary Address for EntityType "CH" - Card

#	Field Name	Data type	Parent	Occurrence	Field Description
37	AddressPurpose ID	C (1)	AddressPurpose	Required (max: 1)	Specifies the possible address usage values, namely: • S: Statement address • C: Card address • P: Pin address • L: Litigation address. • G: Gift address
38	AddressType	C (2)	AddressPurpose	Required (max: 1)	Address type based on the intended purpose of use. This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This code must be already provided as part of the card PrimaryAddress or SecondaryAddress
39	SendingMode	C (1)	AddressPurpose	Required (max: 1)	Default Sending mode Describes what each entity's address type will be used for: • '01': Mail. • '02': Loyalty. • '03': Dispute. • '04': Statement.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the

#	Field Name	Data type	Parent	Occurrence	Field Description
					requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	19401	Invalid Entity Type

#	Error code	Error Description
2	19402	Invalid Client Code
3	19403	Invalid Card Number/Token ID
4	19404	Invalid Address Indicator
5	19405	Invalid corporate ID
6	19406	Invalid Address Line 1
7	19407	Invalid Address Line 2
8	19408	Invalid Address
9	19409	Invalid Address Line 3
10	19410	Invalid Address Line 4
11	19411	Invalid ZipCode
12	19412	Invalid CareOf
13	19413	Invalid City Code
14	19414	Invalid Region Code
15	19415	Invalid Country Code
16	19416	Invalid Address Purpose ID
17	19417	Invalid Purpose Address Type
18	19418	Invalid Sending Mode
19	19419	Validity Start Date is accepted only in case of Secondary Address
20	19420	Validity Start Date must not be in the past
21	19421	Validity Start Date cannot be greater than Validity End Date
22	19422	Validity End Date is accepted only in case of Secondary Address
23	19423	Validity End Date must not be in the past
24	19424	Validity End Date cannot be lesser than Validity Start Date
26	19426	City Code or City Free Text is required
27	19427	Region Code or Region Free Text is required
32	19432	AddressPurpose not required in case of Primary Address for cards
33	19433	AddressPurpose not allowed for EntityType "CL" – Client.
36	19436	Card number unexpected for entity type "CL" - CLIENT
37	19437	Client code unexpected for entity type "CH" - CARD

#	Error code	Error Description
38	19438	Invalid Address Type

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateAddressWebService/V2	POST	
/Clients/V2/:clientCode/Addresses Or /Cards/V2/:aliasPAN Or /Corporates/V2/:corporateld	PATCH	Required clientCode=[C(22)] aliasPAN=[C(22)] corporateld=[C(15)]

REST sample call

```
{
  "keyValues": [
    {
```

```

        "key": "TOKEN",
        "data": "Y",
        "type": ""
    },
    ],
    "requestInfo": {
        "requestUID": "",
        "requestDate": "",
        "userID": ""
    },
    "bankCode": "",
    "entityType": "",
    "clientCode": "",
    "cardNumber": "",
    "corporateId": "",
    "address": {
        "addressIndicator": "",
        "addressType": "",
        "sendingMode": "",
        "validityRangeDate": {
            "startDt": "",
            "endDt": ""
        }
    },
    "careOf": "",
    "addressLine1": "",
    "addressLine2": "",
    "addressLine3": "",
    "addressLine4": "",
    "zipCode": "",
    "cityCode": "",
    "cityFreeText": "",
    "regionCode": "",
    "regionFreeText": "",
    "countryCode": "",
    "addressPurpose": "",
    "addressPurposeID": ""
    }
}

```

4.32 activateContactless API

API overview

This web service serves to activate the contactless feature on the card. It is important to note that the web service requires as input the Card Number or the Card Token ID.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

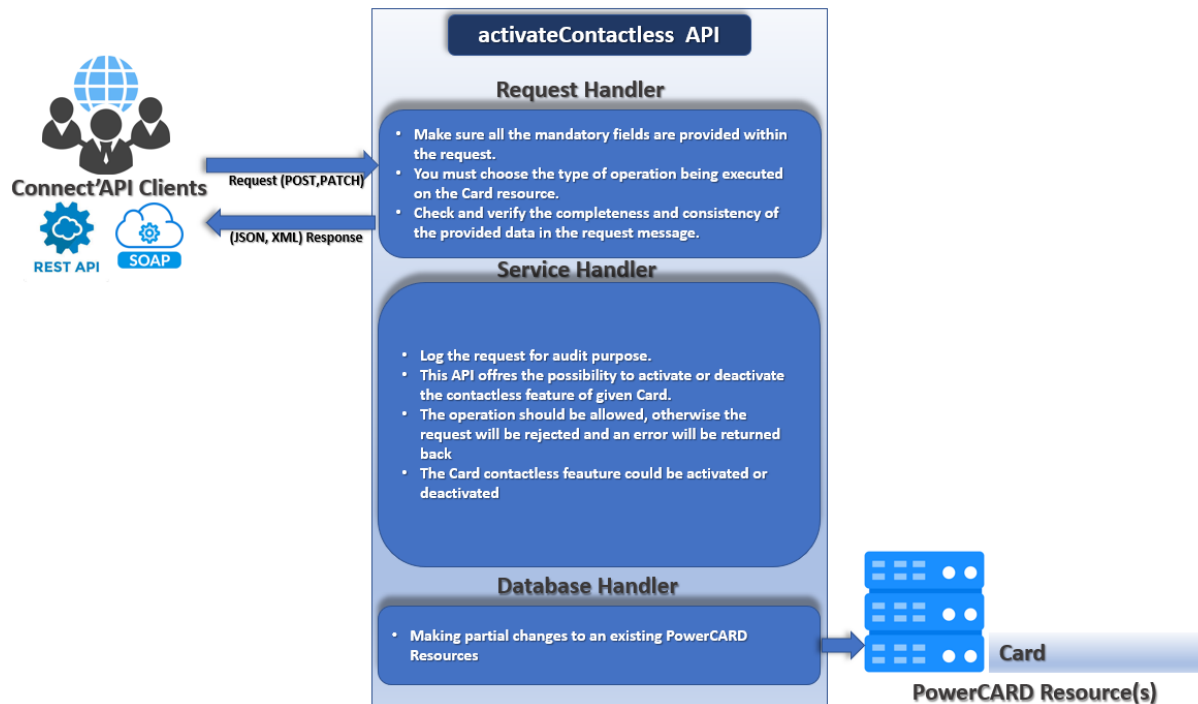


Figure 32: activateContactless process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with requestmessages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
5	ExpiryDate	Date	-	Required (max: 1)	Refers to the expiry date of the card
6	ActivationFlag	C (1)	-	Required (max: 1)	This one character flag indicates whether the operation request concerns an activation or a deactivation. Possible values are: - A: Activation - D: Deactivation

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.

#	Field Name	Data type	Parent	Occurrence	Field Description
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	19101	Invalid Card Number / Card Token ID
2	19102	Operation Not Permitted
3	19103	Contactless Card Feature is already Activated

#	Error code	Error Description
4	19104	Contactless Card Feature is already Deactivated
5	19105	Card Is Cancelled
6	19106	Card Is suspended
7	19107	Card Is Replaced
8	19108	Card Is Expired
9	19109	Operation Not Permitted,the Card Must Have At Least 1 Transaction EMV

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ActivateContactLessWebService/V2	POST	
/Cards/V2/:aliasPAN/ ContactLess/:activationFlag	GET	Required: aliasPAN=[alphanumeric] activationFlag=[A/D]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "expiryDate": "",
  "activationFlag": ""
}
```

4.33 searchAuthorization API

API overview

This web service is used to search for authorizations in the system. Third parties applications developers may use different criteria to search and display authorizations performed by customer, card.... Criteria's could include amount ranges, dates, Account id, transaction types, network... for more details, please refer the technical section of this API.

Functional description

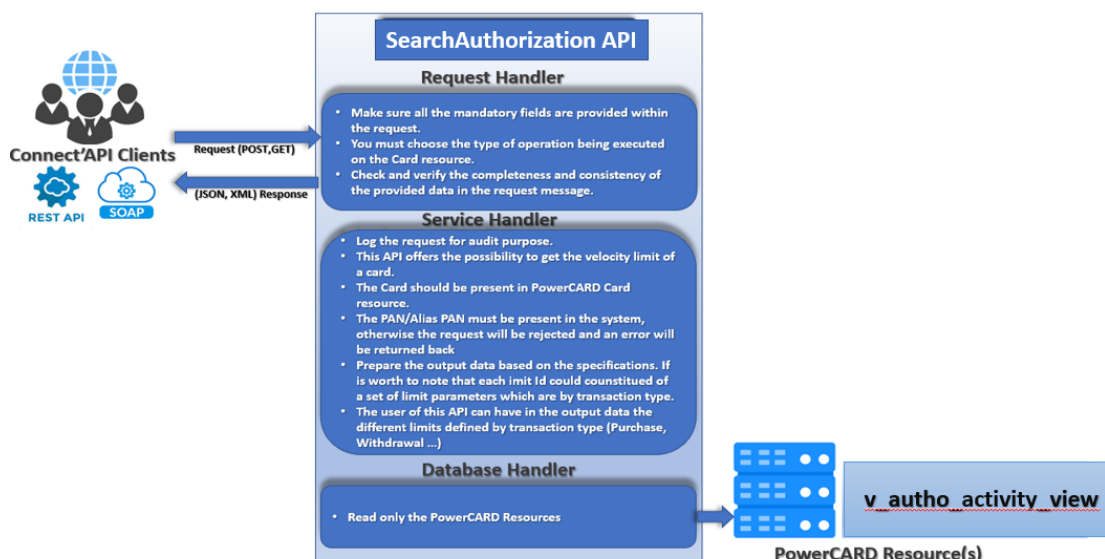


Figure 33: searchAuthorization process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all

#	Field Name	Data type	Parent	Occurrence	Field Description
					the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
12	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
-	RangeDate	Complex	-	Optional (max: 1)	This can contain the range of the dates when the authorizations transmitted. The aggregate is constituted of two items (Start Date & End Date) In case of No Date range specified, the interval will start from the first day

#	Field Name	Data type	Parent	Occurrence	Field Description
					of the month to the request date.
4	StartDT	DateTime	RangeDate	Optional (max:1)	Start Date Format should be YYYYMMDD Data Mapping: Table name: AUTHO _ACTIVITY _ADM Field name: RESPONSE _DATE_ AND _TIME
5	EndDT	DateTime	RangeDate	Optional (max:1)	End Date Format should be YYYYMMDD Data Mapping: Table name: AUTHO _ACTIVITY _ADM Field name: RESPONSE _DATE_ AND _TIME
-	DataSource	Complex	-	Optional (max:1)	
6	SourceIndicator	ClosedEnum	DataSource	Optional (max:1)	Refers to the source indicator, possible values are: • ADM: Dual Message authorization movements. • TMP: Temporary authorization movement used for batch processing. • PRE: Pre-authorizations waiting completion • MTH: Dual Message authorization movements waiting for matching with transaction.

#	Field Name	Data type	Parent	Occurrence	Field Description
					• MSC: Miscellaneous authorization movements. • INC: Incremental authorization movements. • FEE: Authorization fees. • ATM: ATM financial transaction movements. • ASM: Single Message - Financial transaction movements. • SAF: Authorization movements that cannot be inserted in any of other Auto-activity tables. • TRN: Transactions extracted from ATM and ASM tables • HST: Authorization activity history data.
7	SourceAccountId	C(24)	-	Optional (max:1)	Source Account Id
8	Destination AccountId	C(24)	-	Optional (max:1)	Destination Account Id
9	Reference Number	C(12)	-	Optional (max:1)	Contains the reference number as defined in ISO-8583 norm (Field 37)
10	InternalSTAN	C(6)	-	Optional (max:1)	Internal STAN (System Trace Audit Number).
11	ExternalSTAN	C(6)	-	Optional (max:1)	External STAN (System Trace Audit Number). This number is generated

#	Field Name	Data type	Parent	Occurrence	Field Description
					by the sender. It is a unique identification of the transaction. This number should remain unchanged in all the messages related to this transaction.
12	TransmissionTime	DateTime	-	Optional (max:1)	Transmission date sent by the initiator
13	MessageType	C(4)	-	Optional (max:1)	Message type as defined in the ISO8583 protocol. This code must be previously defined in the PowerCARD MESSAGE_LIST table under the general parameter section.
14	ActionCode	C(3)	-	Optional (max:1)	Refers to the desired action, possible values are: • A: Add Bank Account • U: Update Bank Account • D: Delete Bank Account
15	ProcessingCode	C(2)	-	Optional (max:1)	Must contains a code that identifies the cardholder transaction type. Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
-	AmountFrom	Complex	-	Optional (max:1)	This aggregate contains the amount as well as the transaction currency.
16	Amt	Long	AmountFrom	Optional (max:1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code

#	Field Name	Data type	Parent	Occurrence	Field Description
					sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
17	CurCode	C(3)	AmountFrom	Optional (max:1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	AmountTo	Complex	-	Optional (max:1)	This aggregate contains the amount as well as the transaction currency.
18	Amt	Long	AmountTo	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
19	CurCode	C(3)	AmountTo	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
20	AuthoType	C(1)	-	Optional (max:1)	Refers to the authorization type, possible values are: • I: Issuing • A: Acquiring • B: Both issuing & acquiring

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system

#	Field Name	Data type	Parent	Occurrence	Field Description
					error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	Authorizations	Complex	-	Optional infinite list	
5	ProcessingCode	C(2)	Authorizations	Optional (max:1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
6	ReferenceNumber	C(12)	Authorizations	Required (max: 1)	Contains the reference number as defined in ISO-8583 norm (Field 37)
7	InternalSTAN	C(11)	Authorizations	Required (max: 1)	Internal STAN (System Trace Audit Number).
8	ExternalSTAN	C(11)	Authorizations	Required (max: 1)	External STAN (System Trace Audit Number). This number is generated by the sender. It is a unique identification of the transaction. This number should remain unchanged in all the messages related to this transaction.
9	SourceAccountId	C(24)	Authorizations	Optional (max:1)	Source institution identification code

#	Field Name	Data type	Parent	Occurrence	Field Description
10	Destination AccountId	C(24)	Authorizations	Optional (max:1)	Destination institution identification code
11	MessageType	C(4)	Authorizations	Optional (max:1)	Message type as defined in the ISO8583 protocol. This code must be previously defined in the PowerCARD MESSAGE_LIST table under the general parameter section.
12	CardNumber	C(22)	Authorizations	Optional (max:1)	Represents the Card Number or the Card Token Id. It should contain a valid value.
13	CardAcceptoreId	C(15)	Authorizations	Optional (max:1)	Identification code of the card acceptor
14	TransmissionTime	DateTime	Authorizations	Optional (max:1)	Transmission date sent by the initiator
15	LocaleDate	DateTime	Authorizations	Optional (max:1)	Local transaction date and time
-	Transaction Amount	Complex	Authorizations	Optional (max:1)	
16	Amt	Long	Transaction Amount	Optional (max:1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
17	CurCode	C(3)	Transaction Amount	Optional (max:1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD
18	ActionCode	C(3)	Authorizations	Optional (max:1)	Indicates the action taken or to take and the reason
19	AuthorizationCode	C(6)	Authorizations	Optional (max:1)	Authorization code (Code assigned by the authorization body and indicating the agreement)
20	CaptureCode	C(6)	Authorizations	Optional (max:1)	Forwarding institution identification code
21	RoutingCode	C(6)	Authorizations	Optional (max:1)	Acquiring institution identification code
22	SourceIndicator	C(80)	Authorizations	Optional (max:1)	Source Indicator

Specific error codes

#	Error code	Error Description
1	18401	Invalid Range Date
2	18402	Invalid Data Source
3	18404	Invalid Action Code
4	18405	Invalid Processing Code
5	18406	Invalid Transaction Amount
6	18407	Invalid Message Type
7	18408	Invalid Authorization Type

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is		x

Status Code	Description	Success Response	Error Response
	probably valid but needs to be requested again later.		
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchAuthorization WebService/V2	POST	
/Authorizations/V2/:aliasPAN/DeliveryAddress/	GET	Required: aliasPAN=[alphanumeric]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": ""
}
```

4.34 GetParameterList API

API overview

This web service is used to search the referential lists in the system. Third-party application developers can use different criteria to find and view the desired lists as below:



The API support two modes for searching:

Parameter: the query must include search filters and the column to search.

Referential: the query must include the table name which data must be returned.

Functional description

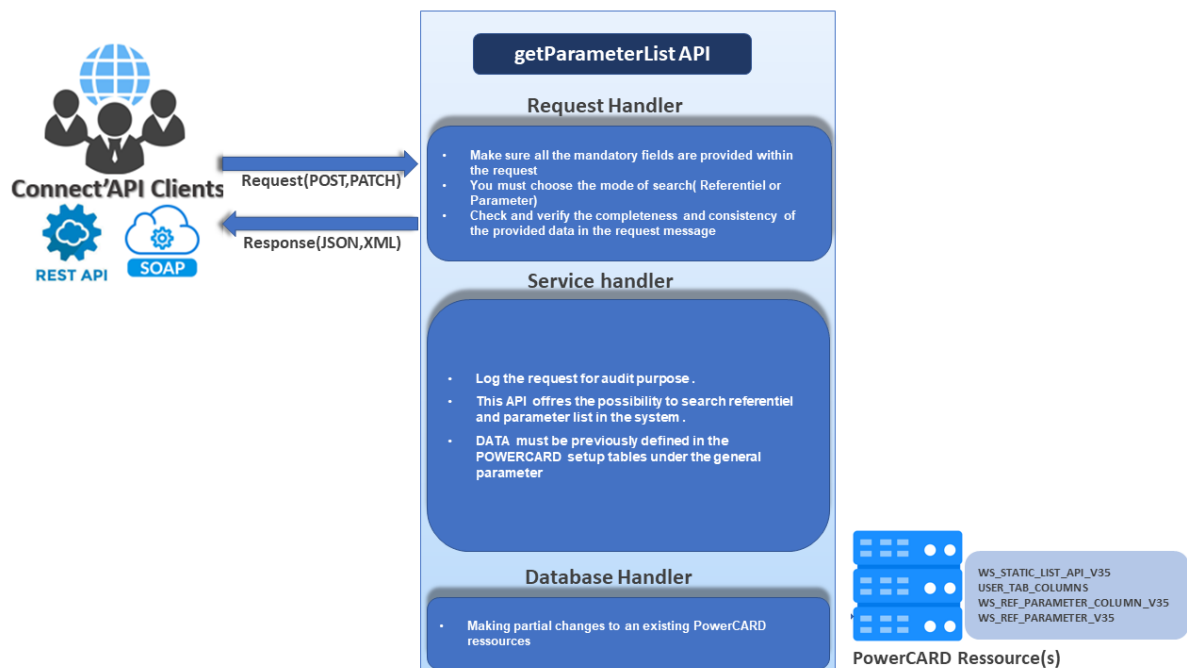


Figure 34: GetParameterList process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
-	ParameterList	Complex	-	Required infinite list	
4	parametermode	C(1)	ParameterList	Optional (max:1)	Refers to the parameter Mode, possible values are: - R: Referentiel - P: Parameter

#	Field Name	Data type	Parent	Occurrence	Field Description
					Defaulted to 'R' when not provided.
5	ParameterName	C(36)	ParameterList	Required (max : 1)	The name of the parameter This is the name of the table to return which must be defined in the PowerCARD table WS_REF_PARAMETER_V35
6	Parameter ColumnList	C(128)	ParameterList	Required infinite list	List of parameter columns This is the name of the columns to return which must be defined in the PowerCARD table WS_REF_PARAMETER_COLUMN_V35 This field is mandatory in case of parameter mode.
-	Parameter ConditionList	Complex	ParameterList	Optional infinite list	
7	ConditionCode	C(36)	Parameter ConditionList	Required (max : 1)	Represents the name of the column to filter with
8	ConditionValue	C(1000)	Parameter ConditionList	Required (max : 1)	Represents the value of the column to filter with
9	ConditionType	C(36)	Parameter ConditionList	Optional (max : 1)	

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should

#	Field Name	Data type	Parent	Occurrence	Field Description
					contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	Parameter ResultList	Complex	-	Required infinite list	The name of the parameter This is the name of the table to return which must be defined in the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD table WS_REF_PARAMETER_V35
8	ParameterName	C(36)	Parameter ResultList	Required (max : 1)	
-	ParameterRecordList	Complex	Parameter ResultList	Required infinite list	
-	ColumnResultList	Complex	Parameter RecordList	Required infinite list	The name of the parameter. This is the name of the table to return which must be defined in the PowerCARD table WS_REF _PARAMETER _V35
9	ColumnName	C(1000)	ColumnResult List	Required (max : 1)	Represents the value of the column.
10	ColumnValue	C(1000)	ColumnResult List	Required (max : 1)	
11	ColumnType	C(1000)	ColumnResult List	Required (max : 1)	
12	ColumnFormat	C(1000)	ColumnResult List	Required (max : 1)	

Specific error codes

#	Error code	Error Description
1	30000	Invalid Parameter Name
2	30001	Invalid Parameter Mode
3	30002	Invalid Parameter Condition List
4	30003	Invalid Parameter Column List Or Parameter Mode
5	30004	Invalid Parameter List

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x

Status Code	Description	Success Response	Error Response
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetParameterListWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "parameterList": {
    "parametermode": "",
    "parameterName": "",
    "parameterColumnList": [""],
    "parameterConditionList": [
      {
        "conditionCode": "",
        "conditionValue": "",
        "conditionType": ""
      }
    ]
  }
}
```

4.35 orderStockCards API

API overview

Branches that have an prepaid card inventory and sell cards usually have what is called a stock. A stock order business function simply means the act of ordering of new stock to refill the inventory.

Each time the agent in the branch capture a new order, the system will make sure that cards with the appropriate product and plastic will be produced. Then shipped to increase the inventory level of the branch that made the request

Functional description

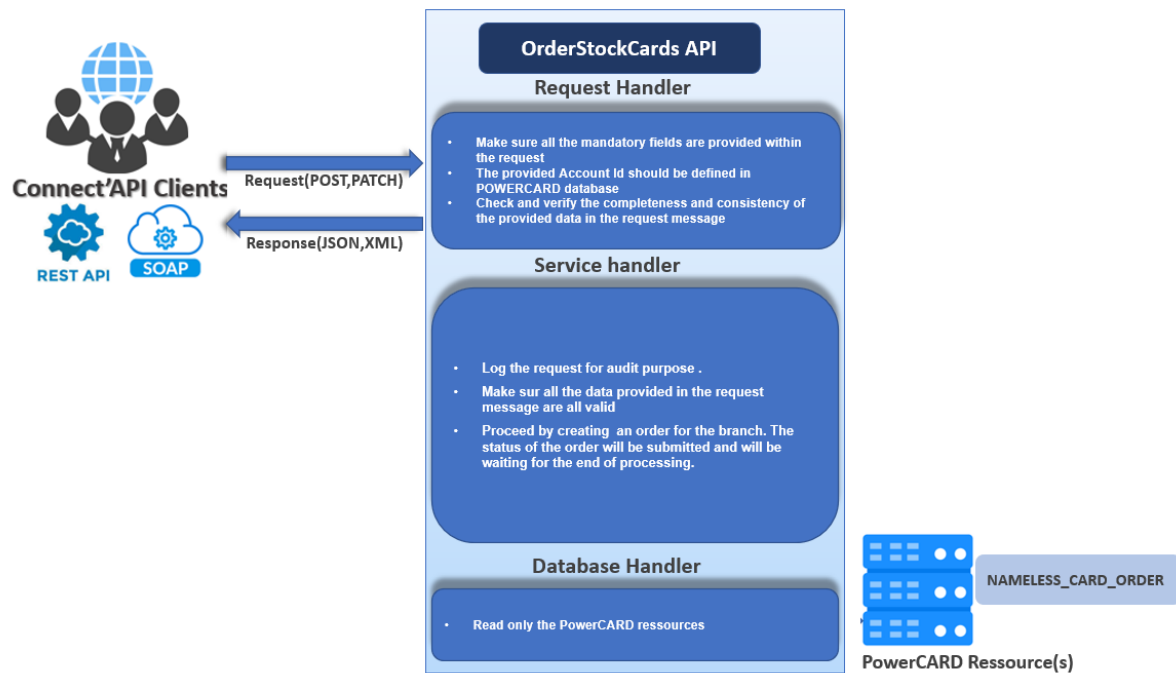


Figure 35: orderStockCards process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	BranchCode	C (6)	-	Optional (max : 1)	Represent the branch of the financial institution.
5	CatPIProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code being requested in the application. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
6	CatPIProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
7	CatAccount Product	C (6)	-	Optional (max : 1)	It is a 6 digit code identifying the account product code being requested in the application. Valid values must be retrieved from PowerCARD CATALOGUE

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PRODUCT table with product type = A
8	CatAccount ProductVersion	C (10)	-	Optional (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
9	PlasticCode	C (3)	-	Optional (max : 1)	It refers to the plastic of the card to be personalized. Valid values must be previously defined in the PowerCARD PLASTIC_ LIST table and could be populated using
10	NbrOfCards	Long	-	Required (max : 1)	This field indicates the number of cars to be personalized
11	EmbossedName	C(26)	-	Optional (max : 1)	This field indicates the embossed name of the cards.
-	PreLoadedAmnt	Complex	-	Optional (max : 1)	It refers to the amount of money that needs to be available of the prepaid card during the first issuance.
12	Amt	Long	PreLoadedAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
13	CurCode	C (3)	PreLoadedAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	BatchNumber	C (25)	-	Required (max : 1)	Must contain the batch number generated internally by PowerCARD and assigned to the order being submitted by the agent. This number could be used later on for follow up purpose.

Specific error codes

#	Error code	Error Description
1	17601	Invalid Bank Code
2	17602	Invalid Branch Code
3	17603	Invalid Card product Code
4	17604	Invalid Credit profile code
5	17605	Invalid Plastic Code
6	17606	Invalid Pre-loaded Amount
7	17607	Invalid Currency Code

Response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	

Status Code	Description	Success Response	Error Response
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/OrderStokCardsWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "requestUID": "",
  "branchCode": "",
  "catPIProduct": "",
  "catPIProductVersion": "",
  "catAccountProduct": "",
  "catAccountProductVersion": "",
  "plasticCode": "",
  "nbrOfCards": "",
  "embossedName": "",
  "preLoadedAmnt": {
    "amt": "",
    "curCode": ""
  }
}
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "parameterList": {
    "parametermode": "",
    "parameterName": "",
    "parameterColumnList": [""],
```

```

    "parameterConditionList":[
      {
        "conditionCode":"","
        "conditionValue":"","
        "conditionType"
      }
    ]
  }
}

```

4.36 GetStockCardsOverview API

API overview

The current service may be used by any third party application to retrieve all the pre-paid batches with different set of criteria including Bank, Branch, Batch Number... the answer must contain the list of batches along with a set of details that could help the end user during its analysis or decision making.

Functional description

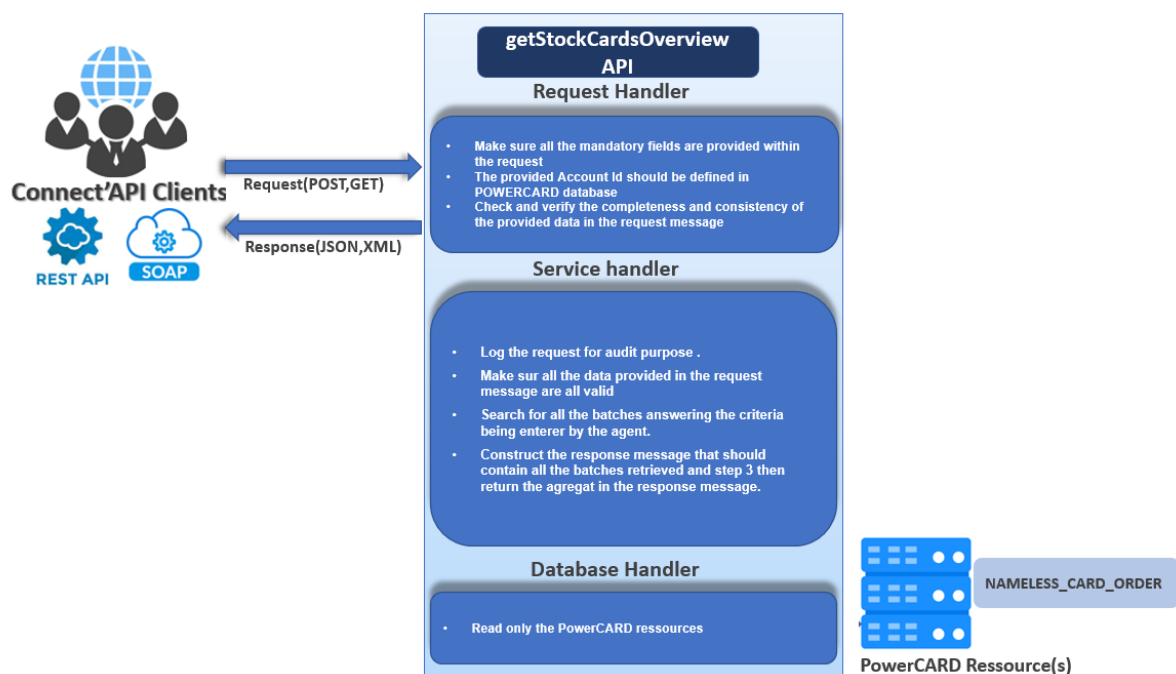


Figure 36: GetStockCardsOverview process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information

#	Field Name	Data type	Parent	Occurrence	Field Description
					necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	BranchCode	C (6)	-	Optional (max : 1)	The branch id of the institution.
5	BatchNumber	C (25)	-	Optional (max : 1)	May contain the batch number generated internally by PowerCARD and assigned to the order being submitted by the agent. A batch number is a identifying a set of cards requested in the same order.
-	RequestedDate	Complex	-	Optional (max : 1)	The selection range date aggregate. System will look for all orders made

#	Field Name	Data type	Parent	Occurrence	Field Description
					between start & End date.
6	StartDt	DateTime	RequestedDate	Optional (max : 1)	Start date
7	EndDt	DateTime	RequestedDate	Optional (max : 1)	End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.

#	Field Name	Data type	Parent	Occurrence	Field Description
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	BatchStock Record	Complex	-	Optional infinite list	Refers to all batches with the criteria being entered
5	BatchNumber	C (25)	BatchStock Record	Required (max : 1)	Refer to a unique identification number generated internally by PowerCARD and assigned to the order being submitted by the agent. A batch number is a identifying a set of cards requested in the same order.
6	BatchStatus	ClosedEnum	BatchStock Record	Required (max : 1)	Status of the batch. Possible values are: • P: Order Pending • B: Order Processed • R: Order Rejected • A: Order Received • C: Order Cancelled
7	BatchStatusDate	DateTime	BatchStock Record	Required (max : 1)	Status date Last update date of the batch
8	BatchOrderDate	DateTime	BatchStock Record	Required (max : 1)	Date when the order took place
9	BatchRequester	C (15)	BatchStock Record	Required (max : 1)	Refers to the Id of the user who made the order
10	BankCode	C (22)	BatchStock Record	Required (max : 1)	The bank code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be previously defined in the PowerCARD BANK table under the general parameter section. It should be retrieved from PowerCARD
11	BranchCode	C (15)	BatchStock Record	Optional (max : 1)	Branch code. 6 digits code of the branch to which the card belongs Valid values must be downloaded from PowerCARD BRANCH table
12	CatPIProduct	C (6)	BatchStock Record	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code being requested in the application. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
13	CatPIProduct Version	C (10)	BatchStock Record	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
14	CatAccount Product	C (6)	BatchStock Record	Optional (max : 1)	It is a 6 digit code identifying the account product code being requested in the application. Valid values must be retrieved from PowerCARD CATALOGUE

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PRODUCT table with product type = A
15	CatAccount ProductVersion	C (10)	BatchStock Record	Optional (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
16	PlasticCode	PlasticCode _Type	BatchStock Record	Required (max : 1)	It refers to the plastic of the card to be personalized. Valid values must be previously defined in the PowerCARD PLASTIC _LIST table
17	NbrOfCards Ordered	Long	BatchStock Record	Required (max : 1)	This field indicates the number of cards to be personalized
18	NbrOfSoldCards	Long	BatchStock Record	Required (max : 1)	This field indicates the number of cards being sold to consumers
19	NbrOfRejected Cards	Long	BatchStock Record	Required (max : 1)	This field indicates the number of cards being rejected for invalidity
20	NbrOfAvailable Cards	Long	BatchStock Record	Required (max : 1)	This field indicates the number of cards not yet sold.
21	EmbossedName	C(26)	BatchStock Record	Optional (max : 1)	This field represents the embossed name of the PAN.
-	PreLoadedAmnt	Complex	BatchStock Record	Required (max : 1)	Refers to the amount that should be pre loaded in the account
22	Amt	Long ()	PreLoadedAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10n; where n is the currency exponent of the currency code sent in the field

#	Field Name	Data type	Parent	Occurrence	Field Description
					below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
23	CurCode	NC (3)	PreLoadedAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD

Specific error codes

#	Error code	Error Description
1	17701	Invalid Bank Code
2	17702	Invalid Branch Code
3	17703	Batch Number Not Found
4	17704	Invalid range dates

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x

Status Code	Description	Success Response	Error Response
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetStockCardsOverviewWebService/V2	POST	
/StockCards/V2? branchCode={branchCode}&batchNumber={batchNumber}&startDt={startDt}&endDt={endDt}	GET	Optional: clientCode=[C(6)] batchNumber=[C(25)] startDt=[dateTime] endDt=[dateTime]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "expiryDate": "",
  "branchCode": "",
  "batchNumber": "",
  "requestedDate": {
    "startDt": "",
    "endDt": ""
  }
}
```

4.37 GetStatementHistory API

API overview

A client or agent may use this web service to retrieve account information for a statement period or multiple statement periods. They may optionally provide a date range to limit the number of Account Statement aggregates that are returned in the response. Note that the statement is based on the normal cutoff cycles and the date range must return statements that ended during the specified range.

Functional description

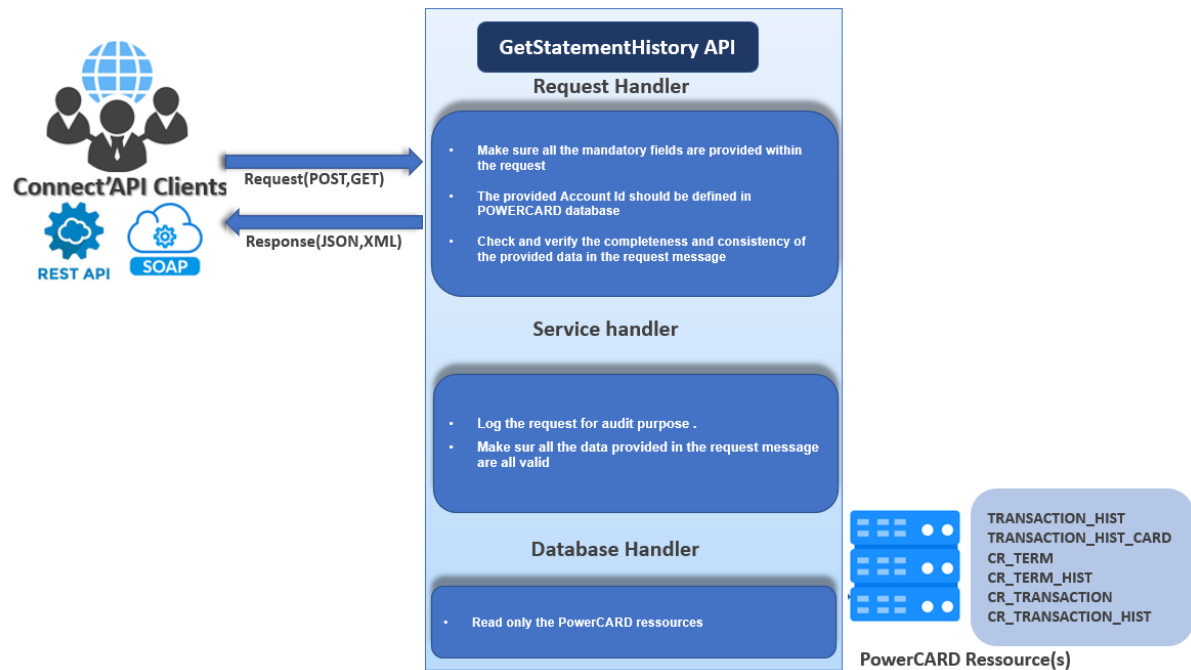


Figure 37: GetStatementHistory process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountID	C (24)	-	Required (max : 1)	The CMS account identifier as defined in PowerCARD.
-	StatSelRangeDt	Complex	-	Required (max : 1)	The selection range date aggregate. It must contain the range date for which statements should be returned.
5	StartDt	DateTime	StatSelRangeDt	Optional (max : 1)	Start date
6	EndDt	DateTime	StatSelRangeDt	Optional (max : 1)	End date

Response table

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					• Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	MaxRowNbr	Long ()	-	Optional (max : 1)	This will depict the maximum number of row that can be included in the response message. Note that if the result exceeds this number, only this number of statement will be returned. This mechanism is used to limit the number of statement that can be fetched in one call.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Statement	Complex	-	Required infinite list	Statement Records This aggregate is intended to contain the customer statement (list of transactions for a given period of time)
6	StatementId	C (20)	Statement	Required (max : 1)	Statement identifier A unique statement id within the search result.
7	StatementDate	DateTime	Statement	Optional (max : 1)	Statement date Date when the account was cutoff/ ended
-	AccountBalance	Complex	-	Optional infinite list	This aggregate contains data related to the balance of the accounts linked to the card transmitted in the request message.
8	CMSAccountId	C (24)	AccountBalance	Required (max : 1)	The CMS account identifier as defined in PowerCARD.
9	AccountType	C (40)	AccountBalance	Required (max : 1)	Account type Refers to the type of the account. Possible values can be downloaded from the PowerCARD ACCOUNT_TYPE_LIST table
10	ClientCode	C(24)	AccountBalance	Optional (max : 1)	The client code present in the table CLIENT in PowerCARD.
11	ClientHostID	C(24)	AccountBalance	Optional (max : 1)	External Identified of the customer in the host
-	CrdAcctInfo	Complex	AccountBalance	Optional (max : 1)	The credit account information This aggregate contain information with regards to the credit account profile, account credit limit...

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid only in case of credit card account.
12	CatAccountProduct	C (6)	CrdAcctInfo	Required (max : 1)	It is a 6 digit code identifying the account product code assigned to the account. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
13	CatAccount ProductVersion	C (10)	CrdAcctInfo	Required (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
14	CreditLimit	Long ()	CrdAcctInfo	Optional (max : 1)	Sets the maximum amount that can be borrowed on the cardholder's account. The value assigned to this field can be used to set initial credit limits for newly created cards in the PowerCARD system. In order to be valid, the value of this field should not exceed the ceiling amount defined in the credit account profile. If empty, the system will inherit the default value defined in the credit account profile.
15	CashLimit	Long ()	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of cash that can

#	Field Name	Data type	Parent	Occurrence	Field Description
					be obtained on the cardholder's account (ATM withdrawal, cash advance).
16	Loanlimit	Long ()	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of loan transactions that can be performed on the cardholder's account. The value of this field should not exceed the total credit limit permitted on the cardholder's account.
17	paymentMode AutoDebitInd	C (1)	CrdAcctInfo	Required (max : 1)	Should contain the automatic debit payment mode indicator: • 'M': Automatic debit is mandatory. • 'P': Automatic debit is allowed. • 'F': Automatic debit is not allowed.
18	CycleCode	C (20)	CrdAcctInfo	Optional (max : 1)	The credit cycle code The cycle is the period of time that elapses between 2 consecutive billing statements. The value of this field indicates the code that refers to the new billing cycle that should be assigned to the account. Possible values must be retrieved from the PowerCARD CYCLE _CUTOFF _PARAMETERS table

#	Field Name	Data type	Parent	Occurrence	Field Description
19	DefaultRate	Decimal	CrdAcctInfo	Optional (max : 1)	It refers to the default interest rate that could be charged by credit card issuer. Unless it is specified in the credit account profile, this rate will be used by PowerCARD for all type of account balances
20	StatementCode	C (2)	CrdAcctInfo	Optional (max : 1)	Refers to the setting of statement parameters in PowerCARD that enables to specify all the data relative to the statement that will be sent to the customer. It includes periodicity, fees amount... This setting is defined as an index that must be retrieved from PowerCARD STATEMENT_PARAMETERS table using
-	RequestedCredit Amount	Complex	CrdAcctInfo	Optional (max : 1)	The requested credit amount. The captured limit is made by PowerCARD.
21	Amt	Long ()	RequestedCredit Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
22	CurCode	NC (3)	RequestedCredit Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction

#	Field Name	Data type	Parent	Occurrence	Field Description
					amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	DebitOrder Amount	Complex	CrdAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
23	Amt	Long ()	DebitOrder Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
24	CurCode	NC (3)	DebitOrder Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	DebitOrder Percentage	Complex	CrdAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
25	Amt	Long ()	DebitOrder Percentage	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n

#	Field Name	Data type	Parent	Occurrence	Field Description
					digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	NC (3)	DebitOrder Percentage	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	CurrentBalance	Complex	Account Balance	Required (max : 1)	Must include the amount value of the account balance. - For deposit Accounts, it refers to Current available balance (typically, ledger balance less outstanding authorizations). - For Credit Card Accounts, it refers to the cardholder outstanding balance during the cycle present cycle.
27	Amt	Long ()	CurrentBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
28	CurCode	NC (3)	CurrentBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
29	BalanceStatus Date	DateTime	AccountBalance	Optional (max : 1)	Refers to the last update date of the concerned balance.
-	CurrentDebit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the debit transactions performed by the cardholder during the present cycle. Valid only for credit accounts
30	Amt	Long ()	CurrentDebit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
31	CurCode	NC (3)	CurrentDebit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD
-	CurrentCredit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the credit transactions performed by the cardholder during the present cycle. (payment, refund...) Valid only for credit accounts
32	Amt	Long ()	CurrentCredit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
33	CurCode	NC (3)	CurrentCredit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	ClosingBalance	Complex	AccountBalance	Optional (max : 1)	The closing balance is the positive or negative outstanding amount in an account at the conclusion of the last cycle. Valid only for credit accounts
34	Amt	Long ()	ClosingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the

#	Field Name	Data type	Parent	Occurrence	Field Description
					currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
35	CurCode	NC (3)	ClosingBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	MinimumDue	Complex	AccountBalance	Optional (max : 1)	The last minimum amount calculated by the system that the cardholder must pay by the due date. Valid only for credit accounts
36	Amt	Long ()	MinimumDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
37	CurCode	NC (3)	MinimumDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section. Valid values must be retrieved from PowerCARD
38	LastCutoff	DateTime	AccountBalance	Optional (max : 1)	Refers to the last Account statement date Valid only for credit accounts
39	NextDueDate	DateTime	AccountBalance	Optional (max : 1)	It refers to the date the minimum monthly payment is due Valid only for credit accounts
40	Status	ClosedEnum	AccountBalance	Optional (max : 1)	This field must contain the status of the account: - For Deposit accounts, Valid values are: - N = Normal - C = Canceled - S = Suspended - W = Waiting for validation - V = To be sent for validation - R = Rejected. - For Credit Card Accounts: 0 = Normal 1 = Supervision 2 = Pre legal 3 = Legal 4 = Insurance management

#	Field Name	Data type	Parent	Occurrence	Field Description
					• 5 = Lost / Closed • 6 = Lost / Death • 7 = Closed / Canceled • 8 = Write Off • 9 = Untraceable
-	withdrawal Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the account that may be withdrawn.
41	Amt	Long ()	withdrawal Available	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
42	CurCode	NC (3)	withdrawal Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	purchase Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the account that may be used for purchase transactions.
43	Amt	Long ()	purchase Available	Optional (max : 1)	Amount.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
44	CurCode	NC (3)	purchase Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	withdrawalLimit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of cash that can be obtained on the cardholder's account (ATM withdrawal, cash advance).
45	Amt	Long ()	withdrawalLimit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
46	CurCode	NC (3)	withdrawalLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	purchaseLimit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of money that can be used by the cardholder to make purchase transactions.
47	Amt	Long ()	purchaseLimit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
48	CurCode	NC (3)	purchaseLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
-	PendingBalance	Complex	AccountBalance	Optional (max : 1)	This is the total amount of all pending authorizations.
49	Amt	Long	PendingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field

#	Field Name	Data type	Parent	Occurrence	Field Description
					below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
50	CurCode	C (3)	PendingBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD
48	DueDt	Date	Statement	Optional (max : 1)	Payment Due Date It indicates the date when the payment is due. Valid only in case of credit card account.
-	MinAmtDue	Complex	Statement	Optional (max : 1)	Minimum Amount due It is the minimum payment amount that the customer is required to pay by the due date. This field is valid only in case of credit card account.
49	Amt	Long ()	MinAmtDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
50	CurCode	NC (3)	MinAmtDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD using the Web Service: getReferentialList
-	DelinqAging	Complex	Statement	Optional infinite list	Delinquency Aging Aggregate. This aggregate shows how long due amounts from a customer have been outstanding
51	Deduplicate Comment	C (2048)	DelinqAging	Required (max : 1)	A description field that contains the result of the deduplication process execution
52	Deduplicate Result	ClosedEnum	DelinqAging	Required (max : 1)	It refers to the result of the execution of the rule. Possible values are: - N: No deduplication found - D: Deduplication found
53	StartDt	DateTime	Statement	Optional (max : 1)	Selection Start Date time provided in the request message to allow client to use these dates to define the cycle start date.
54	EndDt	DateTime	Statement	Optional (max : 1)	End Date time. Provided to allow client to use these dates to define the cycle end date
-	StatTrans	Complex	Statement	Required infinite list	The card transaction aggregate details.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	transaction Common	Complex	StatTrans	Required (max : 1)	This aggregate contains data which may be used in both contexts (Issuing and acquiring). For instance the transaction date, Amount, Card number, VAT Amount... Present all the time.
55	TransId	C (23)	transaction Common	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
56	RefSeq	N (4)	transaction Common	Required (max : 1)	Refers to the transaction sequence Number
-	TrnType	Complex	transaction Common	Required (max : 1)	This aggregate must contain the transaction code as defined in PowerCARD TRANSACTIONS_DEFINITION table as well as the description of the transaction. Note that the valid transaction codes must be retrieved from PowerCARD
57	Code	C (20)	TrnType	Required (max : 1)	Must contain the code of the parameter being shared
58	Label	C (52)	TrnType	Required (max : 1)	The label of the parameter being shared
59	ReversalFlag	C (1)	transaction Common	Optional (max : 1)	Nature of the transaction. - N: Normal - R: Reversal
60	TransSign	C (1)	transaction Common	Required (max : 1)	Transaction sign

#	Field Name	Data type	Parent	Occurrence	Field Description
					Contains a code that indicates the sign of the transaction. Possible values are: - D: Debit - C: Credit
61	TransDate	DateTime	transaction Common	Required (max : 1)	The date when the transaction was made.
62	TransPostDate	Date	transaction Common	Required (max : 1)	Refers to the date when the concerned transaction was posted to the cardholder account
63	TransDateEvent	DateTime	transaction Common	Optional (max : 1)	The date when the transaction was proceed by the card management system.
-	TransAmt	Complex	transaction Common	Required (max : 1)	Identified the transaction amount in the currency used during the transaction. The entry must be a numeric greater than zero.
64	Amt	Long ()	TransAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
65	CurCode	NC (3)	TransAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid values must be retrieved from PowerCARD
66	AuthCode	C(6)	transaction Common	Optional (max : 1)	Authorization aggregate Contains the Authorization code and date. Mandatory only in case an issuer, an authorizing processor, or Stand-In Processing (STIP) provides an approval of a transaction.
67	CardNumber	C (22)	transaction Common	Optional (max : 1)	The card number used during the transaction.
-	Country	Complex	transaction Common	Required (max : 1)	Country where the transaction took place.
68	CountryCode	NC (3)	Country	Required (max : 1)	Identify the country code ISO-3166 1-numeric
69	CountryName	C (250)	Country	Optional (max : 1)	Identify the country name Both the country code and the country name are downloaded from the PowerCARD COUNTRY table present under the general parameter section.
70	Network	ClosedEnum ()	transaction Common	Required (max : 1)	The network field allows you to specify the network where transactions are made. It is to mention that this code should be previously defined in the NETWORK table under the general parameter section. To populate the valid values from PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	MCC	Complex	transaction Acquiring	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the PowerCARD MCC table
71	MccCode	C (4)	MCC	Optional (max : 1)	MCC code or Merchant category code. This code must be previously defined in the PowerCARD MCC table under the general parameter section.
72	MccLabel	C (100)	MCC	Optional (max : 1)	MCC description as stored in PowerCARD.
73	PosEntryMode	ClosedEnum	transaction Acquiring	Required (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : • 00: Unknown or terminal not used • 01: Manual (key entry) • 02: Magnetic stripe read • 03: Integrated circuit card read • 04: Contactless device-read
74	MerchantAcronym	C (40)	transaction Acquiring	Optional (max : 1)	The Merchant Acronym

#	Field Name	Data type	Parent	Occurrence	Field Description
					Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or member's imprinter plate.
-	transactionIssuing	Complex	StatTrans	Optional (max : 1)	Data which are more related to the issuing context are shared in this aggregate. Note that the transaction aggregate must contain the transactionCommon aggregate and one of the two other aggregate (transactionAcquiring or transactionIssuing)
75	Description	C (40)	transaction Issuing	Optional (max : 1)	Short description of the merchant where the transaction occurred.
76	Territoriality	ClosedEnum	transaction Issuing	Optional (max : 1)	Valid values : • ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. • NAT: National, apply only to transactions made in the same country as the issuer country. • INT: International, apply only to transactions that took place outside the

#	Field Name	Data type	Parent	Occurrence	Field Description
					cardholder country.
-	BillCurAmt	Complex	transaction Issuing	Optional (max : 1)	This aggregate will contain the billing amount in the issuer's billing currency.
77	Amt	Long ()	BillCurAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
78	CurCode	NC (3)	BillCurAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
79	TransAccountId	C (24)	transaction Issuing	Optional (max : 1)	The Account Number where the transaction is posted.
-	StatTransTotal	Complex	Statement	Optional (max : 1)	Contains a summary of the transactions. Note that this aggregate should not be used in a multi-currency context.
80	NbTransactions	Long	StatTransTotal	Required (max : 1)	Indicates the number of transactions returned.
-	TotalCredit	Complex	StatTransTotal	Optional (max : 1)	Refers to the sum total of transaction amounts of transactions with Credit sign.

#	Field Name	Data type	Parent	Occurrence	Field Description
81	Amt	Long ()	TotalCredit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
82	CurCode	NC (3)	TotalCredit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	TotalDebit	Complex	StatTransTotal	Optional (max : 1)	Refers to the sum total of transaction amounts of transactions with debit sign.
83	Amt	Long ()	TotalDebit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
84	CurCode	NC (3)	TotalDebit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.

Specific error codes

#	Error code	Error Description
1	02801	Invalid range date
2	02802	Invalid credit card number
3	02803	Invalid Currency Code
4	02804	Invalid Account Status
5	02805	Invalid Transaction Code
6	02806	Invalid Account Id
7	02807	Shaddow Account Error
8	02808	Invalid Account Number
9	02809	Card Expired
10	02810	Cr Profile Error

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetStatementHistoryWebService/V2	POST	
/StockCards/V2? branchCode={branchCode}&batchNumber={batchNumber}&startDt={startDt}&endDt={endDt}	GET	Optional: clientCode=[C(6)] batchNumber =[C(25)] startDt =[dateTime] endDt=[dateTime]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "accountId": "",
  "statSelRangeDt": {
    "startDt": "",
    "endDt": ""
  }
}
```

4.38 updateCustomerName API

API overview

Connect'API offers a selection of web services with regards to customer detail maintenance, the update person name web services is one of those services that can be used by both agents and customers in order to change the person name of an existing customer in the system.

Functional description

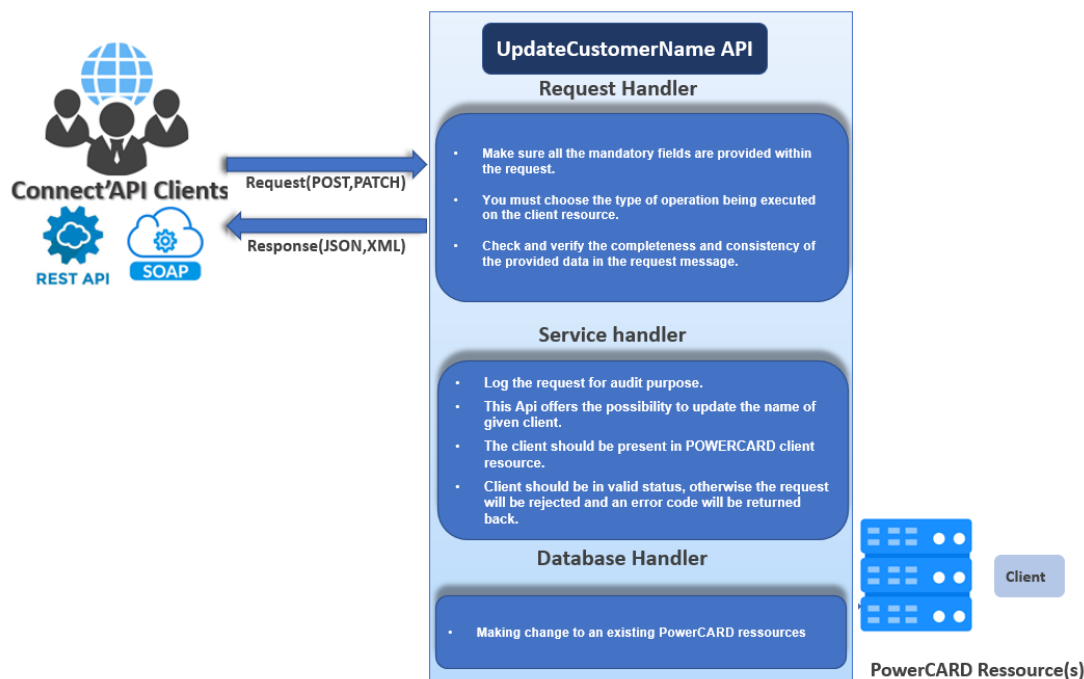


Figure 38: updateCustomerName process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CustomerId	C (22)	-	Required (max : 1)	Used to transmit the customer identification code in PowerCARD. If provided only data belonging to this customer will be returned. If present in the message, this code must be already defied in the PowerCARD CLIENT table
5	TitleCode	C (2)	-	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
6	FamilyName	C (40)	-	Required (max: 1)	Cardholder's last name.
7	SecondFamilyName	C (40)	-	Optional (max: 1)	Cardholder's second family name if compound names are in use
8	FirstName	C (40)	-	Required (max: 1)	Cardholder's given name.

#	Field Name	Data type	Parent	Occurrence	Field Description
9	SecondFirstName	C (40)	-	Optional (max: 1)	Cardholder's second first name if compound names are in use.
10	MaidenName	C (40)	-	Optional (max: 1)	Mother's Maiden name

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system

#	Field Name	Data type	Parent	Occurrence	Field Description
					error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	02301	Invalid Customer Id
2	02302	Invalid Title Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/v2	POST	
/Clients/v2/:customerId/TitleCode/:titleCode/ FamilyName/:familyName/ SecondFamilyName/:secondFamilyName/ FirstName/:firstName/ SecondFirstName/:secondFirstName/ MaidenName/:maidenName	PATCH	Required customerId=[C(22)] Optional TitleCode FamilyName SecondFamilyName FirstName SecondFirstName MaidenName

REST sample call

```

    "keyValues": [
      {
        "key": "TOKEN",
        "data": "Y",
        "type": "",
      }
    ],
    "requestInfo": {
      "requestUID": "",
      "requestDate": "",
      "userID": ""
    },
    "customerId": "",
    "TitleCode": "",
    "FamilyName": "",
    "SecondFamilyName": "",
    "FirstName": "",
    "SecondFirstName": "",
    "MaidenName": ""}

```

4.39 updateCustomerCategory

API overview

The current web service could be used as a way to change some characteristics of an existing customer in PowerCARD system. It is mainly used by the agent in the bank desiring to apply a change a customer characteristic like its customer category.

Functional description

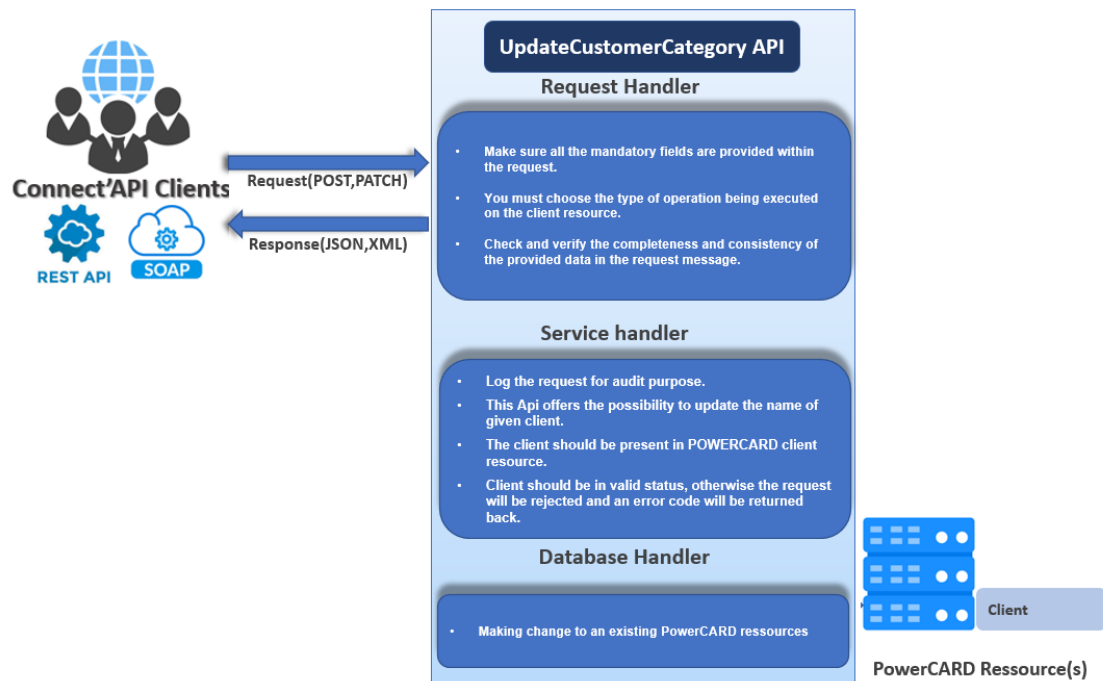


Figure 39: updateCustomerCategory

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CustomerId	C (22)	-	Required (max : 1)	Used to transmit the customer identification code in PowerCARD. If provided only data belonging to this customer will be returned. If present in the message, this code must be already defied in the PowerCARD CLIENT table
5	CustomerSegment	C (3)	-	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the

#	Field Name	Data type	Parent	Occurrence	Field Description
					description of the error encountered

Specific error codes

#	Error code	Error Description
1	03601	Invalid Customer Id
2	03602	Invalid Customer Segment

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/v2	POST	
/Clients/v2/:customerId/CustomerSegment/:customerSegment	PATCH	Required customerId=[C(22)]

PowerCARD Resource URI	Method	URI Params
		customerSegment=[C(3)]

REST sample call

```
"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"customerId": "",
"customerSegment": ""}
```

4.40 lostStolenDeclaration API

API overview

This web services is used to initiate and update a lost stolen case. The lost and stolen cases can be initiated in two states. Confirmed or Waiting for confirmation. Unconfirmed cases can be subsequently confirmed or canceled. Based on the lost and stolen case reason authorizations can be rejected and transaction posting can be suspended for the concerned PAN.

Note that updating the declaration is only authorized if the declaration is awaiting confirmation.

In case of replacement the new PAN and expiry date will be returned on the response.

As output the web service returns the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

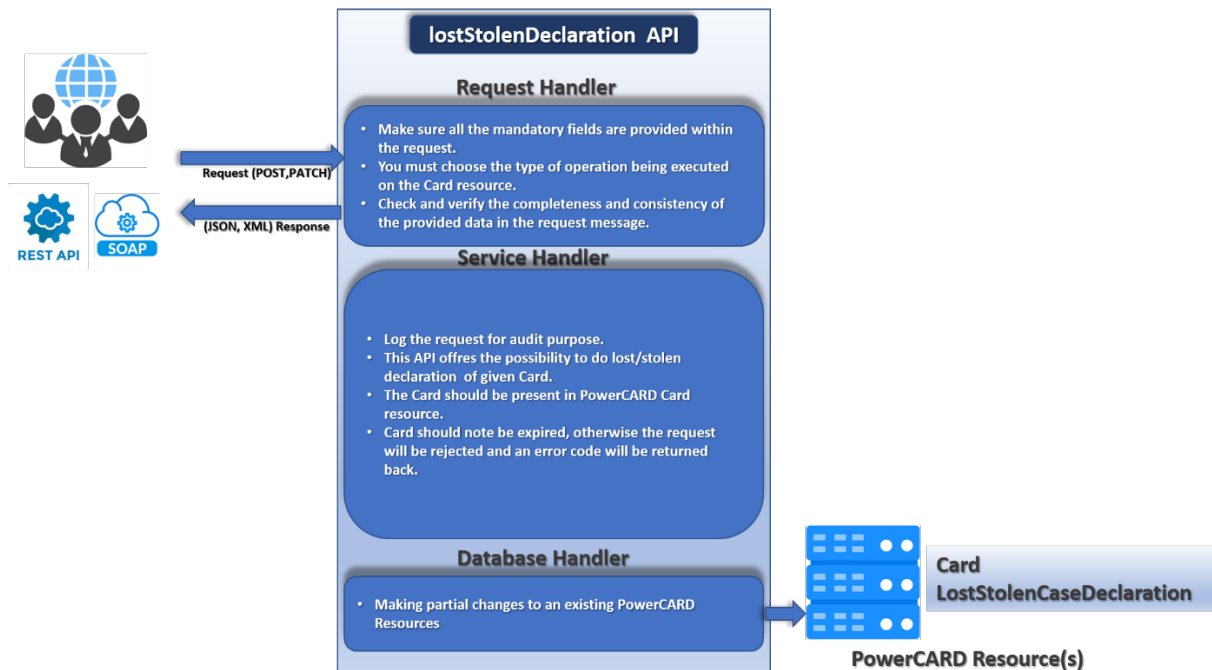


Figure 40: lostStolenDeclaration process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	KeyValues	Complex	-	Optional infinite list	The additional key values protocol extension. The supplementary Information related to the Lost/Stolen declaration will be added using the following keys: Reported Origin: - Key : REPORT_ORIGIN - Data: The report origin. Location: - Key : LOCATION

#	Field Name	Data type	Parent	Occurrence	Field Description
					- Data : The location Loss reason: - Key : LOSS_REASON - Data : The report origin. Sample: <pre>keyValues": [{ "key": "REPORT_ORIGIN", "data": "ZZZZ" }, { "key": " LOCATION", "data": "YYYYYY" }, { "key": " LOSS_REASON", "data": "2222" }]</pre>
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: 2222 2222 -2222 -2222 -2222 -222222 222222 Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was

#	Field Name	Data type	Parent	Occurrence	Field Description
					submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max : 1)	This field indicates the card number of the Lost/ Stolen declaration concerned card
5	ActionType	C(1)	-	Required (max : 1)	A: Add a lost Stolen record U: Update a lost and stolen record
6	DeclarationStatus	C(1)	-	Required (max : 1)	C : "Confirmed". Can be used with action A and U (Add or Update). The cardholder confirms that the card is lost/stolen. In this case the card status will be set to 'Canceled'. When action is U (Update), a lost and stolen case with status W must be existing for the PAN. W : "Waiting confirmation". Can be used with action A (Add) only. The cardholder is not sure that his card is lost or stolen.. In this case the card status will be set to 'Suspended' D : "Cancelled". Can be used with action U(Update) only. A lost and stolen case with status W (Waiting for confirmation) must

#	Field Name	Data type	Parent	Occurrence	Field Description
					be existing for the PAN. The cardholder confirms back that the card is found, then the case is closed and cards are reinstated to the original status
7	CardStatusReason	C (2)	-	Required (max : 1)	This field indicates the card status reason after submitting the declaration. This code must be a valid value that must previously defined in the STATUS_REASON_LIST table under the general parameter section.
-	StopListRangeDate	Complex	-	Optional (max : 1)	The stop list range date. This aggregate is used to transmit the period of the concerned card's stop list reason.
9	StartDt	DateTime	LostStolen RangeDate	Required (max : 1)	This field indicates the start date of the concerned card's stop list reason.
10	EndDt	DateTime	LostStolen RangeDate	Optional (max : 1)	This field indicates the end date of the concerned card's stop list reason.
11	replacePanFlag	C(1)	-	Optional (max: 1)	If the "replacement option" defined in card status reason is set to "ALLOW" (in the table STATUS_REASON_LIST for the user reason), the value of this flag will specify whether a new PAN will be generated or not. The possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					'Y':Yes Replace PAN 'N': No Do not replace PAN This flag triggers replacement only when the declaration status is confirmed.
12	IncidentDate	Date	-	Optional (max : 1)	This field indicates the date when the card has been lost or stolen. If not specified this date gets systematically the business date as default date during the card Lost/Stolen case declaration
12	DeclarationDate	Date	-	Optional (max : 1)	This field indicates the date when the card has been declared as lost or stolen. If not specified this date gets systematically the business date as default date during the card Lost/Stolen case declaration.
13	CountryCode	C(3)	-	Optional (max : 1)	This field represents the country where the card has been lost or found. This value must be present in the PowerCARD COUNTRY table under the general parameters section.
14	City	C (40)	-	Optional (max: 1)	This field represents the city where the card has been lost. This is a free text city name
15	StateProvince	C (40)	-	Optional (max: 1)	Free text indicating the state or the

#	Field Name	Data type	Parent	Occurrence	Field Description
					province where the card has been lost
16	ReportedByCardholder	C(1)	-	Optional (max: 1)	A flag to indicate if the lost stolen declaration is done by the cardholder or not. The possible values are: • 'Y': Yes • 'N': No
17	ReporterName	C (40)	-	Optional (max: 1)	The reporter name if the declaration isn't done by the cardholder.
18	ReporterComment	C (300)	-	Optional (max: 1)	The reporter comment.
19	PoliceDeclarationInd	C(1)	-	Required (max : 1)	Y/N indicator Y: Police declaration done N: Police declaration not done.
-	PoliceDeclaration	Complex	-	Optional (max: 1)	This aggregate is used if the cardholder has done a police declaration
20	Police DeclarationId	C(24)	PoliceDeclaration	Required (max : 1)	This field is the unique code identifying the police report.
21	Police DeclarationDate	Date	PoliceDeclaration	Required (max : 1)	This field indicates the date of police report.
22	Location	C(40)	PoliceDeclaration	Required (max : 1)	This field represents the place where the police incident has been reported.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.

#	Field Name	Data type	Parent	Occurrence	Field Description
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered.
5	CardNumber	C (22)	-	Optional (max : 1)	Card Number

#	Field Name	Data type	Parent	Occurrence	Field Description
					Represents the card number generated by PowerCARD in case of the lost / stolen declaration is confirmed and the status reason chosen allows the replacement of the card.
6	ExpiryDt	DateTime	-	Optional (max : 1)	Refers to the expiry date of the card being returned back in the response message.

Specific error codes

#	Error code	Error Description
1	22001	Invalid Card Number
2	22102	Card Expired
3	22103	Invalid Declaration Status
4	22104	Declaration update not allowed
5	22105	Invalid Range Date
6	22106	Invalid City Code
7	22107	Invalid Country Code
8	22108	Invalid Police declaration Indicator
9	22109	Invalid Replace Pan Flag
10	22110	Invalid Action Type
11	22111	Invalid Declaration Status
12	22112	When Action Is U, a lost and stolen Case With Status W Must Be Existing For The Pan
13	22113	Card Account Link Not Founded
14	22114	Multi Cards Account Link Founded
15	22115	Invalid Card Status Reason
16	22116	Police Declaration Date Not Founded
17	22117	Location Police Declaration Not Founded
18	22118	Invalid Bank Code Customer

RESTful response codes

4.41 updateCustomerDemogData

API overview

PowerCARD offers a selection of web services with regards to customer detail maintenance, the update customer demographic data web service is one of those web services that can be used by both agents and customers in order to change the customer information.

Functional description

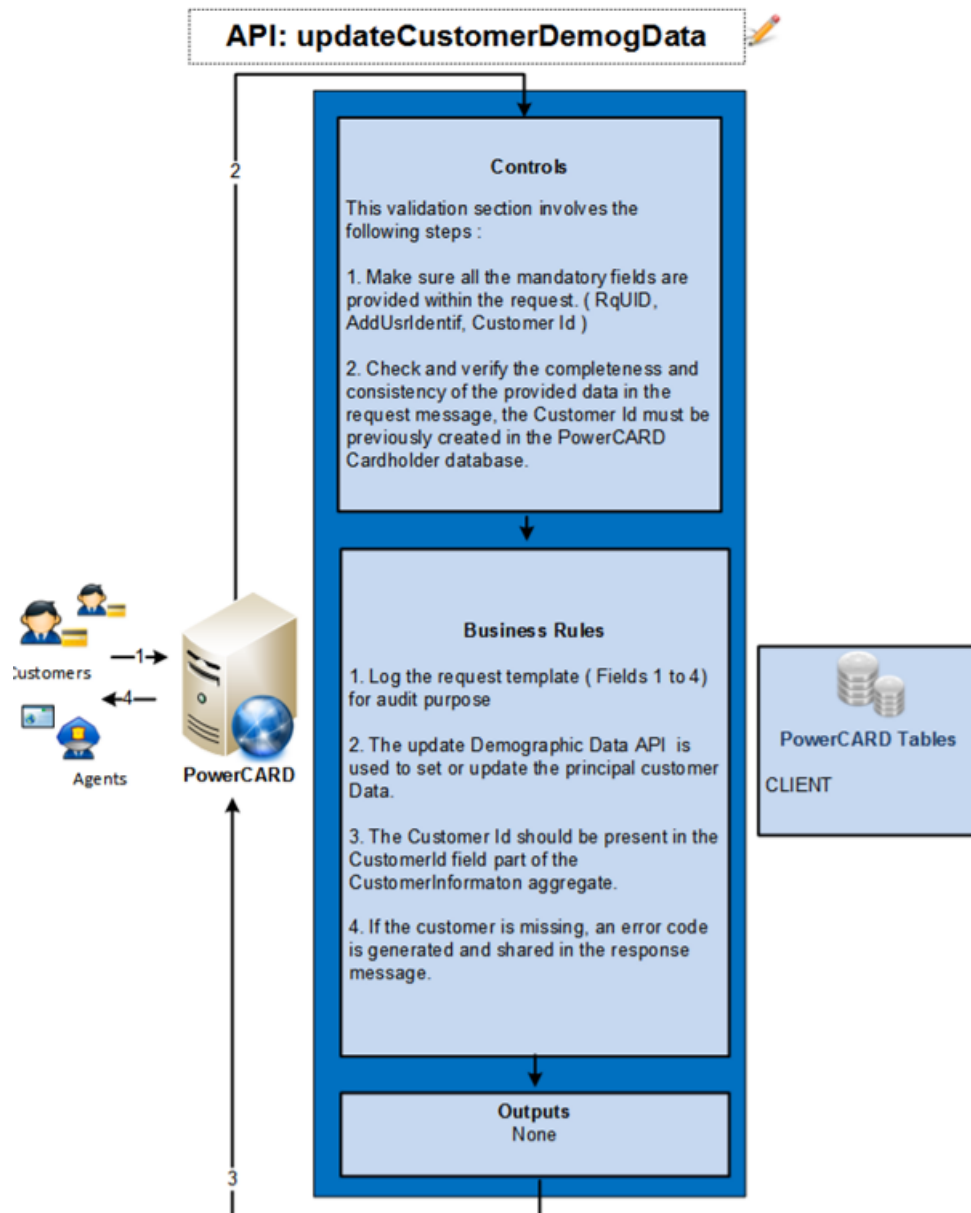


Figure 41: updateCustomerDemogData process

Technical specification

Version 1

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RqUID	C (36)	-	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message.
-	AddUsrIdentif	Complex	-	Required (max: 1)	This aggregate describes the caller context and the technical user identification. Notice that this user has to be declared in the PowerCARD compliance module for being able to call and execute a WebService
2	UserId	C (15)	AddUsrIdentif	Optional (max: 1)	Represents the user Id who made the request.
3	RequestDate	DateTime	AddUsrIdentif	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
4	UserLanguage	C (5)	AddUsrIdentif	Required (max: 1)	The language used by the current user. This language is taken into consideration while preparing the answer. The possible values are the following: • en_US: for English • es_ES: for Spanish • fr_FR: for French
-	ClientDemog	Complex	-	Required (max: 1)	Refers to the demographic information of

#	Field Name	Data type	Parent	Occurrence	Field Description
					the client such as names, professional activity, job details ...
10	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
11	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client. Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.
12	BranchCode	C (6)	ClientDemog	Optional (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs.
13	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
14	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field doesn't apply to corporate cardholders. Only positive value will be accepted.
15	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable. Defaulted to 'N' when not provided.
16	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
17	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
18	TaxRegistryOffice	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
19	BirthDate	Date	ClientDemog	Optional (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
20	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
21	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
22	VIPLevel	C (1)	ClientDemog	Optional (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP

#	Field Name	Data type	Parent	Occurrence	Field Description
					LIST table under the general parameter section.
23	NationalityCode	C(3)	ClientDemog	Optional (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
24	LanguageCode	C (3)	ClientDemog	Optional (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
25	TitleCode	C (2)	ClientDemog	Optional (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
26	FamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's last name.
27	SecondFamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
28	FirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's given name.
29	SecondFirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if compound names are in use.
30	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
31	OwnerCode	C (2)	ClientDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section. SIA configuration IN : INDIVIDUAL ST : STAFF
32	SocioProfessional Code	C (4)	ClientDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
33	ActivityCode	C (4)	ClientDemog	Optional (max: 1)	Helps to categorize cardholder based

#	Field Name	Data type	Parent	Occurrence	Field Description
					on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
34	CustomerSegment	C (3)	ClientDemog	Optional (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
35	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
36	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
37	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
38	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner.

#	Field Name	Data type	Parent	Occurrence	Field Description
					D: Dependent.
39	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached.
40	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
41	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
42	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
43	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxnnnnnnnn (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached.
44	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
45	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached.
46	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
47	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00 xxxxxxxxxxxx (xx based on the country, e.g 39 – Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					[no format check is foreseen in WS processing]
48	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address.
49	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address.
-	Documents	Complex	-	Optional (infinite list)	To provide details about the document that might be requested by the bank: passport details, civil ID details ...
50	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table under the general parameters section.
51	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
52	DocumentIssueDate	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
53	DocumentExpiryDate	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD
54	DocumentIssueCity	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
55	DocumentIssue Country	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	ClientCustomFields	Complex	-	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that

#	Field Name	Data type	Parent	Occurrence	Field Description
					can be provided by a given bank.
56	FieldName	C (30)	ClientCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
57	FieldValue	C (256)	ClientCustomFields	Required (max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RqUID	C (36)			Request Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message. This value must be used to match response with request messages
2	ResultId	C (80)	-	Required (max: 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will

#	Field Name	Data type	Parent	Occurrence	Field Description
					be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. - SystemError: An unknown system error occurred during the process. Example: <ResultId> ProceedWith Success </ResultId>
3	ErrorCode	C (80)	-	Required (max: 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request. They are defined in the "General error Definition" table or in the "business rules" section of each specific web service.
4	ErrorDesc	C (250)	-	Required (max: 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	19301	Invalid Bank Code
2	19302	Invalid Client Code
3	19303	Invalid Branch Code
4	19304	Invalid Family Status
6	19306	Invalid Gender Code
10	19310	Invalid Birth Date
11	19311	Invalid Birth Country Code
12	19312	Invalid Birth City Code

#	Error code	Error Description
13	19313	Invalid VIP Level
14	19314	Invalid Nationality Code
15	19315	Invalid Language Code
16	19316	Invalid Title Code
22	19322	Invalid Owner Code
23	19323	Invalid Socio-professional Code
24	19324	Invalid Activity Code
25	19325	Invalid Customer Segment
29	19329	Invalid Lodging Mode
30	19330	Invalid Channel Number
31	19331	Invalid Phone Type 1
33	19333	Invalid Phone Type 2
35	19335	Invalid Phone Type 3
37	19337	Invalid Phone Type 4
42	19342	Invalid Document Code
44	19344	Invalid Document Issue Date
45	19345	Invalid Document Expiry Date
48	19349	Invalid Client Custom Field name
49	19305	Invalid Client Demog
50	19350	Invalid Number Of Dependents

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is		x

Status Code	Description	Success Response	Error Response
	probably valid but needs to be requested again later.		
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerDemogWebService/V2	POST	
/Clients/V2/	PATCH	Required Optional

REST sample call

```

    "keyValues": [
      {
        "key": "TOKEN",
        "data": "Y",
        "type": "",
      }
    ],
    "requestInfo": {
      "requestUID": "",
      "requestDate": "",
      "userID": ""
    },
    "ClientDemog": {
      "BankCode": "",
      "ClientCode": "",
      "BranchCode": "",
      "FamilyStatus": "",
      "NumberOfDependents": "",
      "Gender": "",
      "LegalId": "",
      "TaxNumber": "",
      "TaxRegistryOffice": "",
      "BirthDate": "",
      "BirthCountry": "",
      "BirthCity": "",
      "VIPLevel": "",
      "NationalityCode": "",
      "LanguageCode": "",
      "TitleCode": "",
      "FamilyName": "",
      "SecondFamilyName": "",
      "FirstName": "",
      "SecondFirstName": "",
      "MaidenName": "",
      "OwnerCode": "",
      "SocioProfessionalCode": "",
      "ActivityCode": "",
      "CustomerSegment": "",
      "TakingOnDate": "",
      "EmployerName": "",
      "StaffID": "",
      "LodgingMode": "",
      "ChannelNumber": "",

```

```

"Phone1": {
  "PhoneType": "",
  "PhoneNumber": ""
},
"Phone2": {
  "PhoneType": "",
  "PhoneNumber": ""
},
"Phone3": {
  "PhoneType": "",
  "PhoneNumber": ""
},
"Phone4": {
  "PhoneType": "",
  "PhoneNumber": ""
},
"Email1": "",
"Email2": "",
},
"Documents": [
  {
    "DocumentCode": "",
    "DocumentId": "",
    "DocumentIssueDate": "",
    "DocumentExpiryDate": "",
    "DocumentIssueCity": "",
    "DocumentIssueCountry": ""
  },
  ""
],
"#text": [
  "...\\n...",
  "...\\n..."
],
"ClientCustomFields": [
  {
    "FieldName": "",
    "FieldValue": ""
  },
  ""
]

```

4.42 checkCorporateContactInformation API

API overview

This web service can be used to check if the corporate administrator is valid and logged within the corporate database.

As output the web service returns the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

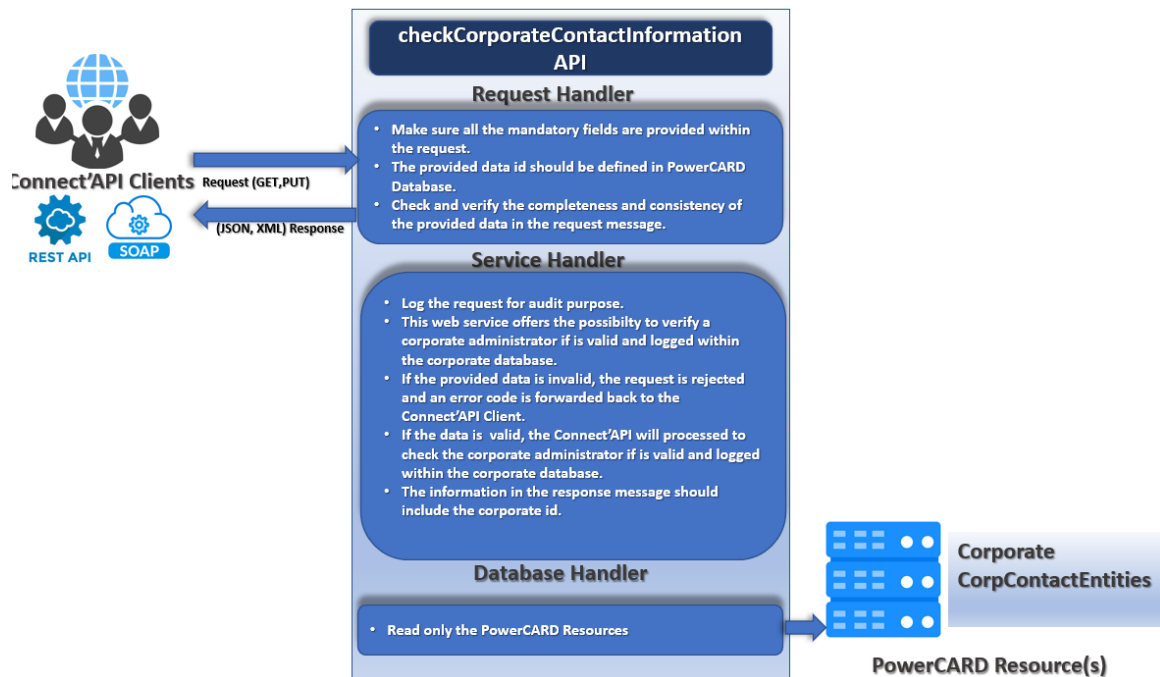


Figure 42: checkCorporateContactInformation

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD-4E04-8842-7668D9C 001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CorporateId	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	-	Required (max : 1)	The unique employee ID with the corporate.
6	FirstName	C(250)	-	Required (max : 1)	The first name of the employee
7	LastName	C(250)	-	Required (max : 1)	The family name of the employee
8	TitleCode	C(2)	-	Required (max : 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
9	Email	C(64)	-	Required (max : 1)	The email address of the employee

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table

Specific error codes

#	Error code	Error Description
1	30101	Invalid Corporate Id
2	30102	Invalid Employee Id
3	30103	Invalid Employee Email
4	30104	Invalid First Name
5	30105	Invalid Last Name
6	30106	Invalid Title Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ GetCorporateContactInformationWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "corporateId": "",
  "employeeInformation": "",
  "employeeId": "",
  "firstName": "",
  "lastName": "",
  "email": ""
}
```

4.43 getCardCharacteristics API

API overview

This web services can be used to get the card characteristics that have been already negotiated for a card.

Functional description

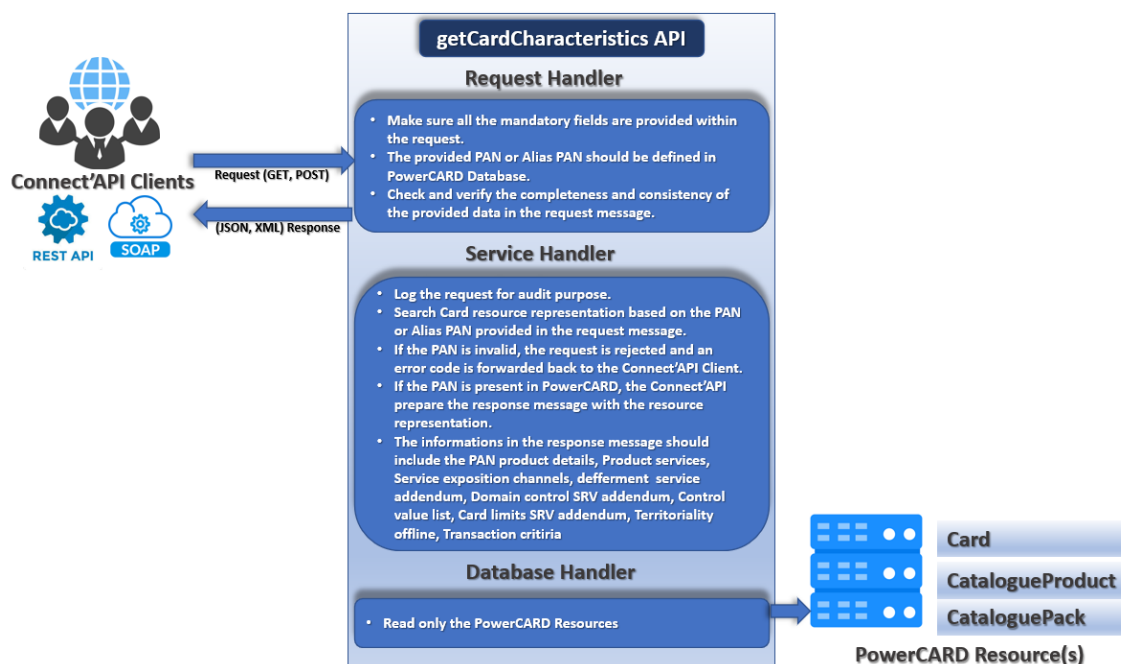


Figure 43: getCardCharacteristics

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
	CardNumber	C(22)	-	Required (max : 1)	The generated PAN number in when the status of the request status is C and Card createion status is A

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	PANProductDetails	Complex	-	Required (max: 1)	This aggregate contains details of each subscription request
6	CatPIProduct	C (6)	PANProduct Details	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
7	CatPIProduct Version	C (10)	PANProduct Details	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
8	PackCode	C (6)	PANProduct Details	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles. Must be a valid value defined in CATALOGUE _PACK
9	productCurrency	C	PANProduct Details	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar).

#	Field Name	Data type	Parent	Occurrence	Field Description
					The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
10	cycleCutoff	C(3)	PANProduct Details	Required (max : 1)	The account cycle cutoff assigned to the account: Must be defined in cycle_cutoff_parameter
-	ProductServices	Complex	PANProduct Details	Required (Infinite)	This aggregate contains details of the services to which the collaborator has applied
	ServiceCode	C(6)	ProductServices	Required (max : 1)	Must be defined in SERVICES_DEFINITION table List of values example: • CTLESS: Contactless. • SECECO: Secured ecommerce • ECOM00: Ecommerce • CORSOL: Corporate Solidarity • DOMSTP: Domain Setup • VELSTP: Velocity setup • DDDEFR: Direct debit deferment.
9	ServiceStatus	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: • Y: Service is enabled. • N: Service is disabled.
	IsServiceMandatory	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: • Y: Service is mandatory

#	Field Name	Data type	Parent	Occurrence	Field Description
					and can not be disabled. N: Service is optional and can be disabled.
10	ServiceExposition Channels	Complex	ProductServices	Optional (Infinite)	This is aggregate is used to list the service exposition channels along with channel activation status
11	ChannelType	C(3)	ServiceExposition Channels	Required (max : 1)	BAG: Bank agent channel COW: Corporate portal channel CHM: Cardholder Mobile app CHW: Cardholder portal
12	ServiceToBeExposed InChannel	C(1)	ServiceExposition Channels	Required (max : 1)	Y: Service can be visible in the channel. N: Service might be visible or invisible in the channel. In case visible, the service should be greyed out.
-	DeffermentService Addendum	Complex	ProductServices	Conditional (Infinite)	Mandatory if service code is DDDEFR
34	DeferredDays Number	Float	Defferment ServiceAddendum	Required (max : 1)	The number of days for debit deferment. The list of value must be defined in the table static_list with collection name = 'PaymentDDDefermentDays'.
	DefermentStartDate	Date	Defferment ServiceAddendum	Required (max : 1)	The start date of the deferment. Must be defaulted to current business date
	DefermentEnddate	Date	Defferment ServiceAddendum	Optional (max : 1)	The date where the deferment will stop.
-	DomainControl SRVAddendum	Complex	ProductServices	Conditional infinite list	Mandatory if service code is

#	Field Name	Data type	Parent	Occurrence	Field Description
					DOMSTP
5	ControlType	ClosedEnum	DomainControl SRVAddendum	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: - CTR: Country - MCC: MCC - PRS: Processing code - MER: Merchant - CUR: Currency
6	ControlOperator	ClosedEnum	DomainControl SRVAddendum	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValueList
-	ControlValueList	Complex	DomainControl SRVAddendum	Optional infinite list	Contains various values, such as country codes, processing codes ...
7	ControlValue	C (15)	ControlValueList	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section: - COUNTRY table for ControlType = 'CTR'. - MCC table for ControlType = 'MCC'. - PROCESSING_LIST table for ControlType = 'PRS'. - MERCHANT table for ControlType = 'MER'.

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY_ TABLE table for ControlType = 'CUR'. If not, the request will be rejected.
-	CardLimitSRV Addendum	Colmplex	-	Conditional infinite list	Mandatory if service code is VELSTP
5	LimitID	C (4)	CardLimitSRV Addendum	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.
6	LimitDescription	C (64)	CardLimitSRV Addendum	Optional (max : 1)	Description of the limit
7	LimitAbv Wording	C (32)	CardLimitSRV Addendum	Optional (max : 1)	A brief description of the limit
-	ValidityRangeDate	Complex	CardLimitSRV Addendum	Required (max: 1)	A limit can be temporary and restricted to a range date
8	StartDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
9	EndDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
-	Period	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	The Period of limit that should be set on the card.
10	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: Fixed period - W: Rolling period
11	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off
12	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: for daily - W: for Weekly - M: for Monthly - Y: for Yearly
-	TerritorialityOnline	Colmplex	CardLimitSRV Addendum	Optional (max: 3)	The Territoriality Online
13	TerritorialityType	C (3)	Territoriality Online	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of

#	Field Name	Data type	Parent	Occurrence	Field Description
					transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
14	TransactionAmount	Decimal (18,3)	Territoriality Online	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
14	TransactionAmount MAX	Decimal (18,3)	Territoriality Online	Required (max : 1)	The max total amount can be negotiated for the selected period and territoriality
15	NumberOfTransactions	Number (4)	Territoriality Online	Required (max : 1)	Must contain the target the number of transactions of the limit.
15	NumberOf TransactionsMAX	Number (4)	Territoriality Online	Required (max : 1)	The max number of transactions that can be negotiated for the selected period and territoriality
-	Territoriality Offline	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	Territoriality Offline
16	TerritorialityType	C (3)	Territoriality Offline	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country.

#	Field Name	Data type	Parent	Occurrence	Field Description
					INT: International, apply only to transactions that took place outside the cardholder country.
17	TransactionAmount	Decimal (18,3)	Territoriality Offline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
18	NumberOfTransactions	Number (4)	Territoriality Offline	Required (max : 1)	Must contain the target the number of transactions of the limit.
19	TotalTransaction AmountOnline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
20	TotalNumberOf TransactionsOnline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
21	TotalTransaction AmounOffline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
22	TotalNumberOf TransactionsOffline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	TransactionCriteria	Colmplex	CardLimitSRV Addendum	Optional (max : 1)	This aggregate is used to define the transaction criteria
23	SequenceNumber	Number (9)	TransactionCriteria	Optional (max : 1)	Should contain Sequence Number
24	ProcessingCode	C (2)	TransactionCriteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING _LIST table under the general parameter section.

#	Field Name	Data type	Parent	Occurrence	Field Description
25	CountryCode	C (3)	Transaction Criteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
26	MCC	C (4)	Transaction Criteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the PowerCARD MCC table
27	PaymentType	C (2)	Transaction Criteria	Optional (max : 1)	Should contain the payment type
28	CurrencyCode	C (3)	Transaction Criteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
29	MaxAmountPerTrx	Long	Transaction Criteria	Optional (max : 1)	Used to set the maximum amount per transaction.
30	CardEntryMode	ClosedEnum	Transaction Criteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : 00: Unknown or terminal not used

#	Field Name	Data type	Parent	Occurrence	Field Description
					01: Manual (key entry) 02: Magnetic stripe read 03: Integrated circuit card read 04: Contactless device-read

Specific error codes

#	Error code	Error Description
1	30301	Invalid Card Number/ Token Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/ GetProductDetailsWebService/V2	POST	
	GET	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  }
}
```

4.44 getCardNumberFromToken API

API overview

When used in credit, debit or prepaid card transactions, tokens, sometimes referred to as surrogate value are created to replace the card number. A token, rather than the actual credit/debit card number, would be used to complete the purchases a customer is making with its card. In this context, this web service is mainly used to retrieve the clear card number when providing a token value. PowerCARD is in fact a tokenization system that accepts as entry token value and proceeds by the reverse process of redeeming a token for its associated PAN value.

Users are also required to pass the card expiry date in the list of input data to process the de-tokenization process and return back the clear card number.

Functional description

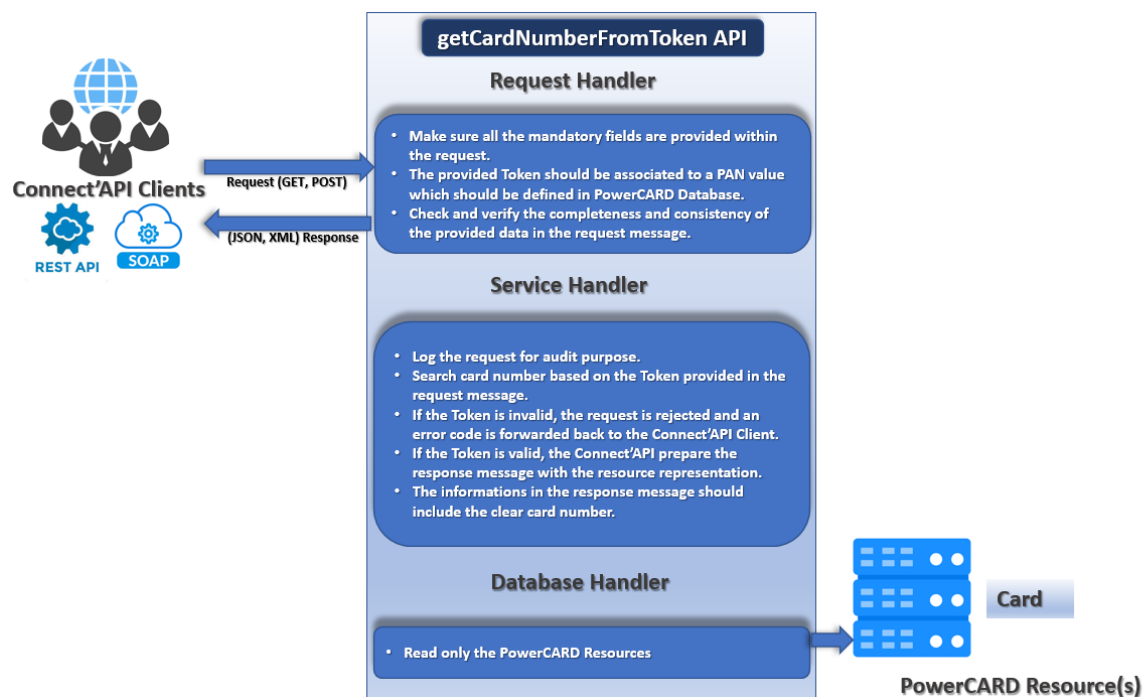


Figure 44: getCardNumberFromToken process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9C 001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Token	C (30)	-	Required (max : 1)	This field must contain the token value for which a card number must be returned.
5	ExpDt	DateTime	-	Required (max : 1)	Refers to the card expiry date.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
5	CardNumber	C (22)	-	Required (max : 1)	Refers to the card number associated to the token value provided in the request message

Specific error codes

#	Error code	Error Description
1	17901	No card number found for this request, please send a valid Token value
2	17902	Invalid expiry date

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/getCardNumberFromTokenWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",

```

```

    "requestDate": "",
    "userID": ""
  },
  "token": "",
  "expDt": ""
}

```

4.45 getInstallmentPlan API

API overview

The get installment plan web service is an API that can be used to get the installment plan of a given loan transaction provided by the loan company to the customer. Whether it is an ongoing loan or an amortized one, this service provides the ability to search for a specific loan in the system using the transaction reference Number and sequence, two fields used internally by PowerCARD to identify uniquely a transaction in the system.

Functional description

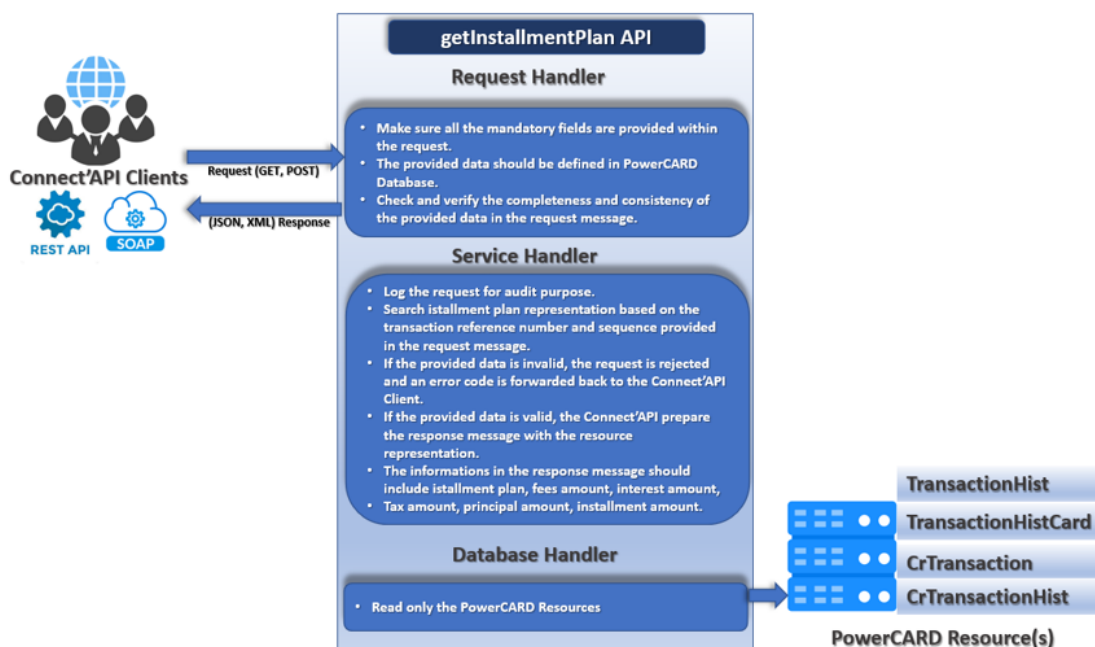


Figure 45: getInstallmentPlan process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
5	RefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
-	InstallmentPlan	Complex	-	Optional (max : 1)	The installment Plan aggregate. For more details refers to the aggregate description.
5	InstallmentNumber	C (23)	Loan	Required (max : 1)	Sequence number for the installment plan. A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
-	FeesAmount	Complex	Loan	Required (max : 1)	Fees amount aggregate.
6	Amt	Long ()	FeesAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
7	CurCode	C (3)	FeesAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	InterestAmount	Complex	Loan	Required (max : 1)	Interest amount aggregate.
8	Amt	Long ()	InterestAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent

#	Field Name	Data type	Parent	Occurrence	Field Description
					of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
9	CurCode	C (3)	InterestAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	TaxAmount	Complex	Loan	Required (max : 1)	Tax amount aggregate.
10	Amt	Long ()	TaxAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	C (3)	TaxAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	PrincipalAmount	Complex	Loan	Required (max : 1)	Principal amount aggregate.
12	Amt	Long ()	PrincipalAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent

#	Field Name	Data type	Parent	Occurrence	Field Description
					of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
13	CurCode	C (3)	PrincipalAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	InstallmentAmount	Complex	Loan	Required (max : 1)	Installment Amount aggregate.
14	Amt	Long ()	InstallmentAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
15	CurCode	C (3)	InstallmentAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
16	Status	C (1)	Loan	Required (max : 1)	B: Billed U: Unbilled
17	BillingDate	Date	Loan	Required (max : 1)	Installment billing date.

Specific error codes

#	Error code	Error Description
1	18101	Installment plan / Loan not found in the system

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/getInstallmentPlanWebService/V2	POST	
/InstallmentPlan/V2/transId/:transId/refSeq/:refSeq	GET	Required transId =[C(23)] refSeq =[C(4)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "transId" : "",
  "refSeq" : ""
}
```

}

4.46 searchLoan API

API overview

The search loan web service is an API that can be used to retrieve all the loans provided by the loan company to the customer. Whether it is an ongoing loan or an amortized one, this service provides the ability to search for all existing loans in the system using different criteria like Customer Id, Account Id or also transaction date. For the last criteria, users must specify a range of dates values to get the list of loans submitted between those two dates, for instance search for the list of loans submitted today, last week, last months...

Functional description

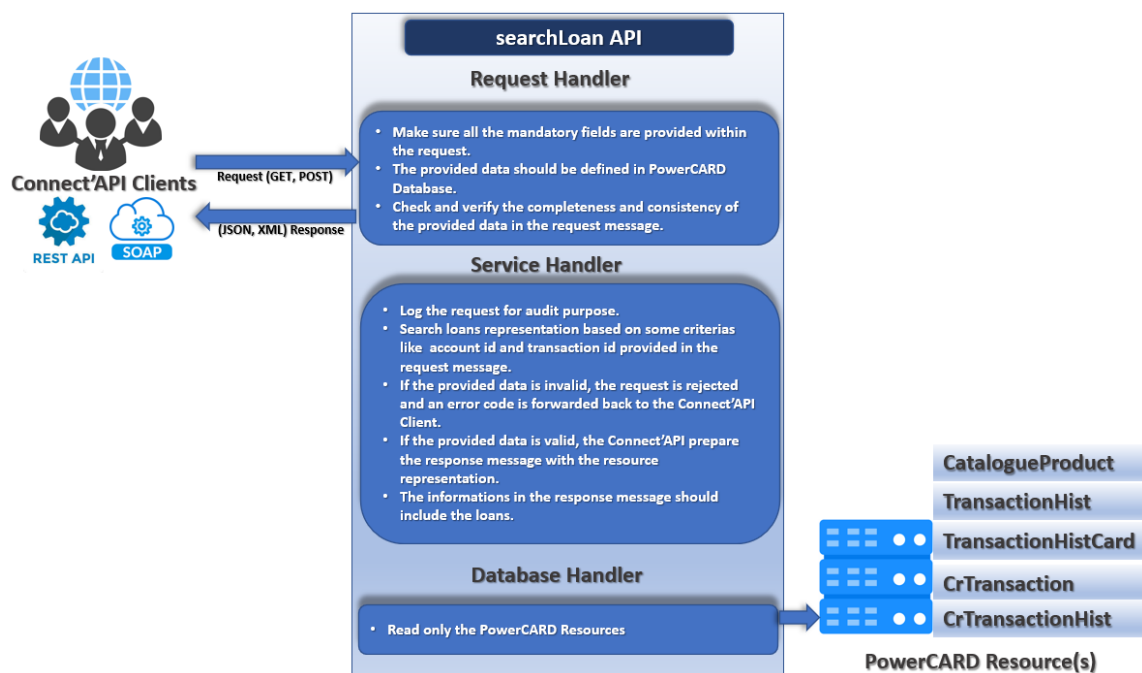


Figure 46: searchLoan process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a

#	Field Name	Data type	Parent	Occurrence	Field Description
					universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Optional (max : 1)	Refers to the PowerCARD card number. This value corresponds to the card to which the loan is posted. Required if AccountId is empty.
5	AccountId	C (40)	-	Optional (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the account to which the loan is posted. Required if CardNumber is empty.
6	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					system to each transaction. The TransId is unique by transaction.
7	RefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number
8	LoanStatus	ClosedEnum	-	Optional (max : 1)	Loan Status This field refers to the loan status as stored in the system. Possible values are as follow: • O: Occurred • R: Reversed • A: Amortized
-	RangeDate	Complex	-	Required (max : 1)	This aggregate is mandatory and should contain the range of the dates when the transactions happened. The aggregate is constituted of two items (Start Date & End Date)
9	StartDt	DateTime	RangeDate	Optional (max : 1)	Start date
10	EndDt	DateTime	RangeDate	Optional (max : 1)	End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.

#	Field Name	Data type	Parent	Occurrence	Field Description
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	MaxRowNbr	Long	-	Optional (max : 1)	This will depict the maximum number of row that can be included in the response message. Note that if the result exceeds this number, only this number of cards will be returned.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This mechanism is used to limit the number of cards that can be fetched in one call.
-	Loans	Complex	-	Optional infinite list	The loan aggregate. For more details refers to the aggregate description.
6	LoanRefNumber	C (23)	Loans	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
7	LoanRefSeq	N (4)	Loans	Required (max : 1)	Refers to the transaction sequence Number
8	AccountId	C (24)	Loans	Required (max : 1)	The Account Number where the transaction is posted.
9	CatLoanProduct	C (6)	Loans	Optional (max : 1)	It is a 6 digit code identifying the loan product. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
10	CatLoanProduct Version	C (10)	Loans	Optional (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
11	LoanStatus	ClosedEnum	Loans	Required (max : 1)	Loan Status

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field refers to the loan status as stored in the system. Possible values are as follow: • O: Occurred • R: Reversed • A: Amortized
12	LoanStartDate	DateTime	Loans	Required (max : 1)	Loan start date. Date of generation of the first installment in PowerCARD
13	LoanEndDate	DateTime	Loans	Required (max : 1)	Loan end date. Date of generation of the last installment in PowerCARD
14	NbrOfTerms	Long	Loans	Required (max : 1)	Number terms due. The total number of installment for that loan
15	NbrOfTermsDue	N (4)	Loans	Required (max : 1)	Number terms due. Remaining number of installment still to be generated.
16	InterestRate	Long	Loans	Required (max : 1)	Interest rate applied for the loan
-	LoanDue	Complex	Loans	Required (max : 1)	Loan due. The total loan amount not yet requested from the customer to be paid
17	Amt	Long ()	LoanDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
18	CurCode	NC (3)	LoanDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	LoanAmt	Complex	Loans	Required (max : 1)	Transaction amount. Total amount and currency of the initial transaction
19	Amt	Long	LoanAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
20	CurCode	C (3)	LoanAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
21	CustomerId	C (22)	Loans	Required (max : 1)	Person identifier Refers to the unique customer identification in the system.
22	Acronym	C (40)	Loans	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or

#	Field Name	Data type	Parent	Occurrence	Field Description
					member's imprinter plate.
23	LoanFinancingType	C (1)	Loans	Required (max : 1)	I : Financed by credit limit E: Financed by additional loan limit
24	LoanCatalogName	C(50)	Loans	Optional (max : 1)	Loan catalog product name.

Specific error codes

#	Error code	Error Description
1	30501	Invalid Account Number
2	30502	Invalid Card Number
3	30503	Invalid Loan Status
4	30504	Invalid Range Date

RESTful response code

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchLoanWebService/V2	POST	
/Cards/V2? accountId={ accountId}&aliasPAN ={aliasPAN}& transId ={transId}& refSeq ={refSeq}&startDt={startDt}& endDt ={endDt}&pageSize ={pageSize}&pageIndex ={pageIndex}	GET	aliasPan=[C(24)] accountId=[C(22)] transId=[C(15)] refSeq=[C(6)] startDt=[DateTime] endDt=[DateTime] pageSize=[PositiveInteger] pageIndex=[PositiveInteger]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "N"
    },
    {
      "key": "pageSize",
      "data": ""
    },
    {
      "key": "pageIndex",
      "data": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "cardNumber": "",
  "accountId": "",
  "transId": "",
  "refSeq": "",
  "loanStatus": "",
  "rangeDate": {
    "startDt": "",
    "endDt": ""
  }
}
```

4.47 changeAccountBillingCycle API

API overview

For a credit account, a billing cycle is defined as the time between the last bill and the most current one.

The purpose of this web service is to allow changing the account billing cycle date by providing a code identifying the cycle date in PowerCARD. Many reasons can be given to justify the billing cycle change, for instance the customer might need to simplify his payment schedule by setting a new date for his billing cycle.

It is important to note that the change will takes effect only starting from the next cycle. This Web Service can be called by a customer or an agent.

Functional description

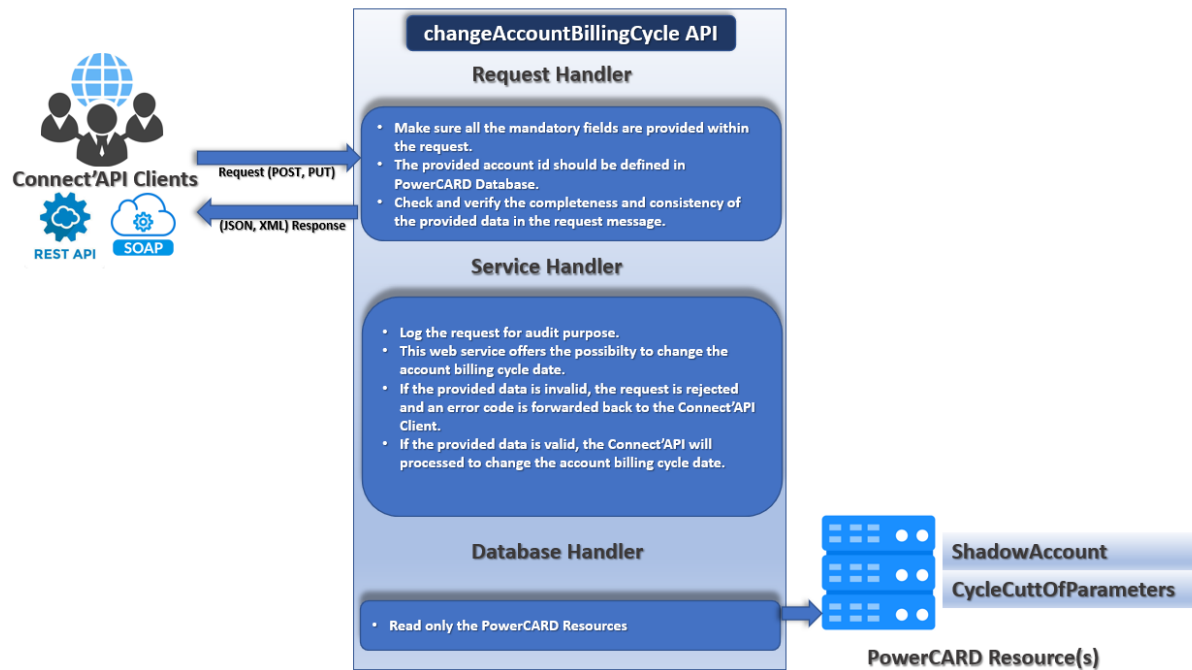


Figure 47: changeAccountBillingCycle process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
5	BillingCycleCode	C (20)	-	Required (max : 1)	The billing cycle code. The billing cycle is the period of time that elapses between 2 consecutive billing statements. The value of this field indicates the code that refers to the new billing cycle that should be assigned to the account. Possible values must be retrieved from the PowerCARD CYCLE _CUTOFF

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PARAMETERS table

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell

#	Field Name	Data type	Parent	Occurrence	Field Description
					you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	09801	Invalid Account Number
2	09802	Invalid Account Status
3	09803	Invalid Billing Cycle Code
4	09804	Change not allowed

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/changeAccountBillingCycleWebService/v2	POST	
/Accounts/v2/changeBillingCycle?accountId={accountId}&billingCycleCode={billingCycleCode}	PATCH	Required accountId=[C(24)] billingCycleCode=[C(2)]

REST sample call

```
"requestInfo": {
  "requestUID" : "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"billingCycleCode": ""
```

4.48 checkCustomFieldValue API

API overview

This web service is used to check the existence of custom field value for specific entity, when the entity is card, token value can be provided instead of card number. PowerCARD is in fact a tokenization system that accepts as entry token value and proceeds by the reverse process of redeeming a token for its associated PAN value.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

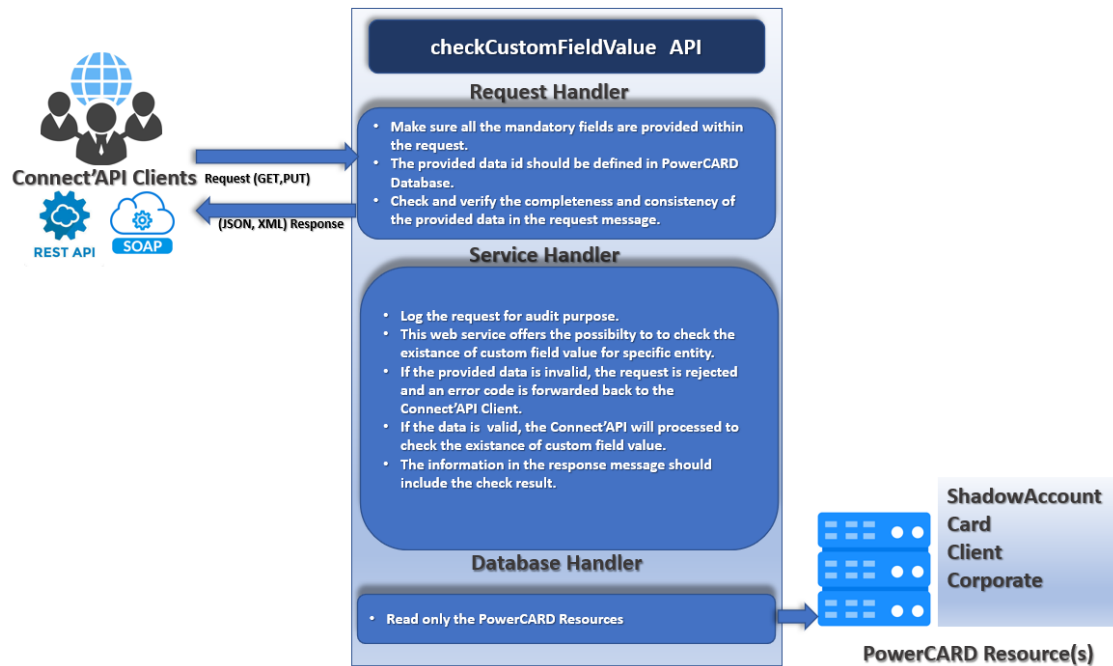


Figure 48: checkCustomFieldValue process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
12	EntityCode	C (2)	-	Required (max: 1)	Entity code: • CH : Card • CL : Client • SA : Account • CO : Corporate
13	EntityID	C (24)	-	Required (max: 1)	Request identifier depending on the entity code: EntityCode = CH : Card number (or token if TOKEN key value present with the value Y), must be present in CARD table EntityCode = CL : Client code, must be present in CLIENT table EntityCode = SH : Shadow account number, must be present in SHADOW_ACCOUNT table EntityCode = CO : Corporate Id , must be present in CORPORATE table
14	FieldName	C (30)	-	Required (max: 1)	Custom field name, must be present in the table USER_DEFINED_FIELDS_PARAM
15	FieldValue	C (256)	-	Required (max: 1)	Custom field value to be checked

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CheckResult	C(2)	-	Required (max: 1)	Field to indicate the existence of the input custom field value: OK : Custom field value is correct KO : Custom field value is incorrect

Specific error codes

#	Error code	Error Description
1	21401	Invalid Entity Code
2	21402	Invalid Card Number/Token Id
3	21403	Invalid Client code
4	21406	Invalid Corporate Id
5	21404	Invalid Shadow account number
6	21405	Invalid Field Name (refers to the scenario where a valid Field Name is referenced but for the wrong entity (f.i. a valid Card Custom Fields but the Entity Code passed is "CL – Client"))

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x

Status Code	Description	Success Response	Error Response
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/checkCustomFieldValueWebService/v2	POST	

REST sample call

```

"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"entityCode": "",
"entityID": "",
"fieldName": "",
"fieldValue": ""

```

4.49 updateAccStatementCode API

API overview

Users may request to change their statement specification by changing for instance the type of statement being received. Meantime the financial institution may request to change some other parameters on the customer statement moving the customer from a category to another one with different statement requirements. To do that PowerCARD offers this web service that only by providing the account number as well as the statement code can adjust that in the PowerCARD database hence all the statements generated will follow the characteristics of the statement code.

Functional description

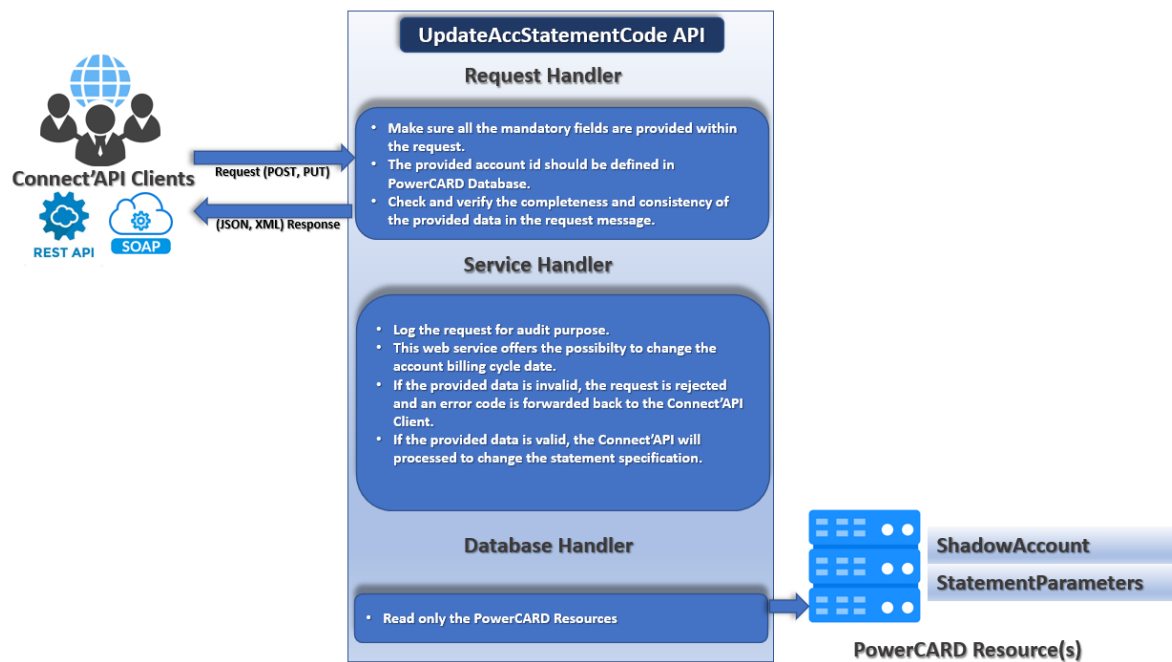


Figure 49: updateAccStatementCode process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	Should contain the credit account number subject to this statement code change This value should be already defined in the PowerCARD SHADOW _ACCOUNT table.
5	StatementCode	C (2)	-	Required (max : 1)	The statement Code that need to be assigned to the account Valid values are stored in STATEMENT_PARAMETERS table.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.

#	Field Name	Data type	Parent	Occurrence	Field Description
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	17301	Invalid Account number
2	17302	Invalid Statement Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/updateAccStatementCodeWebService/v2	POST	
/Accounts/v2/changeStatementCode?accountId={accountId}&statementCode={statementCode}	PATCH	Required accountId=[C(24)] StatementCode=[C(2)]

REST sample call

```

"requestInfo": {
  "requestUID" : "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"statementCode": ""

```

4.50 updateBankAccountStatus API

API overview

The aim of this web service is to update the status of an existing bank account, this web service can be invoked by customers and agents. The change will take place immediately in the system and all the next communication with that account will take the new status.

Functional description

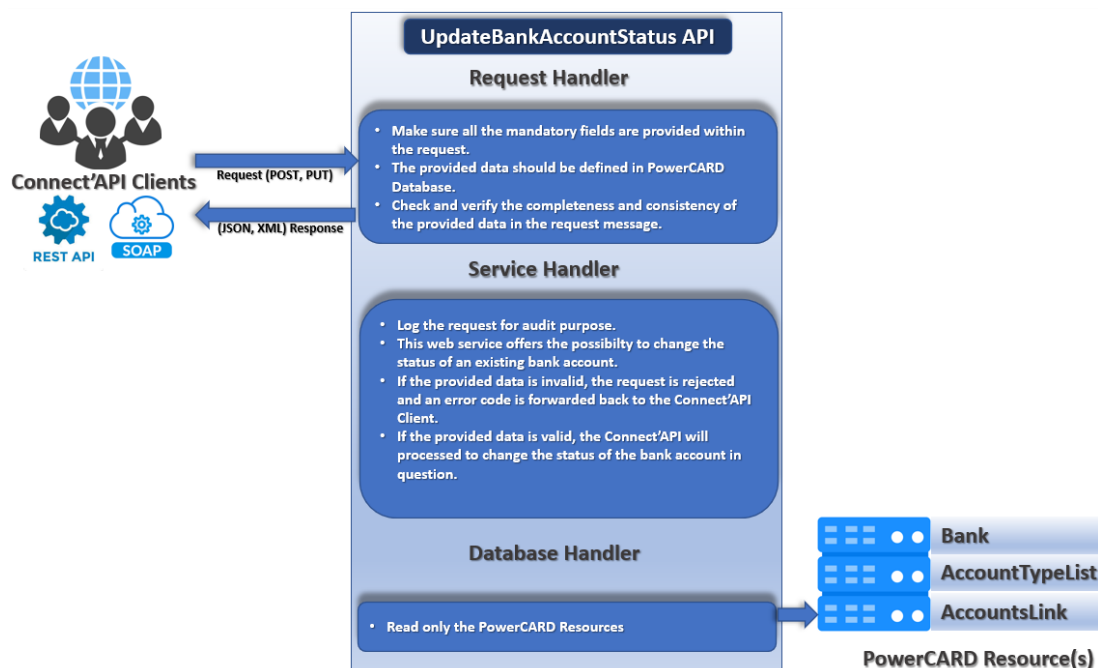


Figure 50: updateBankAccountStatus process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
10	AcctType	C (2)	-	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT_TYPE_LIST table under the general parameter section.
11	AcctSequence	C (2)	-	Optional (max :1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
12	BankCode	C (6)	-	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the

#	Field Name	Data type	Parent	Occurrence	Field Description
					general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
13	BankCodeFreetext	C (6)	-	Optional (max :1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
14	BranchCode	C (6)	-	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
15	BranchCodeFreetext	C (6)	-	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
16	AccountNumber	C (24)	-	Optional (max: 1)	It refers to the bank account number part of the IBAN. It should be defined in ACCOUNTS_LINK table. Requested only in case the IBAN is not provided.
17	AccountCurrency	C (3)	-	Optional (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined

#	Field Name	Data type	Parent	Occurrence	Field Description
					in the PowerCARD CURRENCY TABLE under the general parameters section.
18	IBAN	C (34)	-	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Requested only in case the AccountNumber is not provided.
19	SwiftCode	C (11)	-	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages. Can be provided only in case the IBAN is provided.
20	AccountStatus	C (1)	-	Required (max: 1)	Account Status Refers to the status of the bank Account. Valid values are: • N = Normal • C = Cancelled • S = Suspended • W = Waiting for validation • V = To be sent for validation

#	Field Name	Data type	Parent	Occurrence	Field Description
					R = Rejected.
21	AccountStatusReason	C (2)	-	Optional (max: 1)	Status reason code. It must contain a valid value that is previously defined in the STATUS_REASON_LIST table under the general parameter section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - Proceed With Success: Call was ended successfully. - Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred

#	Field Name	Data type	Parent	Occurrence	Field Description
					during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	19501	Invalid Bank Code
2	19502	Invalid Branch Code
3	19503	Invalid Account Type
4	19504	BankCode missing but AccountId valued
6	19506	BranchCode is missing when AccountNumber is a valued field
7	19507	AccountNumber is missing when BranchCode is a valued field
8	19508	IBAN is missing when SwiftCode is a valued field
9	19509	Both AccountNumber and IBAN are missing
10	19510	Invalid Account Number
11	19511	Invalid IBAN
12	19512	Invalid Swift Code
13	19513	Invalid Account Status
14	19514	Invalid Status Reason Code
15	19515	Invalid Combination Status Code – Status Reason Code
16	19516	Operation not permitted

#	Error code	Error Description
17	19517	An error occurred while inserting or updating a Bank Account
18	19518	Invalid Transition Requested
19	19519	Invalid Account Sequence
20	19520	Invalid Bank Code Free Text
21	19521	Invalid Branch Code Free Text
22	19522	Invalid Account Currency

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/updateAccStatementCodeWebService/v2	POST	
/BankAccount/v2/AccountStatus?acctType={ acctType }&	PATCH	Required acctType=[C(2)]

PowerCARD Resource URI	Method	URI Params
acctSequence={ acctSequence }& bankCode={ bankCode }& bankCodeFreetext ={ bankCodeFreetext }& branchCode={ branchCode }& branchCodeFreetext ={ branchCodeFreetext }& accountNumber={ accountNumber }& accountCurrency ={ accountCurrency }& accountCurrency ={ accountCurrency }& iban ={ iban }& swiftCode={ swiftCode }& accountStatus={ accountStatus }& accountStatusReason ={ accountStatusReason }		acctSequence=[C(2)] bankCode=[C(6)] bankCodeFreetext=[C(6)] branchCode=[C(6)] branchCodeFreetext=[C(6)] accountNumber=[C(24)] accountCurrency=[C(3)] iban=[C(34)] swiftCode=[C(11)] accountStatus=[C(1)] accountStatusReason=[C(2)]

REST sample call

```

{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "acctType": "",
  "acctSequence": "",
  "bankCode": "",
  "bankCodeFreetext": "",
  "branchCode": "",
  "branchCodeFreetext": "",
  "accountNumber": "",
  "accountCurrency": "",
  "iban": "",
  "swiftCode": "",
  "accountStatus": "",
  "accountStatusReason": ""
}

```

4.51 getLoanDetails API

API overview

The get loan web service is an API that can be used to retrieve the loan provided by the loan company to the customer. Whether it is an ongoing loan or an amortized one, this service provides the ability to get details for the existing loan in the system.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

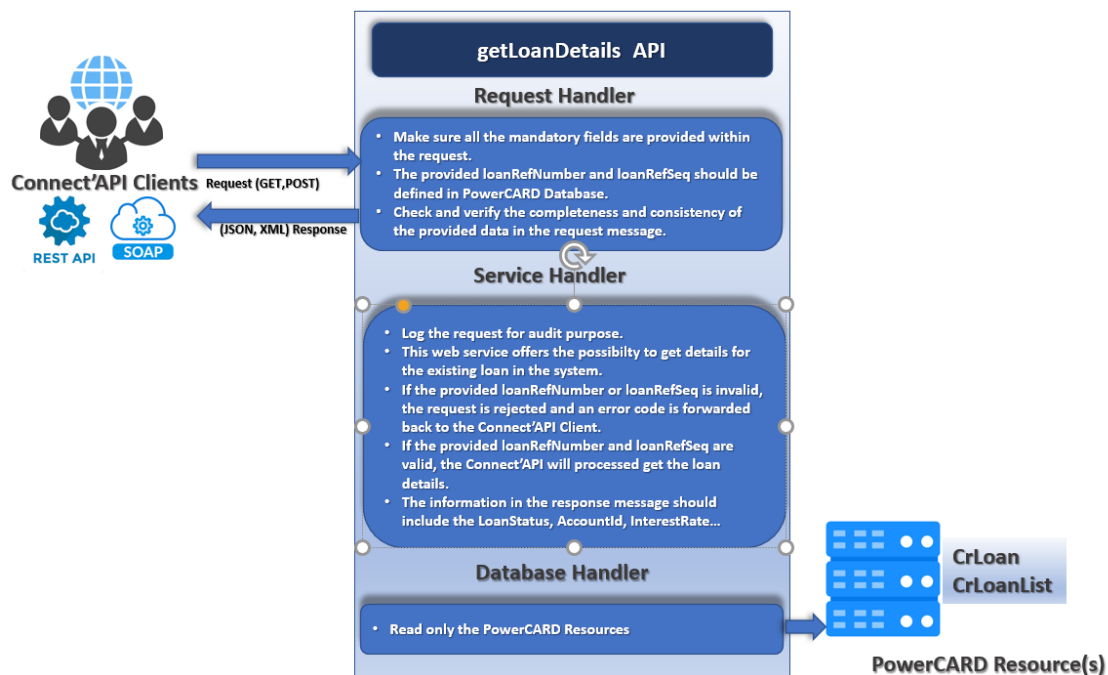


Figure 51: getLoanDetails process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX- XXXXXXXX XXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842- 7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	LoanRefNumber	C (23)	Loans	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
5	LoanRefSeq	N (4)	Loans	Required (max : 1)	Refers to the transaction sequence Number

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					- Proceed With Success: Call was ended successfully. - Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	LoanRefNumber	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
6	LoanRefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number
7	AccountId	C (24)	-	Required (max : 1)	The Account Number where

#	Field Name	Data type	Parent	Occurrence	Field Description
					the transaction is posted.
8	CatLoanProduct	C (6)	-	Optional (max : 1)	It is a 6 digit code identifying the loan product. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
9	CatLoanProduct Version	C (10)	-	Optional (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
10	LoanStatus	ClosedEnum	-	Required (max : 1)	Loan Status This field refers to the loan status as stored in the system. Possible values are as follow: • O: Occurred • R: Reversed • A: Amortized
11	LoanStartDate	DateTime	-	Required (max : 1)	Loan start date. Date of generation of the first installment in PowerCARD
12	LoanEndDate	DateTime	-	Required (max : 1)	Loan end date. Date of generation of the last installment in PowerCARD
13	NbrOfTerms	Long	-	Required (max : 1)	Number terms due. The total number of installment for that loan
14	NbrOfTermsDue	N (4)	-	Required (max : 1)	Number terms due.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Remaining number of installment still to be generated.
15	InterestRate	Long	-	Required (max : 1)	Interest rate applied for the loan
-	LoanDue	Complex	Loans	Required (max : 1)	Loan due. The total loan amount not yet requested from the customer to be paid
16	Amt	Long ()	LoanDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
17	CurCode	NC (3)	LoanDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	LoanAmt	Complex	Loans	Required (max : 1)	Transaction amount. Total amount and currency of the initial transaction
18	Amt	Long	LoanAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
19	CurCode	C (3)	LoanAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
20	CustomerId	C (22)	Loans	Required (max : 1)	Person identifier Refers to the unique customer identification in the system.
21	Acronym	C (40)	Loans	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or member's imprinter plate.
22	LoanFinancingType	C (1)	Loans	Required (max : 1)	I : Financed by credit limit E: Financed by additional loan limit
23	LoanCatalogName	C(50)	Loans	Optional (max : 1)	Loan catalog product name.

Specific error codes

#	Error code	Error Description
1	30201	Invalid Loan Reference Number
2	30202	Invalid Loan Reference Sequence

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x

Status Code	Description	Success Response	Error Response
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetLoanDetailsWebService/V2	POST	
/Loans/V2? loanRefNumber={loanRefNumber}& loanRefSeq={loanRefSeq}	GET	Required loanRefNumber=[C(23)] loanRefSeq=[C(4)]

REST sample call

```

"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"loanRefNumber": "",
"loanRefSeq": ""
}

```

4.52 personalizeCard API

API overview

PowerCARD offers through its web service interface a flexible way to personalize a non-personalized card.

It is important to note that a non-personalized card is a card in which a generic identifier is embossed or printed in place of the cardholder name along with the card number and the expiration date. This service was introduced to help any financial institution wishing to

transform non-personalized card into a personalized card. This process serve to link a card to an existing/new client and (optionally) to activate it.

Note that No card production process will be triggered, the present API should be used to provide the cardholder with a card from the available stock at the bank branches. Renewal or replacement will be personalized.

Functional description

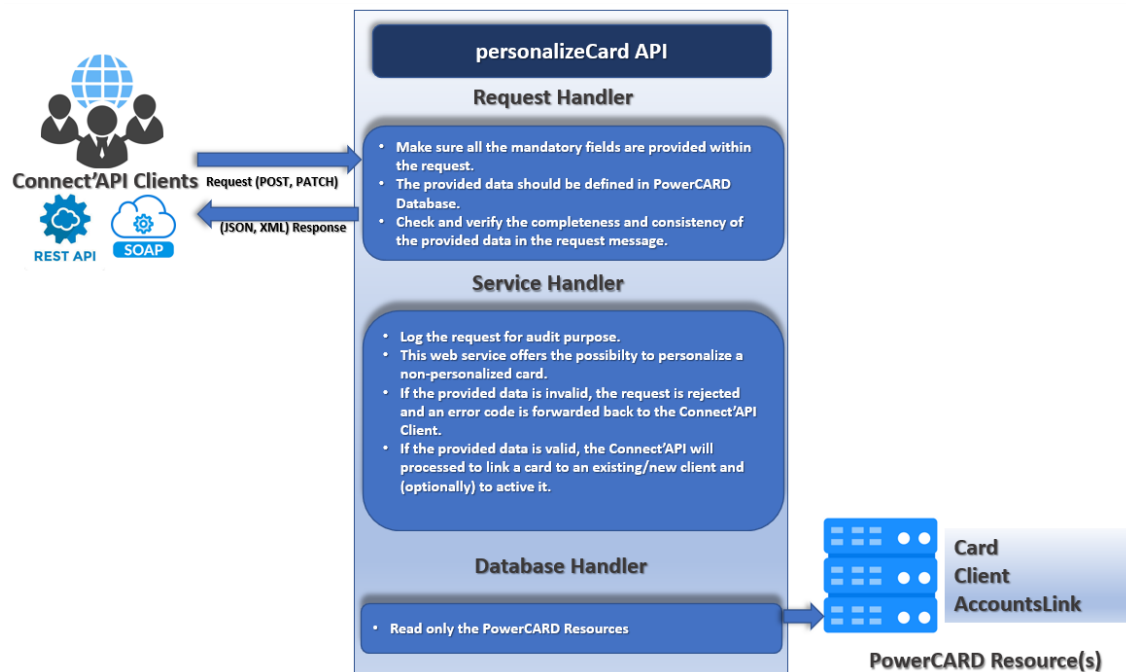


Figure 52: personalizeCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
10	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
11	CardActivationFlag	C (1)	-	Optional (max: 1)	This field indicates whether the card will be activated or not. Possible values are the following: - Y: Card activated - N: Card not activated If this information is not provided, by default, the card is not activated.
-	CustomerDetails	Complex	-	Required (max: 1)	It must contain the details information of the cardholder.
-	ExistingClient	Complex	CustomerDetails	Optional (max: 1)	To provide the existing client code to which all the entities within the boarding request should be linked. When the sub block "ExistingClient" is not provided,

#	Field Name	Data type	Parent	Occurrence	Field Description
					the sub blocks ClientDemog and PrimaryAddress will become mandatory
12	BankCode	C (6)	ExistingClient	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
13	ClientCode	C (24)	ExistingClient	Required (max: 1)	The Client Code of the existing client
-	ClientDemog	Complex	CustomerDetails	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if ExistingClient block is not provided
14	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
15	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client. Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.
16	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which

#	Field Name	Data type	Parent	Occurrence	Field Description
					the cardholder belongs. This value will be propagated at card level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
17	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
18	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support. This field doesn't apply to corporate cardholders. Only positive value will be accepted.
19	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Defaulted to 'N' when not provided.
20	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
21	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
22	TaxRegistryOffice	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
23	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
24	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
25	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
26	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP LIST table under the general parameter section.
27	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the

#	Field Name	Data type	Parent	Occurrence	Field Description
					general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
28	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
29	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
30	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
31	SecondFamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
32	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
33	SecondFirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if

#	Field Name	Data type	Parent	Occurrence	Field Description
					compound names are in use.
34	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
35	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section SIA configuration: IN : INDIVIDUAL ST : STAFF
36	SocioProfessional Code	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
37	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
38	CustomerSegment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing

#	Field Name	Data type	Parent	Occurrence	Field Description
					segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
39	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
40	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
41	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
42	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner. • D: Dependent.
43	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					There is no propagation of this aggregate from client to card.
44	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
45	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
46	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.

#	Field Name	Data type	Parent	Occurrence	Field Description
47	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
48	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
49	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					There is no propagation of this aggregate from client to card.
50	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
51	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
52	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
53	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address. There is no propagation of this field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder

#	Field Name	Data type	Parent	Occurrence	Field Description
					Becomes mandatory when sub block ExistingClient is not provided.
54	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
55	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
56	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
57	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
58	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
59	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
60	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.
61	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.

#	Field Name	Data type	Parent	Occurrence	Field Description
62	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
63	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
64	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
65	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's

#	Field Name	Data type	Parent	Occurrence	Field Description
					region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
66	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Secondary Address	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
67	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
68	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid only in case of Address Type "Temporary"
69	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
70	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
71	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
72	AddressLine1	C (64)	Secondary Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
73	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
74	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
75	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
76	ZipCode	C (10)	Secondary Address	Required (max: 1)	Recipient's postal code. This field should not be validated.
77	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					When allowed, one of the two fields CityCode or CityFreeText should be provided.
78	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
79	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
80	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
81	CountryCode	C (3)	Secondary Address	Required (max: 1)	Country code. Represents the country of the address being sent or received.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport details, civil ID details ...
82	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passeport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
83	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
84	DocumentIssue Date	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
85	DocumentExpiry Date	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD
86	Document IssueCity	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
87	Document IssueCountry	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Client	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
89	FieldName	C (30)	ClientCustom Fields	Required(max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
90	FieldValue	C (256)	ClientCustom Fields	Required (max: 1)	The value of the user defined field.
-	CardDetails	Complex	-	Optional (max: 1)	It must contain the details information of card and bank accounts data of the requested card. Not mandatory in case of One Shot card
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
91	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
92	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD EMBOSSING PRIORITY table under the card parameters section.
93	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
94	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
95	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the Client if not provided at card level.
96	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
97	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
98	NationalityCode	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
99	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be

#	Field Name	Data type	Parent	Occurrence	Field Description
					previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
100	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
101	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.
102	SecondFamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
103	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the Client if not provided at card level.
104	SecondFirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
105	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
106	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. SIA configuration : IN "INDIVIDUAL" for retail cards BU "BUSINESS" for Business/Corporate cards.
107	SocioProfessional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
108	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional

#	Field Name	Data type	Parent	Occurrence	Field Description
					occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
109	CustomerSegment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
110	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
111	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.
112	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder

#	Field Name	Data type	Parent	Occurrence	Field Description
					section. It will be inherited from the Client if not provided at card level.
113	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the last one is also not available.
114	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.
118	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.

#	Field Name	Data type	Parent	Occurrence	Field Description
119	SecondEmbossed Name	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
120	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
121	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
122	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
123	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
124	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxnnnnnnnn (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
125	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
126	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
127	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
128	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the

#	Field Name	Data type	Parent	Occurrence	Field Description
					cardholder can be reached. The phone number where the cardholder can be reached. SI A will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
129	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no propagation of this field from client to card.
130	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
131	PlasticDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
132	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail,

#	Field Name	Data type	Parent	Occurrence	Field Description
					Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
133	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.
-	PrimaryAddress	Complex	CardDetails	Required (max: 1)	To provide the primary address of the card
136	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
137	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
138	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
139	LinkFlag	C(1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's

#	Field Name	Data type	Parent	Occurrence	Field Description
					address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
140	LinkEntityCode	C(2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
141	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
142	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
143	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
144	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Not required when the address is linked
145	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
146	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
147	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
148	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level

#	Field Name	Data type	Parent	Occurrence	Field Description
149	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
150	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
151	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Secondary Address	Complex	CardDetails	Optional infinite list	To provide the secondary address of the cardholder.
152	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
153	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRange Date	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
154	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
155	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
156	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
157	LinkFlag	C(1)	Secondary Address	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field should be used for card addresses only.
158	LinkEntityCode	C(2)	Secondary Address	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
159	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
160	AddressLine1	C (64)	Secondary Address	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
161	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
162	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
163	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4

#	Field Name	Data type	Parent	Occurrence	Field Description
					4th line of the recipient address. Not required when the address is linked
164	ZipCode	C (10)	Secondary Address	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
165	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
166	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
167	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
168	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
169	CountryCode	C (3)	Secondary Address	Required (max: 1)	The country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Communication Events	Complex	-	Optional infinite list	To list all the communication event defined for the institution. (card delivery,

#	Field Name	Data type	Parent	Occurrence	Field Description
					pin delivery, statement....). Each event might be linked to 1 or many communication channels
	EventCode	C(4)	Communication Events	Required (max: 1)	This is the event code that can be triggered by the system. The list of event codes is defined in the following table: EVENT_TABLE
	EventActivation Status	C(1)	Communication Events	Required (max: 1)	This field defines if the event is enabled or disable. Y: Event active N: Event disabled
	IsEvent Mandatory	C(1)	Communication Events	Required (max: 1)	This field indicates if the event is defined as mandatory: Y: event is mandatory and can not be disabled. N: Event is optional and can be disabled.
-	Communication Channels	Complex	Communication Events	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
	ChannelType	C(2)	Communication Channels	Required (max: 1)	This field indicates the channel type that is defined for the a particular event: Value 1: SMS channel Value 2: Email channel Value 3: Physical file channel

#	Field Name	Data type	Parent	Occurrence	Field Description
					Value 4: Push channel
	Channel ActivationStatus	C(1)	Communication Channels	Required (max: 1)	This field indicate if the communication channel is enabled or disabled: Y: channel active N: channel disabled
	IsChannel Mandatory	C(1)	Communication Channels	Required (max: 1)	This field indicates if the channel is defined as mandatory: Y: channel is mandatory and can not be disabled. N: channel is optional and can be disabled.
	AllowsCustomPreference	C(1)	Communication Channels	Required (max: 1)	This field indicates if the PANholder is allowed to define permanent communication channel phone number, or email or address (for channel types SMS/Push or Email or physical file respectively).
	ChannelPhone Type	C(2)	Communication Channels	Optional (max: 1)	The channel phone type
	ChannelAddressFrom Entity	C(2)	Communication Channels	Optional (max: 1)	The channel address from entity
	ChannelAddress Type	C(2)	Communication Channels	Optional (max: 1)	The channel address type
	ChannelEmail Number	C(1)	Communication Channels	Optional (max: 1)	The channel email number
	ChannelAllows TempPreference	C(1)	Communication Channels	Required (max: 1)	The channel temp preference
	TemporaryCustom PreferenceSet	C(1)	Communication Channels	Required (max: 1)	The temporary custom preference set
	TMPPreference StartDate	Date	Communication Channels	Optional (max: 1)	TMP Preference Start Date
	TMPPreference EndDate	Date	Communication Channels	Optional (max: 1)	TMP Preference End Date

#	Field Name	Data type	Parent	Occurrence	Field Description
	TMPPhoneType	C(2)	Communication Channels	Optional (max: 1)	TMP Phone Type
	TMPAddressFrom Entity	C(2)	Communication Channels	Optional (max: 1)	TMP Address From Entity
	TMPAddress Type	C(2)	Communication Channels	Optional (max: 1)	TMP Address Type
	TMPEmailNumber	C(2)	Communication Channels	Optional (max: 1)	TMP Email Number
-	AuxiliaryProduct	Complex	CardDetails	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)
173	ProductIdentifier	C (6)	AuxiliaryProduct	Required (max: 1)	ID of the auxiliary product linked to the card. This code must be defined in the PowerCARD LINKED PRODUCT 1DESC table under the cardholder parameters section.
174	ContractNumber	C (16)	AuxiliaryProduct	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.
175	MembershipDate	Date	AuxiliaryProduct	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD
176	BeneficiaryAge	Long (3)	AuxiliaryProduct	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.
177	BeneficiaryName	C (40)	AuxiliaryProduct	Optional (max: 1)	Full name of the product's beneficiary.

#	Field Name	Data type	Parent	Occurrence	Field Description
178	StatusCode	C (1)	AuxiliaryProduct	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Cancelled Products with a 'Cancelled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
179	StatusDate	Date	AuxiliaryProduct	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
180	InsuranceBranch Code	C (6)	AuxiliaryProduct	Optional (max: 1)	Applies to insurance products solely. Specifies the branch code of the insurance company. Must be defined in the PowerCARD BRANCH table under the general parameters section.
181	InsuranceAge	Long (4)	AuxiliaryProduct	Optional (max: 1)	Specifies in number of years the amount of time that has elapsed since the membership date. Only positive value will be accepted.
182	AuxiliaryFee Amount	Decimal (18,3)	AuxiliaryProduct	Optional (max: 1)	Fees amount charged to the cardholder as remuneration for the auxiliary product.
183	PolicyNumber	C (32)	AuxiliaryProduct	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents to the insurance company.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	BankAccount	Complex	CardDetails	Optional infinite list	To provide information regarding bank accounts to which the card is linked. Mandatory for debit cards. Not required for credit cards.
184	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT _TYPE_LIST table under the general parameter section.
185	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
186	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
187	BankCodeFreetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
188	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the

#	Field Name	Data type	Parent	Occurrence	Field Description
					cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
189	BranchCodeFree text	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
190	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
191	AccountCurrency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
192	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for

#	Field Name	Data type	Parent	Occurrence	Field Description
					the bank account number. Becomes mandatory if SwiftCode is provided.
193	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
-	CardCustomFields	Complex	CardDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
194	FieldName	C (30)	CardCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
195	FieldValue	C (256)	CardCustomFields	Required(max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	09601	Invalid Card Number/Token Id
2	09602	Invalid Card Activation Flag
3	09603	Card is already registered to another Client
4	09604	Invalid Client Code
5	09605	Error while generating the Client Code
6	09606	Invalid Bank Code in Customer Details segment
7	09607	Invalid Branch Code in Customer Details Segment
8	09608	Invalid Family Status
10	09610	Invalid Cardholder's Gender
14	09614	Invalid Cardholder's Birth Date
17	09617	Invalid VIP Level
18	09618	Invalid Nationality Code
19	09619	Invalid Language Code
20	09620	Invalid Title Code
26	09626	Invalid Owner Code
27	09627	Invalid Socio-Professional Code
28	09628	Invalid Activity Code
29	09629	Invalid Customer Segment
33	09633	Invalid Lodging Mode
34	09634	Invalid Channel Number
35	09635	Invalid Phone Type 1
37	09637	Invalid Phone Type 2
39	09639	Invalid Phone Type 3
41	09641	Invalid Phone Type 4
45	09645	Invalid Primary Address Type
46	09646	Invalid Primary Address Sending Mode

#	Error code	Error Description
55	09655	Invalid Client Primary Address City Code
57	09657	Invalid Client Primary Address Region Code
59	09659	Invalid Client Primary Address Country Code
67	09667	City Code or City Free Text is required for Card Secondary Address
68	09668	Region Code or Region Free Text is required for Card Secondary Address
75	09675	Invalid Client Secondary Address Type
76	09676	Invalid Client Secondary Address Sending Mode
85	09685	Invalid Client Secondary Address City Code
87	09687	Invalid Client Secondary Address Region Code
89	09689	Invalid Client Secondary Address Country Code
90	09690	Validity Start Date must not be in the past in case Client Secondary Address
91	09691	Validity Start Date cannot be greater than Validity End Date in case Client Secondary Address
92	09692	Validity End Date must not be in the past in case Client Secondary Address
95	09695	City Code or City Free Text is required for Card Secondary Address
96	09696	Region Code or Region Free Text is required for Card Secondary Address
103	09703	Invalid Client Document Code
111	09711	Invalid Card Branch Code
112	09712	Invalid Card Priority Code
113	09713	Invalid Card Gender
115	09715	Invalid Card Birth Date
117	09717	Invalid Card VIP Level
118	09718	Invalid Card Nationality Code
119	09719	Invalid Card Language Code
120	09720	Invalid Card Title Code
126	09726	Invalid Card Owner Code

#	Error code	Error Description
127	09727	Invalid Card Socio Professional Code
128	09728	Invalid Card Activity Code
129	09729	Invalid Card Customer Segment
133	09733	Invalid Card Status Code
134	09734	Invalid Card Status Reason
136	09736	Invalid Card Card Fees Code
137	09737	Invalid Card Velocity Limit Index
141	09741	Invalid Card Corporate Id
142	09742	Invalid Card Phone Type 1
144	09744	Invalid Card Phone Type 2
146	09746	Invalid Card Phone Type 3
148	09748	Invalid Card Phone Type 4
152	09752	Invalid Card Plastic Delivery Method
153	09753	Invalid Card Pin Delivery Method
154	09754	Invalid Card Promotion Code
159	09759	Invalid Card Primary Address Type
160	09760	Invalid Card Primary Address Sending Mode
161	09761	Invalid Card Primary Address Purpose ID
169	09769	Invalid Card Primary Address City Code
171	09771	Invalid Card Primary Address Region Code
173	09773	Invalid Card Primary Address Country Code
181	09781	City Code or City Free Text is required for Card Secondary Address
182	09782	Region Code or Region Free Text is required for Card Secondary Address
189	09789	Invalid Card Secondary Address Type
190	09790	Invalid Card Secondary Address Sending Mode
191	09791	Invalid Card Secondary Address Purpose ID
199	09799	Invalid Card Secondary Address City Code
201	09801	Invalid Card Secondary Address Region Code
203	09803	Invalid Card Secondary Address Country Code

#	Error code	Error Description
204	09804	Validity Start Date must not be in the past in case Card Secondary Address
205	09805	Validity Start Date cannot be greater than Validity End Date in case Card Secondary Address
206	09806	Validity End Date must not be in the past in case Card Secondary Address
209	09809	City Code or City Free Text is required for Card Secondary Address
210	09810	Region Code or Region Free Text is required for Card Secondary Address
215	09815	Region Free Text is required in case of Region not valued for Client Secondary Address
232	09832	Invalid Insurance Branch Code
236	09836	Invalid Account Type
238	09838	Invalid Bank Code in Bank Account Segment
240	09840	Invalid Branch Code in Bank Account Segment
244	09844	BranchCode is missing when AccountNumber is a valued field
245	09845	AccountNumber is missing when BranchCode is a valued field
246	09846	IBAN is missing when SwiftCode is a valued field
248	09848	Invalid Account Number
249	09849	Invalid Account Currency
252	09852	Invalid Card Field Name
254	09854	Duplicate AddressType, more Address with the same AddressType
255	09855	Duplicate AddressPurpose, more AddressPurposeID with the same purpose
256	09856	An error occurred while inserting or updating a Bank Account
257	09858	Client Demog is Mandatory when Existing Client is not provided
258	09859	Primary Address is mandatory when Existing Client is not provided
259	09860	Invalid Link Flag
260	09861	Invalid Link Entiy Code
261	09862	Invalid Link Address Type

#	Error code	Error Description
262	09863	Invalid Purpose Address Type
263	09864	Invalid Purpose Address Type Sending Mode
264	09865	Document Expiry Date must not be in the past
265	09866	Document Expiry Date cannot be lesser than document issue date
266	09867	Document Issue Date can not be in the future

RESTful Response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/personalizeCardWebService/v2	POST	

REST sample call

```

"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"CardNumber": "",
"CardActivationFlag": "",
"CustomerDetails": {
  "ExistingClient": {
    "BankCode": "",
    "ClientCode": ""
  },
  "ClientDemog": {
    "BankCode": "",
    "ClientCode": "",
    "BranchCode": "",
    "FamilyStatus": "",
    "NumberOfDependents": "",
    "Gender": "",
    "LegalId": "",
    "TaxNumber": "",
    "TaxRegistryOffice": "",
    "BirthDate": "",
    "BirthCountry": "",
    "BirthCity": "",
    "VIPLLevel": "",
    "NationalityCode": "",
    "LanguageCode": "",
    "TitleCode": "",
    "FamilyName": "",
    "SecondFamilyName": "",
    "FirstName": "",
    "SecondFirstName": "",
    "MaidenName": "",
    "OwnerCode": "",
    "SocioProfessionalCode": "",
    "ActivityCode": "",
    "CustomerSegment": "",
    "TakingOnDate": "",
    "EmployerName": "",
    "StaffID": "",
    "LodgingMode": "",
    "ChannelNumber": "",
    "Phone1": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Phone2": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Phone3": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Phone4": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Email1": "",
    "Email2": ""
  },
  "PrimaryAddress": {
    "AddressType": "",
    "SendingMode": "",
    "CareOf": "",
    "AddressLine1": "",
    "AddressLine2": "",
    "AddressLine3": "",
    "AddressLine4": "",
    "ZipCode": "",
    "CityCode": "",
    "CityFreeText": "",
    "RegionCode": ""
  }
}

```

```

    "RegionFreeText": "",
    "CountryCode": ""
  },
  "Secondary Address": {
    "AddressType": "",
    "SendingMode": "",
    "ValidityRangeDate": {
      "StartDt": "",
      "EndDt": ""
    },
    "CareOf": "",
    "AddressLine1": "",
    "AddressLine2": "",
    "AddressLine3": "",
    "AddressLine4": "",
    "ZipCode": "",
    "CityCode": "",
    "CityFreeText": "",
    "RegionCode": "",
    "RegionFreeText": "",
    "CountryCode": ""
  },
  "Documents": {
    "DocumentCode": "",
    "DocumentId": "",
    "DocumentIssueDate": "",
    "DocumentExpiryDate": "",
    "DocumentIssueCity": "",
    "DocumentIssueCountry": ""
  },
  "ClientCustomFields": {
    "FieldName": "",
    "FieldValue": ""
  }
},
"CardDetails": {
  "CardDemog": {
    "BranchCode": "",
    "PriorityCode": "",
    "Gender": "",
    "LegalId": "",
    "BirthDate": "",
    "BirthCity": "",
    "VIPLevel": "",
    "NationalityCode": "",
    "LanguageCode": "",
    "TitleCode": "",
    "FamilyName": "",
    "SecondFamilyName": "",
    "FirstName": "",
    "SecondFirstName": "",
    "MaidenName": "",
    "OwnerCode": "",
    "SocioProfessionalCode": "",
    "ActivityCode": "",
    "CustomerSegment": "",
    "EmployerName": "",
    "StaffID": "",
    "ChannelNumber": "",
    "StatusCode": "",
    "StatusReason": "",
    "CardFeesCode": "",
    "VelocityLimitIndex": "",
    "ServicesSetupIndex": "",
    "EmbossedName": "",
    "SecondEmbossedName": "",
    "CorporateId": "",
    "Phone1": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Phone2": {
      "PhoneType": "",
      "PhoneNumber": ""
    },
    "Phone3": {

```

```

        "PhoneType": "",
        "PhoneNumber": ""
    },
    "Phone4": {
        "PhoneType": "",
        "PhoneNumber": ""
    },
    "Email1": "",
    "Email2": ""
},
"PlasticDeliveryMethod": "",
"PinDeliveryMethod": "",
"PromotionCode": "",
"TransactionFeeIndex": "",
"AuthorizarionFeeIndex": "",
"PrimaryAddress": {
    "AddressType": "",
    "SendingMode": "",
    "CareOf": "",
    "LinkFlag": "",
    "LinkEntityCode": "",
    "LinkAddressType": "",
    "AddressLine1": "",
    "AddressLine2": "",
    "AddressLine3": "",
    "AddressLine4": "",
    "ZipCode": "",
    "CityCode": "",
    "CityFreeText": "",
    "RegionCode": "",
    "RegionFreeText": "",
    "CountryCode": ""
},
"Secondary Address": {
    "AddressType": "",
    "SendingMode": "",
    "ValidityRangeDate": {
        "StartDt": "",
        "EndDt": ""
    },
    "CareOf": [
        ""
    ],
    "LinkFlag": "",
    "LinkEntityCode": "",
    "AddressLine1": "",
    "AddressLine2": "",
    "AddressLine3": "",
    "AddressLine4": "",
    "ZipCode": "",
    "CityCode": "",
    "CityFreeText": "",
    "RegionCode": "",
    "RegionFreeText": "",
    "CountryCode": ""
},
"AddressPurpose": {
    "AddressPurposeID": "",
    "AddressType": "",
    "SendingMode": ""
},
"AuxiliaryProduct": {
    "ProductIdentifier": "",
    "ContractNumber": "",
    "MembershipDate": "",
    "BeneficiaryAge": "",
    "BeneficiaryName": "",
    "StatusCode": "",
    "StatusDate": "",
    "InsuranceBranchCode": "",
    "InsuranceAge": "",
    "AuxiliaryFeeAmount": "",
    "PolicyNumber": ""
},
"BankAccount": {
    "AcctType": "",

```



```

    "AcctSequence": "",
    "BankCode": "",
    "BankCodeFreetext": "",
    "BranchCode": "",
    "BranchCodeFreetext": "",
    "AccountNumber": "",
    "AccountCurrency": "",
    "IBAN": "",
    "SwiftCode": ""
  },
  "CardCustomFields": {
    "FieldName": "",
    "FieldValue": ""
  }
}

```

4.53 transferAccountToAccount API

API overview

Transfer Account to Account web service enables customers, agents or any third party to transfer money instantly from a given account to another account in the appropriate currency. The services can be enabled only by providing a limited but important set of details including the source account, destination account, amount, currency...

The operation will debit the source account and credit the destination account with the transfer amount in the right account currency.

The system run some controls on the input data and may reject the operation in case of any data inconsistency.

Functional description

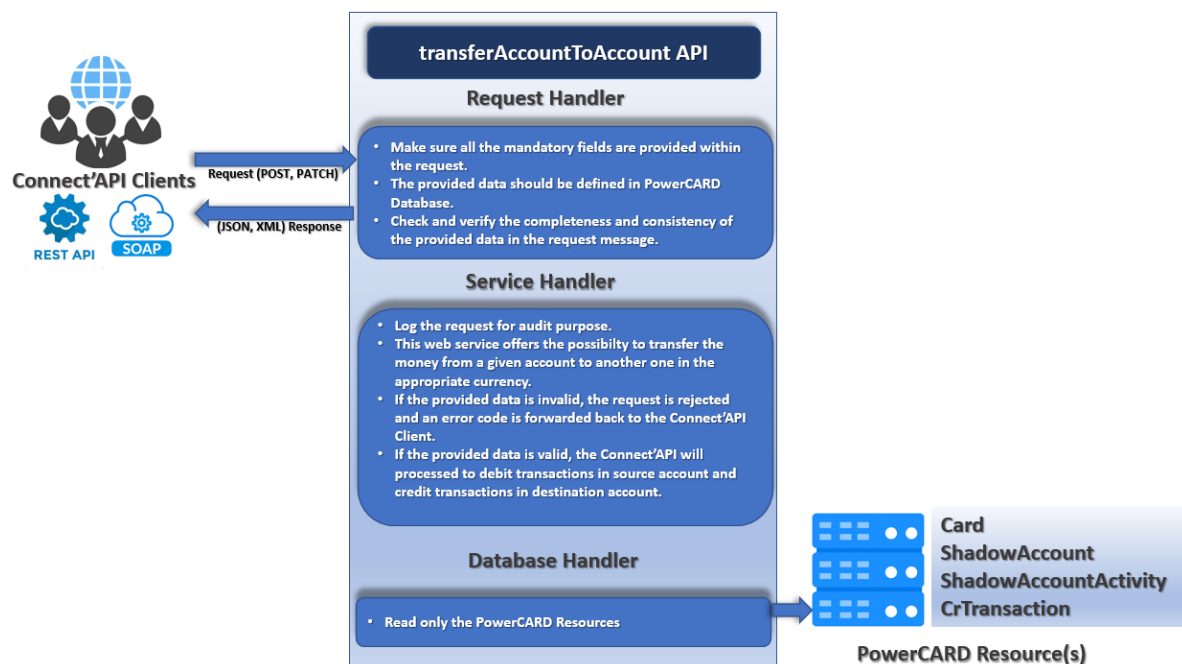


Figure 53: transferAccountToAccount process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
v2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	SrcAccountNbr	C (40)	-	Required (max : 1)	The source account number: In case of credit Account, It should refer to the internal account number. This value should be already defined in the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					SHADOW _ACCOUNT table.
5	DestAccountNbr	C (40)	-	Required (max : 1)	The destination account number: In case of credit Account, It should refer to the internal account number. This value should be already defined in the PowerCARD SHADOW _ACCOUNT table.
6	ReversalFlag	ClosedEnum	-	Required (max : 1)	Indicates whether it is a normal or reversal transaction. - N: Normal. - R: Reversal.
7	CaptureReasonCode	ClosedEnum	-	Optional (max : 1)	This field holds the transaction capture reason code, or transaction reversal reason code. Values assigned to this field must be defined in the TRANS _REASON_CODE table under the PowerCARD general parameters section.
-	TransferAmount	Complex	-	Required (max : 1)	The transaction amount and currency. Amount and currency of the transaction that will be posted to the customer account (Source & Destination).
8	Amt	Long	TransferAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
9	CurCode	C (3)	TransferAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
10	TransactionDate	DateTime	-	Required (max : 1)	The transaction date. Date and time when the transaction took place.
11	ValueDate	DateTime	-	Optional (max : 1)	The transaction value date. Date when the transaction will be considered the accounts.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: Proceed With Success: Call

#	Field Name	Data type	Parent	Occurrence	Field Description
					was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	16801	Invalid Source Account Number
2	16802	Invalid destination Account Number
3	16803	Invalid Amount
4	16804	Invalid Currency
5	16805	Invalid Capture Reason Code
6	16806	Invalid Transaction Code, Please make sure that transaction Codes DT & CT are both defined
7	16807	Invalid Reversal Flag
8	16808	Invalid Source Account Status
9	16809	Invalid destination Account Status

Response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/transferAccountToAccountWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "srcAccountNbr": "",
  "destAccountNbr": "",
  "reversalFlag": "",
  "captureReasonCode": "",
  "transferAmount": {
    "amt": "",
    "curCode": ""
  },
  "transactionDate": "",
  "valueDate": ""
}
```

4.54 searchFinancialAccount

API overview

The search loan web service is an API that can be used to retrieve all the financial accounts provided by the the system. this service provides the ability to search for all existing accounts in the system using different criteria's like Corporate Id, Account Id or also Parent Account Id.

The system run some controls on the input data and may reject the operation in case of any data inconsistency.

Functional description

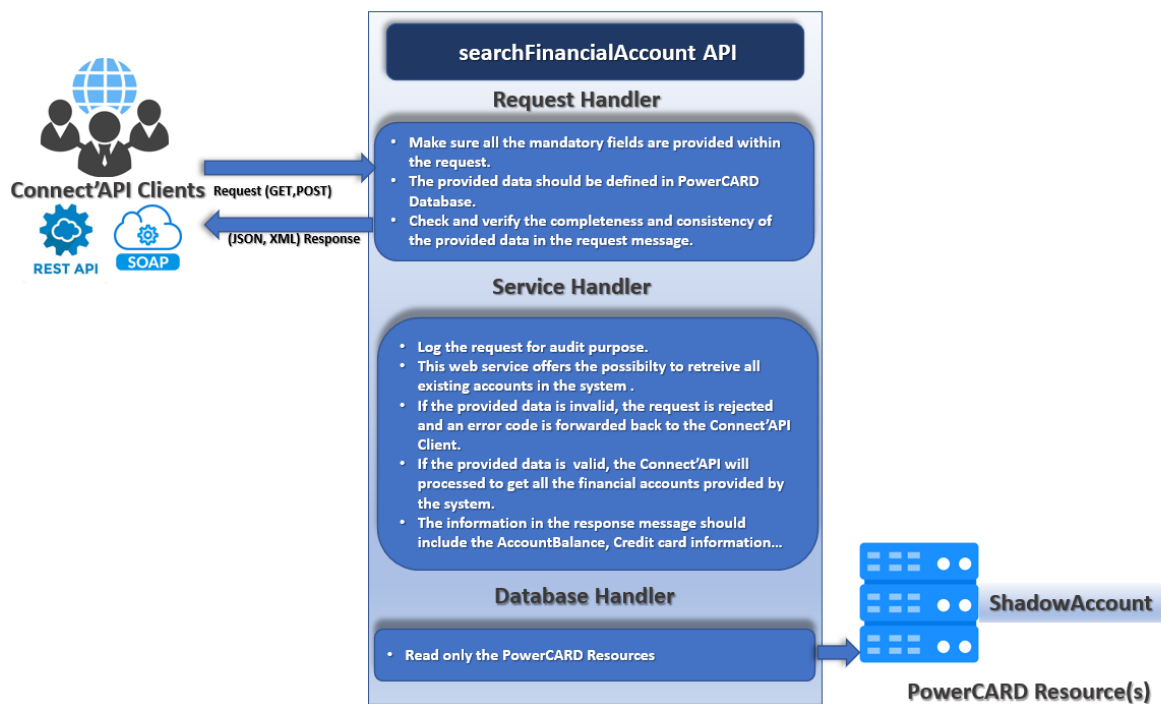


Figure 54: searchFinancialAccount process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a

#	Field Name	Data type	Parent	Occurrence	Field Description
					universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	ClientCode	C (24)	-	Optional (max : 1)	Represent the customer identifier. This field is required if CorporateId and CardNumber is empty.
5	CorporateId	C (15)	-	Optional (max : 1)	The corporate identifier. This field is required if ClientCode and CardNumber is empty.
6	AccountId	C (40)	-	Optional (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the financial account ID
7	ParentAccountId	C (40)	-	Optional (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the parent financial

#	Field Name	Data type	Parent	Occurrence	Field Description
					account ID in case of a corporate or family structure.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	FinancialAccount	Complex	-	Optional infinite list	This aggregate contains the list of financial accounts.
-	AccountBalance	Complex	FinancialAccount	Optional infinite list	This aggregate contains data related to the balance of the accounts linked to the card transmitted in the request message.
6	CMSAccountId	C (24)	AccountBalance	Required (max : 1)	The CMS account identifier as defined in PowerCARD.
7	AccountType	C (40)	AccountBalance	Required (max : 1)	Account type Refers to the type of the account. Possible values can be downloaded from the PowerCARD ACCOUNT_TYPE_LIST table
8	ClientCode	C(24)	AccountBalance	Optional (max : 1)	The client code present in the table CLIENT in PowerCARD.
9	ClientHostID	C(24)	AccountBalance	Optional (max : 1)	External Identified of the customer in the host
-	CrdAcctInfo	Complex	AccountBalance	Optional (max : 1)	The credit account information This aggregate contain information with regards to the credit account profile, account credit limit... Valid only in case of credit card account.

#	Field Name	Data type	Parent	Occurrence	Field Description
10	CatAccount Product	C (6)	CrdAcctInfo	Required (max : 1)	It is a 6 digit code identifying the account product code assigned to the account. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
11	CatAccount ProductVersion	C (10)	CrdAcctInfo	Required (max : 1)	It is a 10 character fields identifying the account product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = A
12	CreditLimit	Long	CrdAcctInfo	Optional (max : 1)	Sets the maximum amount that can be borrowed on the cardholder's account. The value assigned to this field can be used to set initial credit limits for newly created cards in the PowerCARD system. In order to be valid, the value of this field should not exceed the ceiling amount defined in the credit account profile. If empty, the system will inherit the default value defined in the credit account profile.
13	CashLimit	Long	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of cash that can be obtained on the cardholder's account

#	Field Name	Data type	Parent	Occurrence	Field Description
					(ATM withdrawal, cash advance).
14	Loanlimit	Long	CrdAcctInfo	Optional (max : 1)	Used to set the maximum amount of loan transactions that can be performed on the cardholder's account. The value of this field should not exceed the total credit limit permitted on the cardholder's account.
15	paymentMode AutoDebitInd	C (1)	CrdAcctInfo	Required (max : 1)	Should contain the automatic debit payment mode indicator: • 'M': Automatic debit is mandatory. • 'P': Automatic debit is allowed. • 'F': Automatic debit is not allowed.
16	CycleCode	C (20)	CrdAcctInfo	Optional (max : 1)	The credit cycle code The cycle is the period of time that elapses between 2 consecutive billing statements. The value of this field indicates the code that refers to the new billing cycle that should be assigned to the account. Possible values must be retrieved from the PowerCARD CYCLE _CUTOFF _PARAMETERS table
17	DefaultRate	Decimal	CrdAcctInfo	Optional (max : 1)	It refers to the default interest

#	Field Name	Data type	Parent	Occurrence	Field Description
					rate that could be charged by credit card issuer. Unless it is specified in the credit account profile, this rate will be used by PowerCARD for all type of account balances
18	StatementCode	C (2)	CrdAcctInfo	Optional (max : 1)	Refers to the setting of statement parameters in PowerCARD that enables to specify all the data relative to the statement that will be sent to the customer. It includes periodicity, fees amount... This setting is defined as an index that must be retrieved from PowerCARD STATEMENT_PARAMETERS table
-	RequestedCredit Amount	Complex	CrdAcctInfo	Optional (max : 1)	The requested credit amount. The captured limit is made by PowerCARD.
19	Amt	Long	RequestedCredit Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
20	CurCode	NC (3)	Requested CreditAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined

#	Field Name	Data type	Parent	Occurrence	Field Description
					in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	DebitOrder Amount	Complex	CrđAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
21	Amt	Long ()	DebitOrder Amount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
22	CurCode	NC (3)	DebitOrder Amount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	DebitOrder Percentage	Complex	CrđAcctInfo	Optional (max : 1)	Fixed Amount debited each month (credit cards only).
23	Amt	Long ()	DebitOrder Percentage	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
24	CurCode	C (3)	DebitOrder Percentage	Optional (max : 1)	Currency code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	CurrentBalance	Complex	AccountBalance	Required (max : 1)	Must include the amount value of the account balance. - For deposit Accounts, it refers to Current available balance (typically, ledger balance less outstanding authorizations. - For Credit Card Accounts, it refers to the cardholder outstanding balance during the cycle present cycle.
25	Amt	Long ()	CurrentBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	NC (3)	CurrentBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
27	BalanceStatus Date	DateTime	AccountBalance	Optional (max : 1)	Refers to the last update date of the concerned balance. Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
-	CurrentDebit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the debit transactions performed by the cardholder during the present cycle. Valid only for credit accounts
28	Amt	Long ()	CurrentDebit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
29	CurCode	C (3)	CurrentDebit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	CurrentCredit	Complex	AccountBalance	Optional (max : 1)	It refers to the sum of all the credit transactions performed by the cardholder during the present cycle. (payment, refund...)

#	Field Name	Data type	Parent	Occurrence	Field Description
					Valid only for credit accounts
30	Amt	Long ()	CurrentCredit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
31	CurCode	NC (3)	CurrentCredit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	ClosingBalance	Complex	AccountBalance	Optional (max : 1)	The closing balance is the positive or negative outstanding amount in an account at the conclusion of the last cycle. Valid only for credit accounts
32	Amt	Long	ClosingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
33	CurCode	C (3)	ClosingBalance	Optional (max : 1)	Currency code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	MinimumDue	Complex	AccountBalance	Optional (max : 1)	The last minimum amount calculated by the system that the cardholder must pay by the due date. Valid only for credit accounts
34	Amt	Long	MinimumDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999
35	CurCode	NC (3)	MinimumDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
36	LastCutoff	DateTime	AccountBalance	Optional (max : 1)	Refers to the last Account statement date Valid only for credit accounts Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.

#	Field Name	Data type	Parent	Occurrence	Field Description
37	NextDueDate	DateTime	AccountBalance	Optional (max : 1)	It refers to the date the minimum monthly payment is due Valid only for credit accounts Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
38	Status	ClosedEnum	AccountBalance	Optional (max : 1)	This field must contain the status of the account: - For Deposit accounts, Valid values are : N = Normal C = Cancelled S = Suspended W = Waiting for validation V = To be sent for validation R = Rejected. - For Credit Card Accounts: 0 = Normal 1 = Supervision 2 = Pre legal 3 = Legal 4 = Insurance management 5 = Lost /Closed 6 = Lost / Death 7 = Closed / Cancelled 8 = Write Off 9 = Untraceable
-	withdrawal Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the

#	Field Name	Data type	Parent	Occurrence	Field Description
					account that may be withdrawn.
39	Amt	Long ()	withdrawaAvailable	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
40	CurCode	NC (3)	withdrawal Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	purchase Available	Complex	AccountBalance	Optional (max : 1)	This is the total amount of cash present in the account that may be used for purchase transactions.
41	Amt	Long	purchase Available	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
42	CurCode	C (3)	purchase Available	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY _TABLE table under the general parameters section.
-	withdrawal Limit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of cash that can be obtained on the cardholder's account (ATM withdrawal, cash advance).
43	Amt	Long ()	withdrawal Limit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10n; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
44	CurCode	NC (3)	withdrawalLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	purchaseLimit	Complex	AccountBalance	Optional (max : 1)	This is the maximum amount of money that can be used by the cardholder to make purchase transactions.
45	Amt	Long ()	purchaseLimit	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10n; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
46	CurCode	NC (3)	purchaseLimit	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	PendingBalance	Complex	AccountBalance	Optional (max : 1)	This is the total amount of all pending authorizations.
47	Amt	Long ()	PendingBalance	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999
48	CurCode	C (3)	PendingBalance	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	BankAccount	Complex	FinancialAccount	Required infinite list	To provide information regarding bank accounts to which the card is linked.
49	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already

#	Field Name	Data type	Parent	Occurrence	Field Description
					be defined in PowerCARD ACCOUNT _TYPE_LIST table under the general parameter section.
50	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
51	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
52	BankCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
53	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
54	BranchCode Freetext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Required if "direct debit validation" set to NO at Bank level
55	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
56	AccountCurrency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
57	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
58	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money

#	Field Name	Data type	Parent	Occurrence	Field Description
					between banks. Banks also use these codes to exchanging messages.
59	DefaultFlag	C(1)	BankAccount	Required (max:1)	Flag to indicate the default bank account: Y : Default bank account N : Not the default bank account

Specific error codes

#	Error code	Error Description
1	30101	Invalid Account Number
2	30102	Invalid Card Number
3	30103	Invalid Loan Status
4	30104	Invalid Range Date

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested		x

Status Code	Description	Success Response	Error Response
	is not supported for the PowerCARD Resource.		
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchFinancialAccountWebService/V2	POST	
/Accounts/V2? clientCode={clientCode}& corporateId={corporateId}&parentAccountId={parentAccountId}&pageSize={pageSize}&pageIndex={pageIndex}	GET	clientCode=[C(24)] corporateId =[C(22)] accountId =[C(15)] parentAccountId =[C(6)] pageSize=[PositiveInteger] pageIndex=[PositiveInteger]

REST sample call

```
{
  "keyValues": [
    {
      "key": "pageSize",
      "data": ""
    },
    {
      "key": "pageIndex",
      "data": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "clientCode": "",
  "corporateId": "",
  "accountId": "",
  "parentAccountId": ""
}
```

4.55 addSupplementaryCard API

API overview

This web service is intended to be used by both agents and customers to set up an additional cardholder like a family member on an existing card.

Any new supplementary card created in the system is issued under the same account as the primary cardholder. They will together share the same account balance.

Supplementary cardholders have no legal liability to pay the charges that they incur when they use their cards. However, the primary account holder is responsible for paying all the charges posted to his account.

Functional description

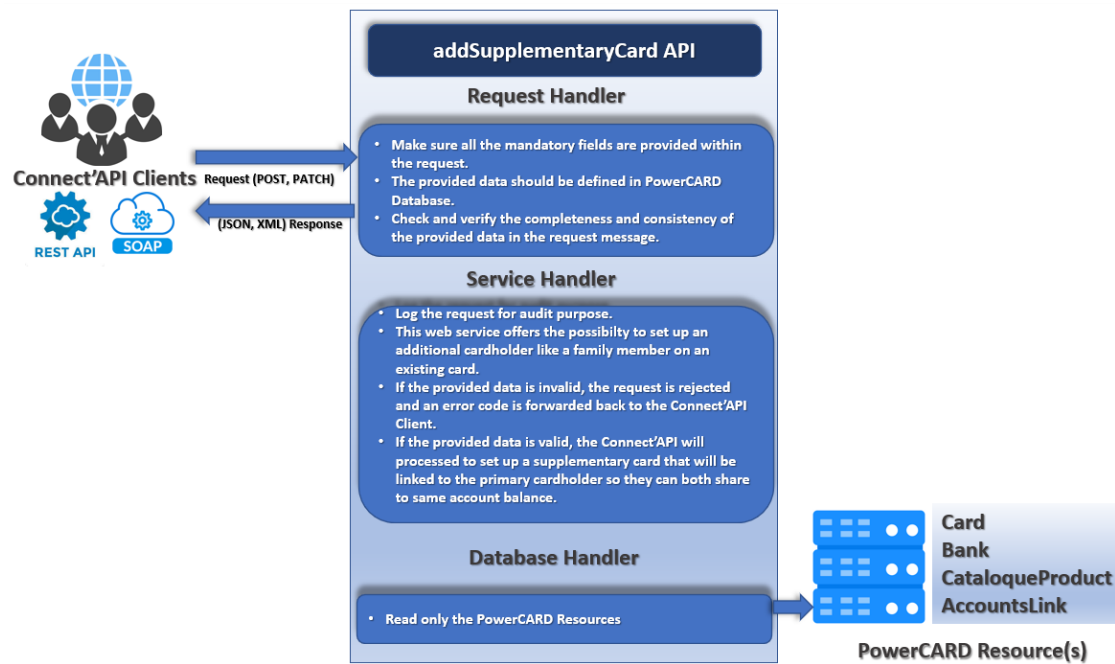


Figure 55: addSupplementaryCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
-	SupplementaryCard	Complex	-	Required (max : 1)	The supplementary card details.
4	BankCode	C (22)	SupplementaryCard	Required (max : 1)	The bank code. This code must be previously defined in the PowerCARD BANK table under the general parameter section. It should be retrieved from PowerCARD using the Web Service: getReferentialList
5	BranchCode	C (15)	SupplementaryCard	Optional (max : 1)	Branch code. 6 digits code of the branch to which the card belongs Valid values must be downloaded from PowerCARD BRANCH table using the Web Service: getReferentialList
6	ParentCard	C(23)	SupplementaryCard	Required (max : 1)	Parent card.
7	CatPIProduct	C (6)	SupplementaryCard	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code being requested in the application. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					with product type = P using the Web Service: getReferentialList
8	CatPIProductVersion	C (10)	SupplementaryCard	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P using the Web Service: getReferentialList
9	PlasticCode	C (3)	SupplementaryCard	Required (max : 1)	It refers to the plastic of the card to be personalized. Valid values must be previously defined in the PowerCARD PLASTIC _LIST table and could be populated using the Web Service: getReferentialList
11	CardActivationFlag	C (1)	-	Optional (max: 1)	This field indicates whether the card will be activated or not. Possible values are the following: • Y: Card activated • N: Card not activated If this information is not provided, by default, the card is not activated.
-	CustomerDetails	Complex	-	Required (max: 1)	It must contain the details information of the cardholder.
-	ExistingClient	Complex	CustomerDetails	Optional (max: 1)	To provide the existing client code to which all the entities within the boarding request should be linked.

#	Field Name	Data type	Parent	Occurrence	Field Description
					When the sub block "ExistingClient" is not provided, the sub blocks ClientDemog and PrimaryAddress will become mandatory
12	BankCode	C (6)	ExistingClient	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
13	ClientCode	C (24)	ExistingClient	Required (max: 1)	The Client Code of the existing client
-	ClientDemog	Complex	CustomerDetails	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if ExistingClient block is not provided
14	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
15	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client. Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
16	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs. This value will be propagated at card level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
17	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
18	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support. This field doesn't apply to corporate cardholders. Only positive value will be accepted.
19	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male.

#	Field Name	Data type	Parent	Occurrence	Field Description
					- F: Female. - N: Not applicable. Defaulted to 'N' when not provided.
20	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
21	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
22	TaxRegistryOffice	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
23	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
24	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
25	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
26	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP LIST table under the general parameter section.
27	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined

#	Field Name	Data type	Parent	Occurrence	Field Description
					in the PowerCARD COUNTRY table under the general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
28	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
29	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table under the general parameters section.
30	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
31	SecondFamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use

#	Field Name	Data type	Parent	Occurrence	Field Description
32	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
33	SecondFirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if compound names are in use.
34	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
35	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section SIA configuration: IN : INDIVIDUAL ST : STAFF
36	SocioProfessional Code	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
37	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation. This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
38	Customer Segment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
39	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
40	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
41	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
42	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner. • D: Dependent.
43	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
44	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
45	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
46	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
47	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
48	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
49	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					[no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
50	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
51	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
52	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
53	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					There is no propagation of this field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder Becomes mandatory when sub block ExistingClient is not provided.
54	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.
55	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
56	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
57	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
58	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
59	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
60	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.

#	Field Name	Data type	Parent	Occurrence	Field Description
61	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
62	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
63	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
64	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or

#	Field Name	Data type	Parent	Occurrence	Field Description
					RegionFreeText should be provided.
65	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
66	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Secondary Address	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
67	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
68	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ValidityRange Date	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
69	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
70	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
71	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
72	AddressLine1	C (64)	Secondary Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
73	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
74	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
75	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
76	ZipCode	C (10)	Secondary Address	Required (max: 1)	Recipient's postal code. This field should not be validated.
77	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
78	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
79	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
80	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
81	CountryCode	C (3)	Secondary Address	Required (max: 1)	Country code. Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport details, civil ID details ...
82	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passeport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
83	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.
84	DocumentIssue Date	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
85	DocumentExpiry Date	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD

#	Field Name	Data type	Parent	Occurrence	Field Description
86	DocumentIssue City	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
87	DocumentIssue Country	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	Client	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
89	FieldName	C (30)	ClientCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
90	FieldValue	C (256)	ClientCustom Fields	Required (max: 1)	The value of the user defined field.
-	CardDetails	Complex	-	Optional (max: 1)	It must contain the details information of card and bank accounts data of the requested card. Not mandatory in case of One Shot card
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as

#	Field Name	Data type	Parent	Occurrence	Field Description
					names, card product type, plastic type...
91	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
92	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD EMBOSSING PRIORITY table under the card parameters section.
93	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
94	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time. The Legal ID will be inherited from the Client if not provided at card level.
95	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
96	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
97	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
98	NationalityCode	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999"

#	Field Name	Data type	Parent	Occurrence	Field Description
					in case of value not available
99	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
100	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
101	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.
102	SecondFamilyName	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
103	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
104	SecondFirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
105	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
106	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. SIA configuration : IN "INDIVIDUAL" for retail cards BU "BUSINESS" for Business/Corporate cards.
107	SocioProfessional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be

#	Field Name	Data type	Parent	Occurrence	Field Description
					inherited from the Client if not provided at card level.
108	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
109	Customer Segment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
110	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the Client if not provided at card level.
111	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
112	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
113	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the last one is also not available.
114	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.

#	Field Name	Data type	Parent	Occurrence	Field Description
118	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.
119	SecondEmbossedName	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
120	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
121	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
122	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
123	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
124	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached.

#	Field Name	Data type	Parent	Occurrence	Field Description
					There is no propagation of this aggregate from client to card.
125	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
126	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
127	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					2: Home phone. 3: Personal mobile phone. 4: Others.
128	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
129	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no propagation of this field from client to card.
130	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
131	PlasticDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					default parameter at bank and product.
132	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
133	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.
-	PrimaryAddress	Complex	CardDetails	Required (max: 1)	To provide the primary address of the card
136	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
137	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
138	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity.

#	Field Name	Data type	Parent	Occurrence	Field Description
					An entity could be, depending on the context, a customer, a merchant, a chain...
139	LinkFlag	C(1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
140	LinkEntityCode	C(2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
141	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
142	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
143	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2

#	Field Name	Data type	Parent	Occurrence	Field Description
					2nd line of the recipient address. Not required when the address is linked
144	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
145	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
146	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
147	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
148	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is

#	Field Name	Data type	Parent	Occurrence	Field Description
					not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
149	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
150	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
151	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Secondary Address	Complex	CardDetails	Optional infinite list	To provide the secondary address of the cardholder.
152	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
153	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
154	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
155	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
156	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...

#	Field Name	Data type	Parent	Occurrence	Field Description
157	LinkFlag	C(1)	Secondary Address	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
158	LinkEntityCode	C(2)	Secondary Address	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
159	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
160	AddressLine1	C (64)	Secondary Address	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
161	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked

#	Field Name	Data type	Parent	Occurrence	Field Description
162	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
163	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
164	ZipCode	C (10)	Secondary Address	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
165	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
166	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise,

#	Field Name	Data type	Parent	Occurrence	Field Description
					it can be provided only when the free capture of city is allowed at bank level
167	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
168	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
169	CountryCode	C (3)	Secondary Address	Required (max: 1)	The country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Not required when the address is linked.
-	Communication Events	Complex	-	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
	EventCode	C(4)	Communication Events	Required (max: 1)	This is the event code that can be triggered by the system. The list of event codes is defined in the following table: EVENT_TABLE
	EventActivation Status	C(1)	Communication Events	Required (max: 1)	This field defines if the event is enabled or disable. - Y: Event active - N: Event disabled
	IsEventMandatory	C(1)	Communication Events	Required (max: 1)	This field indicates if the event is defined as mandatory: - Y: event is mandatory and can not be disabled. - N: Event is optional and can be disabled.
-	Communication Channels	Complex	Communication Events	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels

#	Field Name	Data type	Parent	Occurrence	Field Description
	ChannelType	C(2)	Communication Channels	Required (max: 1)	This field indicates the channel type that is defined for the a particular event: - Value 1: SMS channel - Value 2: Email channel - Value 3: Physical file channel - Value 4: Push channel
	ChannelActivation Status	C(1)	Communication Channels	Required (max: 1)	This field indicate if the communication channel is enabled or disabled: - Y: channel active - N: channel disabled
	IsChannel Mandatory	C(1)	Communication Channels	Required (max: 1)	This field indicates if the channel is defined as mandatory: - Y: channel is mandatory and can not be disabled. - N: channel is optional and can be disabled.
	AllowsCustom Preference	C(1)	Communication Channels	Required (max: 1)	This field indicates if the PANholder is allowed to define permanent communication channel phone number, or email or address (for channel types SMS/Push or Email or physical file respectively).
	ChannelPhone Type	C(2)	Communication Channels	Optional (max: 1)	The channel phone type
	ChannelAddressFrom Entity	C(2)	Communication Channels	Optional (max: 1)	The channel address from entity
	Channel AddressType	C(2)	Communication Channels	Optional (max: 1)	The channel address type

#	Field Name	Data type	Parent	Occurrence	Field Description
	ChannelEmail Number	C(1)	CommunicationChannel	Optional (max: 1)	The channel email number
	ChannelAllowsTempPreference	C(1)	CommunicationChannel	Required (max: 1)	The channel temp preference
	TemporaryCustomPreference Set	C(1)	CommunicationChannel	Required (max: 1)	The temporary custom preference set
	TMPPreferenceStartDate	Date	CommunicationChannel	Optional (max: 1)	TMP Preference Start Date
	TMPPreferenceEndDate	Date	CommunicationChannel	Optional (max: 1)	TMP Preference End Date
	TMPPhoneType	C(2)	CommunicationChannel	Optional (max: 1)	TMP Phone Type
	TMPAddressFromEntity	C(2)	CommunicationChannel	Optional (max: 1)	TMP Address From Entity
	TMPAddress Type	C(2)	CommunicationChannel	Optional (max: 1)	TMP Address Type
	TMPEmail Number	C(2)	CommunicationChannel	Optional (max: 1)	TMP Email Number
-	AuxiliaryProduct	Complex	CardDetails	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)
173	ProductIdentifier	C (6)	AuxiliaryProduct	Required (max: 1)	ID of the auxiliary product linked to the card. This code must be defined in the PowerCARD LINKED PRODUCT 1DESC table under the cardholder parameters section.
174	ContractNumber	C (16)	AuxiliaryProduct	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.
175	MembershipDate	Date	AuxiliaryProduct	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD

#	Field Name	Data type	Parent	Occurrence	Field Description
176	BeneficiaryAge	Long (3)	AuxiliaryProduct	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.
177	BeneficiaryName	C (40)	AuxiliaryProduct	Optional (max: 1)	Full name of the product's beneficiary.
178	StatusCode	C (1)	AuxiliaryProduct	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Cancelled Products with a 'Cancelled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
179	StatusDate	Date	AuxiliaryProduct	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
180	InsuranceBranch Code	C (6)	AuxiliaryProduct	Optional (max: 1)	Applies to insurance products solely. Specifies the branch code of the insurance company. Must be defined in the PowerCARD BRANCH table under the general parameters section.
181	InsuranceAge	Long (4)	AuxiliaryProduct	Optional (max: 1)	Specifies in number of years the amount of time that has elapsed since the membership date. Only positive value will be accepted.
182	AuxiliaryFee Amount	Decimal (18,3)	AuxiliaryProduct	Optional (max: 1)	Fees amount charged to the cardholder as

#	Field Name	Data type	Parent	Occurrence	Field Description
					remuneration for the auxiliary product.
183	PolicyNumber	C (32)	AuxiliaryProduct	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents to the insurance company.
-	BankAccount	Complex	CardDetails	Optional infinite list	To provide information regarding bank accounts to which the card is linked. Mandatory for debit cards. Not required for credit cards.
184	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT _TYPE_LIST table under the general parameter section.
185	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
186	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Required if "Direct Debit Validation" set to YES at Bank level
187	BankCodeFreetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
188	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
189	BranchCodeFreetext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
190	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
191	AccountCurrency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
192	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
193	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
-	CardCustomFields	Complex	CardDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
194	FieldName	C (30)	CardCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional

#	Field Name	Data type	Parent	Occurrence	Field Description
					fields that can be provided by a given bank. The provided name should exist in the above parameter table.
195	FieldValue	C (256)	CardCustomFields	Required (max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing.

#	Field Name	Data type	Parent	Occurrence	Field Description
					SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CardNumber	C (22)	-	Optional (max: 1)	Represents the Card Number or the Card Token Id generated by PowerCARD in case of successful processing. The Card Token ID is returned if the complex KeyValue with key "TOKEN" is included in the WS call.

Specific error codes

#	Error code	Error Description
1	19601	Invalid Card Number/Token Id
2	19602	Invalid Card Activation Flag
3	19603	Card is already registered to another Client
4	19604	Invalid Client Code
5	19605	Error while generating the Client Code
6	19606	Invalid Bank Code in Customer Details segment
7	19607	Invalid Branch Code in Customer Details Segment
8	19608	Invalid Family Status
10	19610	Invalid Cardholder's Gender
14	19614	Invalid Cardholder's Birth Date
17	19617	Invalid VIP Level

#	Error code	Error Description
18	19618	Invalid Nationality Code
19	19619	Invalid Language Code
20	19620	Invalid Title Code
26	19626	Invalid Owner Code
27	19627	Invalid Socio-Professional Code
28	19628	Invalid Activity Code
29	19629	Invalid Customer Segment
33	19633	Invalid Lodging Mode
34	19634	Invalid Channel Number
35	19635	Invalid Phone Type 1
37	19637	Invalid Phone Type 2
39	19639	Invalid Phone Type 3
41	19641	Invalid Phone Type 4
45	19645	Invalid Primary Address Type
46	19646	Invalid Primary Address Sending Mode
55	19655	Invalid Client Primary Address City Code
57	19657	Invalid Client Primary Address Region Code
59	19659	Invalid Client Primary Address Country Code
67	19667	City Code or City Free Text is required for Card Secondary Address
68	19668	Region Code or Region Free Text is required for Card Secondary Address
75	19675	Invalid Client Secondary Address Type
76	19676	Invalid Client Secondary Address Sending Mode
85	19685	Invalid Client Secondary Address City Code
87	19687	Invalid Client Secondary Address Region Code
89	19689	Invalid Client Secondary Address Country Code
90	19690	Validity Start Date must not be in the past in case Client Secondary Address
91	19691	Validity Start Date cannot be greater than Validity End Date in case Client Secondary Address

#	Error code	Error Description
92	19692	Validity End Date must not be in the past in case Client Secondary Address
95	19695	City Code or City Free Text is required for Card Secondary Address
96	19696	Region Code or Region Free Text is required for Card Secondary Address
103	19703	Invalid Client Document Code
111	19711	Invalid Card Branch Code
112	19712	Invalid Card Priority Code
113	19713	Invalid Card Gender
115	19715	Invalid Card Birth Date
117	19717	Invalid Card VIP Level
118	19718	Invalid Card Nationality Code
119	19719	Invalid Card Language Code
120	19720	Invalid Card Title Code
126	19726	Invalid Card Owner Code
127	19727	Invalid Card Socio Professional Code
128	19728	Invalid Card Activity Code
129	19729	Invalid Card Customer Segment
133	19733	Invalid Card Status Code
134	19734	Invalid Card Status Reason
136	19736	Invalid Card Card Fees Code
137	19737	Invalid Card Velocity Limit Index
141	19741	Invalid Card Corporate Id
142	19742	Invalid Card Phone Type 1
144	19744	Invalid Card Phone Type 2
146	19746	Invalid Card Phone Type 3
148	19748	Invalid Card Phone Type 4
152	19752	Invalid Card Plastic Delivery Method
153	19753	Invalid Card Pin Delivery Method
154	19754	Invalid Card Promotion Code
159	19759	Invalid Card Primary Address Type
160	19760	Invalid Card Primary Address Sending Mode
161	19761	Invalid Card Primary Address Purpose ID
169	19769	Invalid Card Primary Address City Code

#	Error code	Error Description
171	19771	Invalid Card Primary Address Region Code
173	19773	Invalid Card Primary Address Country Code
181	19781	City Code or City Free Text is required for Card Secondary Address
182	19782	Region Code or Region Free Text is required for Card Secondary Address
189	19789	Invalid Card Secondary Address Type
190	19790	Invalid Card Secondary Address Sending Mode
191	19791	Invalid Card Secondary Address Purpose ID
199	19799	Invalid Card Secondary Address City Code
201	19801	Invalid Card Secondary Address Region Code
203	19803	Invalid Card Secondary Address Country Code
204	19804	Validity Start Date must not be in the past in case Card Secondary Address
205	19805	Validity Start Date cannot be greater than Validity End Date in case Card Secondary Address
206	19806	Validity End Date must not be in the past in case Card Secondary Address
209	19809	City Code or City Free Text is required for Card Secondary Address
210	09810	Region Code or Region Free Text is required for Card Secondary Address
215	19815	Region Free Text is required in case of Region not valued for Client Secondary Address
232	19832	Invalid Insurance Branch Code
236	19836	Invalid Account Type
238	19838	Invalid Bank Code in Bank Account Segment
240	19840	Invalid Branch Code in Bank Account Segment
244	19844	BranchCode is missing when AccountNumber is a valued field
245	19845	AccountNumber is missing when BranchCode is a valued field

#	Error code	Error Description
246	19846	IBAN is missing when SwiftCode is a valued field
248	19848	Invalid Account Number
249	19849	Invalid Account Currency
252	19852	Invalid Card Field Name
254	19854	Duplicate AddressType, more Address with the same AddressType
255	19855	Duplicate AddressPurpose, more AddressPurposeID with the same purpose
256	19856	An error occurred while inserting or updating a Bank Account
257	19858	Client Demog is Mandatory when Existing Client is not provided
258	19859	Primary Address is mandatory when Existing Client is not provided
259	19860	Invalid Link Flag
260	19861	Invalid Link Entiy Code
261	19862	Invalid Link Address Type
262	19863	Invalid Purpose Address Type
263	19864	Invalid Purpose Address Type Sending Mode
264	19865	Document Expiry Date must not be in the past
265	19866	Document Expiry Date cannot be lesser than document issue date
266	19867	Document Issue Date can not be in the future

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x

Status Code	Description	Success Response	Error Response
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/addSupplementaryCardWebService/v2	POST	

REST sample call

```

"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"supplementaryCard": {
  "bankCode": "",
  "branchCode": "",
  "parentCard": "",
  "catPIProduct": "",
  "catPIProductVersion": "",
  "plasticCode": ""
},
"CardActivationFlag": "",
"CustomerDetails": {
  "ExistingClient": {
    "BankCode": "",
    "ClientCode": ""
  },
  "ClientDemog": {
    "BankCode": "",
    "ClientCode": "",
    "BranchCode": "",
    "FamilyStatus": "",
    "NumberOfDependents": "",
    "Gender": "",
    "LegalId": "",
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    "BirthCountry": "",
    "BirthCity": "",
    "VIPLevel": "",
    "NationalityCode": "",
    "LanguageCode": "",
    "TitleCode": "",
    "FamilyName": "",
    "SecondFamilyName": ""
  }
}

```

```

"FirstName": "",
"SecondFirstName": "",
"MaidenName": "",
"OwnerCode": "",
"SocioProfessionalCode": "",
"ActivityCode": "",
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"TakingOnDate": "",
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},
"Phone4": {
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  "RegionFreeText": "",
  "CountryCode": ""
},
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  },
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  "AddressLine3": "",
  "AddressLine4": "",
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  "CityFreeText": "",
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  "CountryCode": ""
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  "DocumentId": "",
  "DocumentIssueDate": "",
  "DocumentExpiryDate": "",
  "DocumentIssueCity": "",
  "DocumentIssueCountry": ""
},
"ClientCustomFields": {
  "FieldName": "",
  "FieldValue": ""
}

```

```

    },
    "CardDetails": {
      "CardDemog": {
        "BranchCode": "",
        "PriorityCode": "",
        "Gender": "",
        "LegalId": "",
        "BirthDate": "",
        "BirthCity": "",
        "VIPLevel": "",
        "NationalityCode": "",
        "LanguageCode": "",
        "TitleCode": "",
        "FamilyName": "",
        "SecondFamilyName": "",
        "FirstName": "",
        "SecondFirstName": "",
        "MaidenName": "",
        "OwnerCode": "",
        "SocioProfessionalCode": "",
        "ActivityCode": "",
        "CustomerSegment": "",
        "EmployerName": "",
        "StaffID": "",
        "ChannelNumber": "",
        "StatusCode": "",
        "StatusReason": "",
        "CardFeesCode": "",
        "VelocityLimitIndex": "",
        "ServicesSetupIndex": "",
        "EmbossedName": "",
        "SecondEmbossedName": "",
        "CorporateId": "",
        "Phone1": {
          "PhoneType": "",
          "PhoneNumber": ""
        },
        "Phone2": {
          "PhoneType": "",
          "PhoneNumber": ""
        },
        "Phone3": {
          "PhoneType": "",
          "PhoneNumber": ""
        },
        "Phone4": {
          "PhoneType": "",
          "PhoneNumber": ""
        },
        "Email1": "",
        "Email2": ""
      },
      "PlasticDeliveryMethod": "",
      "PinDeliveryMethod": "",
      "PromotionCode": "",
      "TransactionFeeIndex": "",
      "AuthorizarionFeeIndex": "",
      "PrimaryAddress": {
        "AddressType": "",
        "SendingMode": "",
        "CareOf": "",
        "LinkFlag": "",
        "LinkEntityCode": "",
        "LinkAddressType": "",
        "AddressLine1": "",
        "AddressLine2": "",
        "AddressLine3": "",
        "AddressLine4": "",
        "ZipCode": "",
        "CityCode": "",
        "CityFreeText": "",
        "RegionCode": "",
        "RegionFreeText": "",
        "CountryCode": ""
      }
    },
  },

```

```

    "Secondary Address": {
      "AddressType": "",
      "SendingMode": "",
      "ValidityRangeDate": {
        "StartDt": "",
        "EndDt": ""
      },
      "CareOf": [
        ""
      ],
      "LinkFlag": "",
      "LinkEntityCode": "",
      "AddressLine1": "",
      "AddressLine2": "",
      "AddressLine3": "",
      "AddressLine4": "",
      "ZipCode": "",
      "CityCode": "",
      "CityFreeText": "",
      "RegionCode": "",
      "RegionFreeText": "",
      "CountryCode": ""
    },
    "AddressPurpose": {
      "AddressPurposeID": "",
      "AddressType": "",
      "SendingMode": ""
    },
    "AuxiliaryProduct": {
      "ProductIdentifier": "",
      "ContractNumber": "",
      "MembershipDate": "",
      "BeneficiaryAge": "",
      "BeneficiaryName": "",
      "StatusCode": "",
      "StatusDate": "",
      "InsuranceBranchCode": "",
      "InsuranceAge": "",
      "AuxiliaryFeeAmount": "",
      "PolicyNumber": ""
    },
    "BankAccount": {
      "AcctType": "",
      "AcctSequence": "",
      "BankCode": "",
      "BankCodeFreetext": "",
      "BranchCode": "",
      "BranchCodeFreetext": "",
      "AccountNumber": "",
      "AccountCurrency": "",
      "IBAN": "",
      "SwiftCode": ""
    },
    "CardCustomFields": {
      "FieldName": "",
      "FieldValue": ""
    }
  }
}

```

4.56 changeAccountCreditLimit API

Overview

The credit limit is defined as the maximum amount of money that can be spent on a credit card. The Change credit limit web service could be used to either increase or decrease the credit limit on an existing account. The same web service could be also used to set temporary limits with a specified range of date.

This Web Service can be called by a customer or an agent.

Functional description

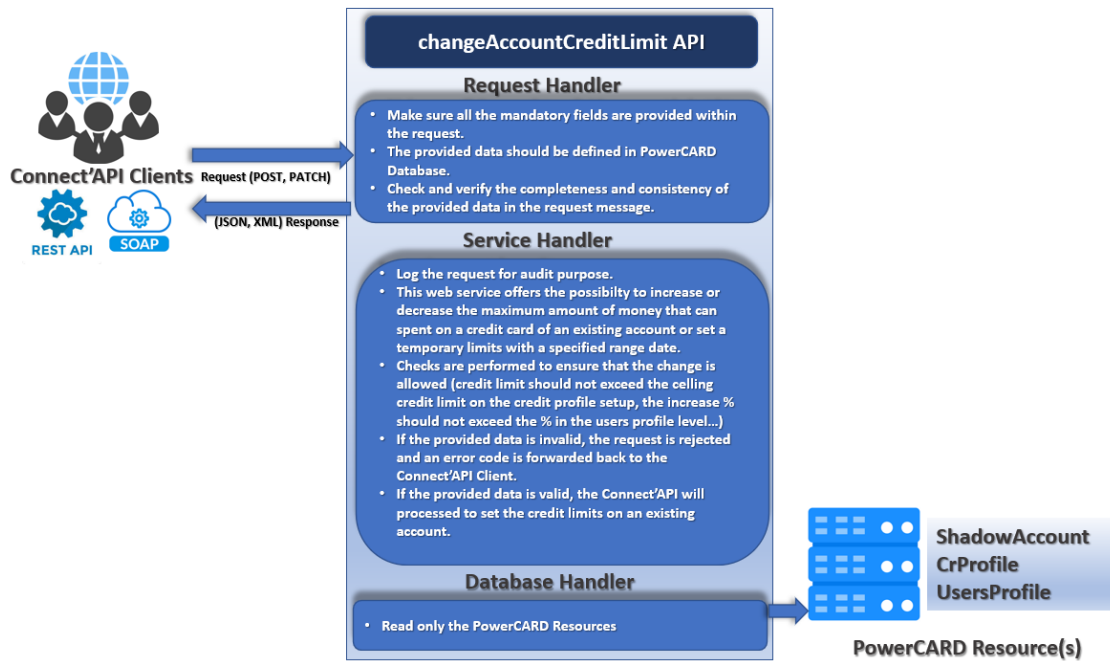


Figure 56: changeAccountCreditLimit process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
-	PermanentLimit	Complex	-	Required (max : 1)	The current/ permanent limits to set on the credit account. It includes the Credit limit, Cash limit and the loan limit)
-	CreditLimitAmt	Complex	PermanentLimit	Required (max : 1)	The maximum amount the cardholder is allowed to use on the credit card account at any time, expressed in the account currency.
5	Amt	Long	CreditLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
6	CurCode	NC (3)	CreditLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	CashLimitAmt	Complex	PermanentLimit	Optional (max : 1)	The maximum amount the cardholder is allowed to use for cash transactions on the credit card account at any time, expressed in the account currency.
7	Amt	Long ()	CashLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
8	CurCode	C (3)	CashLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	LoanLimitAmt	Complex	PermanentLimit	Optional (max : 1)	The maximum amount of loan transactions that can be performed on the cardholder's account. The value of this field should not

#	Field Name	Data type	Parent	Occurrence	Field Description
					exceed the total credit limit permitted on the cardholder's account. The loan amount should be submitted along with the account currency.
9	Amt	Long	LoanLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
10	CurCode	NC (3)	LoanLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	TemporaryLimit	Complex	-	Optional (max : 1)	The temporary limits the customer is allowed to use on the credit account, temporary limit is defined for: Credit limit, Cash limit and Loan limit.
-	CreditLimitAmt	Complex	TemporaryLimit	Required (max : 1)	The maximum amount the cardholder is allowed to use on the credit card account at any time, expressed in the account currency.
11	Amt	Long ()	CreditLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the

#	Field Name	Data type	Parent	Occurrence	Field Description
					currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
12	CurCode	NC (3)	CreditLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	CashLimitAmt	Complex	TemporaryLimit	Optional (max : 1)	The maximum amount the cardholder is allowed to use for cash transactions on the credit card account at any time, expressed in the account currency.
13	Amt	Long ()	CashLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
14	CurCode	NC (3)	CashLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
-	LoanLimitAmt	Complex	TemporaryLimit	Optional (max : 1)	The maximum amount of loan transactions that can be performed on the cardholder's account. The value of this field should not exceed the total credit limit permitted on the cardholder's account. The loan amount should be submitted along with the account currency.
15	Amt	Long ()	LoanLimitAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
16	CurCode	NC (3)	LoanLimitAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	ValidityRangeDate	Complex	SecondaryAddress	Required (max: 1)	Range date validity for a temporary limit.
21	StartDt	DateTime	ValidityRange Date	Required (max: 1)	The Validity Start date
22	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	03501	Invalid Account Number
2	03502	Credit limit is less than the cash limit
3	03503	Cash limit should be equal to or less than cash limit ceiling, contact your administrator
4	03504	Loan limit should be equal to or less than loan limit ceiling, contact your administrator
5	03505	Invalid Currency Code
6	03506	Invalid Shadow Account profile code
7	03507	Not used
8	03508	Not used
9	03509	Not used
10	03510	Not used
11	03511	Not used
12	03512	Not used
13	03513	Not used
14	03514	Not used
15	03515	Not used
16	03516	Not used
17	03517	Not used
18	03518	Credit limit percentage increase exceeds user percentage increase permitted
19	03519	Cash/transfer percentage increase exceeds user percentage increase permitted
20	03520	Loan limit percentage increase exceeds user percentage increase permitted
21	03521	Credit limit percentage increase exceeds profile percentage increase permitted

#	Error code	Error Description
22	03522	Cash/transfer limit percentage increase exceeds profile percentage increase permitted
23	03523	Loan limit percentage increase exceeds profile percentage increase permitted
24	03524	Credit limit exceeds user maximum credit limit increase
25	03525	Cash/transfer limit exceeds user maximum cash/transfer limit increase
26	03526	Loan limit exceeds user maximum loan limit increase
27	03527	Increase limit day is not yet expired
28	03528	Credit limit is greater the customer credit limit
29	03529	Cash limit is greater than the customer cash limit
30	03530	Loan limit is greater than the customer loan limit
31	03531	Temporary credit limit is greater than the customer credit limit
32	03532	Temporary cash limit is greater than the customer cash limit
33	03533	Temporary loan limit is greater than the customer loan limit

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is		x

Status Code	Description	Success Response	Error Response
	probably valid but needs to be requested again later.		
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/changeAccountCreditLimitWebService/v2	POST	

REST sample call

```

"requestInfo": {
  "requestUID" : "",
  "requestDate": "",
  "userID": ""
},
"permanentLimit": {
  "creditLimitAmt" : {
    "amt" : "",
    "curCode" : ""},
  "cashLimitAmt" : {
    "amt" : "",
    "curCode" : ""},
  "loanLimitAmt" : {
    "amt" : "",
    "curCode" : ""}
},
"temporaryLimit" : "",
{
  "creditLimitAmt": {
    "amt" : "",
    "curCode" : ""},
  "cashLimitAmt": {
    "amt" : "",
    "curCode" : ""},
  "loanLimitAmt": {
    "amt" : "",
    "curCode" : ""},
  "validityRangeDate": {
    "startDt" : "",
    "endDt" : ""}
}

```

4.57 convertTransactionToLoan API

API overview

The main purpose of this web service is to convert normal transactions to loans. Loan program, reason and transaction Identifier are one of required fields.

At the end of day processing, manual conversions are treated by PowerCARD using the chosen loan program. During the conversion process, the loan limit will be checked;

Functional description

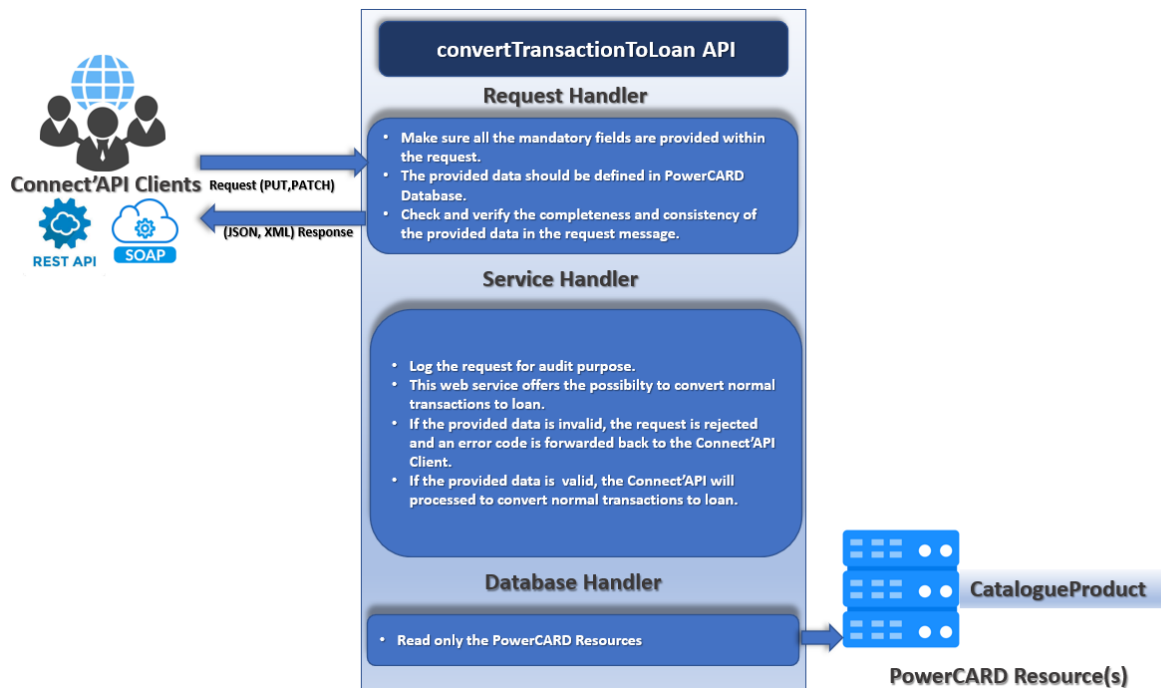


Figure 57: convertTransactionToLoan process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CatalogLoan Product	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loan product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
5	CatalogLoanProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
6	Acronym	C(128)		Optional (max : 1)	Loan acronym
7	TransactionFlag	ClosedEnum	-	Required (max : 1)	This one character flag indicates whether the conversion request concerns an authorization or a transaction. Possible values are: • T: transaction • A: Authorization For the time being, only the option 'T' is supported.

#	Field Name	Data type	Parent	Occurrence	Field Description
8	TransId	C (23)	-	Optional (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction. Mandatory in case of transaction type 'T'
9	RefSeq	N (4)	-	Optional (max : 1)	Refers to the transaction sequence Number Mandatory in case of transaction type 'T'

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will

#	Field Name	Data type	Parent	Occurrence	Field Description
					be inserted in a Maker Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	18901	Invalid Loan Program Code
2	18902	Invalid Reason Code
3	18903	Invalid Transaction Flag
4	18904	Invalid Transaction Id
5	18905	Invalid Reference Sequence
6	18906	Reserved for future use
7	18907	Reserved for future use
8	18908	Reserved for future use
9	18909	Reserved for future use
10	18910	Reserved for future use
11	18911	Transaction cannot be converted
12	18912	Program Code not allowed
13	18913	Internal Error while performing the conversion request
14	18914	Loan limit exceeded

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/convertTransactionToLoanWebService/v2	POST	

REST sample call

```

"requestInfo": {
  "requestUID" : "",
  "requestDate" : "",
  "userID" : ""
},
"permanentLimit": {
  "creditLimitAmt" : {
    "amt" : "",
    "curCode" : ""
  },
  "cashLimitAmt" : {
    "amt" : "",
    "curCode" : ""
  },
  "loanLimitAmt" : {
    "amt" : "",
    "curCode" : ""
  }
},

```

```

"temporaryLimit" : "",
{
  "creditLimitAmt":{
    "amt" : "",
    "curCode"}
  "cashLimitAmt"{
    "amt" : "",
    "curCode" : "",}
  "loanLimitAmt"{
    "amt" : "",
    "curCode" : "",}
  "validityRangeDate"{
    "startDt" : "",
    "endDt" : ""}
}

```

4.58 createCreditCard API

API overview

The current web service is used to allow users creating a credit card in the system. When the card is created, the customer can then start using his card to make purchase, ATM withdrawal....

Many details with regards to the card are shared within the request. Those data include information related to the card product, credit limits, fees, insurances and others. All are then checked by the system before it will proceed by the creation of the credit card.

This Web Service can be called by a Customer or an Agent.

Functional description

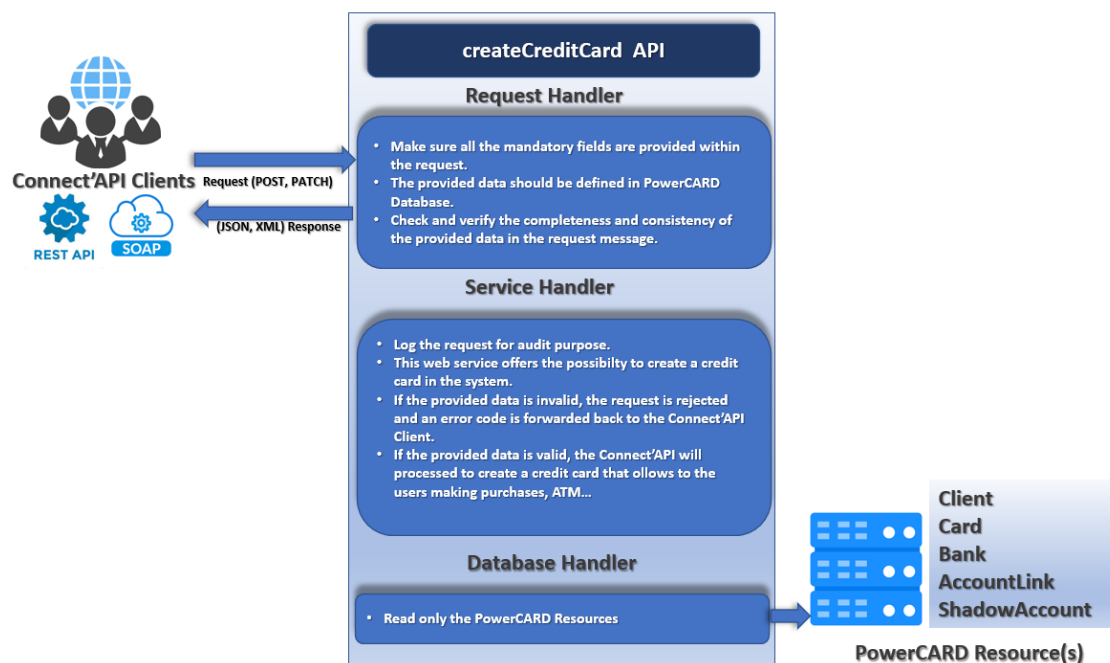


Figure 58: createCreditCard process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
11	CardActivationFlag	C (1)	-	Optional (max: 1)	This field indicates whether the card will be activated or not. Possible values are the following: - Y: Card activated - N: Card not activated If this information is not provided, by default, the card is not activated.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	CustomerDetails	Complex	-	Required (max: 1)	It must contain the details information of the cardholder.
-	ExistingClient	Complex	CustomerDetails	Optional (max: 1)	To provide the existing client code to which all the entities within the boarding request should be linked. When the sub block "ExistingClient" is not provided, the sub blocks ClientDemog and PrimaryAddress will become mandatory
12	BankCode	C (6)	ExistingClient	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
13	ClientCode	C (24)	ExistingClient	Required (max: 1)	The Client Code of the existing client
-	ClientDemog	Complex	CustomerDetails	Optional (max: 1)	Refers to the demographic information of the client such as names, professional activity, job details ... This aggregate is Mandatory if ExistingClient block is not provided
14	BankCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the bank or the institution. This code must be already defined in the PowerCARD BANK table.
15	ClientCode	C (24)	ClientDemog	Required (max: 1)	This aggregate is to provide the client code in case boarding request is for an existing client.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Also, the field can be used when the issuers want to use their own client code, and the client code will not be generated by PowerCARD.
16	BranchCode	C (6)	ClientDemog	Required (max: 1)	This is the 6 digits' code that identifies the branch to which the cardholder belongs. This value will be propagated at card level if not already specified at card. This code must be previously defined in the PowerCARD BRANCH table under the general parameters section. A generic default value can be setup and used by the Bank in case it is not possible to define all the branches.
17	FamilyStatus	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's marital status: • S: Single. • M: Married. • D: Divorced. • W: Widower. • N: Not applicable. Any value other than the above will result in a rejection of the record. Defaulted to 'N' when not provided.
18	NumberOfDependents	Long (2)	ClientDemog	Optional (max: 1)	Number of persons relying on the cardholder for financial support.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field doesn't apply to corporate cardholders. Only positive value will be accepted.
19	Gender	C (1)	ClientDemog	Optional (max: 1)	Specifies the cardholder's gender code: • M: Male. • F: Female. • N: Not applicable. Defaulted to 'N' when not provided.
20	LegalId	C (20)	ClientDemog	Optional (max: 1)	Represents the legal ID number provided by the card applicant at application time
21	TaxNumber	C (30)	ClientDemog	Optional (max: 1)	Represents the tax number provided by the card applicant at application time
22	TaxRegistryOffice	C (30)	ClientDemog	Optional (max: 1)	Represents the tax registry office provided by the card applicant at application time
23	BirthDate	Date	ClientDemog	Required (max: 1)	The year, month, and day of the cardholder's birth. Format: YYYYMMDD
24	BirthCountry	C (30)	ClientDemog	Optional (max: 1)	Country where the cardholder was born.
25	BirthCity	C (30)	ClientDemog	Optional (max: 1)	Cardholder's birth town.
26	VIPLevel	C (1)	ClientDemog	Required (max: 1)	Predefined level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. This code must be previously defined in the PowerCARD VIP

#	Field Name	Data type	Parent	Occurrence	Field Description
					LIST table under the general parameter section.
27	NationalityCode	C(3)	ClientDemog	Required (max: 1)	Country code of the cardholder's citizenship according to the ISO-3166 specification. This code must be previously defined in the PowerCARD COUNTRY table under the general parameters section. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
28	LanguageCode	C (3)	ClientDemog	Required (max: 1)	Code of the cardholder's language according to the ISO 639-1 specification. This code must be previously defined in the PowerCARD LANGUAGE LIST table under the general parameters section.
29	TitleCode	C (2)	ClientDemog	Required (max: 1)	Prefix or suffix added to the customer's name. Titles may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in the PowerCARD TITLE LIST table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section.
30	FamilyName	C (40)	ClientDemog	Required (max: 1)	Cardholder's last name.
31	SecondFamilyName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second family name if compound names are in use
32	FirstName	C (40)	ClientDemog	Required (max: 1)	Cardholder's given name.
33	SecondFirstName	C (40)	ClientDemog	Optional (max: 1)	Cardholder's second first name if compound names are in use.
34	MaidenName	C (40)	ClientDemog	Optional (max: 1)	Mother's Maiden name
35	OwnerCode	C (2)	ClientDemog	Required (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code must be defined in the PowerCARD OWNER LIST table under the general parameters section SIA configuration: IN : INDIVIDUAL ST : STAFF
36	SocioProfessional Code	C (4)	ClientDemog	Required (max: 1)	Used for the segmentation of cardholders based on such criteria as job industry, and professional qualification. This code must be previously defined in the PowerCARD SOCIOPROF LIST table under the general parameters section.
37	ActivityCode	C (4)	ClientDemog	Required (max: 1)	Helps to categorize cardholder based on their professional occupation.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be previously defined in the PowerCARD CLIENT ACTIVITY SET table under the general parameters section.
38	CustomerSegment	C (3)	ClientDemog	Required (max: 1)	Specifies the customer segments to which the cardholder belongs. This field is used mainly for marketing segmentation purposes. Assignable values must be previously defined in the PowerCARD CUSTOMER SEGMENT table under the general parameters section.
39	TakingOnDate	Date	ClientDemog	Optional (max: 1)	Date when the cardholder joined the company. It is the employment start date. Format: YYYYMMDD
40	EmployerName	C (26)	ClientDemog	Optional (max: 1)	Name of the cardholder's employer.
41	StaffID	C (20)	ClientDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization.
42	LodgingMode	C (1)	ClientDemog	Optional (max: 1)	Indicates if the cardholder is a tenant, a dweller, or an owner of his place of residence: • N: Not reported or not applicable. • R: Renter. • O: Owner. • D: Dependent.

#	Field Name	Data type	Parent	Occurrence	Field Description
43	ChannelNumber	C (15)	ClientDemog	Optional (max: 1)	Distinct number of the sales channel used to attract the customer. This code must be previously defined in the PowerCARD CHANNEL table under the cardholder section.
-	Phone1	Complex	ClientDemog	Required (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
44	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
45	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxnnnnnnnn (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	ClientDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate from client to card.
46	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
47	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	ClientDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
48	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
49	PhoneNumber	C (30)	Phone3	Required	The phone number where the

#	Field Name	Data type	Parent	Occurrence	Field Description
				(max: 1)	cardholder can be reached. SIA will ask the following format: 00xxnnnnnnnn (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	ClientDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
50	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
51	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxnnnnnnnn (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]

#	Field Name	Data type	Parent	Occurrence	Field Description
52	Email1	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 1st email address. There is no propagation of this field from client to card.
53	Email2	C (256)	ClientDemog	Optional (max: 1)	Cardholder's 2nd email address. There is no propagation of this field from client to card.
-	PrimaryAddress	Complex	CustomerDetails	Optional (max: 1)	To provide the primary address of the cardholder Becomes mandatory when sub block ExistingClient is not provided.
54	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
55	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
56	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
57	AddressLine1	C (64)	PrimaryAddress	Required (max: 1)	Address Line 1 1st line of the recipient address.
58	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2

#	Field Name	Data type	Parent	Occurrence	Field Description
					2nd line of the recipient address.
59	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
60	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address.
61	ZipCode	C (10)	PrimaryAddress	Required (max: 1)	Recipient's postal code. This field should not be validated.
62	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
63	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
64	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table

#	Field Name	Data type	Parent	Occurrence	Field Description
					under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
65	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
66	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Country aggregate Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Secondary Address	Complex	CustomerDetails	Optional infinite list	To provide the secondary address of the cardholder
67	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE_LIST table under the general parameters section.
68	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
69	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
70	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
71	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
72	AddressLine1	C (64)	Secondary Address	Required (max: 1)	Address Line 1 1st line of the recipient address.
73	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address.
74	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address.
75	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address.
76	ZipCode	C (10)	Secondary Address	Required (max: 1)	Recipient's postal code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field should not be validated.
77	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY table under the general parameters section. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.
78	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. It can be provided only when the free capture of city is allowed at bank level.
79	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
80	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
81	CountryCode	C (3)	Secondary Address	Required (max: 1)	Country code. Represents the country of the address being sent or received. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) Note that the country code and name are stored in the PowerCARD COUNTRY table under the general parameter section.
-	Documents	Complex	CustomerDetails	Optional infinite list	To provide details about the document that might be requested by the bank: passport details, civil ID details ...
82	DocumentCode	C (2)	Documents	Required (max: 1)	The document type (passeport, civil ID...). This code must be previously defined in the PowerCARD DOCUMENT LIST table.
83	DocumentId	C (30)	Documents	Required (max: 1)	The identification serial number of the document.

#	Field Name	Data type	Parent	Occurrence	Field Description
84	DocumentIssue Date	Date	Documents	Optional (max: 1)	Date when the document is issued. Format: YYYYMMDD
85	DocumentExpiry Date	Date	Documents	Optional (max: 1)	Date when the document will expire. Format: YYYYMMDD
86	DocumentIssue City	C (32)	Documents	Optional (max: 1)	The city name where the document is issued.
87	DocumentIssue Country	C (32)	Documents	Optional (max: 1)	The country name where the document is issued.
-	Client	Complex	CustomerDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
89	FieldName	C (30)	ClientCustom Fields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
90	FieldValue	C (256)	ClientCustom Fields	Required (max: 1)	The value of the user defined field.
-	CardDetails	Complex	-	Optional (max: 1)	It must contain the details information of card and bank

#	Field Name	Data type	Parent	Occurrence	Field Description
					accounts data of the requested card. Not mandatory in case of One Shot card
-	CardDemog	Complex	CardDetails	Required (max: 1)	To provide demographic information of the cardholder and card plastic such as names, card product type, plastic type...
91	BranchCode	C (6)	CardDemog	Optional (max: 1)	This is the 6 digits' code that identifies the cardholder's branch. It should be previously defined in PowerCARD BRANCH table under the general parameters section. The branch code will be inherited from the Client if not provided at card level.
92	PriorityCode	C (3)	CardDemog	Optional (max: 1)	This field indicates the priority level of Card embossing. It should be previously defined in PowerCARD EMBOSSING PRIORITY table under the card parameters section.
93	Gender	C (1)	CardDemog	Optional (max: 1)	This field specifies the cardholder's gender code: • M = Male, • F = Female, • N = Not applicable. The gender will be inherited from the Client if not provided at card level.
94	LegalId	C (20)	CardDemog	Optional (max: 1)	This field represents the Legal ID number provided by the card applicant at application time.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The Legal ID will be inherited from the Client if not provided at card level.
95	BirthDate	Date	CardDemog	Optional (max: 1)	This field specifies the cardholder's birth date in the format: YYYYMMDD. The birth date will be inherited from the Client if not provided at card level.
96	BirthCity	C (30)	CardDemog	Optional (max: 1)	Cardholder's birth city. It will be inherited from the Client if not provided at card level.
97	VIPLevel	C (1)	CardDemog	Optional (max: 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.
98	NationalityCode	C (3)	CardDemog	Optional (max: 1)	This is the Country code of the cardholder's citizenship according to the ISO-3166 specification. This code should be previously defined in PowerCARD COUNTRY table under the general parameters section. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) with a "*" value "N.A" in case of value not available #2) 3-numeric ISO (e.g. "380" for Italy) with a "*" value "999" in case of value not available
99	LanguageCode	C (3)	CardDemog	Optional (max: 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
100	TitleCode	C (2)	CardDemog	Optional (max: 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
101	FamilyName	C (40)	CardDemog	Optional (max: 1)	Indicates the Cardholder's last name. It will be inherited from the Client if not provided at card level.

#	Field Name	Data type	Parent	Occurrence	Field Description
102	SecondFamily Name	C (40)	CardDemog	Optional (max: 1)	This is Cardholder's second family name if compound names are in use. It will be inherited from the Client if not provided at card level.
103	FirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's given name. It will be inherited from the Client if not provided at card level.
104	SecondFirstName	C (40)	CardDemog	Optional (max: 1)	This is the Cardholder's second first name if compound names are in use. It will be inherited from the Client if not provided at card level.
105	MaidenName	C (40)	CardDemog	Optional (max: 1)	This is the Mother's maiden name. This field will be inherited from the Client if not provided at card level.
106	OwnerCode	C (2)	CardDemog	Optional (max: 1)	This field is typically used to distinguish between staff, corporate, and individual cardholders. This code should be defined in PowerCARD OWNER LIST table under the general parameters section. It will be inherited from the Client if not provided at card level. SIA configuration : IN "INDIVIDUAL" for retail cards BU "BUSINESS" for Business/Corporate cards.
107	SocioProfessional Code	C (4)	CardDemog	Optional (max: 1)	Used for the segmentation of

#	Field Name	Data type	Parent	Occurrence	Field Description
					cardholders based on such criteria as job industry, and professional qualification. This code should be previously defined in PowerCARD SOCIOPROF LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
108	ActivityCode	C (4)	CardDemog	Optional (max: 1)	This code categorizes the cardholders based on their professional occupation. This code should be previously defined in PowerCARD CLIENT ACTIVITY SET table under the general parameters section. It will be inherited from the Client if not provided at card level.
109	CustomerSegment	C (3)	CardDemog	Optional (max: 1)	This field specifies the Cardholder's customer segment. The customer segment is used mainly for marketing segmentation purposes. Assignable values must be previously defined in PowerCARD CUSTOMER SEGMENT table under the general parameters section. It will be inherited from the Client if not provided at card level.
110	EmployerName	C (26)	CardDemog	Optional (max: 1)	This is the name of the cardholder's employer. It will be inherited from the

#	Field Name	Data type	Parent	Occurrence	Field Description
					Client if not provided at card level.
111	StaffID	C (20)	CardDemog	Optional (max: 1)	Specifies the cardholder's ID number within the employing organization. It will be inherited from the Client if not provided at card level.
112	ChannelNumber	C (15)	CardDemog	Optional (max: 1)	This is a distinct number of the sales channel used to attract the customer. This code should be previously defined in PowerCARD CHANNEL table under the cardholder section. It will be inherited from the Client if not provided at card level.
113	StatusCode	C (1)	CardDemog	Optional (max: 1)	This field indicates the initial card status. The status code should be previously defined in PowerCARD STATUS LIST table under the general parameters section. A StatusCode different than "Normal" cannot be provided without StatusReason. If not provided, it will be defaulted to the status code defined at product level or set to "Normal" if the last one is also not available.
114	StatusReason	C (2)	CardDemog	Optional (max: 1)	This field indicates the initial card status reason. This code should be previously defined in PowerCARD STATUS REASON LIST table under the

#	Field Name	Data type	Parent	Occurrence	Field Description
					general parameters section. StatusReason cannot be provided without StatusCode. If not provided, it will be defaulted to the status reason defined at product level.
118	EmbossedName	C (26)	CardDemog	Optional (max: 1)	Primary name that the cardholder wishes to emboss on the card plastic. If not provided in the file, PowerCARD concatenates the first and last name fields as a default value.
119	SecondEmbossedName	C (26)	CardDemog	Optional (max: 1)	This is a secondary name that the cardholder wishes to emboss on the card plastic underneath the primary name.
120	Corporateld	C (15)	CardDemog	Optional (max: 1)	Indicates the corporate to which the card belongs. This field is mandatory if the value transmitted in the owner code field is related to a corporate (that is set to "BUSINESS")
-	Phone1	Complex	CardDemog	Optional (max: 1)	The phone number 1 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
121	PhoneType	C (2)	Phone1	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone.

#	Field Name	Data type	Parent	Occurrence	Field Description
					2: Home phone. 3: Personal mobile phone. 4: Others.
122	PhoneNumber	C (30)	Phone1	Required (max: 1)	The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone2	Complex	CardDemog	Optional (max: 1)	The phone number 2 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
123	PhoneType	C (2)	Phone2	Required (max: 1)	Type of the phone number, possible values are: 0: Office fixed line phone. 1: Office mobile phone. 2: Home phone. 3: Personal mobile phone. 4: Others.
124	PhoneNumber	C (30)	Phone2	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxx (xx based on the

#	Field Name	Data type	Parent	Occurrence	Field Description
					country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone3	Complex	CardDemog	Optional (max: 1)	The phone number 3 at which the cardholder can be reached. There is no propagation of this aggregate from client to card.
125	PhoneType	C (2)	Phone3	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
126	PhoneNumber	C (30)	Phone3	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
-	Phone4	Complex	CardDemog	Optional (max: 1)	The phone number 4 at which the cardholder can be reached. There is no propagation of this

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate from client to card.
127	PhoneType	C (2)	Phone4	Required (max: 1)	Type of the phone number, possible values are: • 0: Office fixed line phone. • 1: Office mobile phone. • 2: Home phone. • 3: Personal mobile phone. • 4: Others.
128	PhoneNumber	C (30)	Phone4	Required (max: 1)	The phone number where the cardholder can be reached. The phone number where the cardholder can be reached. SIA will ask the following format: 00xxxxxxxxxx (xx based on the country, e.g 39 – Italy) [no format check is foreseen in WS processing]
129	Email1	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's first email address. There is no propagation of this field from client to card.
130	Email2	C (256)	CardDemog	Optional (max: 1)	Indicates the Cardholder's second email address. There is no propagation of this field from client to card.
131	PlasticDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card plastic (For example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
132	PinDelivery Method	C (3)	CardDemog	Optional (max: 1)	This field identifies the kind of mailing method used to deliver the card PIN (For example: Assured mail, Ordinary mail, Airmail...). When provided, it will be validated against the list defined at bank and product level. Otherwise, the field will be automatically defaulted from the default parameter at bank and product.
133	PromotionCode	C (10)	CardDemog	Optional (max: 1)	This field identifies the promotion or the campaign to which the card belongs. When provided, it will be validated against the list defined at bank level.
-	PrimaryAddress	Complex	CardDetails	Required (max: 1)	To provide the primary address of the card
136	AddressType	C (2)	PrimaryAddress	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES_TYPE_LIST table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
137	SendingMode	C (1)	PrimaryAddress	Required (max: 1)	To be set to « M » - Mail.
138	CareOf	C (64)	PrimaryAddress	Optional (max: 1)	Care of Represent the recipient, name that will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
139	LinkFlag	C(1)	PrimaryAddress	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
140	LinkEntityCode	C(2)	PrimaryAddress	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
141	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already

#	Field Name	Data type	Parent	Occurrence	Field Description
					present as part of the client addresses.
142	AddressLine1	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 1 1st line of the recipient address. This field is mandatory in case of address not linked.
143	AddressLine2	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
144	AddressLine3	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
145	AddressLine4	C (64)	PrimaryAddress	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
146	ZipCode	C (10)	PrimaryAddress	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
147	CityCode	C (5)	PrimaryAddress	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level.

#	Field Name	Data type	Parent	Occurrence	Field Description
					When allowed, one of the two fields CityCode or CityFreeText should be provided.
148	CityFreeText	C (32)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
149	RegionCode	C (3)	PrimaryAddress	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
150	RegionFreeText	C(30)	PrimaryAddress	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
151	CountryCode	C (3)	PrimaryAddress	Required (max: 1)	Refers to the ISO country code of

#	Field Name	Data type	Parent	Occurrence	Field Description
					the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy) #2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Secondary Address	Complex	CardDetails	Optional infinite list	To provide the secondary address of the cardholder.
152	AddressType	C (2)	Secondary Address	Required (max: 1)	Address type Address type based on the intended purpose of use. This is a 2 digit code that must be defined in the PowerCARD ADDRESSES _TYPE _LIST table under the general parameters section.
153	SendingMode	C (1)	Secondary Address	Required (max: 1)	To be set to « M » - Mail.
-	ValidityRangeDate	Complex	Secondary Address	Optional (max: 1)	An address can be temporary and restricted to a range date Valid only in case of Address Type "Temporary"
154	StartDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date
155	EndDt	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date
156	CareOf	C (64)	Secondary Address	Optional (max: 1)	Care of Represent the recipient, name that

#	Field Name	Data type	Parent	Occurrence	Field Description
					will be used in any communication with the entity. An entity could be, depending on the context, a customer, a merchant, a chain...
157	LinkFlag	C(1)	Secondary Address	Optional (max: 1)	A flag to indicate if the card address should be linked to another entity's address. The possible values are: • 'Y': Yes • 'N': No This field should be used for card addresses only.
158	LinkEntityCode	C(2)	Secondary Address	Optional (max: 1)	Indicate the entity code to which the card address is being linked. The possible values are: • 'CL': Client • 'BA': Branch This field becomes mandatory when LinkFlag is set to 'Y'.
159	LinkAddressType	C(2)	PrimaryAddress	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field becomes mandatory when LinkFlag is set to 'Y' and LinkEntityCode = 'CL'. This address type must be already present as part of the client addresses.
160	AddressLine1	C (64)	Secondary Address	Optional (max: 1)	Address Line 1 1st line of the recipient address.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field is mandatory in case of address not linked.
161	AddressLine2	C (64)	Secondary Address	Optional (max: 1)	Address Line 2 2nd line of the recipient address. Not required when the address is linked
162	AddressLine3	C (64)	Secondary Address	Optional (max: 1)	Address Line 3 3rd line of the recipient address. Not required when the address is linked
163	AddressLine4	C (64)	Secondary Address	Optional (max: 1)	Address Line 4 4th line of the recipient address. Not required when the address is linked
164	ZipCode	C (10)	Secondary Address	Optional (max: 1)	Recipient's postal code. This field is mandatory in case of address not linked. It should not be validated.
165	CityCode	C (5)	Secondary Address	Optional (max: 1)	The city code of the cardholder's address. This code must be present in the PowerCARD CITY CODE table under the general parameters section. Not required when the address is linked. Mandatory when the free capture of city is not allowed at bank level. When allowed, one of the two fields CityCode or CityFreeText should be provided.

#	Field Name	Data type	Parent	Occurrence	Field Description
166	CityFreeText	C (32)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's city when the city predefined referential is not managed at institution level. Not required when the address is linked. Otherwise, it can be provided only when the free capture of city is allowed at bank level
167	RegionCode	C (3)	Secondary Address	Optional (max: 1)	The region code of the cardholder's address. This code must be present in the PowerCARD REGION table under the general parameters section. Mandatory when the free capture of region is not allowed at bank level. When allowed, one of the two fields RegionCode or RegionFreeText should be provided.
168	RegionFreeText	C(30)	Secondary Address	Optional (max: 1)	A free text to capture the cardholder's region when the region predefined referential is not managed at institution level. It can be provided only when the free capture of region is allowed at bank level
169	CountryCode	C (3)	Secondary Address	Required (max: 1)	The country code of the cardholder's address. Two options are supported: #1) 3-aphanumeric ISO (e.g. "ITA" for Italy)

#	Field Name	Data type	Parent	Occurrence	Field Description
					#2) 3-numeric ISO (e.g. "380" for Italy) This value must be present in the PowerCARD COUNTRY table under the general parameters section. Not required when the address is linked.
-	Communication Events	Complex	-	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
	EventCode	C(4)	Communication Events	Required (max: 1)	This is the event code that can be triggered by the system. The list of event codes is defined in the following table: EVENT_TABLE
	EventActivation Status	C(1)	Communication Events	Required (max: 1)	This field defines if the event is enabled or disable. - Y: Event active - N: Event disabled
	IsEventMandatory	C(1)	Communication Events	Required (max: 1)	This field indicates if the event is defined as mandatory: Y: event is mandatory and can not be disabled. N: Event is optional and can be disabled.
-	Communication Channels	Complex	Communication Events	Optional infinite list	To list all the communication event defined for the institution. (card delivery,

#	Field Name	Data type	Parent	Occurrence	Field Description
					pin delivery, statement....). Each event might be linked to 1 or many communication channels
	ChannelType	C(2)	Communication Channels	Required (max: 1)	This field indicates the channel type that is defined for the a particular event: - Value 1: SMS channel - Value 2: Email channel - Value 3: Physical file channel - Value 4: Push channel
	ChannelActivation Status	C(1)	Communication Channels	Required (max: 1)	This field indicate if the communication channel is enabled or disabled: Y: channel active N: channel disabled
	IsChannel Mandatory	C(1)	Communication Channels	Required (max: 1)	This field indicates if the channel is defined as mandatory: - Y: channel is mandatory and can not be disabled. - N: channel is optional and can be disabled.
	AllowsCustom Preference	C(1)	Communication Channels	Required (max: 1)	This field indicates if the PANholder is allowed to define permanent communication channel phone number, or email or address (for channel types SMS/Push or Email or physical file respectively).
	ChannelPhone Type	C(2)	Communication Channels	Optional (max: 1)	The channel phone type

#	Field Name	Data type	Parent	Occurrence	Field Description
	ChannelAddress FromEntity	C(2)	Communication Channels	Optional (max: 1)	The channel address from entity
	ChannelAddress Type	C(2)	Communication Channels	Optional (max: 1)	The channel address type
	ChannelEmail Number	C(1)	Communication Channels	Optional (max: 1)	The channel email number
	ChannelAllows TempPreference	C(1)	Communication Channels	Required (max: 1)	The channel temp preference
	TemporaryCustomPreference Set	C(1)	Communication Channels	Required (max: 1)	The temporary custom preference set
	TMPPreference StartDate	Date	Communication Channels	Optional (max: 1)	TMP Preference Start Date
	TMPPreference EndDate	Date	Communication Channels	Optional (max: 1)	TMP Preference End Date
	TMPPhoneType	C(2)	Communication Channels	Optional (max: 1)	TMP Phone Type
	TMPAddress FromEntity	C(2)	Communication Channels	Optional (max: 1)	TMP Address From Entity
	TMPAddress Type	C(2)	Communication Channels	Optional (max: 1)	TMP Address Type
	TMPEmail Number	C(2)	Communication Channels	Optional (max: 1)	TMP Email Number
-	AuxiliaryProduct	Complex	CardDetails	Optional infinite list	To provide information regarding the auxiliary products to which the cardholder is registered (member rewards,)
173	ProductIdentifier	C (6)	AuxiliaryProduct	Required (max: 1)	ID of the auxiliary product linked to the card. This code must be defined in the PowerCARD LINKED PRODUCT 1DESC table under the cardholder parameters section.
174	ContractNumber	C (16)	AuxiliaryProduct	Optional (max: 1)	Unique reference of the contractual agreement binding the cardholder to the provider of the auxiliary product.

#	Field Name	Data type	Parent	Occurrence	Field Description
175	MembershipDate	Date	AuxiliaryProduct	Optional (max: 1)	Date when the cardholder's membership becomes effective for the 1st auxiliary product. Format: YYYYMMDD
176	BeneficiaryAge	Long (3)	AuxiliaryProduct	Optional (max: 1)	Age of the product's beneficiary. Only positive value will be accepted.
177	BeneficiaryName	C (40)	AuxiliaryProduct	Optional (max: 1)	Full name of the product's beneficiary.
178	StatusCode	C (1)	AuxiliaryProduct	Optional (max: 1)	Specifies the validity status of the auxiliary product: - N: Normal - C: Cancelled Products with a 'Cancelled' status are not eligible for further processing. Only 'Normal' products are taken into consideration by PowerCARD. Defaulted to 'N' when empty.
179	StatusDate	Date	AuxiliaryProduct	Optional (max: 1)	Date at which the product status was last updated. Format: YYYYMMDD
180	InsuranceBranchCode	C (6)	AuxiliaryProduct	Optional (max: 1)	Applies to insurance products solely. Specifies the branch code of the insurance company. Must be defined in the PowerCARD BRANCH table under the general parameters section.
181	InsuranceAge	Long (4)	AuxiliaryProduct	Optional (max: 1)	Specifies in number of years the amount of time that has

#	Field Name	Data type	Parent	Occurrence	Field Description
					elapsed since the membership date. Only positive value will be accepted.
182	AuxiliaryFeeAmount	Decimal (18,3)	AuxiliaryProduct	Optional (max: 1)	Fees amount charged to the cardholder as remuneration for the auxiliary product.
183	PolicyNumber	C (32)	AuxiliaryProduct	Optional (max: 1)	Distinct number that uniquely identifies the customer and their dependents to the insurance company.
-	BankAccount	Complex	CardDetails	Optional infinite list	To provide information regarding bank accounts to which the card is linked. Mandatory for debit cards. Not required for credit cards.
184	AcctType	C (2)	BankAccount	Required (max: 1)	Refers to the type of the account (Billing, saving ...) The provided value must already be defined in PowerCARD ACCOUNT_TYPE_LIST table under the general parameter section.
185	AcctSequence	C (2)	BankAccount	Required (max: 1)	A static and defined reference to indicate the posting level of the debit account. The provided value should be in synch with the value used at level of transaction posting rules.
186	BankCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank holding the cardholder's debit account.

#	Field Name	Data type	Parent	Occurrence	Field Description
					This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
187	BankCodeFreetext	C (6)	BankAccount	Optional (max: 1)	The Bank Code free text. Required if "Direct Debit Validation" set to NO at Bank level
188	BranchCode	C (6)	BankAccount	Optional (max: 1)	Code of the bank branch holding the cardholder's debit account. This code must be defined in the PowerCARD BANK table under the general parameters section. Required if "Direct Debit Validation" set to YES at Bank level
189	BranchCodeFreetext	C (6)	BankAccount	Optional (max: 1)	The Branch Code free text. Required if "direct debit validation" set to NO at Bank level
190	AccountNumber	C (24)	BankAccount	Required (max: 1)	Specifies the cardholder's debit account number. This field is used in conjunction with debit cards.
191	AccountCurrency	C (3)	BankAccount	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined

#	Field Name	Data type	Parent	Occurrence	Field Description
					in the PowerCARD CURRENCY TABLE under the general parameters section.
192	IBAN	C (34)	BankAccount	Optional (max: 1)	International bank identification number as defined by ISO. An IBAN is used to uniquely identify a customer's bank account. The IBAN consists of an alphabetical country code, followed by two digits, and then up to thirty-five characters for the bank account number. Becomes mandatory if SwiftCode is provided.
193	SwiftCode	C (11)	BankAccount	Optional (max: 1)	It's used to uniquely identify banks and financial institutions globally - it says who and where they are. These codes are used when transferring money between banks. Banks also use these codes to exchanging messages.
-	CardCustomFields	Complex	CardDetails	Optional infinite list	To provide User defined fields. User defined fields will be managed by a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank.
194	FieldName	C (30)	CardCustomFields	Required (max: 1)	The name of the user defined field. User defined fields will be managed by

#	Field Name	Data type	Parent	Occurrence	Field Description
					a new parameter screen at bank level where it will be possible to specify the list of additional fields that can be provided by a given bank. The provided name should exist in the above parameter table.
195	FieldValue	C (256)	CardCustomFields	Required (max: 1)	The value of the user defined field.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWithSuccess: Call was ended successfully. - ProceedWithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	CardNumber	C (22)	-	Optional (max: 1)	Represents the Card Number or the Card Token Id generated by PowerCARD in case of successful processing. The Card Token ID is returned if the complex KeyValue with key "TOKEN" is included in the WS call.

Specific error codes

#	Error code	Error Description
1	19601	Invalid Card Number/Token Id
2	19602	Invalid Card Activation Flag
3	19603	Card is already registered to another Client
4	19604	Invalid Client Code
5	19605	Error while generating the Client Code
6	19606	Invalid Bank Code in Customer Details segment
7	19607	Invalid Branch Code in Customer Details Segment
8	19608	Invalid Family Status
10	19610	Invalid Cardholder's Gender

#	Error code	Error Description
14	19614	Invalid Cardholder's Birth Date
17	19617	Invalid VIP Level
18	19618	Invalid Nationality Code
19	19619	Invalid Language Code
20	19620	Invalid Title Code
26	19626	Invalid Owner Code
27	19627	Invalid Socio-Professional Code
28	19628	Invalid Activity Code
29	19629	Invalid Customer Segment
33	19633	Invalid Lodging Mode
34	19634	Invalid Channel Number
35	19635	Invalid Phone Type 1
37	19637	Invalid Phone Type 2
39	19639	Invalid Phone Type 3
41	19641	Invalid Phone Type 4
45	19645	Invalid Primary Address Type
46	19646	Invalid Primary Address Sending Mode
55	19655	Invalid Client Primary Address City Code
57	19657	Invalid Client Primary Address Region Code
59	19659	Invalid Client Primary Address Country Code
67	19667	City Code or City Free Text is required for Card Secondary Address
68	19668	Region Code or Region Free Text is required for Card Secondary Address
75	19675	Invalid Client Secondary Address Type
76	19676	Invalid Client Secondary Address Sending Mode
85	19685	Invalid Client Secondary Address City Code
87	19687	Invalid Client Secondary Address Region Code
89	19689	Invalid Client Secondary Address Country Code
90	19690	Validity Start Date must not be in the past in case Client Secondary Address

#	Error code	Error Description
91	19691	Validity Start Date cannot be greater than Validity End Date in case Client Secondary Address
92	19692	Validity End Date must not be in the past in case Client Secondary Address
95	19695	City Code or City Free Text is required for Card Secondary Address
96	19696	Region Code or Region Free Text is required for Card Secondary Address
103	19703	Invalid Client Document Code
111	19711	Invalid Card Branch Code
112	19712	Invalid Card Priority Code
113	19713	Invalid Card Gender
115	19715	Invalid Card Birth Date
117	19717	Invalid Card VIP Level
118	19718	Invalid Card Nationality Code
119	19719	Invalid Card Language Code
120	19720	Invalid Card Title Code
126	19726	Invalid Card Owner Code
127	19727	Invalid Card Socio Professional Code
128	19728	Invalid Card Activity Code
129	19729	Invalid Card Customer Segment
133	19733	Invalid Card Status Code
134	19734	Invalid Card Status Reason
136	19736	Invalid Card Card Fees Code
137	19737	Invalid Card Velocity Limit Index
141	19741	Invalid Card Corporate Id
142	19742	Invalid Card Phone Type 1
144	19744	Invalid Card Phone Type 2
146	19746	Invalid Card Phone Type 3
148	19748	Invalid Card Phone Type 4
152	19752	Invalid Card Plastic Delivery Method
153	19753	Invalid Card Pin Delivery Method
154	19754	Invalid Card Promotion Code
159	19759	Invalid Card Primary Address Type
160	19760	Invalid Card Primary Address Sending Mode

#	Error code	Error Description
161	19761	Invalid Card Primary Address Purpose ID
169	19769	Invalid Card Primary Address City Code
171	19771	Invalid Card Primary Address Region Code
173	19773	Invalid Card Primary Address Country Code
181	19781	City Code or City Free Text is required for Card Secondary Address
182	19782	Region Code or Region Free Text is required for Card Secondary Address
189	19789	Invalid Card Secondary Address Type
190	19790	Invalid Card Secondary Address Sending Mode
191	19791	Invalid Card Secondary Address Purpose ID
199	19799	Invalid Card Secondary Address City Code
201	19801	Invalid Card Secondary Address Region Code
203	19803	Invalid Card Secondary Address Country Code
204	19804	Validity Start Date must not be in the past in case Card Secondary Address
205	19805	Validity Start Date cannot be greater than Validity End Date in case Card Secondary Address
206	19806	Validity End Date must not be in the past in case Card Secondary Address
209	19809	City Code or City Free Text is required for Card Secondary Address
210	09810	Region Code or Region Free Text is required for Card Secondary Address
215	19815	Region Free Text is required in case of Region not valued for Client Secondary Address
232	19832	Invalid Insurance Branch Code
236	19836	Invalid Account Type
238	19838	Invalid Bank Code in Bank Account Segment
240	19840	Invalid Branch Code in Bank Account Segment

#	Error code	Error Description
244	19844	BranchCode is missing when AccountNumber is a valued field
245	19845	AccountNumber is missing when BranchCode is a valued field
246	19846	IBAN is missing when SwiftCode is a valued field
248	19848	Invalid Account Number
249	19849	Invalid Account Currency
252	19852	Invalid Card Field Name
254	19854	Duplicate AddressType, more Address with the same AddressType
255	19855	Duplicate AddressPurpose, more AddressPurposeID with the same purpose
256	19856	An error occurred while inserting or updating a Bank Account
257	19858	Client Demog is Mandatory when Existing Client is not provided
258	19859	Primary Address is mandatory when Existing Client is not provided
259	19860	Invalid Link Flag
260	19861	Invalid Link Entiy Code
261	19862	Invalid Link Address Type
262	19863	Invalid Purpose Address Type
263	19864	Invalid Purpose Address Type Sending Mode
264	19865	Document Expiry Date must not be in the past
265	19866	Document Expiry Date cannot be lesser than document issue date
266	19867	Document Issue Date can not be in the future

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh		x

Status Code	Description	Success Response	Error Response
	the token and then try again.		
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/createCreditCardWebService/v2	POST	

REST sample call

```

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  "catPIProductVersion": "",
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    "ClientCode": ""
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}

```

4.59 createLoanApplication API

API overview

The main purpose of this web service is to convert normal transactions to loans. Loan program, reason and transaction Identifier are one of required fields.

At the end of day processing, manual conversions are treated by PowerCARD using the chosen loan program. During the conversion process, the loan limit will be checked;

Functional description

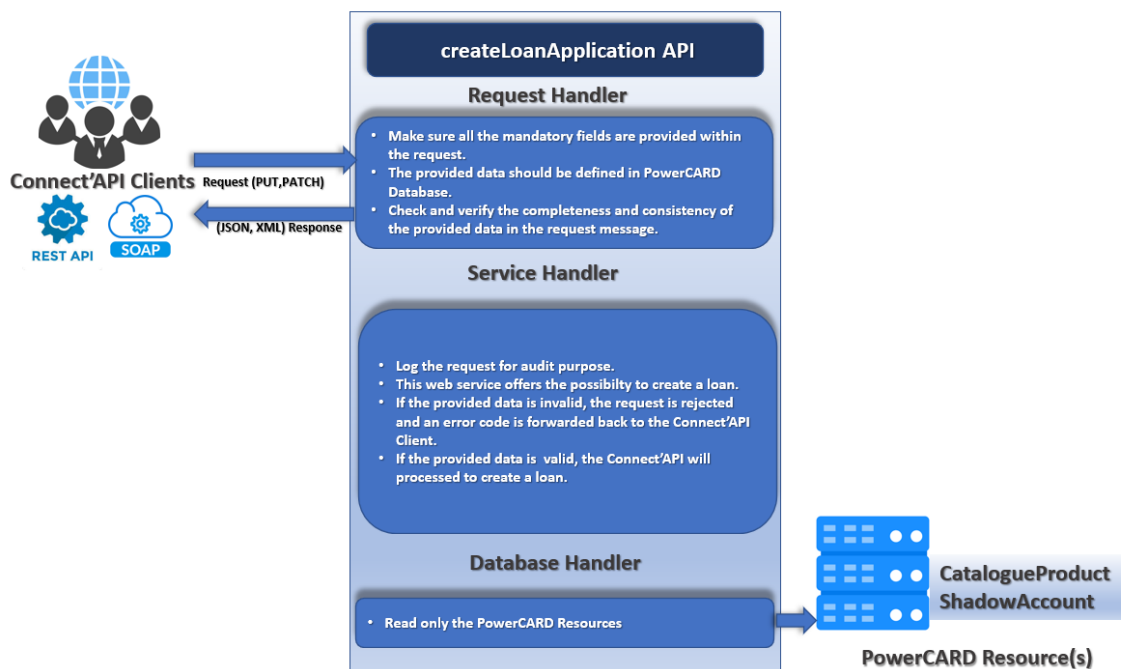


Figure 59: createLoanApplication process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
-	LoanAmount	Complex	-	Required (max : 1)	The Loan amount and currency. Amount and currency of the loan that will be posted to the customer account.
5	Amt	Long ()	LoanAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
6	CurCode	NC (3)	LoanAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY _TABLE table under the general parameters section.
7	CatalogLoan Product	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loan product code. Valid values must be rCatetrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
8	CatalogLoanProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
9	Acronym	C(128)		Optional (max : 1)	Loan acronym

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are:

#	Field Name	Data type	Parent	Occurrence	Field Description
					- Proceed With Success: Call was ended successfully. - Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	ApplicationNumber	C (9)	-	Optional (max : 1)	A 9-digit numeric sequence that should be incremented by one each time the customer is requesting a new loan application.

Specific error codes

#	Error code	Error Description
1	18901	Invalid Loan Program Code
1	18801	Invalid Account Number
2	18802	Invalid Loan Amount

#	Error code	Error Description
3	18803	Invalid Loan Currency
4	18804	Invalid Loan Program
5	18805	Error to generate the application number
6	18806	Error available check

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/convertTransactionToLoanWebService/v2	POST	

REST sample call

```
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
```

```

    "userID": ""
  },
  "accountId" : "",
  "loanAmount" : "",
  "amt" : "",
  "curCode" : "",
  "catalogLoanProduct" : "",
  "catalogLoanProduct Version" : "",
  "acronym" : ""
}

```

4.60 getCorporateHierarchy API

API overview

This web service can be used to get the list of the sub corporates under a particular corporate ID.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

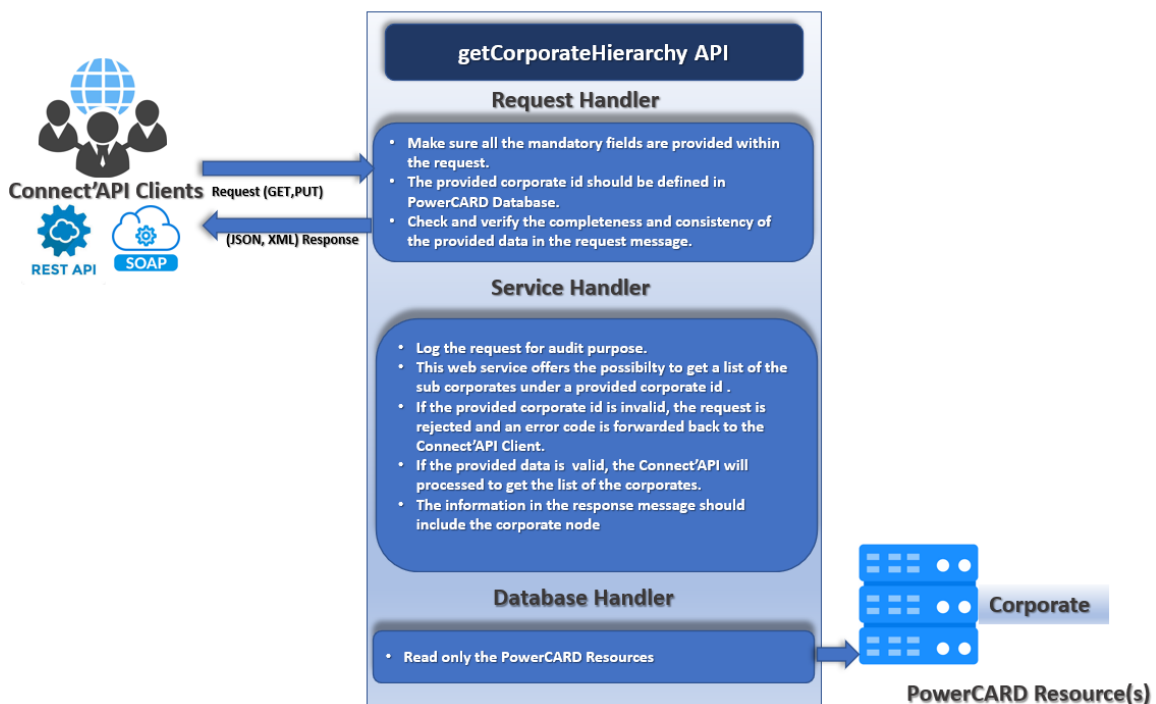


Figure 60: getCorporateHierarchy process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
-	CorporateNode	Complex	-	Required infinite list	The Corporate node
5	CorporateId	C(15)	CorporateNode	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
6	CorporateParentId	C(15)	CorporateNode	Required (max : 1)	The corporate parent ID
7	CorporateName	C(40)	CorporateNode	Required (max : 1)	The corporate name
8	CorporateType	C(3)	CorporateNode	Required (max : 1)	The type of the corporate must be defined in CORPORATE _TYPE.

Specific error codes

#	Error code	Error Description
1	30901	Invalid Corporate Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCorporateHierarchy WebService/ V2	POST	
/Corporates/V2/:corporateId/Hierarchy	GET	Required : corporateId =[C(15)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "corporateId": ""
}
```

4.61 getCorporateSelectedProducts API

API overview

This web service can be used to get the list of products and packs that have been selected by a corporate for its employees.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

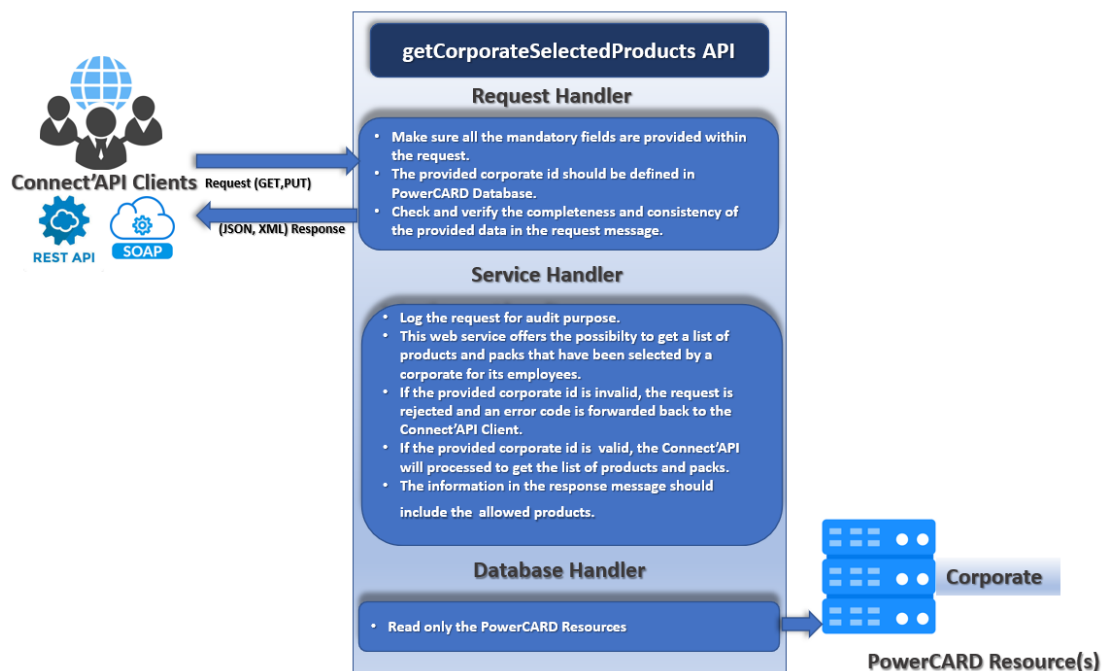


Figure 61: getCorporateSelectedProducts process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT. Date when the request was submitted
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in

#	Field Name	Data type	Parent	Occurrence	Field Description
					the PowerCARD CORPORATE table

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed With Success: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell

#	Field Name	Data type	Parent	Occurrence	Field Description
					you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	AllowedProduct	Complex	-	Required infinite list	The allowed products
5	CatPIProduct	C (6)	AllowedProduct	Conditional (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
6	CatPIProduct Version	C (10)	AllowedProduct	Conditional (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P Mandatory when product type product
7	CatalogParCode	C(6)	AllowedProduct	Conditional (max : 1)	It is a 6 digit character field that identifies the selected product bundles. Must be a valid value defined in CATALOGUE _PACK Mandatory when product type is Pack
8	Description	C (250)	AllowedProduct	Required (max : 1)	Catalogue product description

#	Field Name	Data type	Parent	Occurrence	Field Description
9	Type	C (1)	AllowedProduct	Required (max : 1)	Product type - P: product - C: Pack

Specific error codes

#	Error code	Error Description
1	30901	Invalid Corporate Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCorporateSelectedProductsWebService/V2	POST	
/Corporates/V2/:corporateld/SelectedProducts	GET	Required : corporateld =[C(15)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",

```

```

    "userId": ""
  },
  "corporateId": ""
}

```

4.62 getProductDetails API

API overview

PowerCARD offers a selection of web services with regards to products maintenance, the get product details web service is one of those services that can be used by both agents and customers in order to retrieve more details regarding the concerned Product being sent in the request message.

As output the requester could get information that can include details regarding product details : Limits, Services, Country domain control, MCC domain control, ...

Functional description

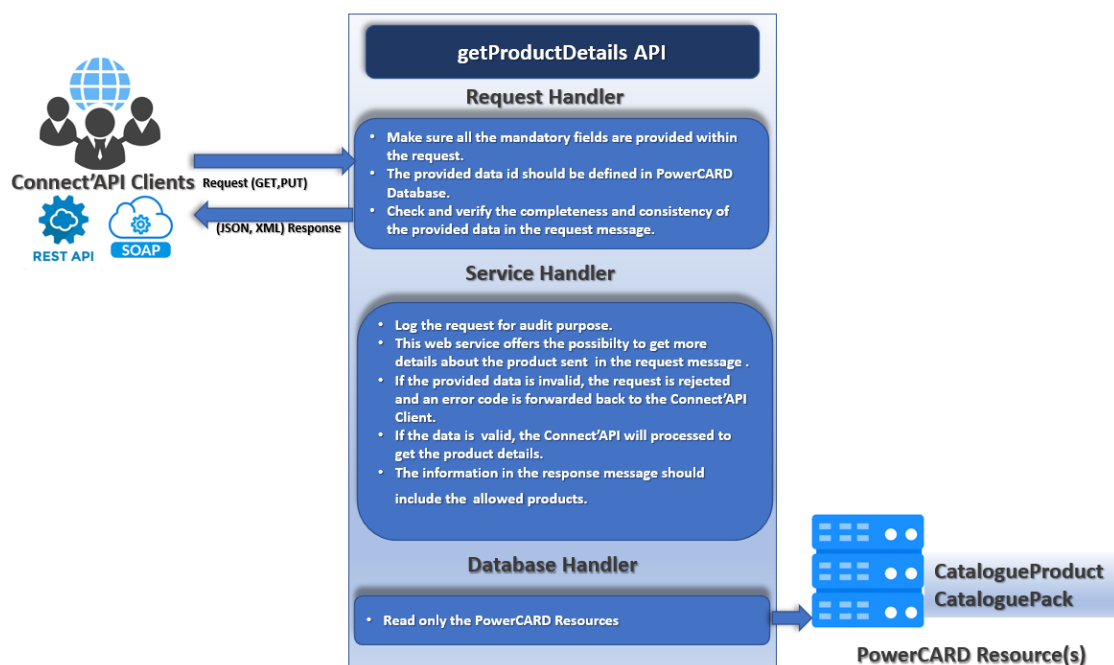


Figure 62: getProductDetails process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXX XX -XXXX-XXXX -XXXX -XXXX XXXXXX Example: 6948DF80-14BD-4E04 -8842-7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CatPIProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
5	CatPIProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					with product type = P
6	PackCode	C (6)	-	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles. Must be a valid value defined in CATALOGUE_PACK

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWithSuccess: Call was ended successfully. - ProceedWithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred

#	Field Name	Data type	Parent	Occurrence	Field Description
					during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	PANProduct Details	Complex	-	Optional (max: 1)	This aggregate contains details of each subscription request
5	CatPIProduct	C (6)	PANProduct Details	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE_PRODUCT table with product type = P
6	CatPIProduct Version	C (10)	PANProduct Details	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE_PRODUCT table with product type = P
7	PackCode	C (6)	PANProduct Details	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Must be a valid value defined in CATALOGUE_PACK
8	productCurrency	C	PANProduct Details	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
9	cycleCutoff	C(3)	PANProduct Details	Required (max : 1)	The account cycle cutoff assigned to the account: Must be defined in cycle_cutoff_parameter
-	ProductServices	Complex	PANProduct Details	Required (Infinite)	This aggregate contains details of the services to which the collaborator has applied
10	ServiceCode	C(6)	ProductServices	Required (max : 1)	Must be defined in SERVICES_DEFINITION table List of values example: CTLESS: Contactless. SECECO: Secured ecommerce ECOM00: E commerce CORSOL: Corporate Solidarity DOMSTP: Domain Setup VELSTP: Velocity setup

#	Field Name	Data type	Parent	Occurrence	Field Description
					DDDEFR: Direct debit deferment.
11	ServiceStatus	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: Y: Service is enabled. N: Service is disabled.
12	IsService Mandatory	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: Y: Service is mandatory and can not be disabled. N: Service is optional and can be disabled.
-	ServiceExposition Channels	Complex	ProductServices	Optional (Infinite)	This is aggregate is used to list the service exposition channels along with channel activation status
13	ChannelType	C(3)	ServiceExposition Channels	Required (max : 1)	BAG: Bank agent channel COW: Corporate portal channel CHM: Cardholder Mobile app CHW: Cardholder portal
14	ServiceToBeExposed InChannel	C(1)	ServiceExposition Channels	Required (max : 1)	Y: Service can be visible in the channel. N: Service might be visible or invisible in the channel. In case visible, the service should be greyed out.
-	DeffermentService Addendum	Complex	ProductServices	Conditional (Infinite)	Mandatory if service code is DDDEFR
15	DeferredDays Number	Float	DeffermentService Addendum	Required (max : 1)	The number of days for debit deferment. The list of value must be defined in the table static_list with collection name

#	Field Name	Data type	Parent	Occurrence	Field Description
					= 'PaymentDD DefermentDays'.
16	Deferment StartDate	Date	DeffermentService Addendum	Required (max : 1)	The start date of the deferment. Must be defaulted to current business date
17	DefermentEnddate	Date	DeffermentService Addendum	Optional (max : 1)	The date where the deferment will stop.
-	DomainControlSRV Addendum	Complex	ProductServices	Conditional infinite list	Mandatory if service code is DOMSTP
18	ControlType	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: CTR: Country MCC: MCC PRS: Processing code MER: Merchant CUR: Currency
19	ControlOperator	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValueList
-	ControlValueList	Complex	DomainControlSRV Addendum	Optional infinite list	Contains various values, such as country codes, processing codes ...
20	ControlValue	C (15)	ControlValue List	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section:

#	Field Name	Data type	Parent	Occurrence	Field Description
					COUNTRY table for ControlType = 'CTR'. MCC table for ControlType = 'MCC'. PROCESSING_LIST table for ControlType = 'PRS'. MERCHANT table for ControlType = 'MER'. CURRENCY_TABLE table for ControlType = 'CUR'. If not, the request will be rejected.
-	CardLimitSRV Addendum	Colmplex	-	Conditional infinite list	Mandatory if service code is VELSTP
21	LimitID	C (4)	CardLimitSRV Addendum	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.
22	LimitDescription	C (64)	CardLimitSRV Addendum	Optional (max : 1)	Description of the limit
23	LimitAbv Wording	C (32)	CardLimitSRV Addendum	Optional (max : 1)	A brief description of the limit
-	ValidityRange Date	Complex	CardLimitSRV Addendum	Required (max: 1)	A limit can be temporary and restricted to a range date
24	StartDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
25	EndDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date Format:

#	Field Name	Data type	Parent	Occurrence	Field Description
					yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
-	Period	Complex	CardLimitSRV Addendum	Optional (max: 4)	The Period of limit that should be set on the card.
26	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: D: Fixed period W: Rolling period
27	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off
28	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: • D: for daily • W: for Weekly • M: for Monthly Y: for Yearly
-	Territoriality Online	Colmplex	CardLimitSRV Addendum	Optional (max: 3)	The Territoriality Online
29	Territoriality Type	C (3)	Territoriality Online	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
30	Transaction Amount	Decimal (18,3)	Territoriality Online	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.

#	Field Name	Data type	Parent	Occurrence	Field Description
31	Transaction AmountMAX	Decimal (18,3)	Territoriality Online	Required (max : 1)	The max total amount can be negotiated for the selected period and territoriality
32	NumberOf Transactions	Number (4)	Territoriality Online	Required (max : 1)	Must contain the target the number of transactions of the limit.
33	NumberOf TransactionsMAX	Number (4)	Territoriality Online	Required (max : 1)	The max number of transactions that can be negotiated for the selected period and territoriality
-	Territoriality Offline	Complex	CardLimitSRV Addendum	Optional (max: 4)	Territoriality Offline
34	Territoriality Type	C (3)	Territoriality Offline	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
35	Transaction Amount	Decimal (18,3)	Territoriality Offline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
36	NumberOf Transactions	Number (4)	Territoriality Offline	Required (max : 1)	Must contain the target the number of transactions of the limit.
37	TotalTransaction AmountOnline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
38	TotalNumberOf TransactionsOnline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of

#	Field Name	Data type	Parent	Occurrence	Field Description
					transactions of the limit.
39	TotalTransaction AmounOffline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
40	TotalNumberOf TransactionsOffline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	Transaction Criteria	Complex	CardLimitSRV Addendum	Optional (max : 1)	This aggregate is used to define the transaction criteria
41	SequenceNumber	Number (9)	Transaction Criteria	Optional (max : 1)	Should contain Sequence Number
42	ProcessingCode	C (2)	Transaction Criteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
43	CountryCode	C (3)	Transaction Criteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
44	MCC	C (4)	Transaction Criteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD MCC table
45	PaymentType	C (2)	Transaction Criteria	Optional (max : 1)	Should contain the payment type
46	CurrencyCode	C (3)	Transaction Criteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
47	MaxAmount PerTrx	Long	Transaction Criteria	Optional (max : 1)	Used to set the maximum amount per transaction.
48	CardEntry Mode	ClosedEnum	Transaction Criteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : 00: Unknown or terminal not used 01: Manual (key entry) 02: Magnetic stripe read 03: Integrated circuit card read 04: Contactless device-read

Specific error codes

#	Error code	Error Description
1	31101	Invalid Catalog Product Id
2	31102	Invalid Catalog Product version
3	31103	Catalog Product Pack Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetProductDetailsWebService/V2	POST	
/Product/V2?catPIProduct={catPIProduct}& catPIProductVersion={catPIProductVersion}&packCode={packCode}	GET	Required : CatPIProduct=[C(6)] CatPIProductVersion=[C(10)] PackCode=[C(6)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "catPIProduct": "",
  "catPIProductVersion": "",
  "packCode": ""
}
```

4.63 updateCardStatusReason API

API overview

This web services was introduced to allow changing the status of an existing card in the system. Agent and customers can ask to change the status of a card by selecting a reason for that operation from a pre-defined list of reasons. It is important to note that the web service requires as input the Card Number or the Card Token ID.

As output the web service send back the result of the processing that could be either successfully or unsuccessfully, in what case, the system will share an error code for investigation.

Functional description

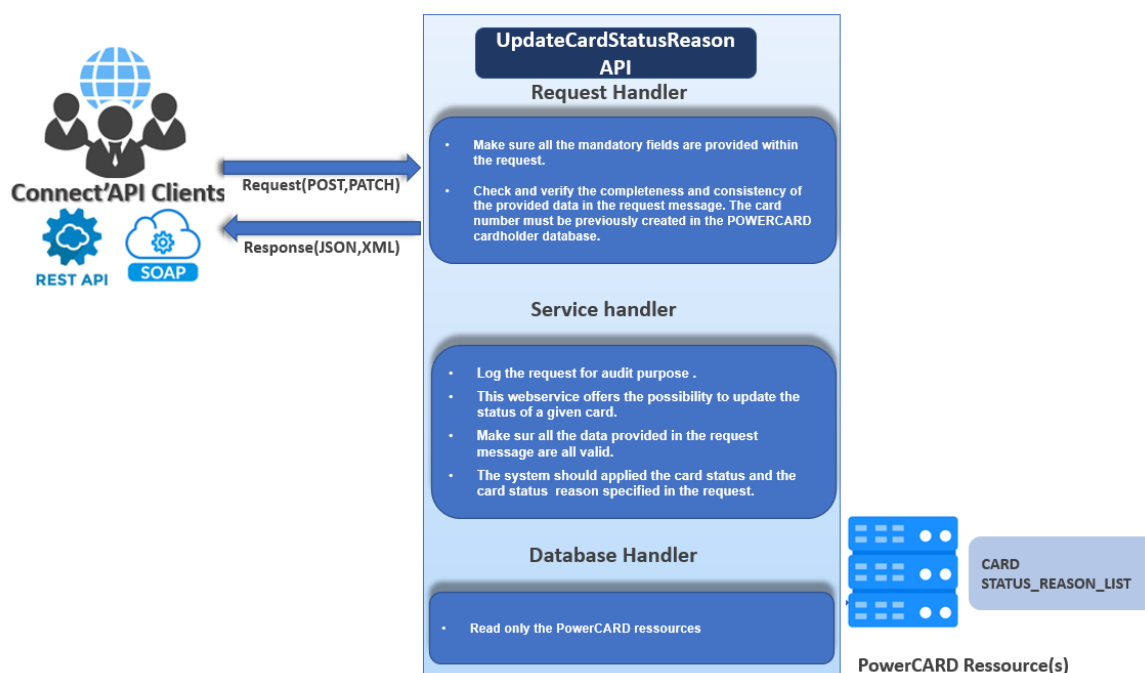


Figure 63: updateCardStatusReason process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a

#	Field Name	Data type	Parent	Occurrence	Field Description
					universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Required (max: 1)	The card number or the Card Token Id. It should contain a valid value
5	NewStatusCode	C (1)	-	Required (max: 1)	Must indicate the new Status Code of an existing Card.
6	StatusReasonCode	C (2)	-	Required (max: 1)	Status Reason code Refers to the reason why the agent or the customer is requesting to change the status of an existing card. It must contain a valid value that is previously defined in the STATUS_REASON_LIST table under the general parameter section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed WithSuccess: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	03601	Invalid Customer Id
1	19201	Invalid Card Number / Card Token ID
2	19202	Invalid Status Code
3	19203	Invalid Reason Code
4	19204	Invalid Combination Status Code - Status Reason Code
5	19205	Invalid Transition Requested
6	19206	Operation Not Permitted

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/v2	POST	
/Cards/v2/:aliasPan/changeStatusReason?newStatusCode={newStatusCode}&statusReasonCode={statusReasonCode}	PATCH	Required aliasPan=[C(22)] newStatusCode =[C(1)] statusReasonCode=[C(2)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"cardNumber": "",
"newStatusCode": "",
"statusReasonCode": ""
}

```

4.64 updateAlertService API

API overview

The aim of this API is to switch-off an alert service or updated the minimum threshold amount.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

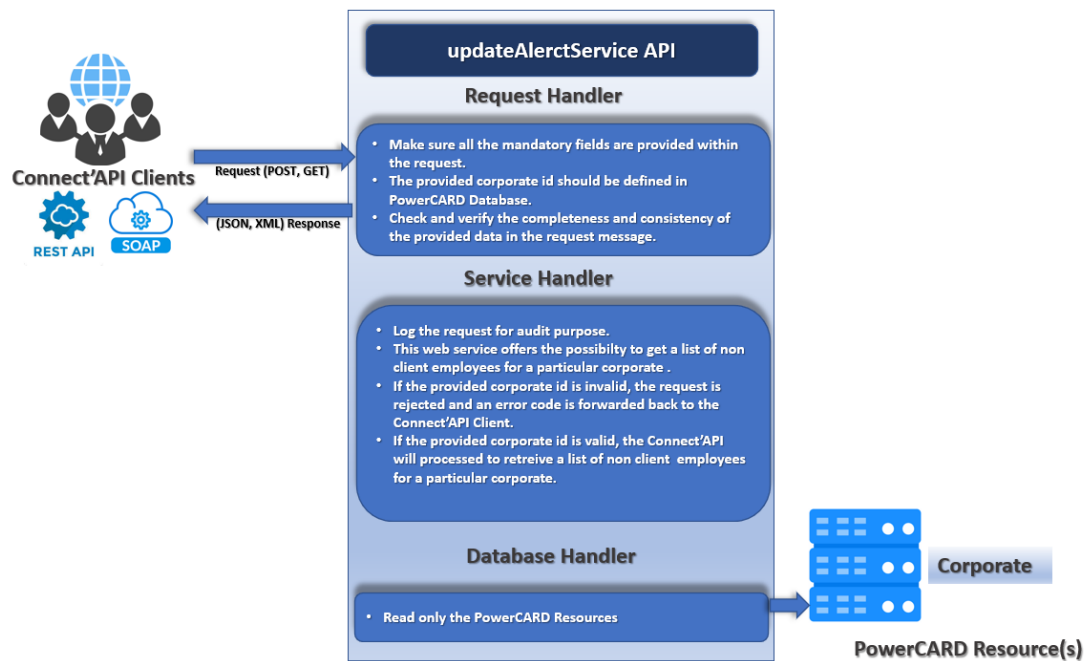


Figure 64: updateService process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
12	CardNumber	C (22)	-	Required (max: 1)	Card Number or Card Token Id
13	ActionCode	C (1)	-	Required (max: 1)	Indicates whether the system is required only to: • F: Switch-off the alert service • U: Updated the minimum threshold amount
14	NotificationCode	C (4)	-	Required (max: 1)	The notification code that should be triggered. This code must be already defined in the PowerCARD EMAIL_PARAMETER, SMS_PARAMETER, LETTER_PARAMETER, PUSH_PARAMETER, _NOTIF_PARAMETER tables depending on notification type.
15	NotificationDestination	C(1)	-	Optional (max: 1)	The notification destination In case of SMS, the phone to use is: • '1' : Phone 1 [*] • '2' : Phone 2 [*] • '3' : Phone 3 [*] • '4' : Phone 4 [*]

#	Field Name	Data type	Parent	Occurrence	Field Description
					In case of Email, the email to be used is: '1' : Email 1 [*] '2' : Email 2 [*] Required for SMS, PUSH and Email notification [*] Defined at Client level in CLIENT table
16	ThresholdAmount	Decimal (18,3)	-	Optional (max: 1)	The threshold amount. If ThresholdAmount is not provided PowerCard will put automatically the threshold amount from the parametrization like a default value. In case of ActionCode set to "U"
17	ThresholdCurrency	C (3)	-	Optional (max: 1)	Specifies the iso currency code of the threshold amount. It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should

#	Field Name	Data type	Parent	Occurrence	Field Description
					contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed WithSuccess: Call was ended successfully. • Proceed WithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	03601	Invalid Customer Id
1	19201	Invalid Card Number / Card Token ID

#	Error code	Error Description
2	19202	Invalid Status Code
3	19203	Invalid Reason Code
4	19204	Invalid Combination Status Code - Status Reason Code
5	19205	Invalid Transition Requested
6	19206	Operation Not Permitted

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/v2	POST	
/AlertService/v2/:aliasPan/update?actionCode={actionCode}¬ificationCode={notificationCode}¬ificationDestination={notificationDestination}&thresholdAmount={thresholdAmount}&thresholdCurrency={thresholdCurrency}	PUT	aliasPan=[C(22)] ActionCode=[C(1)] NotificationCode=[C(4)]

PowerCARD Resource URI	Me	URI Params
		NotificationDestination=[C(1)] ThresholdAmount=[Number] ThresholdCurrency=[C(3)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"cardNumber": "",
"actionCode": "",
"notificationCode": "",
"notificationDestination": "",
"thresholdAmount": "",
"thresholdCurrency": ""
}

```

4.65 updateAccountBalance

API overview

The purpose of the current service is to give the possibility to any allowed third party to update an existing debit account balance in PowerCARD whenever it is required to synchronize the PowerCARD database with data coming from other systems like bank hosts. It is worth to mention that this operation is expected to update the balance information of the account with the data transmitted within the request message. Any issue noticed while trying to update the account will result in a decline of the request, that will be followed by providing an error code back in the response message. For further information, refer the following sections.

Functional description

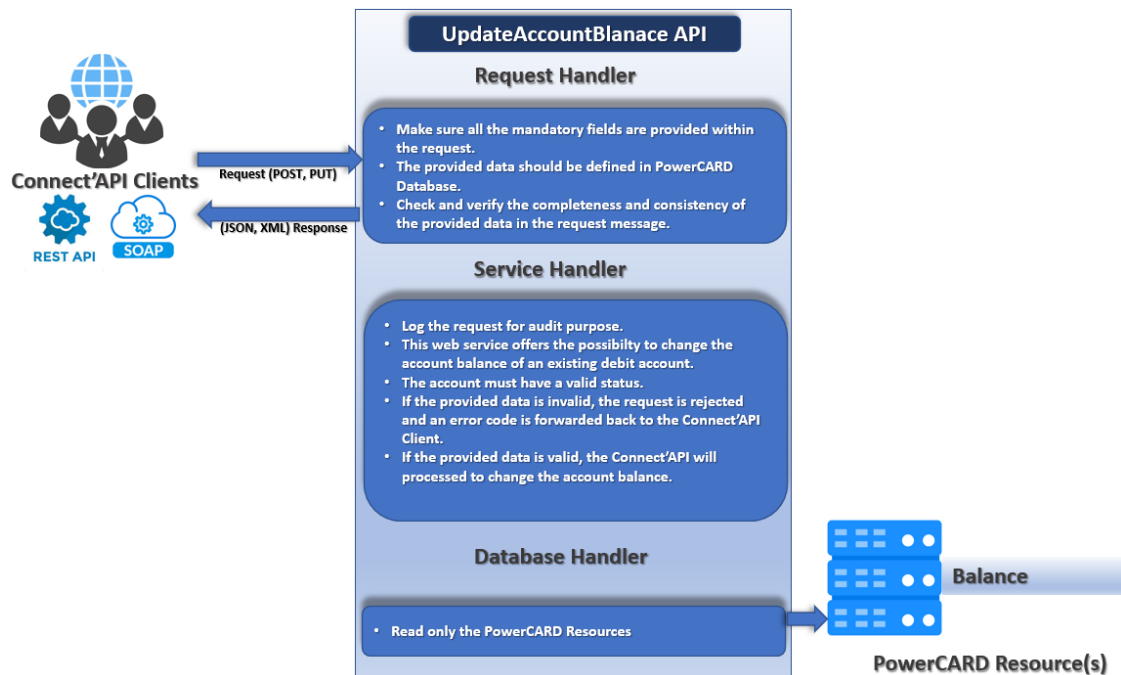


Figure 65: updateAccountBalance process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX -XXXX -XXXX -XXXX - XXXXXXXX XXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
	RequestType	ClosedEnum	-	Required (max : 1)	The possible values for this field are: • UB: Update full balance record • AB: Adjust balance record
	TransactionType	ClosedEnum	-	Optional (max : 1)	Must be present only if the Request Type is equal to AB (Adjust balance record) The value transmitted in this field must be one of the following values: • DP: Deposit • WD: Withdrawal • TR: Transfer Transaction • BF: Bank Fees • OT: Other
	CardNumber	C (22)	-	Required (max : 1)	Card Number The card number to which the account is tied.
	AccountNumber	C (24)	-	Required (max : 1)	It contains a value that identifies an account or customer relationship in the transaction.
-	TransactionAmount	Complex	-	Optional (max : 1)	This field contains the transaction amount that will be used to update

#	Field Name	Data type	Parent	Occurrence	Field Description
					the balance only in case of Deposit; Withdrawal, Transfer Transaction and Bank Fees It should not be part of the message only if RequestType is equal to UB(Update Balance Record).
10	Amt	Long ()	TransactionAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	C (3)	TransactionAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Balance	Complex	-	Optional (max : 1)	Account current balance as available in the host Present only if RequestType is equal to UB(Update Balance Record). Note that the amount is signed. It means that if the amount is negative, it should be prefixed with a '-'
	Sign	ClosedEnum	-	Required (max : 1)	This field is used to specify the sign of the transaction amount only if case

#	Field Name	Data type	Parent	Occurrence	Field Description
					of RequestType equal to AB. Possible Values are: - C: Credit - D: Debit
-	Facilities	Complex	-	Optional (max : 1)	The overdrawn amount the customer is entitled to. This amount will be added from the Balance to get the open to buy. Could be present only if RequestType is equal to UB(Update Balance Record). Note that the amount is signed. It means that if the amount is negative, it should be prefixed with a '-'
10	Amt	Long	TransactionAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	NC (3)	TransactionAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	BlockedAmount	Complex	-	Optional (max : 1)	The blocked amount at the host. This amount is not included in the amounts blocked by

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD (due to authorization) This amount will be deducted from the Balance to get the open to buy. Could be present only if RequestType is equal to UB(Update Balance Record). Note that the amount is signed. It means that if the amount is negative, it should be prefixed with a '-'
10	Amt	Long ()	TransactionAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
11	CurCode	NC (3)	TransactionAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.

#	Field Name	Data type	Parent	Occurrence	Field Description
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • Proceed WithSuccess: Call was ended successfully. • Proceed With SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	15301	Invalid Account Number
2	15302	Invalid Card Number
3	15303	Invalid Request Type
4	15304	Invalid Transaction Type
5	15305	Invalid Transaction Amount
6	15306	Invalid Balance Amount
7	15307	Invalid Sign
8	15308	Invalid Facilities Amount
9	15309	Invalid Blocked amount
10	15310	Inconsistent field Note that the text message will contain the field concerned.
11	15311	Error while updating balance record
12	15312	Operation not permitted, Expired card
13	15313	Operation not permitted, Cancelled card
14	15314	Operation not permitted, Replaced card
15	15315	Card in the stop list
16	15316	Operation not permitted, Suspended card
17	15317	Invalid currency code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x

Status Code	Description	Success Response	Error Response
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCustomerPhoneNumberWebService/v2	POST	
/AlerteService/v2/:aliasPan/update?actionCode={actionCode}¬ificationCode={notificationCode}¬ificationDestination={notificationDestination}&thresholdAmount={thresholdAmount}&thresholdCurrency={thresholdCurrency}	PATCH	Required aliasPan=[C(22)] ActionCode=[C(1)] NotificationCode=[C(4)] NotificationDestination=[C(1)] ThresholdAmount=[Number] ThresholdCurrency=[C(3)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"cardNumber": "",
"actionCode": "",
"notificationCode": "",
"notificationDestination": "",
"thresholdAmount": "",
"thresholdCurrency": ""
}

```

4.66 updateInterestRateIndex API

API overview

Banks or any financial institution can use this service to adjust the interest rate applied to a given account. This operation may result in decreasing or increasing the interest rate

applied. The new interest rate will be effective starting from the effective date part of the request message. Before that date it is the old interest rate index that will still be used until that date.

Functional description

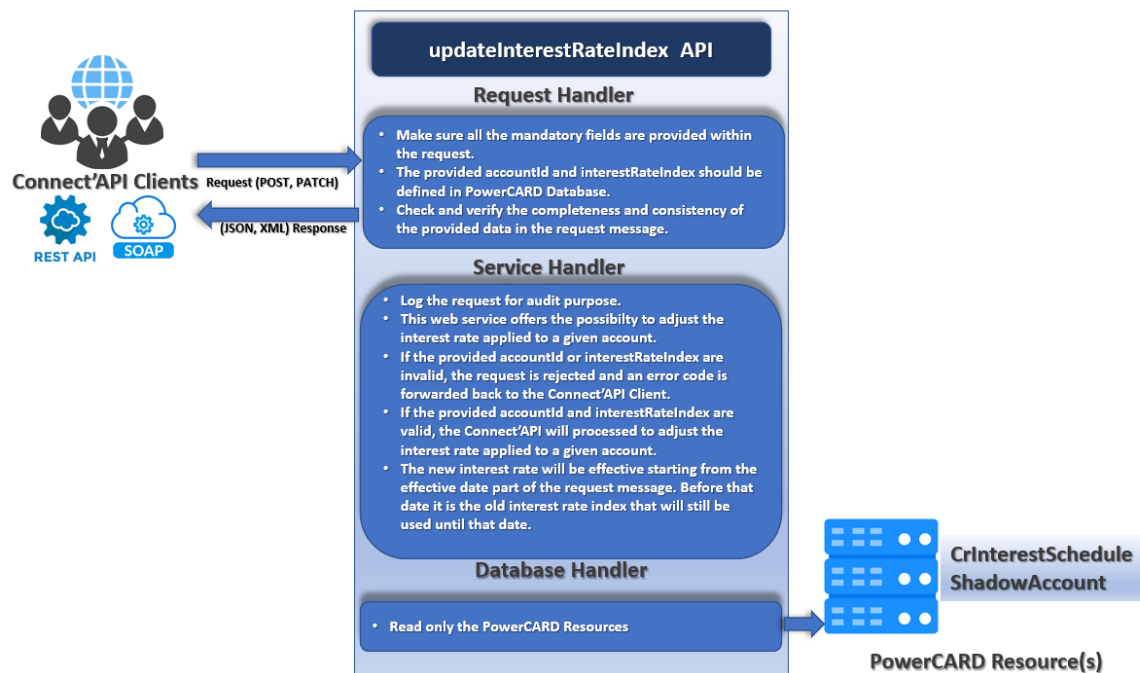


Figure 66: updateInterestRateIndex process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following:

#	Field Name	Data type	Parent	Occurrence	Field Description
					XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	Should contain the credit account number subject to this interest rate change This value should be already defined in the PowerCARD SHADOW _ACCOUNT table.
5	InterestRateIndex	C (4)	-	Required (max : 1)	Refers to the interest rate index as already defined in CR _INTEREST _SCHEDULE table.
6	EffectiveDate	DateTime	-	Required (max : 1)	Effective date starting from when the new interest rate index will be applied

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the

#	Field Name	Data type	Parent	Occurrence	Field Description
					requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	17201	Invalid Interest rate index

#	Error code	Error Description
2	17202	Invalid Account number
3	17203	Invalid Effective Date

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateInterestRateIndexWebService/v2	POST	
/Accounts/v2/:accountId/changeInterestRateIndex?interestRateIndex={interestRateIndex}&effectiveDate={effectiveDate}	PATCH	Required accountId =[C(24)] interestRateIndex =[C(4)] effectiveDate =[dateTime]

REST sample call

```
"keyValues": [
{
```

```

    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "accountId": "",
  "interestRateIndex": "",
  "effectiveDate": ""
}

```

4.67 updateCardCharacteristics API

API overview

The purpose of the current service is to give the possibility to update the card characteristics that have been already negotiated for a card.

Several services to manage are offered (service's domain controls, NFC feature, request deferment of the contributions, Corporate solidarity, ..) and informations are requested depending on the desired service.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

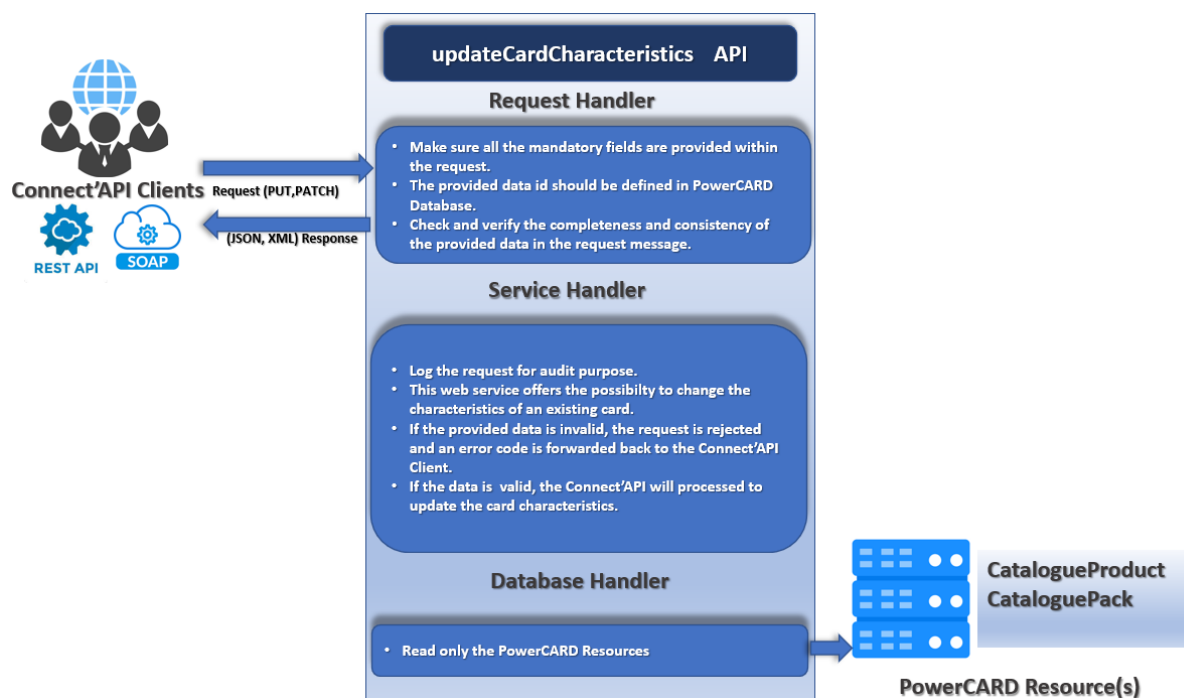


Figure 67: updateCardCharacteristics process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80- 14BD-4E04- 8842-7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C(22)	SubscriptionRequest	Required (max : 1)	The generated PAN number in when the status of the request status is C and Card createion status is A
-	PANProductDetails	Complex	-	Required (max: 1)	This aggregate contains details of each subscription request
5	CatPIProduct	C (6)	PANProductDetails	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD CATALOGUE _PRODUCT table with product type = P
6	CatPIProductVersion	C (10)	PANProductDetails	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
7	PackCode	C (6)	PANProductDetails	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles. Must be a valid value defined in CATALOGUE _PACK
8	productCurrency	C	PANProductDetails	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
9	cycleCutoff	C(3)	PANProductDetails	Required (max : 1)	The account cycle cutoff assigned to the account: Must be defined in cycle_cutoff _parameter
-	ProductServices	Complex	PANProductDetails	Required (Infinite)	This aggregate contains details of the servies to which

#	Field Name	Data type	Parent	Occurrence	Field Description
					the collaborator has applied
10	ServiceCode	C(6)	ProductServices	Required (max : 1)	Refers to list of services on card. The possible values are: CTLESS: Contactless. You can directly use activateContactless API to enable and disable NFC Service STOPLST: Card stop listed You can directly use updateCardStatus API to place the card in stop list RMSTOP: Remove card stop listed. You can directly use updateCardStatus API to remove the card from the stop list queue BLCARD : Block/ Unblock the card You can directly use updateCardStatus API to block or unblock the card. DOMSTP: Domain Setup Use the DomainControlSRV Addendum aggregate to perform the requested action VELSTP: Velocity setup Use the CardLimitSRV Addendum aggregate to perform the requested action DDDEFR: Direct debit deferment

#	Field Name	Data type	Parent	Occurrence	Field Description
					Use the DeffermentService Addendum Aggregate CORSOL: Corporate Solidarity SECECM: Secured ecommerce NSCECM: Not secured ecommerce
11	ServiceStatus	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: - Y: Service is enabled. - N: Service is disabled.
12	IsServiceMandatory	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: - Y: Service is mandatory and can not be disabled. - N: Service is optional and can be disabled.
-	ServiceExposition Channels	Complex	ProductServices	Optional (Infinite)	This is aggregate is used to list the service exposition channels along with channel activation status
13	ChannelType	C(3)	ServiceExposition Channels	Required (max : 1)	Refers to the channel type. The possible values are: - BAG: Bank agent channel - COW: Corporate portal channel - CHM: Cardholder Mobile app - CHW: Cardholder portal
14	ServiceToBeExposed InChannel	C(1)	ServiceExposition Channels	Required (max : 1)	Y: Service can be visible in the channel.

#	Field Name	Data type	Parent	Occurrence	Field Description
					N: Service might be visible or invisible in the channel. In case visible, the service should be greyed out.
-	DeffermentService Addendum	Complex	ProductServices	Conditional (Infinite)	Mandatory if service code is DDDEFR
15	DefferedDaysNumber	Float	DeffermentService Addendum	Required (max : 1)	The number of days for debit deferment. The list of value must be defined in the table static_list with collection name = 'PaymentDDDefermentDays'.
16	DefermentStartDate	Date	DeffermentService Addendum	Required (max : 1)	The start date of the deferment. Must be defaulted to current business date
17	DefermentEnddate	Date	DeffermentService Addendum	Optional (max : 1)	The date where the deferment will stop.
-	DomainControlSRV Addendum	Complex	ProductServices	Conditional infinite list	Mandatory if service code is DOMSTP
18	ControlType	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: - CTR: Country - MCC: MCC - PRS: Processing code - MER: Merchant - CUR: Currency
19	ControlOperator	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or

#	Field Name	Data type	Parent	Occurrence	Field Description
					multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValueList
-	ControlValueList	Complex	DomainControlSRV Addendum	Optional infinite list	Contains various values, such as country codes, processing codes ...
20	ControlValue	C (15)	ControlValueList	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section: - COUNTRY table for ControlType = 'CTR'. - MCC table for ControlType = 'MCC'. - PROCESSING_LIST table for ControlType = 'PRS'. - MERCHANT table for ControlType = 'MER'. - CURRENCY_TABLE table for ControlType = 'CUR'. If not, the request will be rejected.
-	CardLimitSRV Addendum	Colmplex	-	Conditional infinite list	Mandatory if service code is VELSTP
21	LimitID	C (4)	CardLimitSRV Addendum	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.
22	LimitDescription	C (64)	CardLimitSRV Addendum	Optional (max : 1)	Description of the limit
23	LimitAbrvWording	C (32)	CardLimitSRV Addendum	Optional (max : 1)	A brief description of the limit

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ValidityRangeDate	Complex	CardLimitSRV Addendum	Required (max: 1)	A limit can be temporary and restricted to a range date
24	StartDate	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity Start date
25	EndDate	DateTime	ValidityRangeDate	Optional (max: 1)	The Validity End date
-	Period	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	The Period of limit that should be set on the card.
26	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: Fixed period - W: Rolling period
27	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off
28	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: for daily - W: for Weekly - M: for Monthly - Y: for Yearly
-	TerritorialityOnline	Complex	CardLimitSRV Addendum	Optional (max: 3)	The Territoriality Online
29	TerritorialityType	C (3)	TerritorialityOnline	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
30	TransactionAmount	Decimal (18,3)	TerritorialityOnline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.

#	Field Name	Data type	Parent	Occurrence	Field Description
31	TransactionAmount MAX	Decimal (18,3)	TerritorialityOnline	Required (max : 1)	The max total amount can be negotiated for the selected period and territoriality
32	NumberOf Transactions	Number (4)	TerritorialityOnline	Required (max : 1)	Must contain the target the number of transactions of the limit.
33	NumberOf TransactionsMAX	Number (4)	TerritorialityOnline	Required (max : 1)	The max number of transactions that can be negotiated for the selected period and territoriality
-	TerritorialityOffline	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	The Territoriality Offline
34	TerritorialityType	C (3)	TerritorialityOffline	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
35	TransactionAmount	Decimal (18,3)	TerritorialityOffline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
36	NumberOfTransactions	Number (4)	TerritorialityOffline	Required (max : 1)	Must contain the target the number of transactions of the limit.
37	TotalTransaction AmountOnline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
38	TotalNumberOf TransactionsOnline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of

#	Field Name	Data type	Parent	Occurrence	Field Description
					transactions of the limit.
39	TotalTransaction AmounOffline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
40	TotalNumberOf TransactionsOffline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	TransactionCriteria	Colmplex	CardLimitSRV Addendum	Optional (max : 1)	This aggregate is used to define the transaction criteria
41	SequenceNumber	Number (9)	TransactionCriteria	Optional (max : 1)	Should contain Sequence Number
42	ProcessingCode	C (2)	TransactionCriteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
43	CountryCode	C (3)	TransactionCriteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
44	MCC	C (4)	TransactionCriteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD MCC table
45	PaymentType	C (2)	TransactionCriteria	Optional (max : 1)	Should contain the payment type
46	CurrencyCode	C (3)	TransactionCriteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
47	MaxAmountPerTrx	Long	TransactionCriteria	Optional (max : 1)	Used to set the maximum amount per transaction.
48	CardEntryMode	ClosedEnum	TransactionCriteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : • 00: Unknown or terminal not used • 01: Manual (key entry) • 02: Magnetic stripe read • 03: Integrated circuit card read 04: Contactless device-read

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.

#	Field Name	Data type	Parent	Occurrence	Field Description
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	33201	Invalid Catalog Product
2	33202	Invalid Catalog Product Version
3	33203	Invalid Pack Code
4	33204	Invalid Product Currency
5	33205	Invalid Cycle Cutoff
6	33206	Invalid Service Code
7	33207	Invalid Service Status
8	33208	Invalid Service Mandatory Indicator
9	33209	Invalid Channel Type
10	33210	Invalid ServiceTo BeExposed In Channel
11	33211	Invalid Deferred Days Number
12	33212	Invalid Deferment Start Date
13	33213	Invalid Deferment End Date
14	33214	Invalid Control Type
15	33215	Invalid Control Operator
16	33216	Invalid Control ValueList
17	33217	Invalid Control Value
18	33218	Invalid Card Limit SRV Addendum
19	33219	Invalid Limit ID
20	33220	Invalid Limit Description
21	33221	Invalid Limit Abrv Wording
22	33222	Invalid Validity Range Date
23	33223	Invalid Start Date
24	33224	Invalid End Date
25	33225	Invalid Period Type
26	33226	Invalid Period Day Off
27	33227	Invalid Periodicity Flag
28	33228	Invalid Territoriality Type
29	33229	Invalid Transaction Amount
30	33230	Invalid Transaction Amount
31	33231	Invalid Number Of Transactions
32	33232	Invalid Number Of Transactions
33	33233	Invalid Territoriality Type
34	33234	Invalid Total Transaction Amount Online

#	Error code	Error Description
35	33235	Invalid Total Number Of Transactions Online
36	33236	Invalid Total Transaction Amount Offline
37	33237	Invalid Total Number Of Transactions Offline
38	33238	Invalid Sequence Number
39	33239	Invalid Processing Code
40	33240	Invalid Country Code
41	33241	Invalid MCC
42	33242	Invalid Payment Type
43	33243	Invalid Currency Code
44	33244	Invalid Max Amount Per Trx
45	33245	Invalid Card Entry Mode

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCardCharacteristicsWebService/v2	POST	

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"interestRateIndex": "",
"effectiveDate": "",
"pANProductDetails": "",
{
  "catPIProduct": "",
  "catPIProductVersion": "",
  "packCode": "",
  "productCurrency": "",
  "cycleCutoff": ""
},
"productServices": "",
{
  "serviceCode": "",
  "serviceStatus": "",
  "isServiceMandatory": "",
},
"ServiceExposition Channels": "",
{
  "channelType": "",
  "serviceToBeExposedInChannel": ""
},
"DeffermentService Addendum": "",
{
  "deferredDaysNumber": "",
  "defermentStartDate": "",
  "defermentEnddate": ""
},
}
"limitID": "",
"limitDescription": "",
"limitAbrevWording": "", "validityRangeDate": "",
"startDate": "",
"endDate": "",
"period": "", "periodType": "",
"periodDayOff": "",
"periodicityFlag": "",
"territorialityOnline": "",
"territorialityType": "",
"transactionAmount": "",
"transactionAmount": "", "": "",
"numberOfTransactions": "",
"numberOfTransactions": "",
"territorialityOffline": "",
"territorialityType": "",
"transactionAmount": "", "": "",
"numberOfTransactions": "",
"totalTransactionAmountOnline": "",
"totalNumberOfTransactionsOnline": "",
"totalTransactionAmountOffline": "", "": "",
"totalNumberOfTransactionsOffline": "",
"transactionCriteria": "",
"sequenceNumber": "",
"processingCode": "",
"countryCode": "",

```

```

"mcc": "", "": ""
"paymentType": "",
"currencyCode": "",
"maxAmountPerTrx": "",
"cardEntryMode": "",
}

```

4.68 getCollaborationInitiatedSubscription API

API overview

This web service is used to initiate an employee subscription from the corporate portal.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

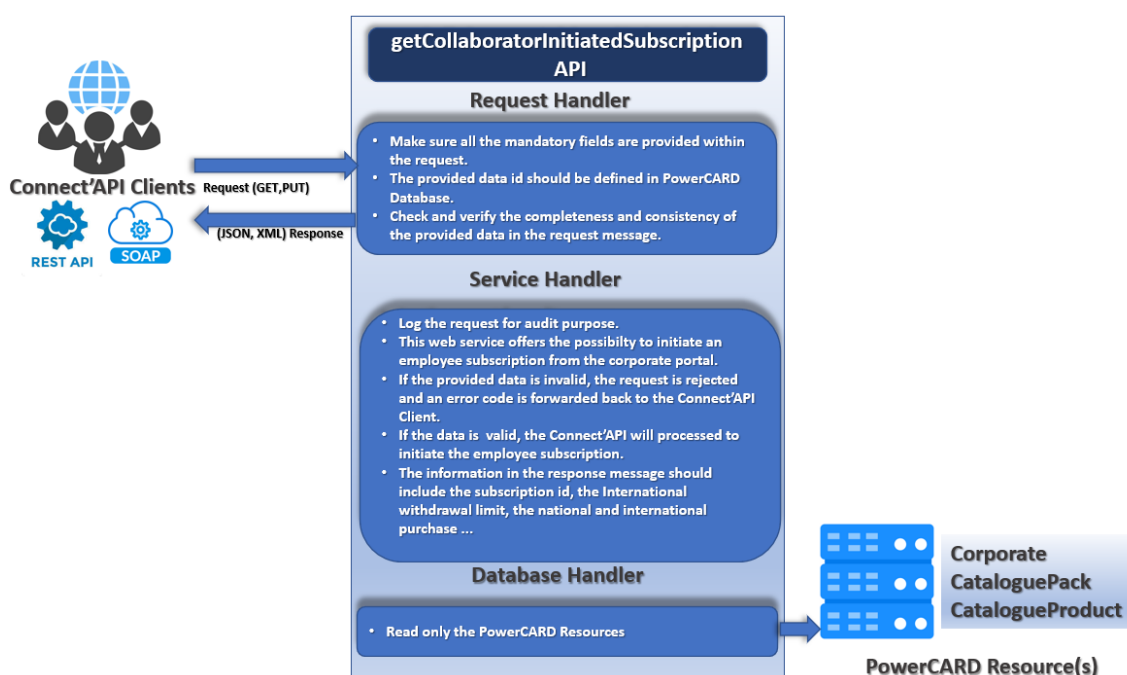


Figure 68: getCollaborationInitiatedSubscription process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	-	Required (max : 1)	The employee ID
6	SubscriptionID	C(22)	-	Optional (max: 1)	The generated subscription ID submitted by corporate portal contact person

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
-	SubscriptionRequest	Complex	-	Required (max: 1)	This aggregate contains details of each subscription request
4	CorporateId	C(15)	SubscriptionRequest	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	SubscriptionRequest	Required (max : 1)	The employee ID
6	SubscriptionID	C(22)	SubscriptionRequest	Optional (max: 1)	The generated subscription ID submitted by corporate portal contact person
6	CatPIProduct	C (6)	SubscriptionRequest	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
7	CatPIProductVersion	C (10)	SubscriptionRequest	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
8	PackCode	C (6)	SubscriptionRequest	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles Must be a valid value defined in CATALOGUE _PACK
9	ContactLessService	C(1)	SubscriptionRequest	Required (max : 1)	A Boolean indicator:

#	Field Name	Data type	Parent	Occurrence	Field Description
					- Y: Contactless service is enabled. - N: Contactless service is disabled.
10	MobileContactLessService	C(1)	SubscriptionRequest	Required (max : 1)	A Boolean indicator: - Y: Mobile contactless service is enabled. - N: Mobile contactless service is disabled.
11	PaymentService	C(1)	SubscriptionRequest	Required (max : 1)	A Boolean indicator: - Y: Payment service is enabled. - N: Payment service is disabled.
12	AcsService	C(1)	SubscriptionRequest	Required (max : 1)	A Boolean indicator: - Y: ACS service is enabled. - N: ACS service is disabled.
32	OrderDate	Date	SubscriptionRequest	Required (max : 1)	The order date
32	CompanySolidarity	C(1)	SubscriptionRequest	Required (max : 1)	The company solidarity
33	DifferedDebit	C(1)	SubscriptionRequest	Required (max : 1)	The differed Debit
34	DifferedDaysNumber	Long	SubscriptionRequest	Required (max : 1)	The number Differed Days
-	AllowedMccList	Complex	SubscriptionRequest	Required infinite list	The allowed MCC list
35	AllowedMcc	C(4)	AllowedMccList	Required (max : 1)	Allowed MCC Must be defined in MCC table
-	ExcludedMccList	Complex	SubscriptionRequest	Required infinite list	The MCCs list to be excluded for domain control.
36	ExcludedMcc	C(4)	ExcludedMccList	Required (max : 1)	Excluded MCC Must be defined in MCC table
-	AllowedCountryList	Complex	SubscriptionRequest	Required infinite list	The ISO countries list to be allowed for domain control

#	Field Name	Data type	Parent	Occurrence	Field Description
37	AllowedCountry	C(3)	AllowedCountryList	Required (max : 1)	The Country Code Must be defined in Country Table
-	ExcludedCountryList	Complex	SubscriptionRequest	Required infinite list	The ISO countries list to be excluded for domain control Must be defined in Country Table
38	ExcludedCountry	C(3)	ExcludedCountryList	Required (max : 1)	The Excluded Country Must be defined in Country Table
-	WithdrawalNational Limit	Complex	SubscriptionRequest	Required (max : 1)	This aggregate holds the national withdrawal limit
24	Amt	Long	WithdrawalNational Limit	Required (max : 1)	The Amount
25	CurCode	C(3)	WithdrawalNational Limit	Required (max : 1)	The Currency Code Must be defined in CURRENCY _TABLE
-	Withdrawal InterationalLimi	Complex	SubscriptionRequest	Required (max : 1)	This aggregate holds the International withdrawal limit
26	Amt	Long	Withdrawal InterationalLimi	Required (max : 1)	The Amount
27	CurCode	C(3)	Withdrawal InterationalLimi	Required (max : 1)	The Currency Code Must be defined in CURRENCY_TABLE
-	PurchaseNationalLimi	Complex	SubscriptionRequest	Required (max : 1)	This aggregate holds the national purchase limit
28	Amt	Long	Purchase NationalLimi	Required (max : 1)	The Amount
29	CurCode	C(3)	Purchase NationalLimi	Required (max : 1)	The Currency Code Must be defined in CURRENCY _TABLE
-	Purchase InterationalLimi	Complex	SubscriptionRequest	Required (max : 1)	This aggregate holds the International withdrawal limit
30	Amt	Long	Purchase InterationalLimi	Required (max : 1)	Amount

#	Field Name	Data type	Parent	Occurrence	Field Description
31	CurCode	C(3)	Purchase InterationalLimi	Required (max : 1)	The Currency Code Must be defined in CURRENCY _TABLE
	RequestStatus	C(1)	-	Required (max : 1)	This is the status of the subscription request that has been initiated by the corporate admin. - S: Submitted not yet accepted by the employee. - O: Ongoing by the employee and not yet completed by him/her - R: Rejected by the employee. - C: Completed by the employee.
	Generated CardNumber	C(22)	-	Required (max : 1)	The generated PAN number in when the status of the request status is C and Card createion status is A
	Generated AccountNumber	C(24)	-	Required (max : 1)	The generated financial account number in when the status of the request status is C and Card createion status is A
	CardCreationStatus	ClosedEnum	-	Required (max : 1)	The Card cration status. Possible values are the following : P: Pending issuer approval. R: Rejected by the issuer. A: Approved by the issuer.

Specific error codes

#	Error code	Error Description
1	31201	Invalid Corporate ID

#	Error code	Error Description
2	31202	Invalid Employee ID
3	31203	Invalid Subscription ID

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetCollaboratorInitiatedSubscriptionWebService/V2	POST	
/Colaborator/V2/InitSubsc?corporateld={corporateld}&employeeId={employeeId}&subscriptionID={subscriptionID}	GET	Required : Corporateld=[C(15)] EmployeeId=[C(15)] SubscriptionID=[C(22)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "corporateld": "",
  "employeeId": "",
  "subscriptionID": ""
}
```

}

4.69 getEventNotificationList API

API overview

This web service is used to get the list of notification events, and their channels along with the activation status for each event and channel preferences. An event might be active or inactive. An event might have multiple channels (Email, SMS, ...). Channel might have permanent and temporary preferences.

Functional description



Figure 69: getEventNotificationList process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique

#	Field Name	Data type	Parent	Occurrence	Field Description
					identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80 -14BD-4E04 -8842-7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	EntityCode	C(2)	-	Required (max: 1)	Indicate the entity code to which the communication strategy is linked. The possible values are: • 'CL': Client • 'CH': PAN • 'SA': Financial account
5	EntityID	C(15)	-	Required (max : 1)	Indicate the entity identification to which the communication strategy is linked. The possible values are: • Should provide the client code where entity code is • Should provide PAN when entity Code is • Should provide the financial

#	Field Name	Data type	Parent	Occurrence	Field Description
					account where the entity code is SA.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	Communication Events	Complex	-	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
4	Communication strategy	C(6)	Communication Events	Required (max: 1)	This is the communication event strategy id that groups all communication events. The list of strategies is defined in the following table: COM_STRATEG
4	EventCode	C(4)	Communication Events	Required (max: 1)	This is the event code that can be triggered by the system. The list of event codes is defined in the following table: EVENT_TABLE
5	EventActivationStatus	C(1)	Communication Events	Required (max: 1)	This field defines if the event is enabled or disable. - Y: Event active - N: Event disabled
6	IsEventMandatory	C(1)	Communication Events	Required (max: 1)	This field indicates if the event is defined as mandatory:

#	Field Name	Data type	Parent	Occurrence	Field Description
					- Y: event is mandatory and can not be disabled. - N: Event is optional and can be disabled. Rule: Events that are mandatory should not be disabled.
-	Communication Channels	Complex	Communication Events	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
7	ChannelType	C(2)	Communication Channels	Required (max: 1)	This field indicates the channel type that is defined for the a particular event: - Value 'SM': SMS channel - Value 'EM': Email channel - Value 'PH': Physical file channel - Value 'MB': Mobile push channel
8	ChannelActivation Status	C(1)	Communication Channels	Required (max: 1)	This field indicate if the communication channel is enabled or disabled: - Y: channel active - N: channel disabled
9	IsChannelMandatory	C(1)	Communication Channels	Required (max: 1)	This field indicates if the channel is defined as mandatory: Y: channel is mandatory

#	Field Name	Data type	Parent	Occurrence	Field Description
					and can not be disabled. N: channel is optional and can be disabled. Rule: Channels that are mandatory should not be disabled.
10	AllowsCustom Preference	C(1)	Communication Channels	Required (max: 1)	This field indicates if the PANholder is allowed to define permanent communication channel phone number, or email or address (for channel types SMS/Push or Email or physical file respectively). - Y: The channel permanent preferences setup is allowed. - N: The channel permanent preferences setup is not allowed.
11	ChannelPhoneType	C(2)	Communication Channels	Optional (max: 1)	This field indicate the phone to be used for the channel. Possible values: - Phone1 - Phone2 - Phone3 - Phone4 This field is used only when the channel type is SMS.
12	ChannelAddress FromEntity	C(2)	Communication Channels	Optional (max: 1)	Indicate the entity code from which the address will be linked. The possible values are: 'CL': Client 'CH': PAN

#	Field Name	Data type	Parent	Occurrence	Field Description
					This field is used only when the channel type is Physical.
32	ChannelAddressType	C(2)	Communication Channels	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field is used only when the channel type is Physical.
32	ChannelEmail Address	C(1)	Communication Channels	Optional (max: 1)	This field indicate the email to be used for the channel. Possible values: - Email1 - Email2 This field is used only when the channel type is Email.
33	ChannelAllows TempPreference	C(1)	Communication Channels	Required (max: 1)	This field indicate if the communication channel is allows defining temporary preferences: - Y: Temporary preferences allowed - N: Temporary preferences not allowed
34	TemporaryCustom PreferenceSet	C(1)	Communication Channels	Required (max: 1)	This field indicates if temporary preferences are set for the channel or not for the concerned entity: - Y: Temporary preferences activated at the entity for the channel - N: Temporary preferences not activated at the

#	Field Name	Data type	Parent	Occurrence	Field Description
					entity for the channel The setup for TMP fields should be allowed only for 'Y' values.
35	TMPPreferenceStartDate	Date	Communication Channels	Optional (max: 1)	Temporary channel preference start date.
36	TMPPreferenceEndDate	Date	Communication Channels	Optional (max: 1)	Temporary channel preference end date. If left blank the preference will be apply forever.
37	TMPPhoneType	C(2)	Communication Channels	Optional (max: 1)	This field indicates the temporary phone to be used for the channel. Possible values: - Phone1 - Phone2 This field is used only when the channel type is SMS.
38	TMPAddressFromEntity	C(2)	Communication Channels	Optional (max: 1)	Indicates the temporary entity code from which the address will be linked. The possible values are: 'CL': Client 'CH': PAN This field is used only when the channel type is Physical.
-	TMPAddressType	C(2)	Communication Channels	Optional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field is used only when the channel type is Physical.

#	Field Name	Data type	Parent	Occurrence	Field Description
24	TMPEmailAddress	C(2)	Communication Channels	Optional (max: 1)	This field indicate the temporary email to be used for the channel. Possible values: - Email1 - Email2 This field is used only when the channel type is Email.

Specific error codes

#	Error code	Error Description
1	31301	Invalid Entity Code
2	31302	Invalid Client Code
	31303	Invalid Card Number/ Token Id
	31304	Invalid Account Id

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/GetEventNotificationListWebService/V2	POST	
/EventNotifications/V2/ entityCode/:entityCode/ entityId/:entityId	GET	Required : entityCode =[C(2)] entityID =[C(15)]

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "entityCode" : "",
  "entityID" : ""
}
```

4.70 searchCollaborators API

API overview

This API is used to get the list of non client employees for a particular corporate.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

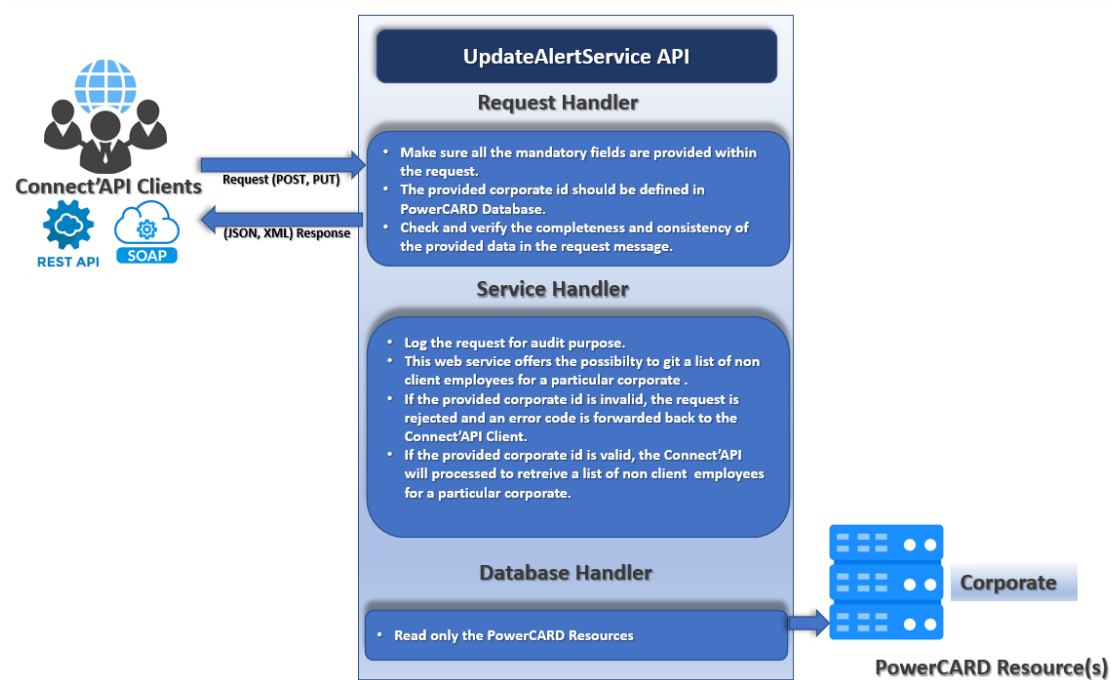


Figure 70: searchCollaborators process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	-	Required (max : 1)	The employee ID

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	EmployeeInformation	Complex	-	Optional infinite list	Employee information
5	CorporateId	C(15)	EmployeeInformation	Required (max : 1)	The organization/ corporate identifier. This code must be already defined in the PowerCARD CORPORATE table
6	EmployeeId	C(15)	EmployeeInformation	Required (max : 1)	The unique employee ID with the corporate.
7	EmployeeStatus	C (1)	EmployeeInformation	Required (max : 1)	The employee status
8	FirstName	C(250)	EmployeeInformation	Required (max : 1)	The first name of the employee
9	LastName	C(250)	EmployeeInformation	Required (max : 1)	The family name of the employee
10	Title code	C(2)	EmployeeInformation	Required (max : 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters section. It will be inherited from the Client if not provided at card level.
11	Email	C(64)	EmployeeInformation	Required (max : 1)	The email address of the employee
12	Language	C(3)	EmployeeInformation	Required (max : 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD

#	Field Name	Data type	Parent	Occurrence	Field Description
					LANGUAGE LIST table under the general parameters section. It will
13	VipLevel	C(1)	EmployeeInformation	Required (max : 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.

Specific error codes

#	Error code	Error Description
1	30101	Invalid Corporate ID
2	30102	Invalid Employee ID

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is		x

Status Code	Description	Success Response	Error Response
	probably valid but needs to be requested again later.		
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchCollaboratorsWebService/V2	POST	
/Colaborators/V2?corporateld={corporateld}&employeeId={employeeId}&pageSize={pageSize}&pageIndex={pageIndex}	GET	corporateld=[C(15)] employeeId=[C(15)]

REST sample call

```
{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "N"
    },
    {
      "key": "pageSize",
      "data": ""
    },
    {
      "key": "pageIndex",
      "data": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "corporateld": "",
  "employeeId": ""
}
```

4.71 ImportCollaborators API

API overview

This API is used to import the list of the employees for a particular corporate.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

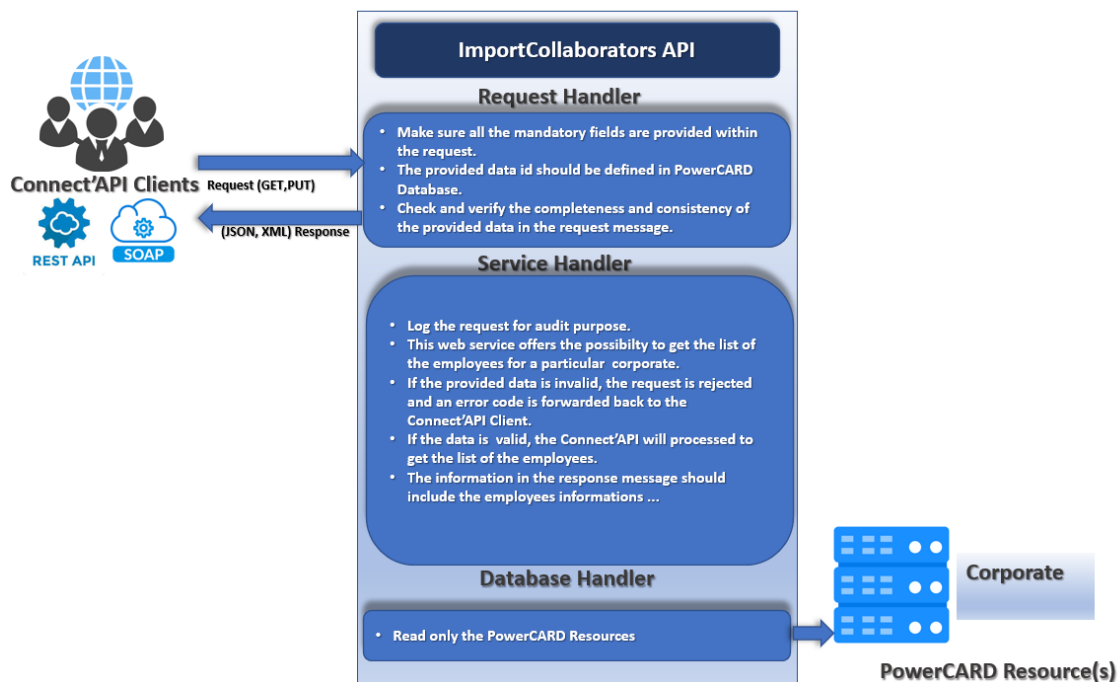


Figure 71: ImportCollaborators process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CorporateId	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
-	EmployeeInformation	Complex	-	Optional infinite list	Employee information
5	CorporateId	C(15)	EmployeeInformation	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
6	EmployeeId	C(15)	EmployeeInformation	Required (max : 1)	The unique employee ID with the corporate.
7	EmployeeStatus	C (1)	EmployeeInformation	Required (max : 1)	The employee status
8	FirstName	C(250)	EmployeeInformation	Required (max : 1)	The first name of the employee
9	LastName	C(250)	EmployeeInformation	Required (max : 1)	The family name of the employee
10	Title code	C(2)	EmployeeInformation	Required (max : 1)	Title code represents a prefix added to the customer's name. It may refer to social honorifics, professional designations, or academic ranks. This code must be previously defined in PowerCARD TITLE LIST table under the general parameters

#	Field Name	Data type	Parent	Occurrence	Field Description
					section. It will be inherited from the Client if not provided at card level.
11	Email	C(64)	EmployeeInformation	Required (max : 1)	The email address of the employee
12	LanguageCode	C(3)	EmployeeInformation	Required (max : 1)	This is the cardholder's language code according to the ISO 639-1 specification. This code should be previously defined in PowerCARD LANGUAGE LIST table under the general parameters section. It will
13	VipLevel	C(1)	EmployeeInformation	Required (max : 1)	This code represents the level of importance associated with the cardholder's professional or social status. It is usually used to grant special privileges to the cardholder. The VIP level should be previously defined in PowerCARD VIP LIST table under the general parameter section. It will be inherited from the Client if not provided at card level.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same

#	Field Name	Data type	Parent	Occurrence	Field Description
					value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	30401	Invalid Employee ID
2	30402	Invalid First Name

#	Error code	Error Description
3	30403	Invalid Last Name
4	30404	Invalid Title Code
5	30405	Invalid Email
6	30406	Invalid Language
7	30407	Invalid Vip Level
8	30408	Invalid Parent Corporate Id
9	30409	Invalid Corporate Id
10	30410	Employee already exists
11	30411	Invalid Employee Status

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SearchCollaboratorsWebService/V2	POST	

REST sample call

```
{
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "corporateId": {
    "employeeId": "",
    "firstName": "",
    "lastName": "",
    "titleCode": "",
    "email": "",
    "language": "",
    "vipLevel": ""
  }
}
```

4.72 updateEventNotification API

API overview

This web service is used activate/deactivate a communication event/channel as well as set communication event channel preferences.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

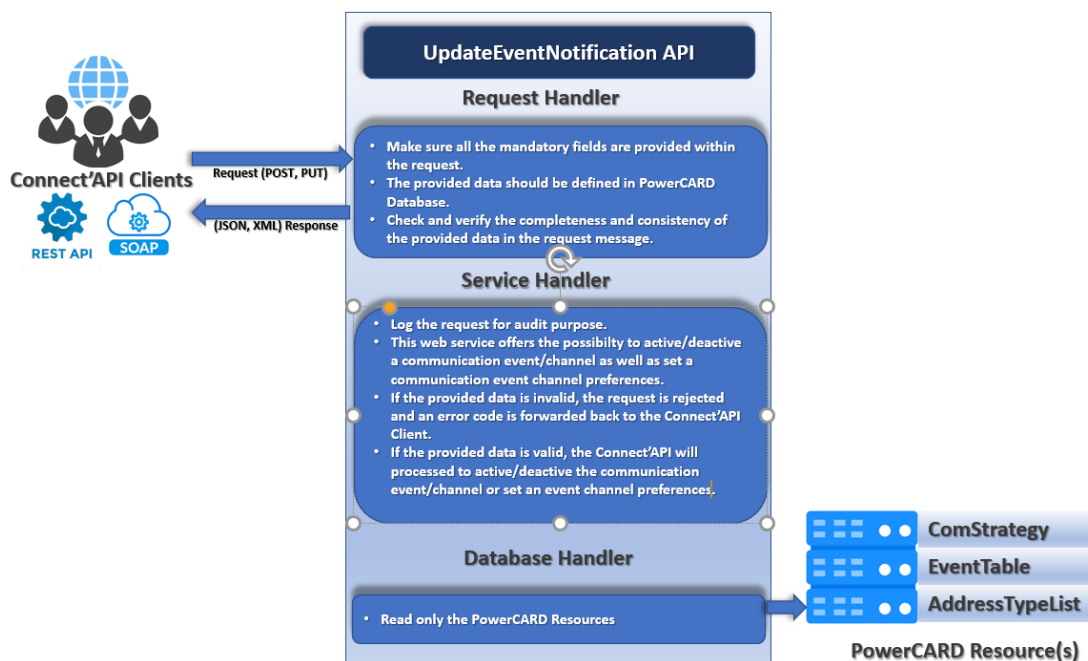


Figure 72: updateEventNotification process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	EntityCode	C(2)	-	Optional (max: 1)	Indicate the entity code to which the communication strategy is linked. The possible values are: • 'CL': Client • 'CH': PAN • 'SA': Financial account
5	EntityID	C(15)	-	Required (max : 1)	Indicate the entity identification to which the communication strategy is linked.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • Should provide the client code where entity code is • Should provide PAN when entity Code is • Should provide the financial account where the entity code is SA.
4	Action type	C(1)	-	Required (max : 1)	This field indicate the update action to be applied for an event or a channel. Possible values: • 1: Activate event • 2: Deactivate event • 3: Activate channel • 4: Deactivate channel • 5: Update permanent communication preference • 6: Set channel temporary preference • 7: Remove channel temporary preference Action 5 Update permanent preferences should be allowed only for channels that allows permanent preferences. Action 6 Update temporary preferences should be allowed only for channels that

#	Field Name	Data type	Parent	Occurrence	Field Description
					allows temporary preferences.
-	Communication Events	Complex	-	Optional infinite list	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels
4	Communication strategy	C(6)	Communication Events	Required (max: 1)	This is the communication event strategy id that groups all communication events. The list of strategies is defined in the following table: COM_STRATEGY
4	EventCode	C(4)	Communication Events	Required (max: 1)	This is the event code that can be triggered by the system. The list of event codes is defined in the following table: EVENT_TABLE
5	EventActivationStatus	C(1)	Communication Events	Conditional (max: 1)	This field defines if the event is enabled or disable. - Y: Event active - N: Event disabled Mandatory when action is: 1: Activate event 2: Deactivate event From end dev rule: Deactivation should be allowed only for events that are not mandatory based on result from

#	Field Name	Data type	Parent	Occurrence	Field Description
					getEventNotificationList services
-	Communication Channels	Complex	Communication Events	Conditional (max: 1)	To list all the communication event defined for the institution. (card delivery, pin delivery, statement....). Each event might be linked to 1 or many communication channels Mandatory when action is: • 3: Activate channel • 4: Deactivate channel • 5: Update permanent communication preference • 6: Set channel temporary preference • 7: Remove channel temporary preference
7	ChannelType	C(2)	Communication Channels	Conditional (max: 1)	This field indicates the channel type that is defined for the a particular event: • Value 'SM': SMS channel • Value 'EM': Email channel • Value 'PH': Physical file channel • Value 'MB': Mobile push channel Mandatory when action is: • 3: Activate channel • 4: Deactivate channel

#	Field Name	Data type	Parent	Occurrence	Field Description
					5: Update permanent communication preference 6: Set channel temporary preference 7: Remove channel temporary preference
8	Channel ActivationStatus	C(1)	Communication Channels	Conditional (max: 1)	This field indicate if the communication channel is enabled or disabled: - Y: channel active - N: channel disabled Mandatory when action is: 3: Activate channel 4: Deactivate channel From end dev Rule: Deactivation should be allowed only for channels that are not mandatory based on result from getEventNotificationList services
11	ChannelPhoneType	C(2)	Communication Channels	Conditional (max: 1)	This field indicate the phone to be used for the channel. Possible values: - Phone1 - Phone2 - Phone3 - Phone4 This field is used only when the channel type is SMS. Mandatory when action is: 5: Update permanent

#	Field Name	Data type	Parent	Occurrence	Field Description
					communication preference And when channel type is: Value 1: SMS channel
12	ChannelAddressFromEntity	C(2)	Communication Channels	Conditional (max: 1)	Indicate the entity code from which the address will be linked. The possible values are: • 'CL': Client • 'CH': PAN This field is used only when the channel type is Physical. Mandatory when action is: • 5: Update permanent communication preference And when channel type is: Value 3: Physical file channel
32	ChannelAddressType	C(2)	Communication Channels	Conditional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field is used only when the channel type is Physical. Mandatory when action is: • 5: Update permanent communication preference And when channel type is: Value 3: Physical file channel

#	Field Name	Data type	Parent	Occurrence	Field Description
32	ChannelEmail Address	C(1)	Communication Channels	Conditional (max: 1)	This field indicate the email to be used for the channel. Possible values: - Email1 - Email2 This field is used only when the channel type is Email. Mandatory when action is: 5: Update permanent communication preference And when channel type is: Value 2: Email channel
35	TMPPreference StartDate	Date	Communication Channels	Conditional (max: 1)	Temporary channel preference start date. Mandatory when action is: 6: Set channel temporary preference
36	TMPPreference EndDate	Date	Communication Channels	Optional (max: 1)	Temporary channel preference end date. If left blank the preference will be apply forever.
37	TMPPhoneType	C(2)	Communication Channels	Conditional (max: 1)	This field indicates the temporary phone to be used for the channel. Possible values: - Phone1 - Phone2 - Phone3 - Phone4 This field is used only when the channel type is SMS. Mandatory when action is:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6: Set channel temporary preference And when channel type is: Value 1: SMS channel
38	TMPAddressFromEntity	C(2)	Communication Channels	Conditional (max: 1)	Indicates the temporary entity code from which the address will be linked. The possible values are: 'CL': Client 'CH': PAN This field is used only when the channel type is Physical. Mandatory when action is: 6: Set channel temporary preference And when channel type is: Value 3: Physical file channel
-	TMPAddressType	C(2)	Communication Channels	Conditional (max: 1)	This is a 2-digit code that must be defined in the PowerCARD ADDRESSES TYPE LIST table under the general parameters section. This field is used only when the channel type is Physical. Mandatory when action is: 6: Set channel temporary preference And when channel type is: Value 3: Physical file channel

#	Field Name	Data type	Parent	Occurrence	Field Description
24	TMPEmailAddress	C(2)	Communication Channels	Conditional (max: 1)	This field indicate the temporary email to be used for the channel. Possible values: - Email1 - Email2 This field is used only when the channel type is Email. Mandatory when action is: 6: Set channel temporary preference And when channel type is: Value 2: Email channel

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWith Success: Call was ended successfully. - ProceedWith SuccessMC:

#	Field Name	Data type	Parent	Occurrence	Field Description
					Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	31401	Communication strategy
2	31402	Invalid Event Code
3	31403	Invalid Event Activation Status
4	31404	Invalid Channel Type
5	31405	Invalid Channel Activation Status
6	31406	Invalid Channel PhoneType
7	31407	Invalid Channel Address From Entity
8	31409	Invalid Channel Address Type
9	31409	Invalid Channel Email Address
10	31410	Invalid TMP Preference Start Date
11	31411	Invalid TMP Prefernce End Date
12	31412	Invalid TMP Phone Type
13	31413	Invalid TMP Address From Entity

#	Error code	Error Description
14	31414	Invalid TMP Address Type
15	31415	Invalid TMP Email Address
16	31416	Invalid Entity Code
17	31417	Invalid Entity ID
18	31418	Invalid Action Type

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateEventNotificationWebService/v2	POST	
/EventNotification/v2/ entityCode/:entityCode / entityId/:entityId/ actionType/:actionType? communicationStrategy	PATCH	Required communicationStrategy=[C(24)] eventCode=[C(24)] eventActivationStatus=[C(24)]

PowerCARD Resource URI	Method	URI Params
={communicationStrategy}& eventCode ={ eventCode }& communicationStrategy ={ eventActivationStatus }& eventActivationStatus ={ channelType }& channelType ={communicationStrategy}& channelActivationStatus ={ channelActivationStatus }& channelPhoneType ={ channelPhoneType }& channelAddressFromEntity ={ channelAddressFromEntity }& channelAddressType ={ channelAddressType }& channelEmailAddress ={ channelEmailAddress }& tmppreferenceStartDate ={ tmppreferenceStartDate }& tmpprefernceEndDate ={ tmpprefernceEndDate }& tmpphoneType ={ tmpphoneType }& tmpaddressFromEntity ={ tmpaddressFromEntity }& tmpphoneType ={ tmpphoneType }& tmpaddressType ={ tmpaddressType }& tmpemailAddress ={ tmpemailAddress }		channelType=[C(24)] channelActivationStatus=[C(24)] channelPhoneType=[C(24)] channelAddressFromEntity=[C(24)] channelAddressType=[C(24)] channelEmailAddress=[C(24)] tmppreferenceStartDate=[C(24)] tmpprefernceEndDate=[C(24)] tmpphoneType=[C(24)] tmpaddressFromEntity=[C(24)] tmpaddressType=[C(24)] tmpemailAddress=[C(24)] entityCode=[C(2)] entityID=[C(15)] actionType=[C(1)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"entityCode" : "",
"entityCode" : "",
"action type": "",
"communicationEvents": {
  "communication strategy": "",
  "eventCode": "",
  "eventActivationStatus": "",
  "communicationChannels": {
    "channelType": "",
    "channelActivationStatus": "",
    "channelPhoneType": "",
    "channelAddressFromEntity": "",
    "channelAddressType": "",
    "channelEmailAddress": "",
    "tmppreferenceStartDate": "",
    "tmpprefernceEndDate": "",
    "tmpphoneType": "",
    "tmpaddressFromEntity": "",
    "tmpaddressType": "",
    "tmpemailAddress": ""
  }
}

```

```

}
}
}

```

4.73 updateCollaboratorSubscriptionStatus API

API overview

This web service is used to update the status of already initiated subscription requests

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

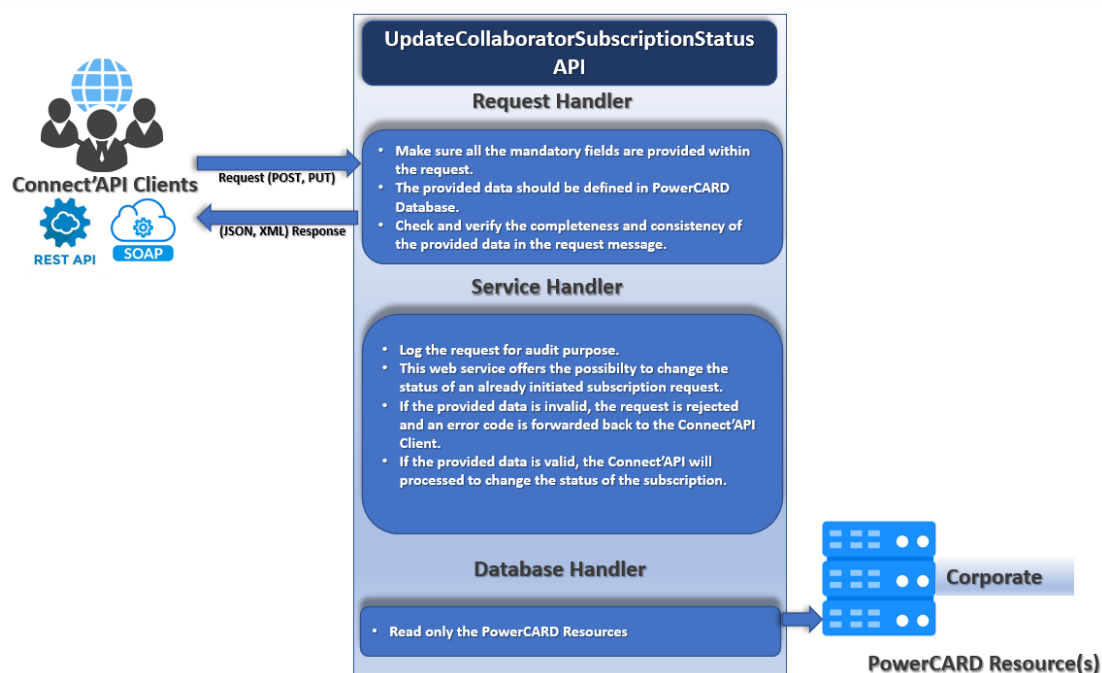


Figure 73: updateCollaboratorSubscriptionStatus process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique

#	Field Name	Data type	Parent	Occurrence	Field Description
					identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXX Example: 6948DF80 -14BD-4E04 -8842-7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	EntityCode	C(2)	-	Optional (max: 1)	Indicate the entity code to which the communication strategy is linked. The possible values are: • 'CL': Client • 'CH': PAN • 'SA': Financial account
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	-	Required (max : 1)	The employee ID
6	SubscriptionID	C(22)	-	Required (max: 1)	The generated subscription ID submitted by corporate portal contact person
7	RequestStatus	C(1)	-	Required (max : 1)	This is the status of the subscription request that has

#	Field Name	Data type	Parent	Occurrence	Field Description
					been initiated by the corporate admin. • O: Ongoing by the employee and not yet completed by him/her. This new status is allowed only in case of request with status 'S' Submitted. • R: Rejected by the employee. This new status is allowed only in case of request with status 'S' Submitted. • C: Completed by the employee. This new status is allowed only in case of request with status 'O' Ongoing.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: • ProceedWithSuccess: Call was ended successfully. • ProceedWithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	31501	Invalid Corporate Id
2	31502	Invalid Employee Id
3	31503	Invalid Subscription Id
4	31504	Invalid Request Status

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCollaboratorSubscriptionStatusWebService/v2	POST	
/Collaborators/v2/SubscriptionStatus?: corporateld=corporateld /: employeeld / employeeld /: subscriptionID? subscriptionID={ requestStatus }	PATCH	Required corporateld=[C(15)] employeeld=[C(15)] subscriptionID=[C(22)] requestStatus=[C(1)]

REST sample call

```
"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
```

```

    "userID": ""
  },
  "entityCode" : "",
  "corporateId" : "",
  "employeeId" : "",
  "subscriptionID" : "",
  "requestStatus" : "" }

```

4.74 initiateCollaboratorSubscription API

API overview

This web service is used to initiate an employee subscription from the corporate portal. The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

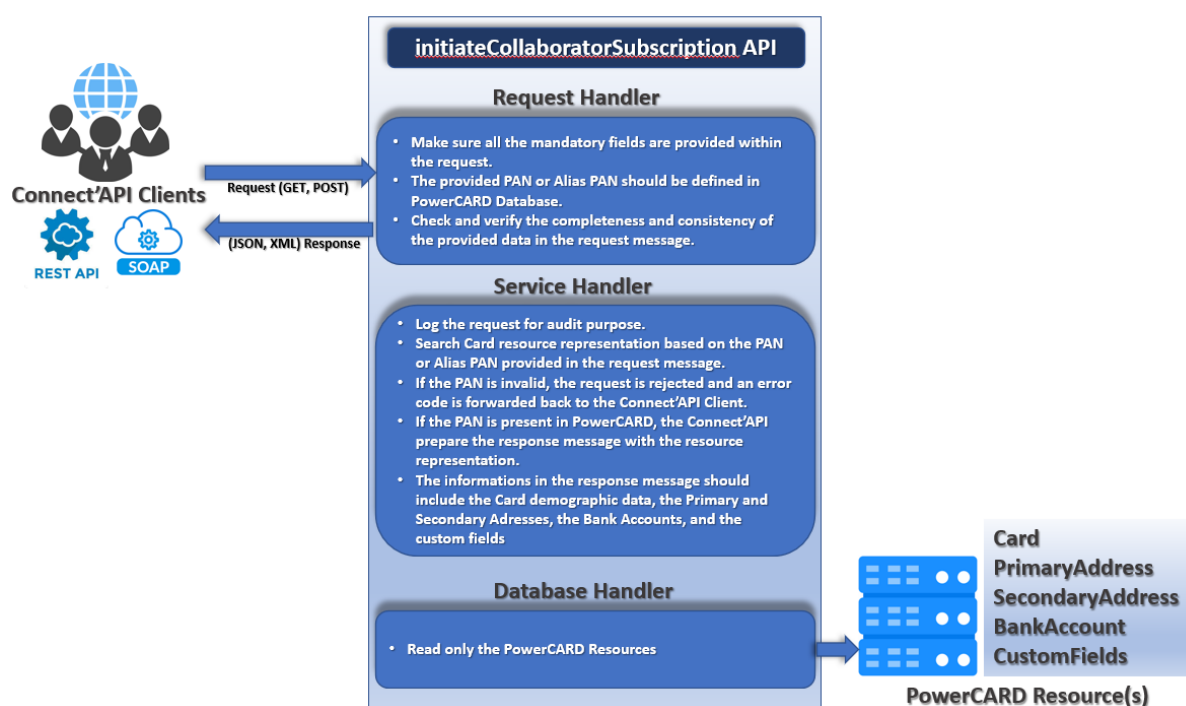


Figure 74: initiateCollaboratorSubscription process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request

#	Field Name	Data type	Parent	Occurrence	Field Description
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX-XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04-8842- 7668D9C001 F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	Corporateld	C(15)	-	Required (max : 1)	The organization/ corporate identifier. This code must be already defied in the PowerCARD CORPORATE table
5	EmployeeId	C(15)	-	Required (max : 1)	The employee ID
-	PANProductDetails	Complex	-	Required (max: 1)	The aggregate contains details of each subscription request.
6	CatPIProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the payment instrument product code assigned to the card. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					with product type = P
7	CatPI ProductVersion	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the payment instrument version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = P
8	PackCode	C (6)	-	Required (max : 1)	It is a 6 digit character field that identifies the selected product bundles. Must be a valid value defined in CATALOGUE _PACK
9	productCurrency	C	PANProductDetails	Required (max: 1)	Specifies the currency code of the cardholder's direct debit account (ISO 4217). It should be the ISO alphanumeric value (e.g. EUR for euro, USD for dollar). The ISO currency code must be previously defined in the PowerCARD CURRENCY TABLE under the general parameters section.
10	cycleCutoff	C(3)	PANProductDetails	Required (max : 1)	The account cycle cutoff assigned to the account: Must be defined in cycle_cutoff _parameter
-	Product service	Complex	PANProductDetails	Required (infinite)	This aggregate contains details of the servies to which the collaborator has applied
11	ServiceCode	C(6)	ProductServices	Required (max : 1)	Refers to list of services on card.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The possible values are: CTLESS: Contactless. You can directly use activateContactless API to enable and disable NFC Service STOPLST: Card stop listed You can directly use updateCard Status API to place the card in stop list RMSTOP: Remove card stop listed. You can directly use updateCard Status API to remove the card from the stop list queue BLCARD : Block/ Unblock the card You can directly use updateCard Status API to block or unblock the card. DOMSTP: Domain Setup Use the DomainControlSRV Addendum aggregate to perform the requested action VELSTP: Velocity setup Use the CardLimitSRV Addendum aggregate to perform the requested action DDDEFR: Direct debit deferment Use the Defferment Service Addendum Aggregate

#	Field Name	Data type	Parent	Occurrence	Field Description
					CORSOL: Corporate Solidarity SECECM: Secured ecommerce NSCECM: Not secured ecommerce
12	AccService	C(1)	-	Required (max : 1)	A Boolean indicator: - Y: ACS service is enabled. - N: ACS service is disabled.
32	OrderDate	Date	-	Required (max : 1)	The order Date
32	Company Soladarity	C(1)	-	Required (max : 1)	The Company solidarity flag
33	DifferedDebit	C(1)	-	Required (max : 1)	The Differed debit flag
34	Differed DaysNumber	Long	-	Required (max : 1)	Differed Days Number
-	SubscriptionRequest	Complex	Subscription Request	Required (max : 1)	The subscription request
-	Allowed MccList	Complex	Subscription Request	Required (max : 1)	The allowed MCC list
13	AllowedMcc	C(4)	Allowed MccList	Required infinite list	Allowed MCC Must be defined in MCC table
-	Excluded MccList	Complex	Subscription Request	Required infinite list	The MCCs list to be excluded for domain control.
14	Excluded Mcc	C(4)	Excluded MccList	Required infinite list	Excluded MCC Must be defined in MCC table
-	Allowed CountryList	Complex	Subscription Request	Required (max : 1)	The ISO countries list to be allowed for domain control
15	Allowed Country	C(3)	Allowed CountryList	Required infinite list	The Country Code Must be defined in Country Table
-	Excluded CountryList	Complex	Subscription Request	Required (max : 1)	The ISO countries list to be excluded for domain control Must be defined in Country Table
16	Excluded Country	C(3)	Excluded CountryList	Required infinite list	The Excluded Country

#	Field Name	Data type	Parent	Occurrence	Field Description
					Must be defined in Country Table
-	Withdrawal National Limi	Complex	-	Required (max : 1)	This aggregate holds the Nation withdrawal limit
24	Amt	Float	Withdrawal National Limi	Required (max : 1)	Amount
25	CurCode	C(3)	Withdrawal National Limi	Required (max : 1)	Currency Must be defined in CURRENCY_TABLE
-	Withdrawal Interational Limi	Complex	-	Required (max : 1)	This aggregate holds the International withdrawal limit
26	Amt	Float	Withdrawal Interational Limi	Required (max : 1)	Amount
27	CurCode	C(3)	Withdrawal Interational Limi	Required (max : 1)	Currency Must be defined in CURRENCY_TABLE
-	Purchase National Limi	Complex	-	Required (max : 1)	This aggregate holds the Nation purchase limit
28	Amt	Float	Purchase National Limi	Required (max : 1)	Amount
29	CurCode	C(3)	Purchase National Limi	Required (max : 1)	Currency Must be defined in CURRENCY_TABLE
-	Purchase Interational Limi	Complex	-	Required (max : 1)	This aggregate holds the International withdrawal limit
30	Amt	Float	Purchase Interational Limi	Required (max : 1)	Amount
12	ServiceStatus	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: Y: Service is enabled. N: Service is disabled.
13	IsService Mandatory	C(1)	ProductServices	Required (max : 1)	A Boolean indicator: Y: Service is mandatory and can not be disabled. N: Service is optional and can be disabled.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ServiceExpositionChannels	Complex	ProductServices	Optional (Infinite)	This is aggregate is used to list the service exposition channels along with channel activation status
14	ChannelType	C(3)	ServiceExposition Channels	Required (max : 1)	Refers to the channel type. The possible values are: BAG: Bank agent channel COW: Corporate portal channel CHM: Cardholder Mobile app CHW: Cardholder portal
15	ServiceToBe ExposedIn Channel	C(1)	Service Exposition Channels	Required (max : 1)	Y: Service can be visible in the channel. N: Service might be visible or invisible in the channel. In case visible, the service should be greyed out.
-	Defferment Service Addendum	Complex	ProductServices	Conditional (Infinite)	Mandatory if service code is DDDEFR
16	DefferedDays Number	Float	Defferment Service Addendum	Required (max : 1)	The number of days for debit deferement. The list of value must be defined in the table static_list with collection name = 'PaymentDD DeffermentDays'.
17	Deferment StartDate	Date	Defferment Service Addendum	Required (max : 1)	The start date of the deferment. Must be defaulted to current business date
18	Deferment Enddate	Date	Defferment Service Addendum	Optional (max : 1)	The date where the deferment will stop.
-	DomainControlSRV Addendum	Complex	ProductServices	Conditional infinite list	Mandatory if service code is

#	Field Name	Data type	Parent	Occurrence	Field Description
					DOMSTP
19	ControlType	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	The ControlType defines the type of control that is to be used throughout the Domain control. Valid Control Types are: CTR: Country MCC: MCC PRS: Processing code MER: Merchant CUR: Currency
20	ControlOperator	ClosedEnum	DomainControlSRV Addendum	Required (max : 1)	Symbol that you use to check or combine values. IN: Allows users to specify one or multiple values in ControlValueList NOT IN: To exclude a list of values from the ControlValueList
-	ControlValueList	Complex	DomainControlSRV Addendum	Optional infinite list	Contains various values, such as country codes, processing codes ...
21	ControlValue	C (15)	ControlValueList	Required (max : 1)	Users must ensure that this code is a valid value defined in the PowerCARD under the general parameter section: COUNTRY table for ControlType = 'CTR'. MCC table for ControlType = 'MCC'. PROCESSING_LIST table for ControlType = 'PRS'. MERCHANT table for ControlType = 'MER'.

#	Field Name	Data type	Parent	Occurrence	Field Description
					CURRENCY _TABLE table for ControlType = 'CUR'. If not, the request will be rejected.
-	CardLimitSRV Addendum	Colmplex	-	Conditional infinite list	Mandatory if service code is VELSTP
22	LimitID	C (4)	CardLimitSRV Addendum	Required (max: 1)	Card limit ID This code is used to identify the limit that will be assigned to the card.
23	LimitDescription	C (64)	CardLimitSRV Addendum	Optional (max : 1)	Description of the limit
24	LimitAbv Wording	C (32)	CardLimitSRV Addendum	Optional (max : 1)	A brief description of the limit
-	ValidityRangeDate	Complex	CardLimitSRV Addendum	Required (max: 1)	A limit can be temporary and restricted to a range date
25	StartDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity Start date Format: yyyy-MM- dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
26	EndDate	DateTime	ValidityRange Date	Optional (max: 1)	The Validity End date Format: yyyy-MM- dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
-	Period	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	The Period of limit that should be set on the card.
27	PeriodType	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: Fixed period - W: Rolling period
28	PeriodDayOff	Number (2)	Period	Optional (max : 1)	Period day off

#	Field Name	Data type	Parent	Occurrence	Field Description
29	PeriodicityFlag	C (1)	Period	Required (max : 1)	Must contain one of the following values: - D: for daily - W: for Weekly - M: for Monthly - Y: for Yearly
-	Territoriality Online	Colmplex	CardLimitSRV Addendum	Optional (max: 3)	The Territoriality Online
30	TerritorialityType	C (3)	Territoriality Online	Required (max: 1)	Valid values: ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
31	Transaction Amount	Decimal (18,3)	Territoriality Online	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
32	Transaction AmountMAX	Decimal (18,3)	Territoriality Online	Required (max : 1)	The max total amount can be negotiated for the selected period and territoriality
33	NumberOf Transactions	Number (4)	Territoriality Online	Required (max : 1)	Must contain the target the number of transactions of the limit.
34	NumberOf Transactions MAX	Number (4)	Territoriality Online	Required (max : 1)	The max number of transactions that can be negotiated for the selected period and territoriality
-	TerritorialityOffline	Colmplex	CardLimitSRV Addendum	Optional (max: 4)	The Territoriality Offline
35	TerritorialityType	C (3)	Territoriality Offline	Required	Valid values:

#	Field Name	Data type	Parent	Occurrence	Field Description
				(max: 1)	ONS: On-us, refers to the criteria of transactions that are conducted in a context where the acquirer is also the issuer. NAT: National, apply only to transactions made in the same country as the issuer country. INT: International, apply only to transactions that took place outside the cardholder country.
36	Transaction Amount	Decimal (18,3)	Territoriality Offline	Required (max : 1)	Must contain the target amount of the limit that should be set on the card.
37	NumberOf Transactions	Number (4)	Territoriality Offline	Required (max : 1)	Must contain the target the number of transactions of the limit.
38	TotalTransaction AmountOnline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
39	TotalNumberOfTransactionsOnline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
40	TotalTransaction AmountOffline	Decimal (18,3)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target amount of the limit that should be set on the card.
41	TotalNumberOf TransactionsOffline	Number (4)	CardLimitSRV Addendum	Optional (max : 1)	Must contain total of the target the number of transactions of the limit.
-	Transaction Criteria	Complex	CardLimitSRV Addendum	Optional (max : 1)	This aggregate is used to define the transaction criteria
42	SequenceNumber	Number (9)	Transaction Criteria	Optional (max : 1)	Should contain Sequence Number

#	Field Name	Data type	Parent	Occurrence	Field Description
43	ProcessingCode	C (2)	Transaction Criteria	Optional (max : 1)	Must contains a code that identifies the cardholder transaction type Any value in this field must be previously defined in the PowerCARD PROCESSING_LIST table under the general parameter section.
44	CountryCode	C (3)	Transaction Criteria	Optional (max : 1)	Country Code The country code is a numerical code identifying nations over the globe. This code should be present in the PowerCARD COUNTRY table under the general parameter section
45	MCC	C (4)	Transaction Criteria	Optional (max : 1)	The Merchant Category Code Distinct number used to classify the merchant business based on the type of services or products it provides. The valid MCC's are stored in the PowerCARD MCC table
46	PaymentType	C (2)	Transaction Criteria	Optional (max : 1)	Should contain the payment type
47	CurrencyCode	C (3)	Transaction Criteria	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
48	MaxAmount PerTrx	Long	Transaction Criteria	Optional (max : 1)	Used to set the maximum amount per transaction.

#	Field Name	Data type	Parent	Occurrence	Field Description
49	CardEntryMode	ClosedEnum	Transaction Criteria	Optional (max : 1)	Field indicating the method by which a point-of-transaction terminal obtains and transmits the cardholder information necessary to complete a transaction Valid values are : • 00: Unknown or terminal not used • 01: Manual (key entry) • 02: Magnetic stripe read • 03: Integrated circuit card read • 04: Contactless device-read

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully.

#	Field Name	Data type	Parent	Occurrence	Field Description
					- ProceedWithSuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	SubscriptionId	C (22)	ResponseInfo	Optional (max : 1)	The subscriptionId is a unique alphanumeric string that identifies your subscription.

Specific error codes

#	Error code	Error Description
1	31601	Invalid CorporateId
2	31602	Invalid Employee Id
3	31603	Invalid Catalog Product
4	31604	Invalid Catalog Product Version
5	31605	Invalid Pack Code
6	31606	Invalid Product Currency
7	31607	Invalid Cycle Cutoff
8	31608	Invalid Service Code

#	Error code	Error Description
9	31609	Invalid Service Status
10	31610	Invalid Service Mandatory Indicator
11	31611	Invalid Channel Type
12	31612	Invalid ServiceTo BeExposed In Channel
13	31613	Invalid Deferred Days Number
14	31614	Invalid Deferment Start Date
15	31615	Invalid Deferment End Date
16	31616	Invalid Control Type
17	31617	Invalid Control Operator
18	31618	Invalid Control ValueList
19	31619	Invalid Control Value
20	31620	Invalid Card Limit SRV Addendum
21	31621	Invalid Limit ID
22	31622	Invalid Limit Description
23	31623	Invalid Limit Abrv Wording
24	31624	Invalid Validity Range Date
25	31625	Invalid Start Date
26	31626	Invalid End Date
27	31627	Invalid Period Type
28	31628	Invalid Period Day Off
29	31629	Invalid Periodicity Flag
30	31630	Invalid Territoriality Type
31	31631	Invalid Transaction Amount
32	31632	Invalid Transaction Amount
33	31633	Invalid Number Of Transactions
34	31634	Invalid Number Of Transactions
35	31635	Invalid Territoriality Type
36	31636	Invalid Total Transaction Amount Online
37	31637	Invalid Total Number Of Transactions Online
38	31638	Invalid Total Transaction Amount Offline
39	31639	Invalid Total Number Of Transactions Offline
40	31640	Invalid Sequence Number
41	31641	Invalid Processing Code

#	Error code	Error Description
42	31642	Invalid Country Code
43	31643	Invalid MCC
44	31644	Invalid Payment Type
45	31645	Invalid Currency Code
46	31646	Invalid Max Amount Per Trx
47	31647	Invalid Card Entry Mode

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/UpdateCollaboratorSubscriptionStatusWebService/v2	POST	
/Collaborators/v2/SubscriptionStatus?: corporateld=corporateld /: employeeld / employeeld /: subscriptionID? subscriptionID={ requestStatus }	PATCH	Required corporateld=[C(15)] employeeld=[C(15)] subscriptionID=[C(22)]

PowerCARD Resource URI	Method	URI Params
		requestStatus=[C(1)]

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"entityCode": "",
"corporateId": "",
"employeeId": "",
"subscriptionID": "",
"requestStatus": "" }

```

4.75 loanReschedule API

Overview

This web service is used to restructuring the terms of an existing loan. The loan can be rescheduled to reduce payment amounts by extending the payment period and increasing the number of payments. The rescheduling is achieved by changing the product of a particular loan.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

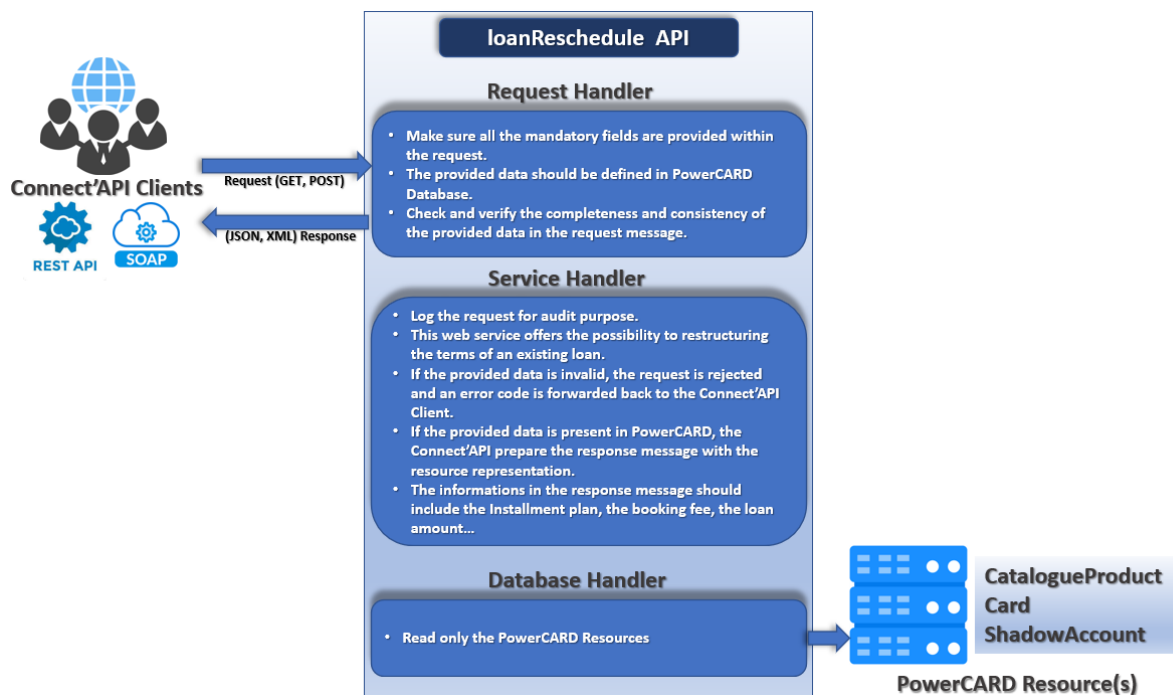


Figure 75: loanReschedule process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX-XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842- 7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	CardNumber	C (22)	-	Optional (max : 1)	Refers to the PowerCARD card number. This value corresponds to the card to which the loan is posted.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Required if AccountId is empty.
5	AccountId	C (24)	-	Optional (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the account to which the loan is posted. Required if CardNumber is empty.
6	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
7	RefSeq	Long (4)	-	Required (max : 1)	Refers to the transaction sequence Number
8	Catalog LoanProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loan product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
	Catalog Loan Product Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
	Acronym	C (128)		Optional (max : 1)	Loan acronym

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	LoanStart Date	DateTime	-	Required (max : 1)	Loan start date. Date of generation of the first installment in PowerCARD
6	LoanEndDate	DateTime	-	Required (max : 1)	Loan end date. Date of generation of the last installment in PowerCARD
7	NbrOfTerms	Long	-	Required (max : 1)	Number terms due. The total number of installment for that loan
8	NbrOfTerms Due	N (4)	-	Required (max : 1)	Number terms due. Remaining number of installment still to be generated.
9	InterestRate	Long	-	Required (max : 1)	Interest rate applied for the loan
-	LoanDue	Complex	-	Required (max : 1)	Loan due. The total loan amount not yet requested from the customer to be paid
10	Amt	Long	LoanDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
11	CurCode	C (3)	LoanDue	Optional (max : 1)	Currency code.

#	Field Name	Data type	Parent	Occurrence	Field Description
					ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	LoanAmt	Complex	-	Required (max : 1)	Transaction amount. Total amount and currency of the initial transaction
12	Amt	Long	LoanAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
13	CurCode	C (3)	LoanAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
14	CustomerId	C (22)	-	Required (max : 1)	Person identifier Refers to the unique customer identification in the system.
15	Acronym	C (40)	-	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or

#	Field Name	Data type	Parent	Occurrence	Field Description
					member's imprinter plate.
16	Loan Financing Type	C (1)	-	Required (max : 1)	I : Financed by credit limit E: Financed by additional loan limit
17	LoanCatalog Name	C (50)	-	Optional (max : 1)	Loan catalog product name.
-	BookingFee	Complex	-	Infinity Optional (max : 1)	Loan booking fee
8	TCCode	C (2)	BookingFee	Required (max : 1)	Booking fee Transaction code
8	TCDesc	C (32)	BookingFee	Required (max : 1)	Booking fee Transaction description
-	FeeAmount	Complex	BookingFee	Required (max : 1)	Loan Booking fee amount
18	Amt	Long	FeeAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
19	CurCode	C (3)	FeeAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Installment Plan	Complex	-	Optional (max : 1)	The installment Plan aggregate. For more details refers to the aggregate description.

#	Field Name	Data type	Parent	Occurrence	Field Description
20	Installment Number	C (23)	Installment Plan	Required (max : 1)	Sequence number for the installment plan. A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
-	FeesAmount	Complex	Installment Plan	Required (max : 1)	Fees amount aggregate.
21	Amt	Long	FeesAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
22	CurCode	C (3)	FeesAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Interest Amount	Complex	Installment Plan	Required (max : 1)	Interest amount aggregate.
23	Amt	Long	Interest Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
24	CurCode	C (3)	Interest Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	TaxAmount	Complex	Installment Plan	Required (max : 1)	Tax amount aggregate.
25	Amt	Long	TaxAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	C (3)	TaxAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Principal Amount	Complex	Installment Plan	Required (max : 1)	Principal amount aggregate.
27	Amt	Long	Principal Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
28	CurCode	C (3)	Principal Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Installment Amount	Complex	Installment Plan	Required (max : 1)	Installment Amount aggregate.
29	Amt	Long	Installment Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
30	CurCode	C (3)	Installment Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
31	Status	C (1)	Installment Plan	Required (max : 1)	B: Billed U: Unbilled
32	BillingDate	Date	Installment Plan	Required (max : 1)	Installment billing date.

Specific error codes

#	Error code	Error Description
1	41101	Invalid Card Number/Token Id
2	41102	Invalid Account Id
3	41103	Invalid Transaction Id

#	Error code	Error Description
4	41104	Invalid Reference Sequence
5	41105	Invalid Catalog Loan Product
6	41106	Invalid Catalog Loan Product Version

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/loanRescheduleWebService/v2	POST	

REST sample call

```
"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
```

```

    "userID": ""
  },
  "cardNumber" : "",
  "accountId" : "",
  "transId" : "",
  "refSeq" : "",
  "catalogLoanProduct" : "",
  "catalogLoanProduct Version" : "",
  "acronym" : ""
}

```

4.76 LoanPartialPayOff API

API overview

This web service is used when the customer wishes to perform a partial payoff of an existing loan. The Partial payoff amount must match with the sum of installments for the N targeted monthly installments.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

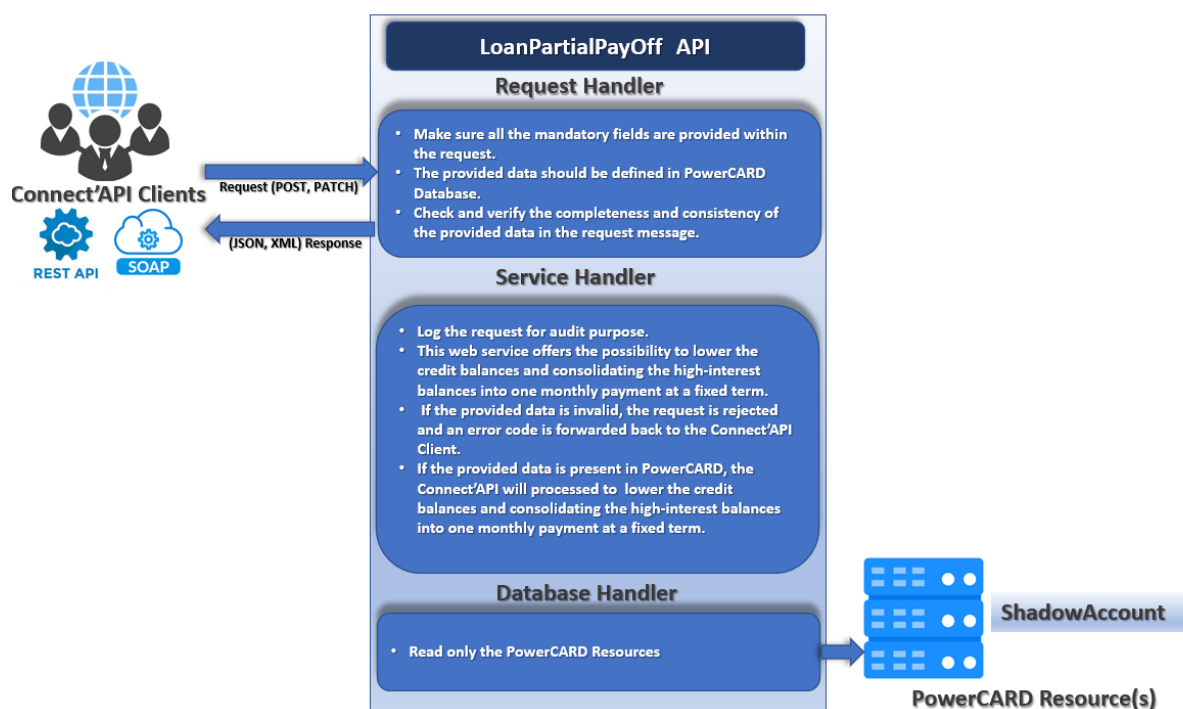


Figure 76: LoanPartialPayOff

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842- 7668D9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
5	Unpaid Installment	C(1)	-	Required (max : 1)	Include unpaid installment in the payoff amount. The possible values are: - Y: Yes - N: No
6	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The TransId is unique by transaction.
7	RefSeq	Decimal (4,0)	-	Required (max : 1)	Refers to the transaction sequence Number.
8	NbTerms	Decimal (4,0)	-	Optional (max : 1)	Number of term to be paid. Required if PayAmnt is empty.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing.

#	Field Name	Data type	Parent	Occurrence	Field Description
					SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	41201	Invalid Account Id
2	41202	Invalid Unpaid Installment
3	41203	Invalid Transaction Id
4	41204	Invalid Reference Sequence
5	41205	Invalid Number of Terms
6	41206	Invalid Partial Pay Off Amount
7	41207	Invalid Partial Pay Off Amount Currency

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x

Status Code	Description	Success Response	Error Response
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/loanPartialPayOffWebService/v2	POST	

REST sample call

```

{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "accountId": "",
  "unpaidInstallment": "",
  "transId": "",
  "refSeq": "",
  "nbTerms": "",
  "partialPayOffAmnt": {
    "amt": "",
    "curCode": ""
  }
}

```

4.77 LoanEarlyPayOff API

API overview

This web service is used when the customer wishes to perform a full payoff of one or more existing loans.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

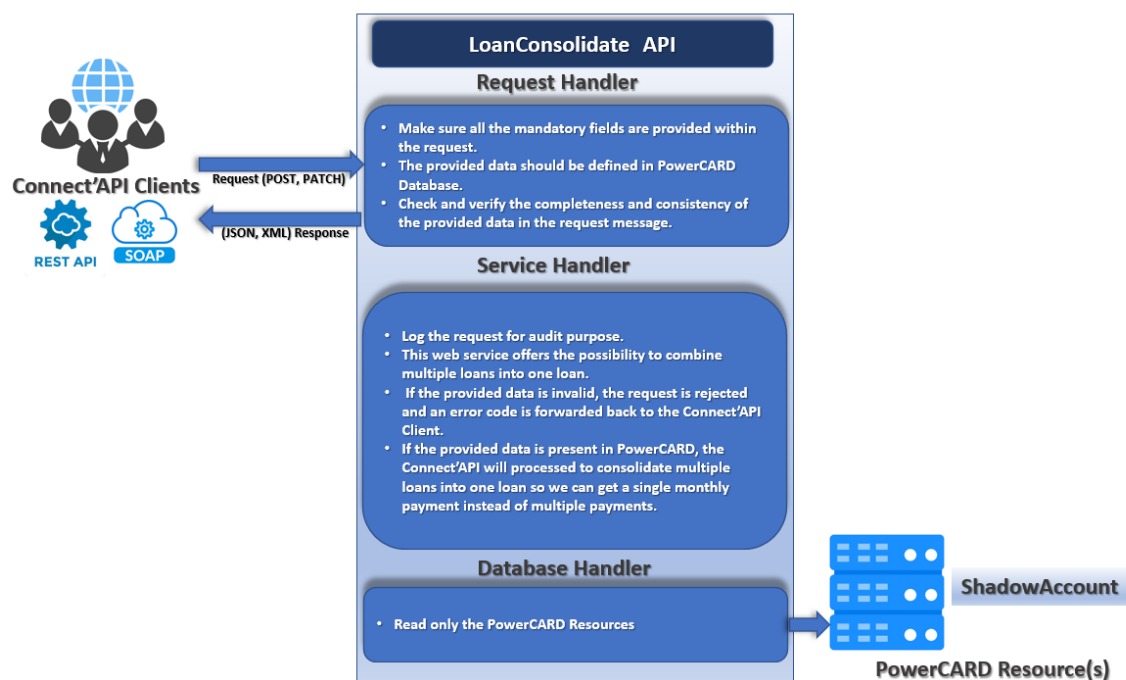


Figure 77: LoanEarlyPayOff process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXXX XXX -XXXX -XXXX -XXXX - XXXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
5	Unpaid Installment	C (1)	-	Required (max : 1)	Include unpaid installment in the payoff amount. The possible values are: - Y: Yes - N: No
-	LoanList	Complex	-	Required (Infinite list)	The Loan list aggregate. For more details refers to the aggregate description.
6	TransId	C (23)	LoanList	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.

#	Field Name	Data type	Parent	Occurrence	Field Description
7	RefSeq	Decimal (4,0)	LoanList	Required (max : 1)	Refers to the transaction sequence Number.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWith Success: Call was ended successfully. - ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	41301	Invalid Account Id
2	41302	Invalid Unpaid Installment
3	41303	Invalid Transaction Id
4	41304	Invalid Reference Sequence

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x

Status Code	Description	Success Response	Error Response
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/loanEarlyPayOffWebService/v2	POST	

REST sample call

```

{
  "keyValues": [
    {
      "key": "TOKEN",
      "data": "Y",
      "type": ""
    }
  ],
  "requestInfo": {
    "requestUID": "",
    "requestDate": "",
    "userID": ""
  },
  "accountId": "",
  "unpaidInstallment": "",
  "transId": "",
  "refSeq": "",
  "loanList": {
    "transId": "",
    "refSeq": ""
  }
}

```

4.78 LoanConsolidate API

API overview

This web service allows you to consolidate (combine) multiple loans into one loan. The result is a single monthly payment instead of multiple payments.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

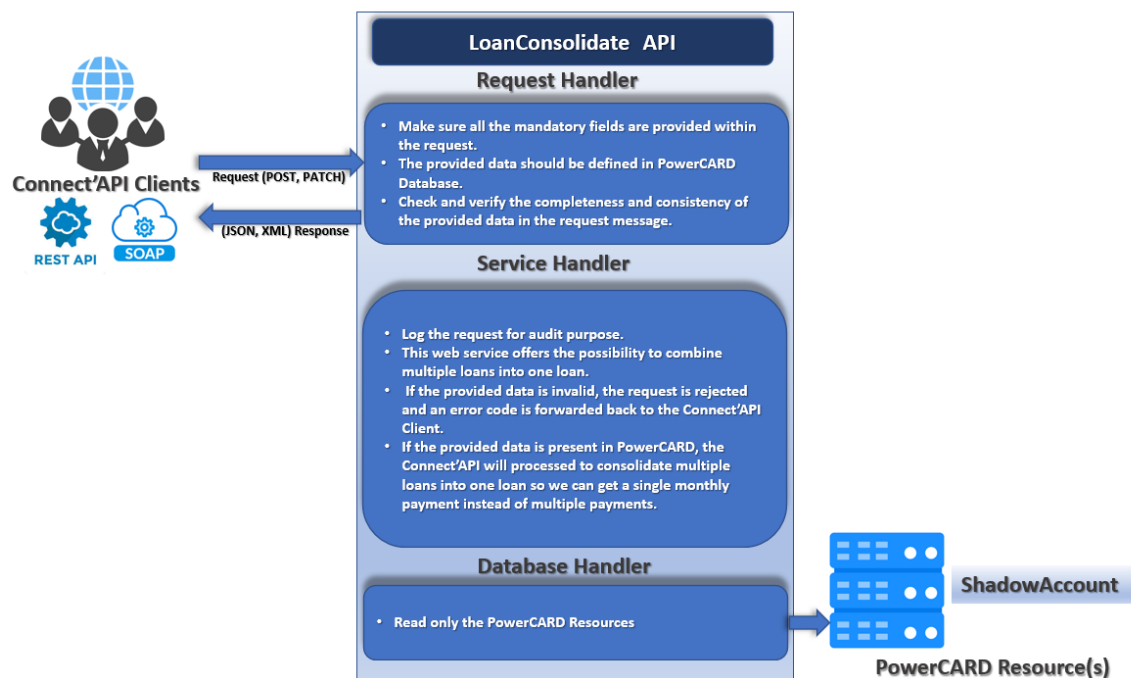


Figure 78: LoanConsolidate process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXXXX XXXXXXXX

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: 6948DF80 -14BD -4E04 -8842-7668D 9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the account to which the loan is posted.
5	Catalog LoanProduct	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loan product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
6	Catalog LoanProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
7	Acronym	C(128)		Optional (max : 1)	Loan acronym
-	Loan List	Complex	-	Required (Infinite List)	The Loan list aggregate. For more details refers to the aggregate description.

#	Field Name	Data type	Parent	Occurrence	Field Description
8	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
9	RefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWith Success: Call was ended successfully. - ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	Loan	Complex	-	Required (max : 1)	The loan aggregate. For more details refers to the aggregate description.
5	LoanRef Number	C (23)	Loan	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
6	LoanRefSeq	N (4)	Loan	Required (max : 1)	Refers to the transaction sequence Number
7	AccountId	C (24)	Loan	Required (max : 1)	The Account Number where the transaction is posted.
8	CatLoanProduct	C (6)	Loan	Optional (max : 1)	It is a 6 digit code identifying the loan product. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table

#	Field Name	Data type	Parent	Occurrence	Field Description
					with product type = N using the Web Service: getReferentialList
9	CatLoan Product Version	C (10)	Loan	Optional (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N using the Web Service: getReferentialList
10	LoanStatus	ClosedEnum	Loan	Required (max : 1)	Loan Status This field refers to the loan status as stored in the system. Possible values are as follow: • O: Occurred • R: Reversed • A: Amortized
11	LoanStart Date	DateTime	Loan	Required (max : 1)	Loan start date. Date of generation of the first installment in PowerCARD
12	LoanEndDate	DateTime	Loan	Required (max : 1)	Loan end date. Date of generation of the last installment in PowerCARD
13	NbrOfTerms	Long	Loan	Required (max : 1)	Number terms due. The total number of installment for that loan
14	NbrOfTerms Due	N (4)	Loan	Required (max : 1)	Number terms due. Remaining number of installment still to be generated.
15	InterestRate	Long	Loan	Required (max : 1)	Interest rate applied for the loan
-	LoanDue	Complex	Loan	Required (max : 1)	Loan due.

#	Field Name	Data type	Parent	Occurrence	Field Description
					The total loan amount not yet requested from the customer to be paid
16	Amt	Long ()	LoanDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
17	CurCode	NC (3)	LoanDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_ TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD using the Web Service: getReferentialList
-	LoanAmt	Complex	Loan	Required (max : 1)	Transaction amount. Total amount and currency of the initial transaction
18	Amt	Long ()	LoanAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.

#	Field Name	Data type	Parent	Occurrence	Field Description
19	CurCode	NC (3)	LoanAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_ TABLE table under the general parameters section. Valid values must be retrieved from PowerCARD using the Web Service: getReferentialList
20	CustomerId	C (22)	Loan	Required (max : 1)	Person identifier Refers to the unique customer identification in the system.
21	Acronym	C (40)	Loan	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name imprinted on the transaction receipt by the merchant's or member's imprinter plate.
22	Loan Financing Type	C (1)	Loan	Required (max : 1)	I : Financed by credit limit E: Financed by additional loan limit

Specific error codes

#	Error code	Error Description
1	41401	Invalid Account Id
2	41402	Invalid Catalog Loan Product
3	41403	Invalid Catalog Loan Product Version
4	41404	Invalid Transaction Id
5	41405	Invalid Reference Sequence

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	

Status Code	Description	Success Response	Error Response
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/LoanConsolidateWebService/v2	POST	

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"unpaidInstallment": "",
"catalogLoanProduct": "",
"catalogLoanProductVersion": "",
"acronym": "",
"loanList": {
  "transId": "",
  "refSeq": ""
}
}

```


4.79 LoanCancellation API

API overview

The loan cancellation web service is an API that can be used to force closure of a loan. The loan is identified by account number / card number , ARN number and sequence .

All billed installment/interest/fees will be cancelled however the unbilled installments will be deactivated and moved to the history and the Loan booking fee will be cancelled.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

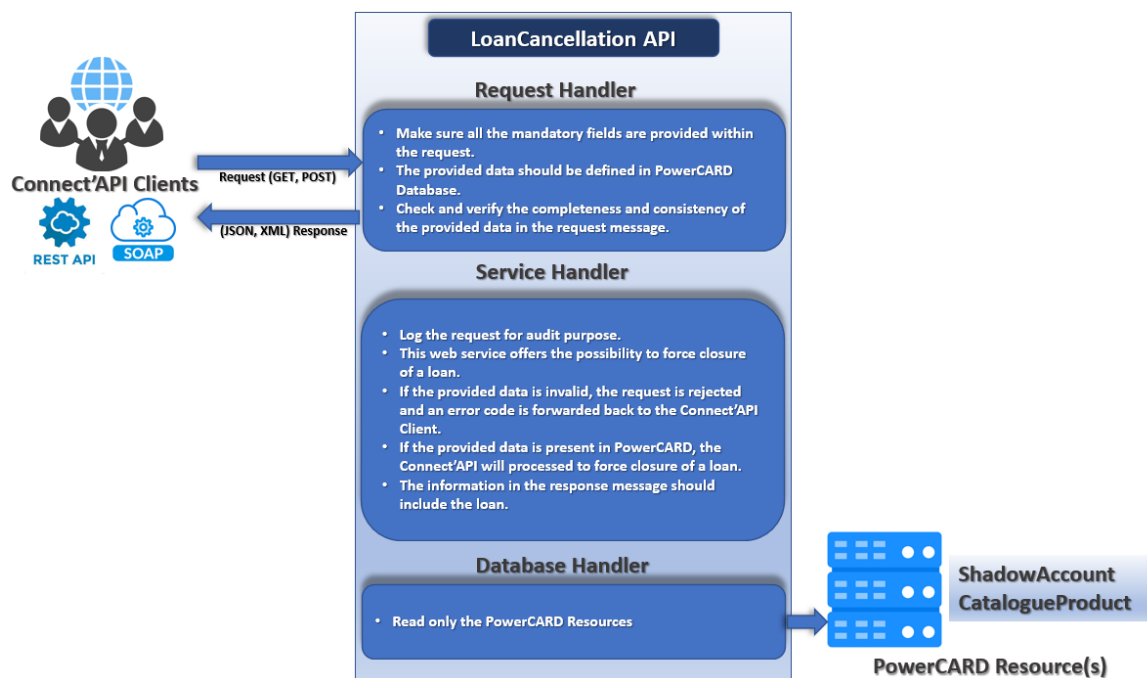


Figure 79: LoanCancellation process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique

#	Field Name	Data type	Parent	Occurrence	Field Description
					identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX -XXXX -XXXX -XXXX - XXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842-7668D 9C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the account to which the loan is posted.
5	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
6	RefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number
7	ReasonCode	C (2)	-	Required (max : 1)	Identifying the closure reason of the loan.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the

#	Field Name	Data type	Parent	Occurrence	Field Description
					processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

Specific error codes

#	Error code	Error Description
1	41501	Invalid Account Id
2	41502	Invalid Transaction Id
3	41503	Invalid Reference Sequence
4	41504	Invalid Reason Code

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/LoanConsolidateWebService/v2	POST	

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"reasonCode": "",
"transId": "",
"refSeq": ""
}

```

4.80 SimulateLoan API

API overview

This web service can be invoked before a loan booking. It returns a loan simulation that can be verified by the agent and agreed with the customer before booking a loan..

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

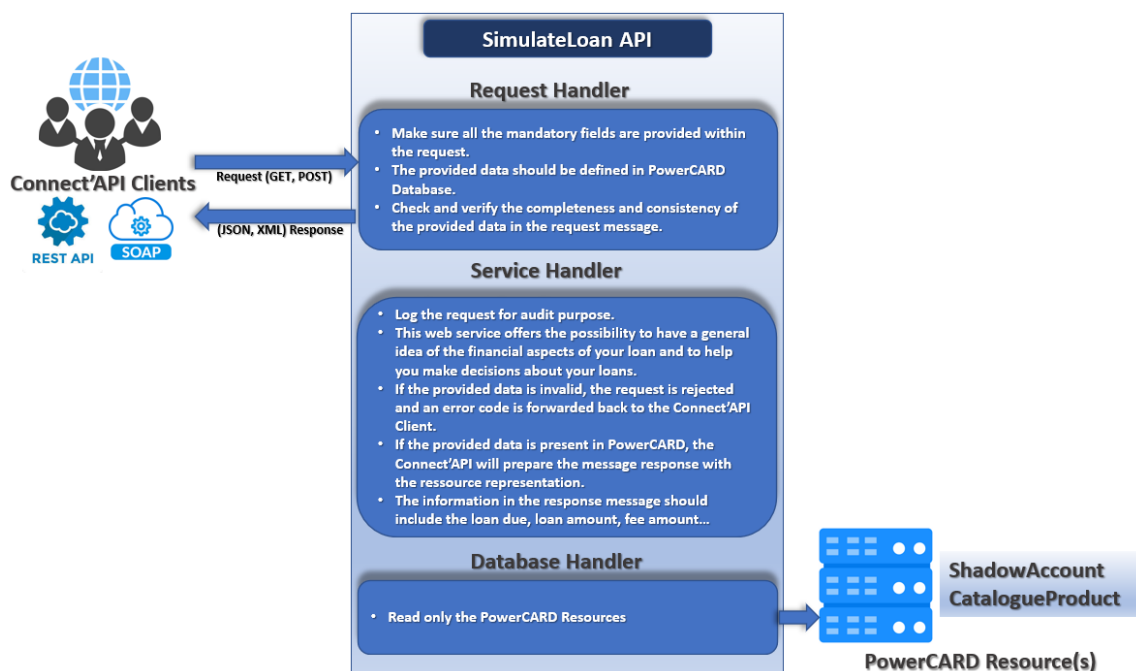


Figure 80: SimulateLoan process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX-XXXX -XXXX -XXXX - XXXXX XXXXXXXX Example: 6948DF80 -14BD -4E04 -8842 -7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Optional (max : 1)	Refers to the PowerCARD Shadow Account Id. This value corresponds to the account to which the loan is posted.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Required if CardNumber is empty.
-	LoanAmount	Complex	-	Required (max : 1)	The Loan amount and currency. Amount and currency of the loan that will be posted to the customer account.
5	Amt	Long	LoanAmount	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
6	CurCode	C (3)	LoanAmount	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
7	CatalogLoan Product	C (6)	-	Required (max : 1)	It is a 6 digit code identifying the loan product code. Valid values must be retrieved from PowerCARD CATALOGUE _PRODUCT table with product type = N
8	Catalog LoanProduct Version	C (10)	-	Required (max : 1)	It is a 10 character fields identifying the loan product version. Valid values must be retrieved from PowerCARD CATALOGUE

#	Field Name	Data type	Parent	Occurrence	Field Description
					_PRODUCT table with product type = N

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: - ProceedWith Success: Call was ended successfully. - ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. - Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the

#	Field Name	Data type	Parent	Occurrence	Field Description
					PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
5	LoanStartDate	DateTime	-	Required (max : 1)	Loan start date. Date of generation of the first installment in PowerCARD
6	LoanEndDate	DateTime	-	Required (max : 1)	Loan end date. Date of generation of the last installment in PowerCARD
7	NbrOfTerms	Long	-	Required (max : 1)	Number terms due. The total number of installment for that loan
8	NbrOf TermsDue	N (4)	-	Required (max : 1)	Number terms due. Remaining number of installment still to be generated.
9	InterestRate	Long	-	Required (max : 1)	Interest rate applied for the loan
-	LoanDue	Complex	-	Required (max : 1)	Loan due. The total loan amount not yet requested from the customer to be paid
10	Amt	Long	LoanDue	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
11	CurCode	C (3)	LoanDue	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	LoanAmt	Complex	-	Required (max : 1)	Transaction amount. Total amount and currency of the initial transaction
12	Amt	Long	LoanAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
13	CurCode	C (3)	LoanAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
14	CustomerId	C (22)	-	Required (max : 1)	Person identifier Refers to the unique customer identification in the system.
15	Acronym	C (40)	-	Optional (max : 1)	The Merchant Acronym Acronym of the merchant in the original transaction. It could be the name

#	Field Name	Data type	Parent	Occurrence	Field Description
					imprinted on the transaction receipt by the merchant's or member's imprinter plate.
16	Loan Financing Type	C (1)	-	Required (max : 1)	I : Financed by credit limit E: Financed by additional loan limit
17	LoanCatalog Name	C (50)	-	Optional (max : 1)	Loan catalog product name.
-	BookingFee	Complex	-	Infinity Optional (max : 1)	Loan booking fee
8	TCCode	C (2)	BookingFee	Required (max : 1)	Booking fee Transaction code
8	TCDesc	C (32)	BookingFee	Required (max : 1)	Booking fee Transaction description
-	FeeAmount	Complex	BookingFee	Required (max : 1)	Loan Booking fee amount
18	Amt	Long	FeeAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
19	CurCode	C (3)	FeeAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Installment Plan	Complex	-	Optional (max : 1)	The installment Plan aggregate. For more details refers to the

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate description.
20	Installment Number	C (23)	Installment Plan	Required (max : 1)	Sequence number for the installment plan. A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
-	FeesAmount	Complex	Installment Plan	Required (max : 1)	Fees amount aggregate.
21	Amt	Long	FeesAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
22	CurCode	C (3)	FeesAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Interest Amount	Complex	Installment Plan	Required (max : 1)	Interest amount aggregate.
23	Amt	Long	Interest Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
24	CurCode	C (3)	Interest Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	TaxAmount	Complex	Installment Plan	Required (max : 1)	Tax amount aggregate.
25	Amt	Long	TaxAmount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	C (3)	TaxAmount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Principal Amount	Complex	Installment Plan	Required (max : 1)	Principal amount aggregate.
27	Amt	Long	Principal Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
28	CurCode	C (3)	Principal Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Installment Amount	Complex	Installment Plan	Required (max : 1)	Installment Amount aggregate.
29	Amt	Long	Installment Amount	Required (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
30	CurCode	C (3)	Installment Amount	Required (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
31	Status	C (1)	Installment Plan	Required (max : 1)	B: Billed U: Unbilled
32	BillingDate	Date	Installment Plan	Required (max : 1)	Installment billing date.

Specific error codes

#	Error code	Error Description
1	41601	Invalid Card Number/Token Id
2	41602	Invalid Account Id

#	Error code	Error Description
3	41603	Invalid Transaction Id
4	41604	Invalid Reference Sequence
5	41605	Invalid Catalog Loan Product
6	41606	Invalid Catalog Loan Product Version

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/simulateLoanWebService/v2	POST	

REST sample call

```
"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  }
],
"requestInfo": {
```

```

    "requestUID" : "",
    "requestDate" : "",
    "userID" : ""
  },
  "cardNumber" : "",
  "accountId" : "",
  "transId" : "",
  "refSeq" : "",
  "catalogLoanProduct" : "",
  "catalogLoanProduct Version" : "",
  "acronym" : ""
}

```

4.81 SimulateLoanPayOff API

API overview

This web service can be invoked before performing a Loan pay off operation for a particular account. Its purpose is to provide payment amount simulation + penalty fees. The agent and the customer can both review the pay off simulation before performing the actual PayOff

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

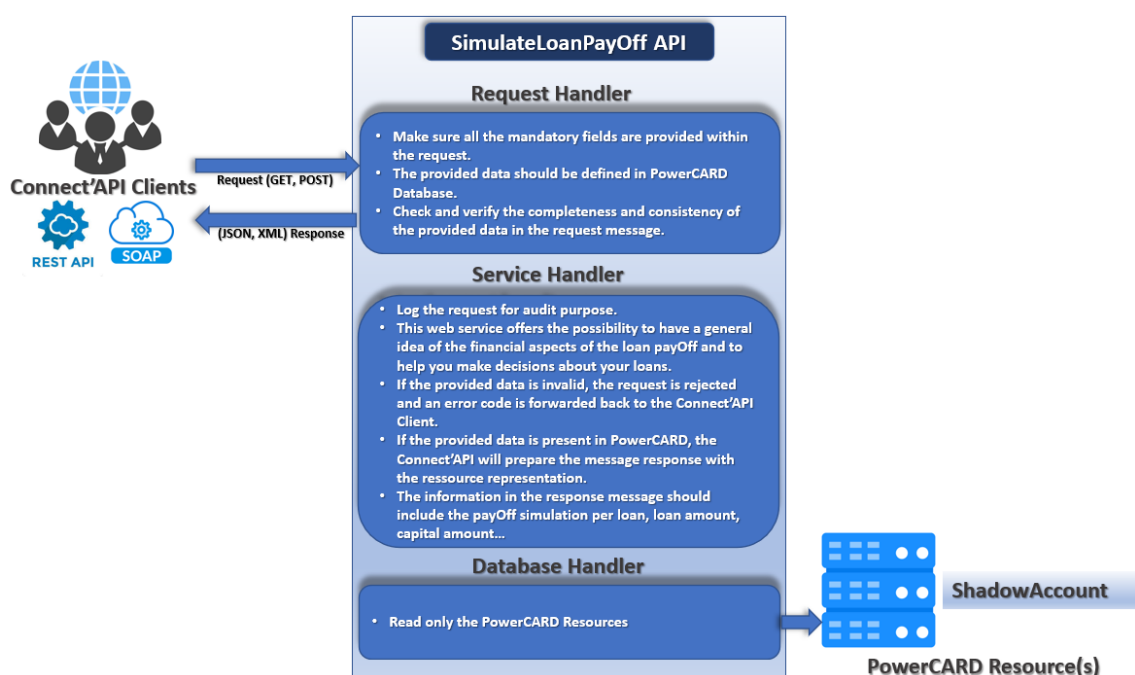


Figure 81: SimulateLoanPayOff process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXXX XXX-XXXX-XXXX-XXXX -XXXX XXXXXXXX Example: 6948DF80 -14BD-4E04 -8842-7668D9 C001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss.SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
4	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
5	Unpaid Installment	C (1)	-	Required (max : 1)	Include unpaid installment in the payoff amount. The possible values are: - Y: Yes - N: No
-	LoanList	Complex	-	Required (Infinite list)	The Loan list aggregate. For more details refers to the

#	Field Name	Data type	Parent	Occurrence	Field Description
					aggregate description.
6	TransId	C (23)	LoanList	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
7	RefSeq	Decimal (4,0)	LoanList	Required (max : 1)	Refers to the transaction sequence Number.

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	ErrorDescription	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered
-	Loan Simulated Payoff	Complex	-	Optional Infinite list	List of payoff simulation per loan
9	TransId	C (23)	Loan Simulated Payoff	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
10	RefSeq	N (4)	Loan Simulated Payoff	Required (max : 1)	Refers to the transaction sequence Number.
-	LoanAmnt	Complex	Loan Simulated Payoff	Required (max : 1)	Loan transaction amount
11	Amt	Long ()	LoanAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
12	CurCode	NC (3)	LoanAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	CapitalAmnt	Complex	Loan Simulated Payoff	Required (max : 1)	Capital balance to be paid
13	Amt	Long ()	CapitalAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
14	CurCode	NC (3)	CapitalAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	InterestAmt	Complex	Loan Simulated Payoff	Required (max : 1)	Interest balance to be paid
15	Amt	Long ()	InterestAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
16	CurCode	NC (3)	InterestAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	FeeAmt	Complex	Loan Simulated Payoff	Required (max : 1)	Fee balance to be paid
17	Amt	Long ()	FeeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
18	CurCode	NC (3)	FeeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	UnpaidInstAmt	Complex	Loan Simulated Payoff	Required (max : 1)	Unpaid installment to be paid
19	Amt	Long ()	UnpaidInstAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$

#	Field Name	Data type	Parent	Occurrence	Field Description
					99,999.99 should be sent as 99999999.
20	CurCode	NC (3)	UnpaidInstAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	PenaltyAmt	Complex	Loan Simulated Payoff	Required (max : 1)	Loan Payoff penalty
21	Amt	Long ()	PenaltyAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
22	CurCode	NC (3)	PenaltyAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.
-	Booking FeeAmt	Complex	Loan Simulated Payoff	Required (max : 1)	Pending Loan booking fee to be paid
23	Amt	Long ()	Booking FeeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
24	CurCode	NC (3)	Booking FeeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Loan PayoffAmnt	Complex	Loan Simulated Payoff	Required (max : 1)	Loan amount to be paid
25	Amt	Long ()	Loan PayoffAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
26	CurCode	NC (3)	Loan PayoffAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	Account PayoffAmnt	Complex	-	Required (max : 1)	Total account amount to be paid
27	Amt	Long ()	Account PayoffAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals.

#	Field Name	Data type	Parent	Occurrence	Field Description
					Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
28	CurCode	C (3)	Account PayoffAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY_TABLE table under the general parameters section.

Specific error codes

#	Error code	Error Description
1	41701	Invalid Account Id
2	41702	Invalid Unpaid Installment
3	41703	Invalid Transaction Id
4	41704	Invalid Reference Sequence

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCard User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested		x

Status Code	Description	Success Response	Error Response
	is not supported for the PowerCARD Resource.		
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SimulateLoanPayOffWebService/v2	POST	

REST sample call

```

"keyValues": [
  {
    "key": "TOKEN",
    "data": "Y",
    "type": "",
  },
],
"requestInfo": {
  "requestUID": "",
  "requestDate": "",
  "userID": ""
},
"accountId": "",
"unpaidInstallment": "",
"transId": "",
"refSeq": "",
"loanList": {
  "transId": "",
  "refSeq": ""
}
}

```

4.82 SimulateLoanPartialPayOff API

API overview

This web service can be invoked before performing a partial Loan pay off operation for a particular account. Its purpose is to provide payment amount simulation + penalty fees. The agent and the customer can both review the pay off simulation before performing the actual partial PayOff. The simulation payment amount will be in function of the number of installments to be paid off.

The following sections are intended to highlight the process as well as the technical specification of the request and response messages.

Functional description

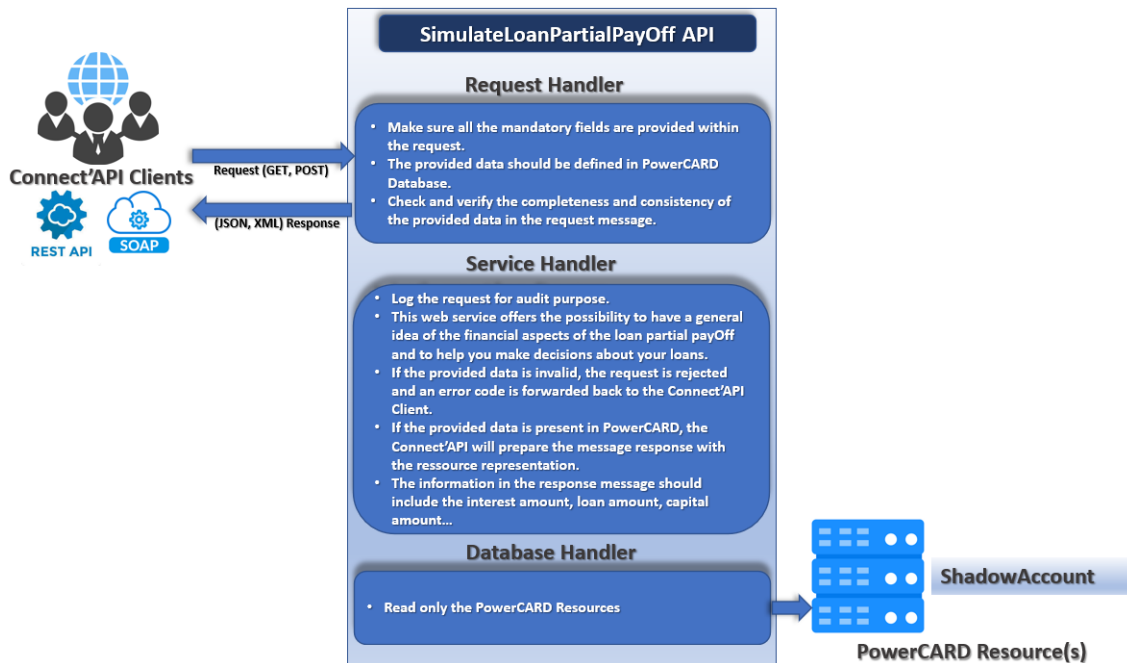


Figure 82: SimulateLoanPartialPayOff process

Technical specification

Version 2

URI version Alias	Initially released in	Deprecated ?
V2	R5.1	No

Request message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	RequestInfo	Complex	-	Required (max: 1)	This aggregate contains all the information necessary to service the request
1	RequestUID	C (36)	RequestInfo	Required (max: 1)	Request Identifier. It is sent by the calling system as a universally unique identifier for the message. Used to match response with request messages. The generation mask should be the following: XXXX XXXX-XXXX -XXXX -XXXX- XXXXX XXXXXXXX Example:

#	Field Name	Data type	Parent	Occurrence	Field Description
					6948DF80 -14BD -4E04-8842 -7668D9C 001F5
2	RequestDate	DateTime	RequestInfo	Required (max: 1)	Date when the request was submitted. . Format: yyyy-MM-dd'T'HH:mm:ss. SSS'Z' The value will be in GMT.
3	UserID	C (15)	RequestInfo	Optional (max: 1)	Represents the user Id who made the request.
8	AccountId	C (24)	-	Required (max : 1)	The CMS account id (shadow account number in PWC). This Id must be previously defined in the PowerCARD SHADOW _ACCOUNT table under the credit revolving section.
9	UnpaidInst	C(1)		Required (max : 1)	Include unpaid installment in payoff amount, value Y/N
10	TransId	C (23)	-	Required (max : 1)	The transaction ID A 23-digit identification number assigned by the system to each transaction. The TransId is unique by transaction.
11	RefSeq	N (4)	-	Required (max : 1)	Refers to the transaction sequence Number.
12	NbTerms	N(4)		Required (max : 1)	Number of term to be paid off

Response message

#	Field Name	Data type	Parent	Occurrence	Field Description
-	ResponseInfo	Complex	-	Required (max: 1)	Through this aggregate PowerCARD Connect'API send a final response status

#	Field Name	Data type	Parent	Occurrence	Field Description
					after the request has been completed.
1	ResponseUID	C (36)	ResponseInfo	Required (max: 1)	Response Identifier. Echoed back by PowerCARD to the requester. It should contain the same value as the one sent in the request message.
2	ResultID	ClosedEnum	ResponseInfo	Required (max : 1)	Refers to the result of processing in PowerCARD. The possible values are: • ProceedWith Success: Call was ended successfully. • ProceedWith SuccessMC: Call made successfully, however the request will be inserted in a Maker Checker queue for further analysis. • Error: An error has occurred during the processing. SystemError: An unknown system error occurred during the process.
3	ErrorCode	C (5)	ResponseInfo	Required (max : 1)	Indicate the error code used by the PowerCARD to tell you that program was experiencing a particular problem during the processing of the request.
4	Error Description	C (250)	ResponseInfo	Optional (max : 1)	If present, this field must contain the description of the error encountered

#	Field Name	Data type	Parent	Occurrence	Field Description
-	LoanAmnt	Complex	-	Required (max : 1)	Initial loan amount
5	Amt	Long	LoanAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
6	CurCode	NC (3)	LoanAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	CapitalAmt	Complex		Required (max : 1)	Capital balance to be paid
7	Amt	Long ()	CapitalAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
8	CurCode	NC (3)	CapitalAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	InterestAmt	Complex		Required (max : 1)	Interest balance to be paid
9	Amt	Long ()	InterestAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
10	CurCode	NC (3)	InterestAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	FeeAmt	Complex		Required (max : 1)	Fee amount to be paid
11	Amt	Long ()	FeeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10 ⁿ ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 99999999.
12	CurCode	NC (3)	FeeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	UnpaidAmt	Complex		Required (max : 1)	Unpaid installment balance to be paid
13	Amt	Long ()	UnpaidAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
14	CurCode	NC (3)	UnpaidAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
-	PenaltyAmt	Complex		Required (max : 1)	Partial payoff Penalty amount
15	Amt	Long ()	PenaltyAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
16	CurCode	NC (3)	PenaltyAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.

#	Field Name	Data type	Parent	Occurrence	Field Description
-	Booking FeeAmt	Complex	-	Required (max : 1)	Loan booking fee to be paid
17	Amt	Long ()	Booking FeeAmt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
18	CurCode	NC (3)	Booking FeeAmt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY _TABLE table under the general parameters section.
19	Interest Covered	N(4)		Required (max : 1)	Number of installment interest due
-	PaymentAmnt	Complex		Required (max : 1)	Total amount to be paid
20	Amt	Long	PaymentAmnt	Optional (max : 1)	Amount. The nominal amount value is multiplied by 10ⁿ; where n is the currency exponent of the currency code sent in the field below; the last n digits are used for decimals. Example: An amount displayed as US\$ 99,999.99 should be sent as 9999999.
21	CurCode	C (3)	PaymentAmnt	Optional (max : 1)	Currency code. ISO currency code of the transaction amount as defined in the PowerCARD CURRENCY

#	Field Name	Data type	Parent	Occurrence	Field Description
					_TABLE table under the general parameters section.

Specific error codes

#	Error code	Error Description
1	41801	Invalid Account Id
2	41802	Invalid Unpaid Installment
3	41803	Invalid Transaction Id
4	41804	Invalid Reference Sequence

RESTful response codes

Status Code	Description	Success Response	Error Response
200 OK	Successful	x	
409 Conflict	The request could not be completed due to a conflict with the current state of the resource		x
401 Unauthorized	The PowerCARD User passed in the token is invalid. Client should refresh the token and then try again.		x
404 Not Found	PowerCARD Resource not found.		x
405 Method Not Allowed	The PowerCARD resource doesn't support the specified HTTP verb.		x
500 Internal Server Error	Servers are not working as expected. The request is probably valid but needs to be requested again later.		x
503 Service Unavailable	Service Unavailable.		x
415 Unsupported Media Type	Returned when the the format you requested is not supported for the PowerCARD Resource.		x
504 Gateway Timeout	A service required to fulfill the request was not available.		x

Consuming a RESTful API

PowerCARD Resource URI	Method	URI Params
/SimulateLoanPartialPayOffWebService/v2	POST	

REST sample call

```
"keyValues": [  
  {  
    "key": "TOKEN",  
    "data": "Y",  
    "type": "",  
  },  
],  
"requestInfo": {  
  "requestUID": "",  
  "requestDate": "",  
  "userID": ""  
},  
"accountId": "",  
"unpaidInstallment": "",  
"transId": "",  
"refSeq": "",  
"loanList": {  
  "transId": "",  
  "refSeq": ""  
}  
}
```

5 Appendix A – Generic error codes

This appendix provides the complete sets of error codes that are used by the system to describe the web services common errors that could be shared in the response messages:

The table below lists rejects codes in numerical order and provides:

- The error code: a numeric code identifying the rejection
- Area: Possible values are the following:
 - Success: No error encountered
 - System Error: In case of database or any other system error that might occur during the processing
 - General Error: common for most of the web services
 - Maker/Checker: when the request is not completely accepted, and should then be subject to maker/checker mechanism.
 - XSD: Any error related to the XSD specification, for instance Invalid message format.
 - Invalid range date
- Business rules and error message: both are dedicated to share the description of the reject code. Used typically to give more details regarding the error encountered.

Error Code	Area.	Business rule	Error message
00000	Success	Request processed successfully, no error encountered.	
00001	System Error	System error, provider must return this code for each system error (ie: database connection error...)	An error description is shared by the system based on the situation
00002	Generic Errors	Invalid RqUID	Invalid RqUID
00003	Generic Errors	Authentication failed	Authentication failed
00004	Generic Errors	Unknown product	Unknown product. The product name is provided in the error message for more details.
00005	Generic Errors	Unknown language	Unknown language
00006	Generic Errors	Invalid customer information	Invalid customer information
00007	Generic Errors	Invalid pagination step	Invalid pagination step
00008	Generic Errors	Access to the bank id data is denied	Access to the bank id data is denied
00010	Generic Errors	Invalid Provider Data	Invalid 'provider name' Data
00011	XSD	Invalid message	Invalid message
00012	Generic Errors	User is blocked	User is blocked
00013	Generic Errors	Account is deactivated	Account is deactivated
00014	Generic Errors	Operation already performed	Operation already performed
00020	Generic Errors	Password Expired	Password Expired
00030	Maker / Checker	Waiting for validation	
00040	Generic Errors	Token management	Error when retrieving interface information

Error Code	Area.	Business rule	Error message
00041	Generic Errors	Token management	Error when retrieving card information
00042	Generic Errors	Token management	Error when retrieving token information
00043	Generic Errors	Token management	Error while generating the token
00044	Generic Errors	Token managment	Error while inserting token information
00099	Invalid range date	Invalid range date	'An error description'
99999	System Error	Database exception error	

6 Appendix B – Field Types

This appendix contains the list of types used by the system; three possible types are available; base, simple and complex:

A complex type or element is an XML element that contains other elements and/or attributes (simple or complex).

However, a simple element is an XML element that contains only text. It cannot contain any other elements or attributes. The text can be of many different base types. It can be one of the types included in the XML Schema definition (Boolean, string, date, etc.), or it can be a custom type that can be defined by the software developers.

6.1 Base types

#	Name	Type	Max len. of value	Restrictions
1	Boolean	boolean		
2	C	string		
3	Decimal	decimal		Decimal indicates a numeric value
4	FileBinary_Type	base64Binary		
5	Long	long		Long is an integer value, which may be positive, negative, or zero, with values ranging from -2147483648 to 2147483647
6	NC	string		Elements of type Narrow Character are elements of character data type with the additional restriction that the only allowable characters are those contained within the ISO Latin-1-character set.
7	Date	date		Formats accepted for a simple date : yyyy-MM-dd or yyyyMMdd
8	DateTime	dateTime		Formats accepted for a date time : yyyy-MM-ddT'HH:mm:ss or yyyy-MM-dd HH:mm:ss
9	DateTimeZone	dateTimeZone		Format accepted for a date time zone : yyyy-MM-ddTHH:mm:ss.SSSZ

7 Document history

Document version	Status	Date	Modification
3.5.01	Draft	16/07/2019	Initial version.
3.5.02	Draft	30/08/2019	Update the objects that constitute the various products in the Issuing Product catalogues Remove the Card Application APIs not compatible with Release 4.1 : • <code>getApplicationDetail</code> • <code>getApplicationDocuments</code> • <code>searchApplication</code> • <code>sendApplication</code> • <code>updateApplicationStatus</code>
3.5.03	Draft	24/09/2019	Adding new web service compatible with Release 4.1 : • <code>updateCardLimits</code>
3.5.04	Draft	27/01/2020	Adaptation of API RESTful resource model with existing Backend Business Web Services. API documentation for sprint #1 – Connect'API Release 5.1 Add a section documenting the security policy used
3.5.05	Draft	01/02/2020	Adding a section documenting the validation resource data Adding a section documenting the protection resource data with Tokenization Adding REST samples call
3.5.06	Draft	10/02/2020	Adding new APIs : • <code>requestPIN</code> • <code>getCardCVC2</code> • <code>getCustomerDetails</code> • <code>feeInquiry</code>
3.5.07	Draft	21/02/2020	Adding new APIs : <code>searchTransactions</code> <code>updateAddress</code> <code>createAccountTransaction</code> <code>updateBankAccount</code> <code>createLoyaltyAccount</code> <code>getLoyaltyAccounts</code>

Document version	Status	Date	Modification
			getAccountBalance updateCustomerEmail UpdateCustomerPhoneNumber getLoyaltyAccountTransactions
3.5.08	Draft	22/02/2020	Adding new APIs : updateLoyaltyAccountStatus getPinHistory updateCardLimits getCardCVC2 getCardLimits getCardDomain setCardDomain changePin verifyPin activateContactless
3.5.09	Draft	25/02/2020	Integration of remarks and comments from the Issuing and online teams (by Radouane Lakhssassi & Ismail Chakour)
3.5.10 3.5.11 3.5.12 3.5.13 3.5.14 3.5.15 3.5.16 3.5.17 3.5.18 3.5.19	Draft	13/04/2020 To 15/06/2020	Adding new APIs : OrderStockCards GetStockCardsOverview GetStatementHistory UpdateCustomerName UpdateCustomerCategory LostStolenDeclaration UpdateCustomerDemogData
3.5.20	Draft	21/06/2020	Review list of APIs: OrderStockCards GetStockCardsOverview GetStatementHistory UpdateCustomerName UpdateCustomerCategory LostStolenDeclaration UpdateCustomerDemogData Adding new APIs : checkCorporateContactInformation getCardCharacteristics
3.5.21	Draft	23/06/2020	Adding new APIs :

Document version	Status	Date	Modification
			checkCorporateContactInformation getCardCharacteristics
3.5.22	Draft	23/06/2020	Adding new APIs : getCardNumberFromToken getInstallmentPlan searchLoan changeaccountbillingcycle checkcustomfieldvalue updateaccstatementcode updatebankaccountstatus getloandetails personalizecard transferaccounttoaccount searchfinancialaccount addsupplementarycard
3.5.23	Draft	24/06/2020	Adding new APIs : changeaccountcreditlimit converttransactiontoloan createcreditcard createloanapplication getcorporatehierarchy getcorporateselectedproducts getproductdetails updatecardstatusreason updatealertservice updateaccountbalance updateinterestrateindex updatecardcharacteristics
3.5.24	Draft	26/06/2020	Adding new APIs : getCollaboratorInitiatedSubscription getEventNotificationList SearchCollaborators ImportCollaborators updateEventNotification UpdateCollaboratorSubscriptionStatus
3.5.25	Draft	01/07/2020	APIs specification updates : updatecardcharacteristics (Overview section update, ProductServices aggregate section update) getProductDetails (Overview section update,

Document version	Status	Date	Modification
			response message update) importCollaborators (request message update) checkCorporateContactInformation (request message update)
3.5.26	Draft	02/07/2020	APIs specification updates : getProductDetails (request message update, response message update)
3.5.27	Draft	09/07/2020	Redesigned API : checkCorporateContactInformation APIs specification updates : getProductDetails (correct occurrence for aggregates AllowedMccList, ExcludedMccList, Allowed ExcludedCountryList) Adding missing schemas Correct chapters numbering problems Adding new APIs : initiateCollaboratorSubscription
3.5.28	Draft	15/07/2020	APIs specification updates : initiateCollaboratorSubscription (response message update) updateBankAccount (Specific Error Codes update) Adding new APIs : LoanReschedule LoanPartialPayOff LoanEarlyPayOff LoanConsolidate LoanCancellation SimulateLoan SimulateLoanPayOff SimulateLoanPartialPayOff
3.5.29	Draft	27/07/2020	APIs specification updates : GetProductDetails (request/response message update, Specific Error Codes update)

Document version	Status	Date	Modification
			InitiateCollaboratorSubscription(request/response message update, Specific Error Codes update). ImportCollaborator (Add EmployeeStatus to EmployeeInformation aggregate, rename Language to LanguageCode, Specific Error Codes update) SearchCollaborators (Add EmployeeStatus to EmployeeInformation aggregate, rename Language to LanguageCode, Specific Error Codes update) SearchFinancialAccounts (response message update, Specific Error Codes update) GetCorporateSelectedProducts (Specific Error Codes update) SearchLoan (Specific Error Codes update)
3.5.30	Draft	29/07/2020	Review all the fields with the Date or DateTime type Adding ISO Date Time Zone format